MightyConnect™ Security Platform

- Mesh P2P VPN
- Powered by WireGuard & FD.io VPP

MightyLink Lab.
(www.mightyconn.com)
Doc. Revision: 1.02

Table of Contents

1. Zero Trust P2P Security Platform
2. WireGuard
3. DPDK & FD.io VPP
4. MightyConnect™ Security Box
5. MightyConnect™ Devices
6. References



(1) "WireGuard" and the "WireGuard" logo are registered trademarks of Jason A. Donenfeld
(2) FD.io VPP and FD.io logo are registered trademarks of The Fast Data Project, a series of LF Projects, LLC

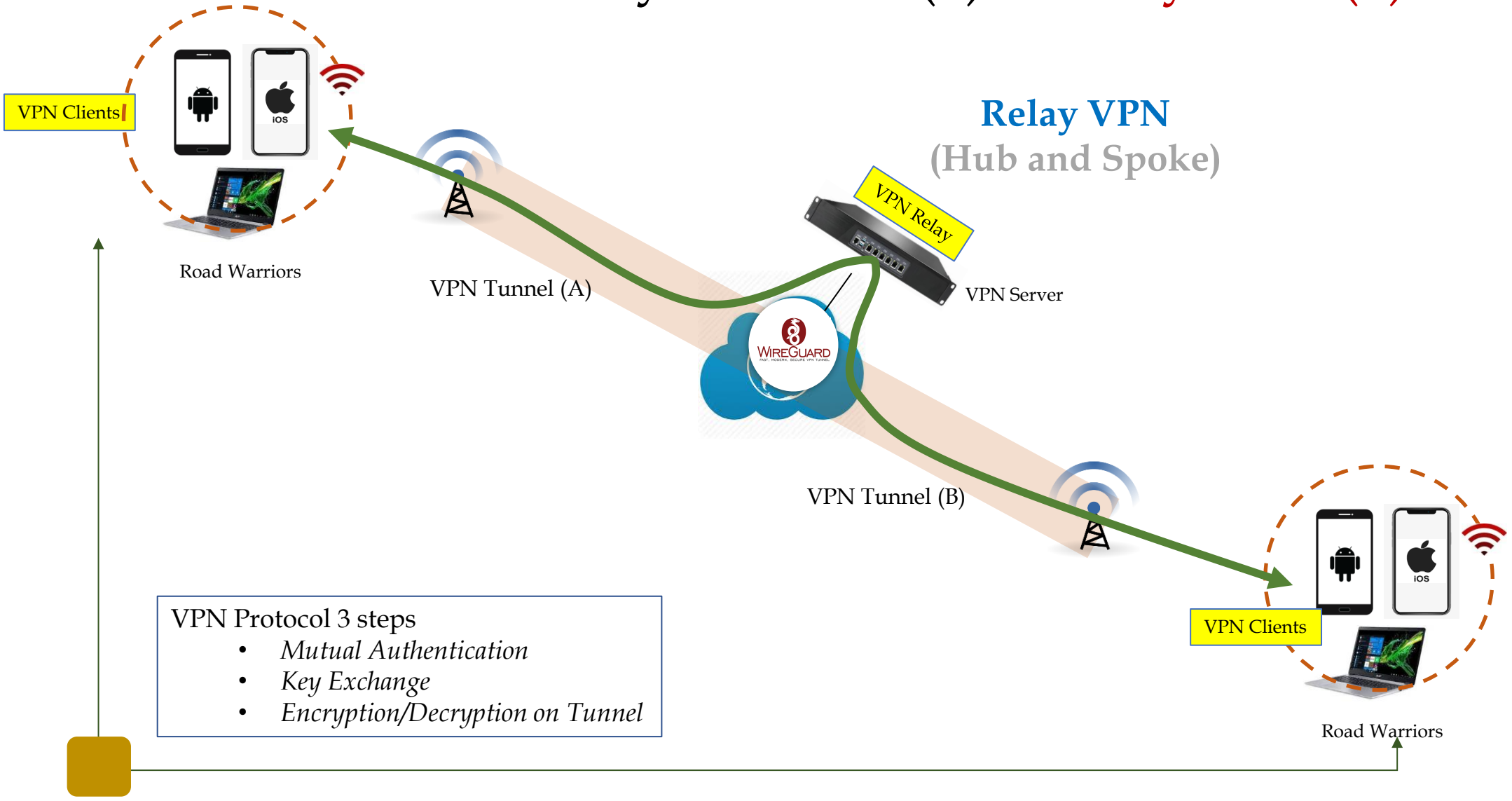
1. Zero Trust P2P Security Platform

Zero Trust P2P Security Platform

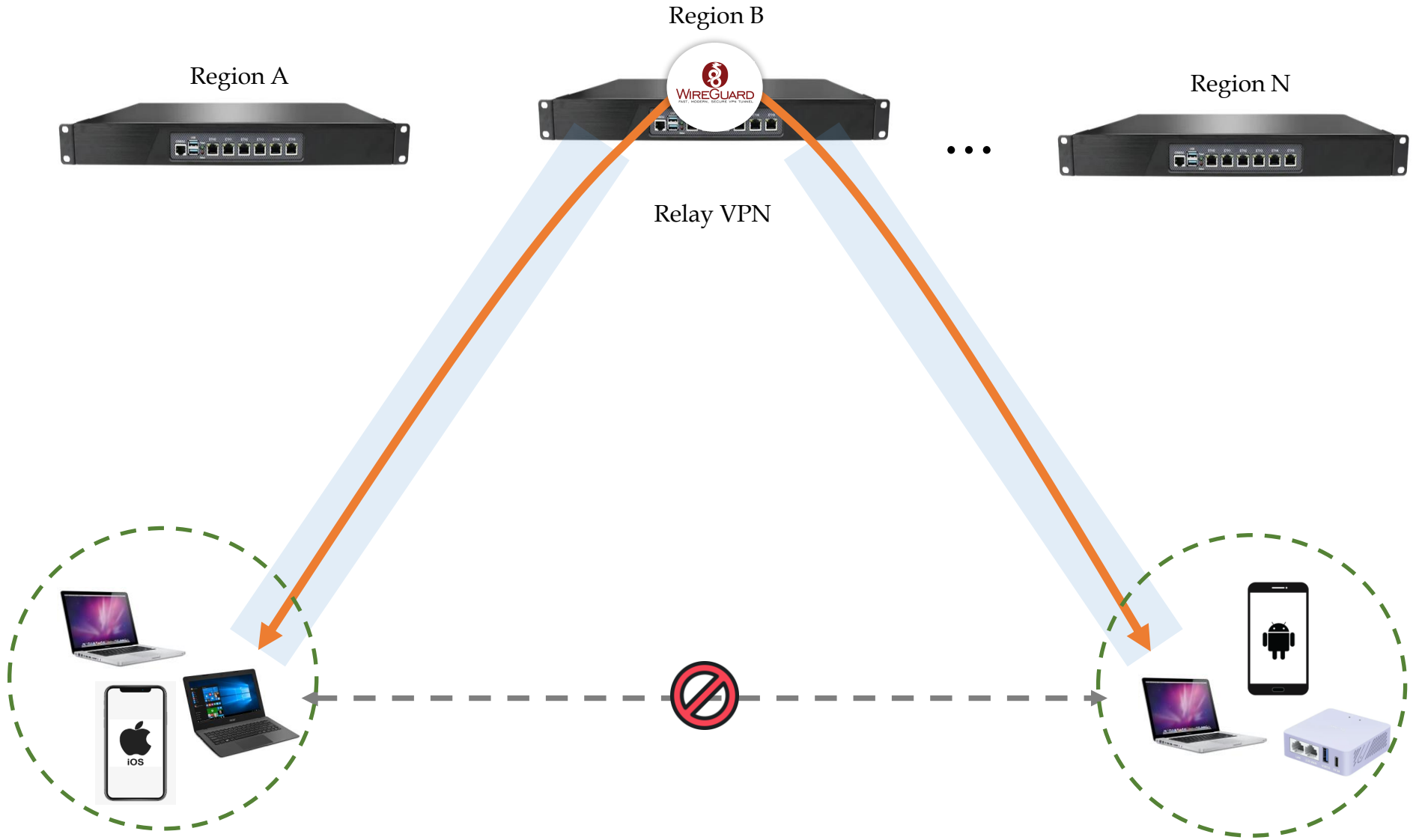
- ZTNA/Secure Network
- No modifications for any applications
- Auto Connect, Direct Connect and Relay Connect



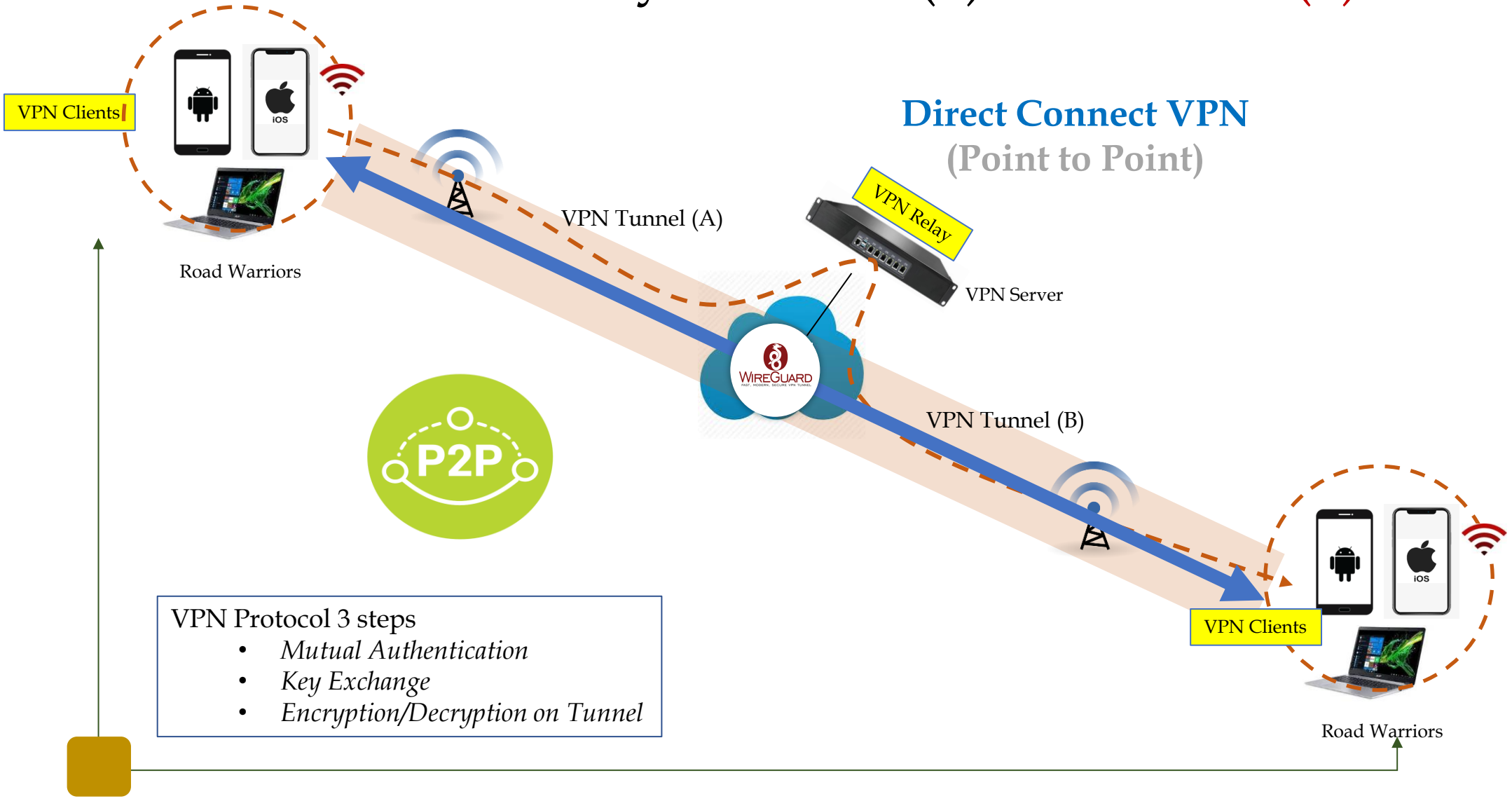
1. Zero Trust P2P Security Platform(1) – Relay VPN(1)



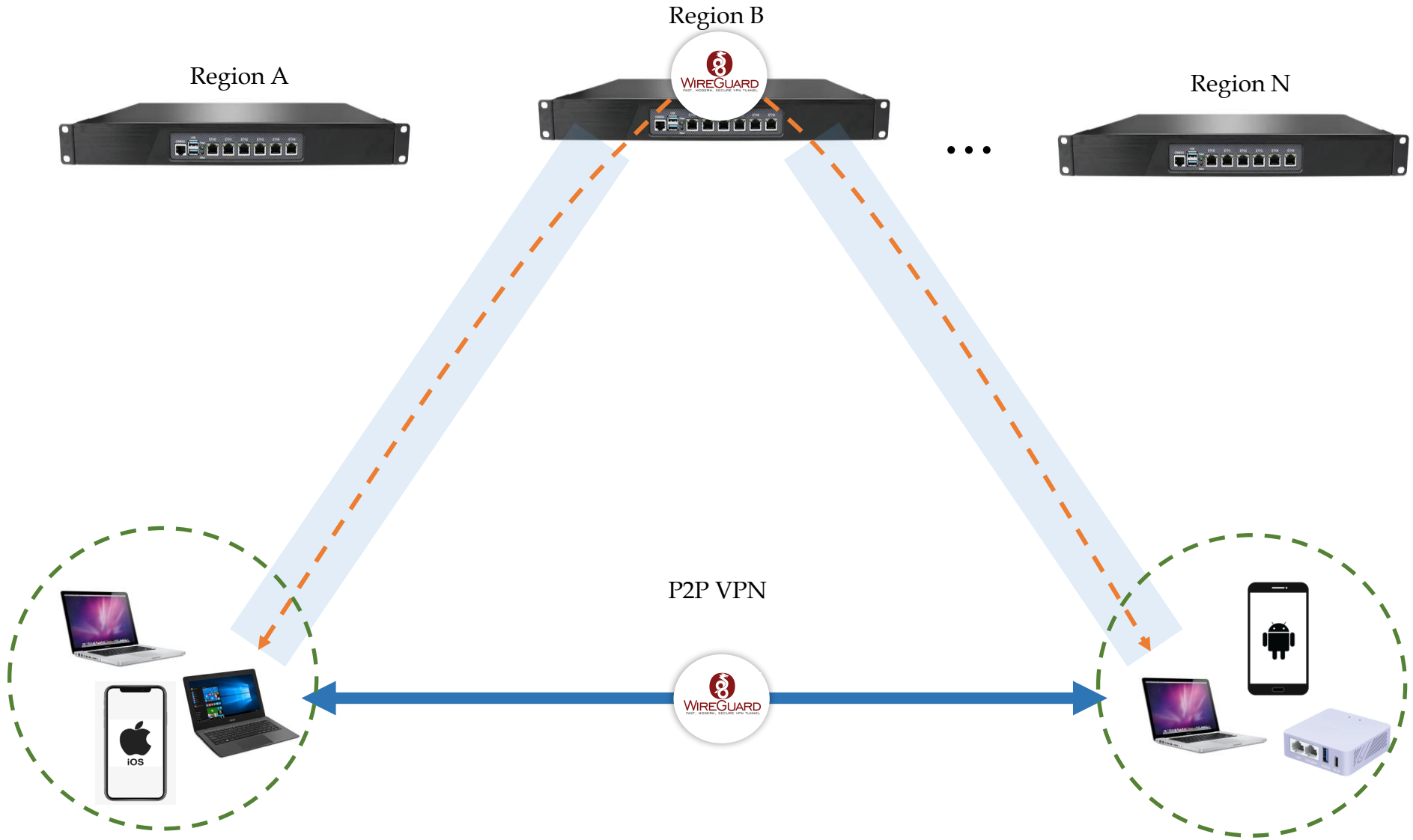
1. Zero Trust P2P Security Platform(1) – Relay VPN(2)



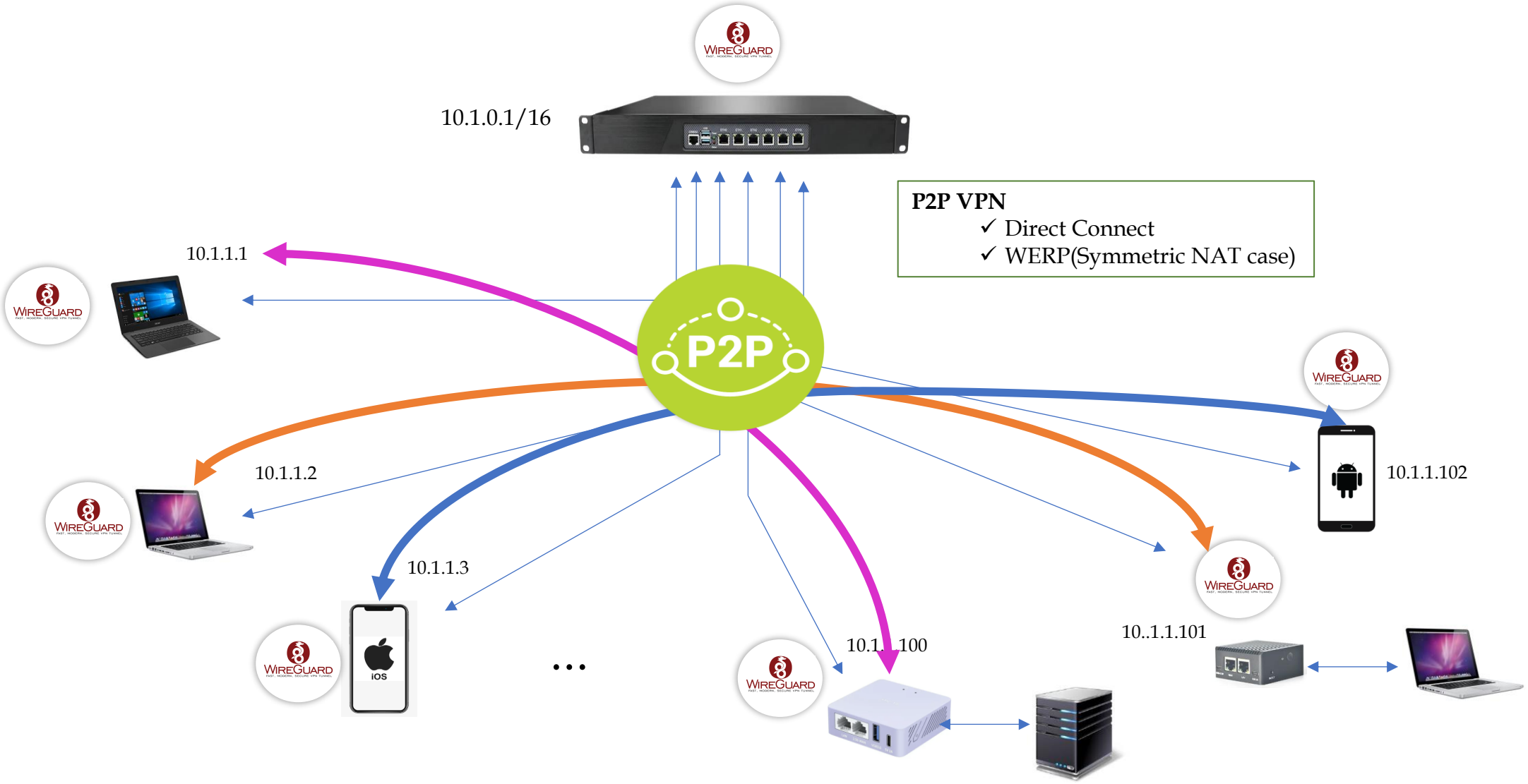
1. Zero Trust P2P Security Platform(2) – P2P VPN(1)



1. Zero Trust P2P Security Platform(2) – P2P VPN(2)



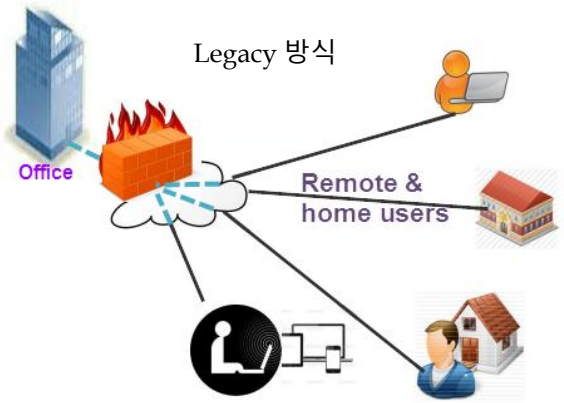
1. Zero Trust P2P Security Platform(2) – P2P VPN(3)



1. Zero Trust P2P Security Platform(3) – ZTNA(1-1)

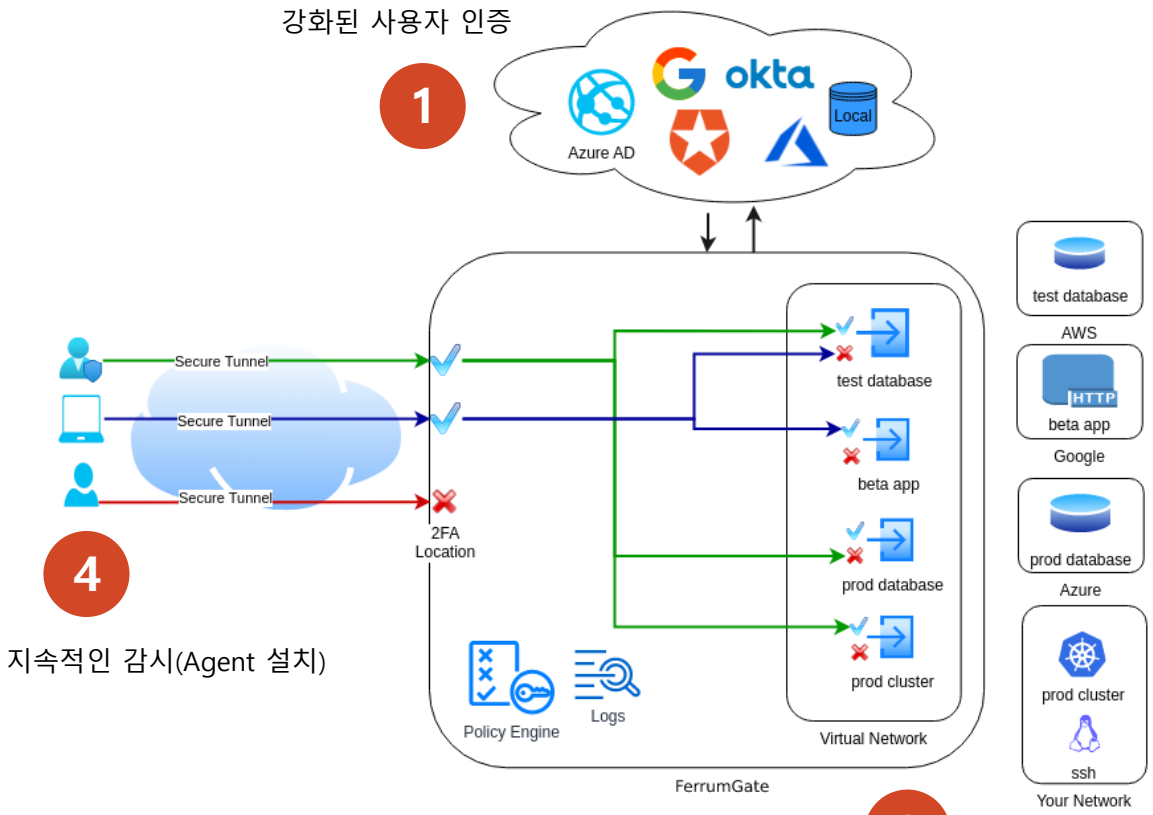
- **Legacy 방식(Firewall/VPN/IPS/UTM)**
  - Internal network(Trusted Area)과 External network(Untrusted Area) 구분
  - Internal network <=> Perimeter Security 제품 ⇔ External network : 안전한 내부망과 불안전한 외부망 경계(Perimeter)에 보안 제품 설치
  - Static Configuration, IP/Port 기반의 통제

VPN is Dead



- **ZTNA or SDP(Software Defined Perimeter) 방식**
  - 환경의 변화: **Cloud, Mobile, BYOD**
    - *이제는 중요한 것들이 Perimeter 바깥에 있는 세상이 되었다.*
  - **Zero Trust**: "Internal, External network 모두 안전하지 않다"에서 출발
  - S/W 방식의 **중앙 Controller**에 의한 접근 통제
    - *실제 network에 접근하기 전에, 인증(Authentication) & 접근 통제(Authorization)을 수행하는 개념*
    - *내부 Resource는 Appliance(Gateway)에 의해 모두 Hidden 상태, 인증을 통과한 후에 외부에 Open되는 구조.* **Dark Network(외부에 보이지 않는 network) 개념 출현**
  - Dynamic Control, IP 기반이 아니라, 사용자 기반의 통제

1. Zero Trust P2P Security Platform(3) – ZTNA(1-2)

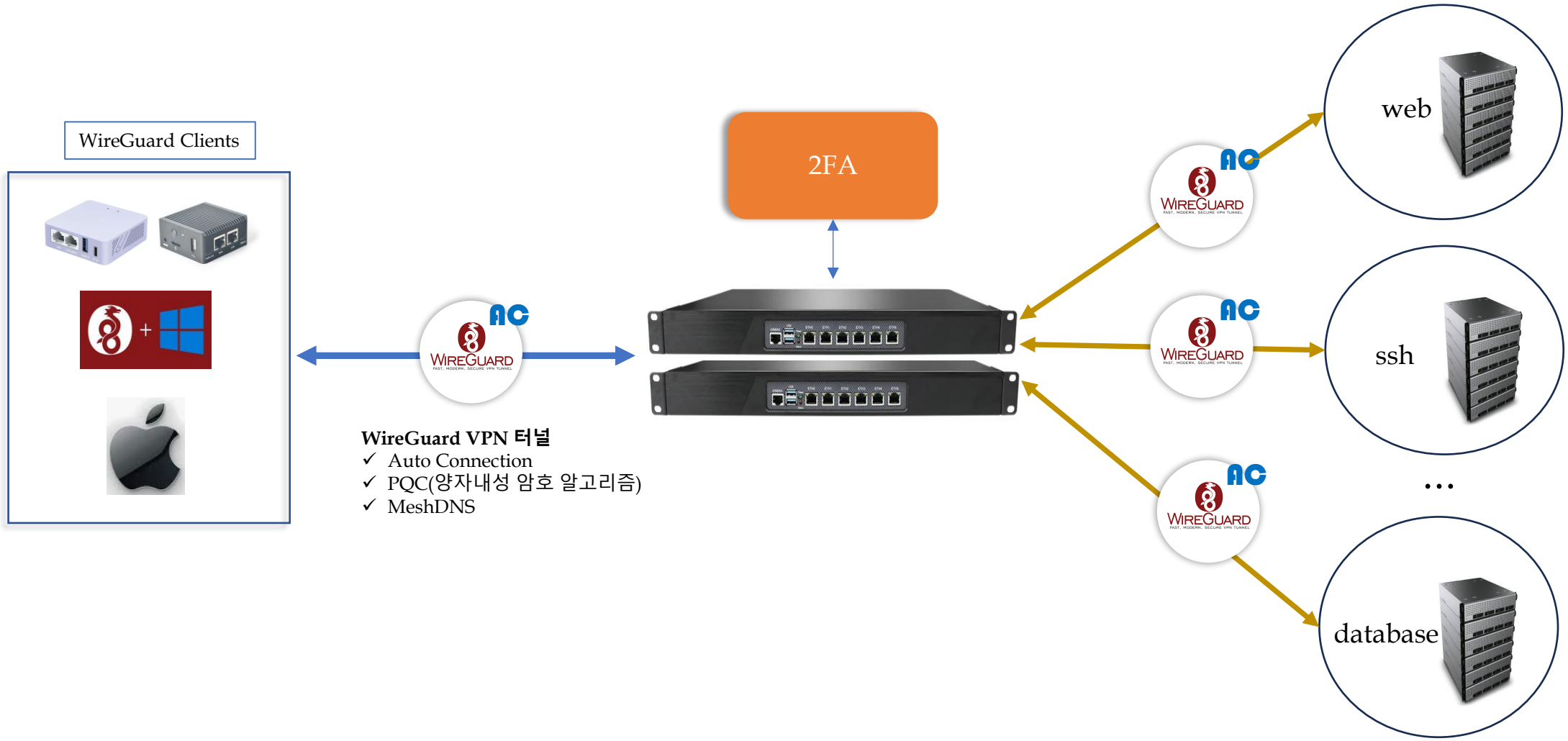


ZTNA의 요건

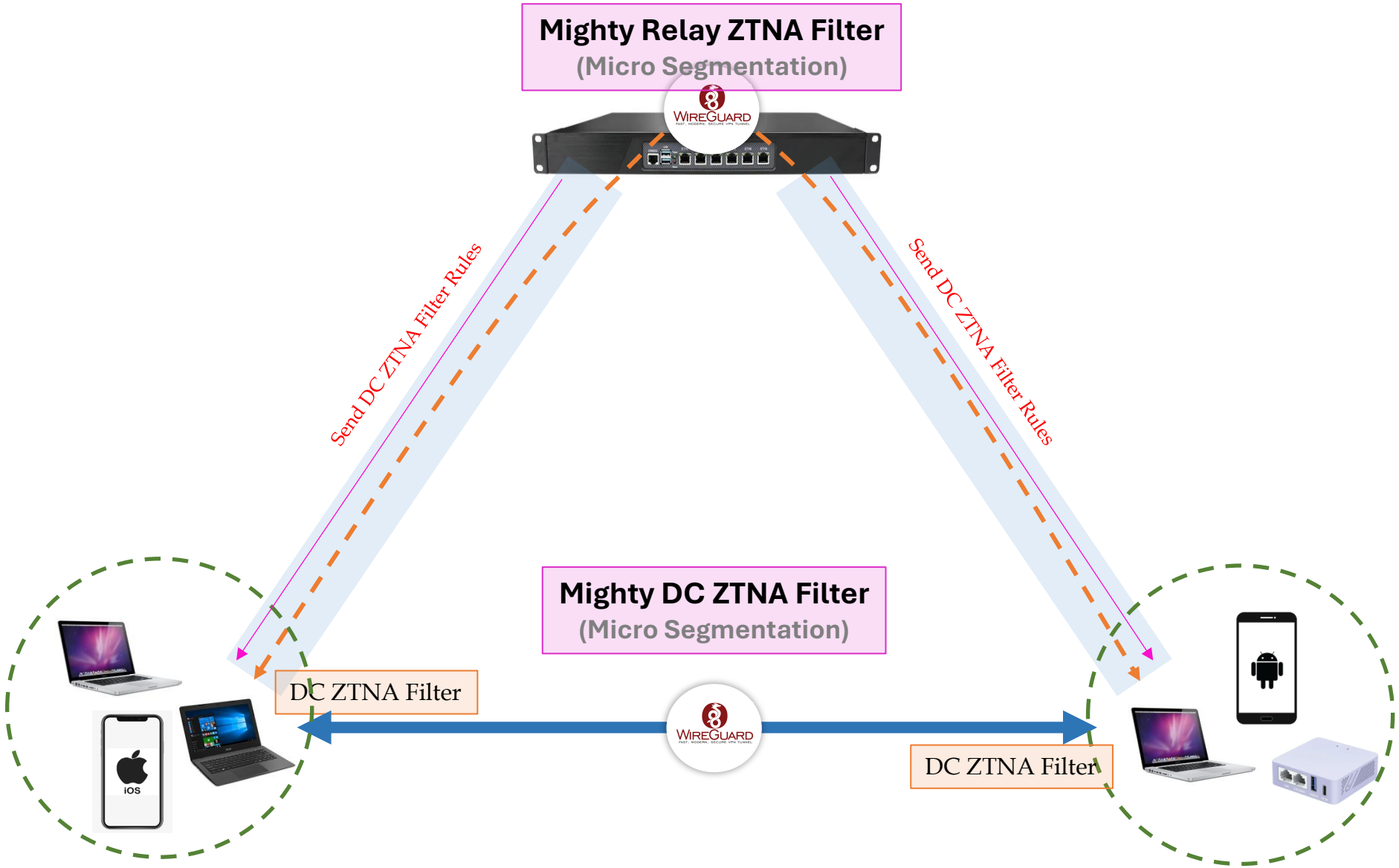
- More Authentication Options
  - MFA: SAML, OAUTH2, LDAP, FIDO2
  - Location Check, Time Check, Device Posture Check
- Micro-segmentation
  - Virtual Application Networks
- Least privilege access
  - RBAC(Role Based Access Control)
- Continuous monitoring
  - Every packet, every flow, every connection

[출처] https://ferrumgate.com/

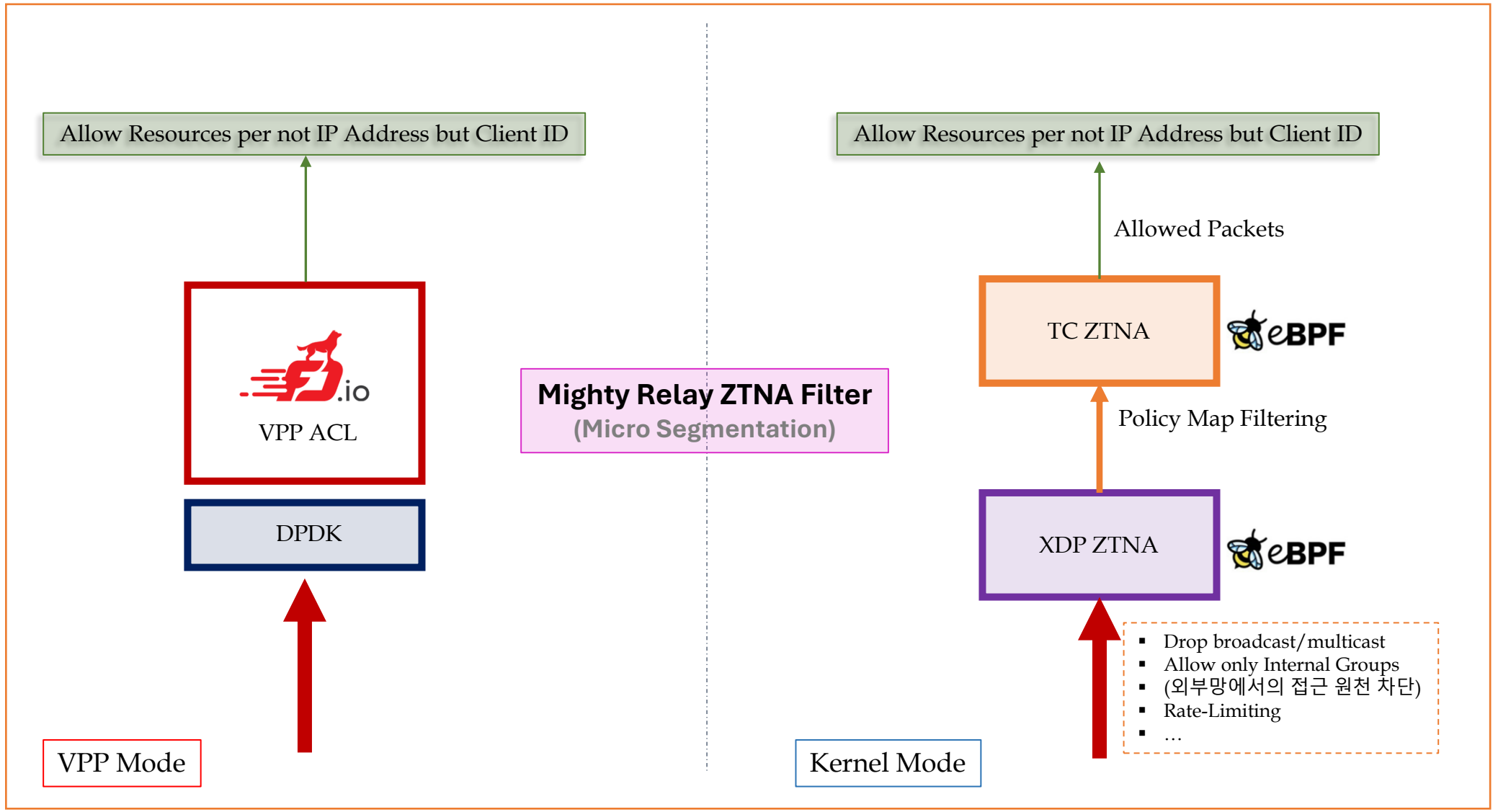
1. Zero Trust P2P Security Platform(3) – ZTNA(2)



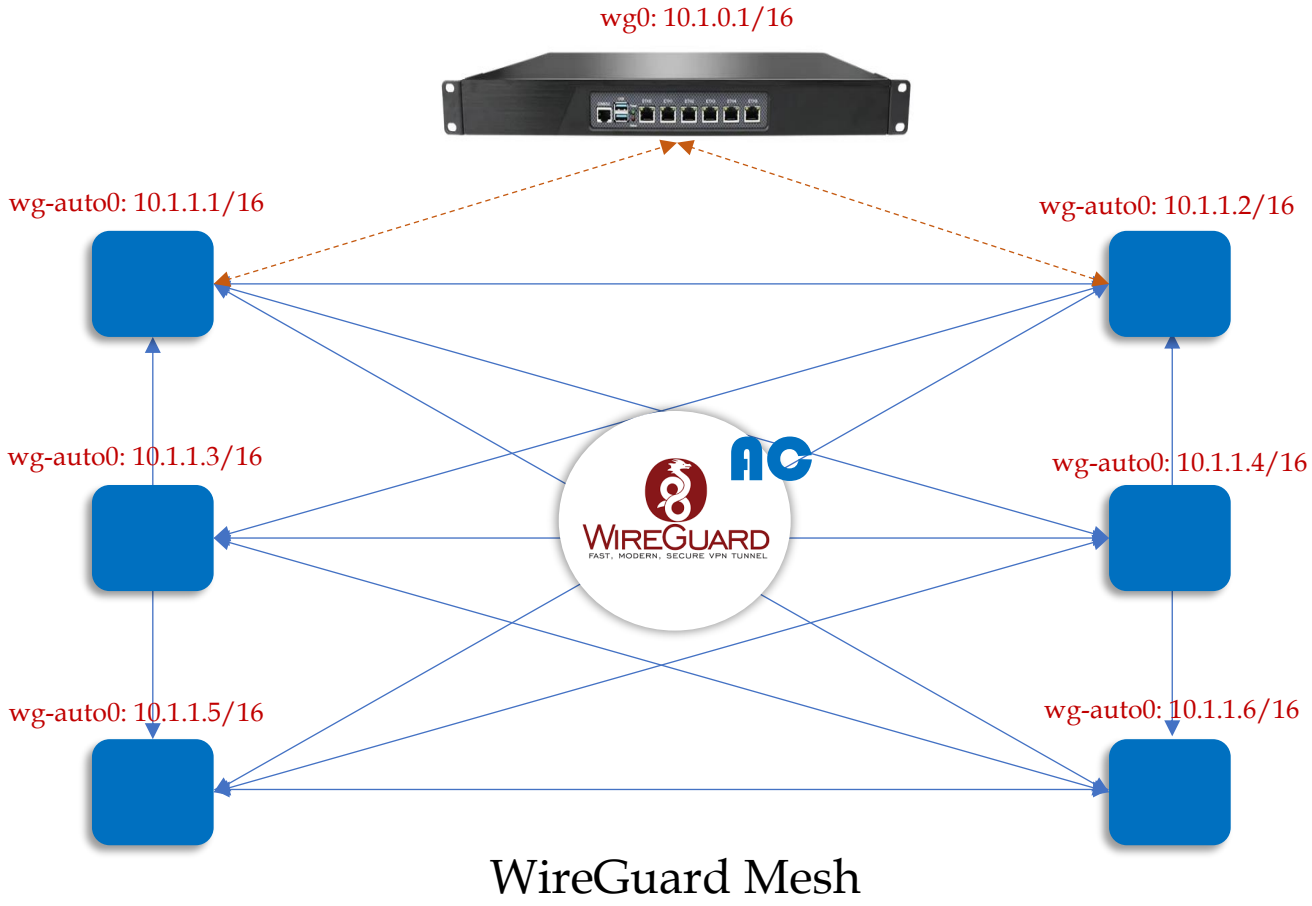
1. Zero Trust P2P Security Platform(3) – ZTNA(3-1)



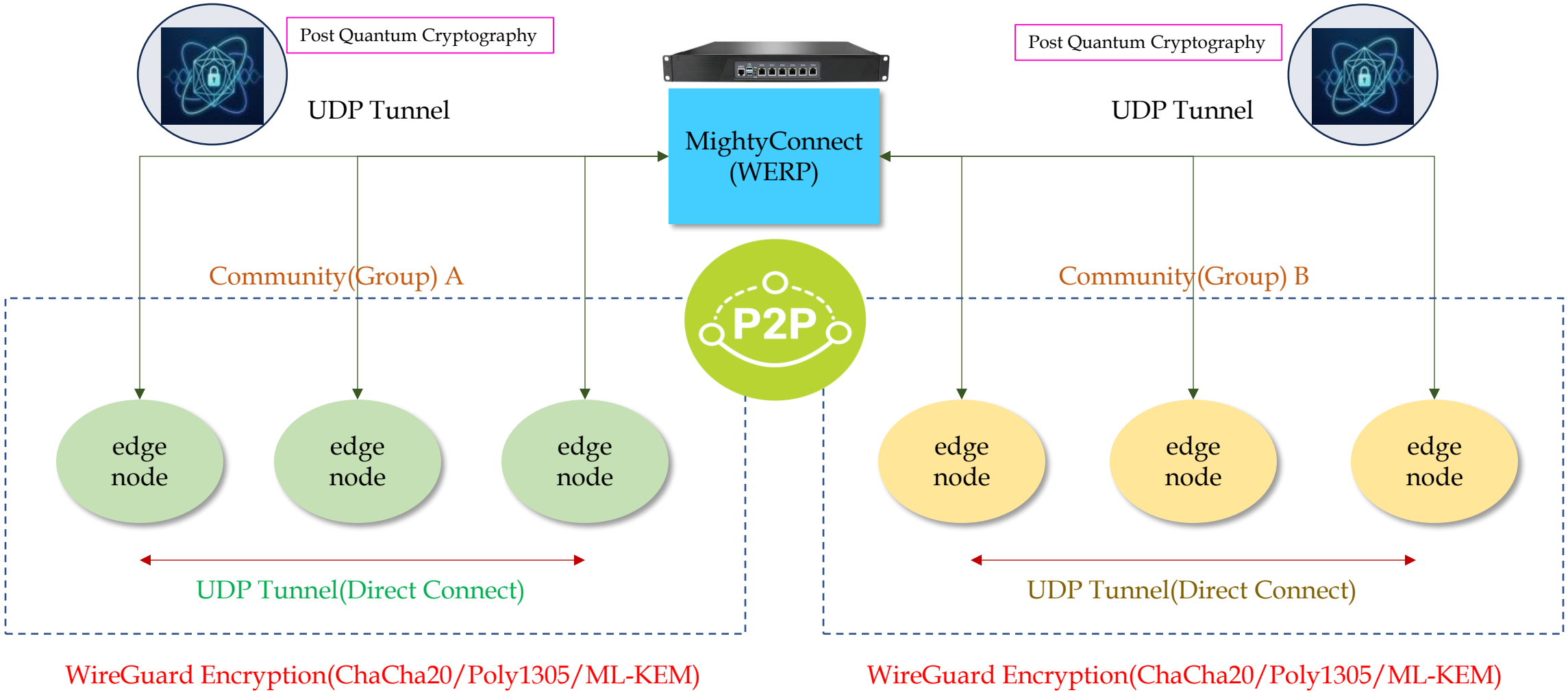
1. Zero Trust P2P Security Platform(3) – ZTNA(3-2)



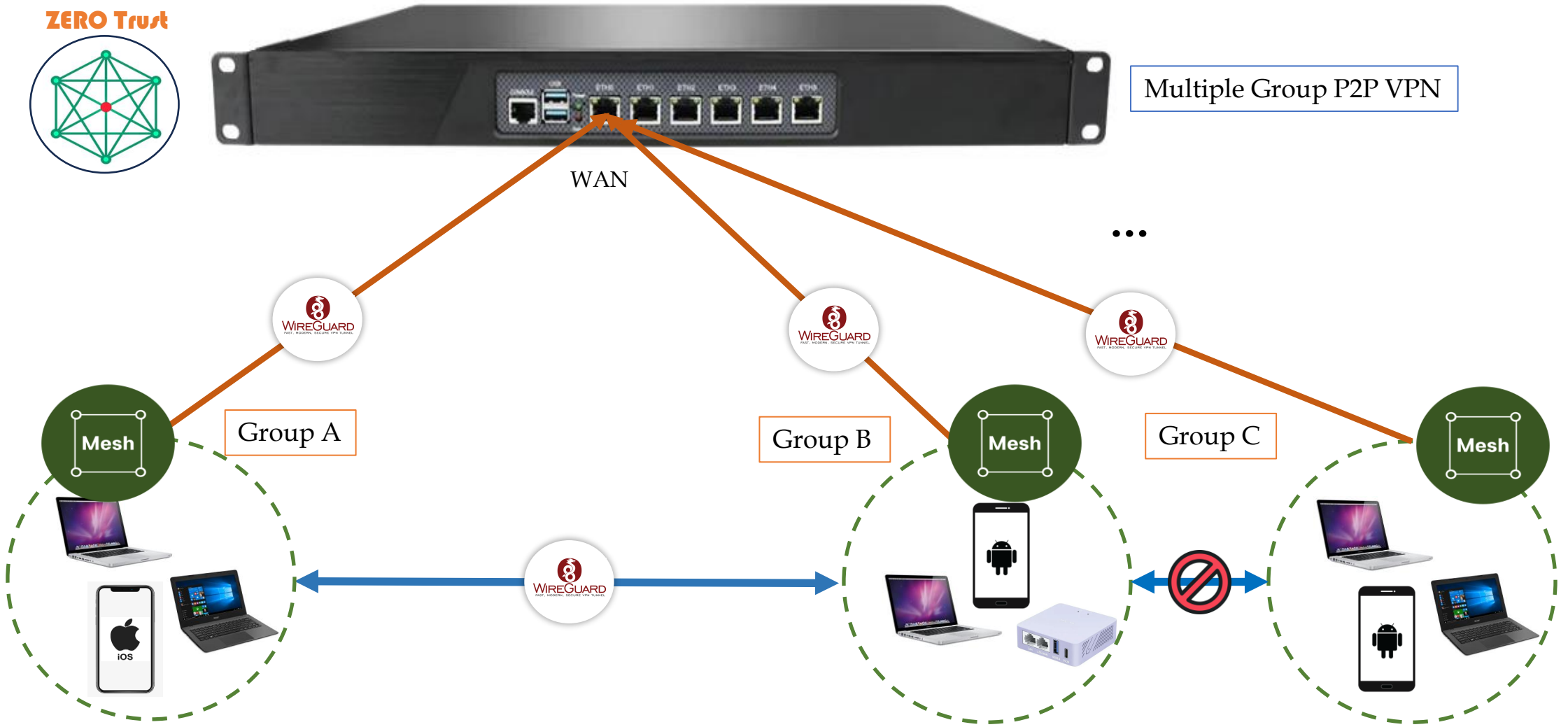
1. Zero Trust P2P Security Platform(4) - Mesh(1)



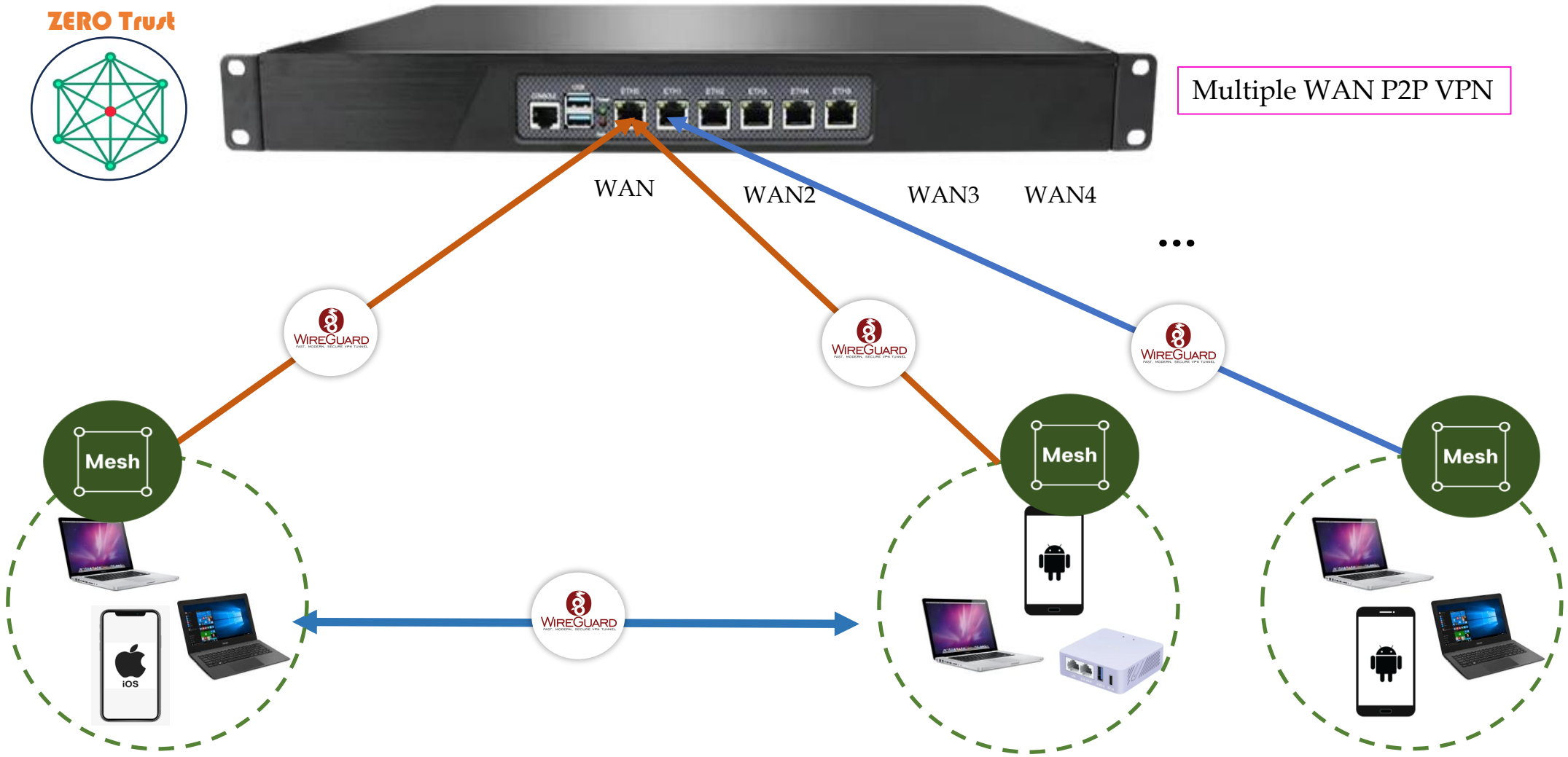
1. Zero Trust P2P Security Platform(4) – Mesh(2)



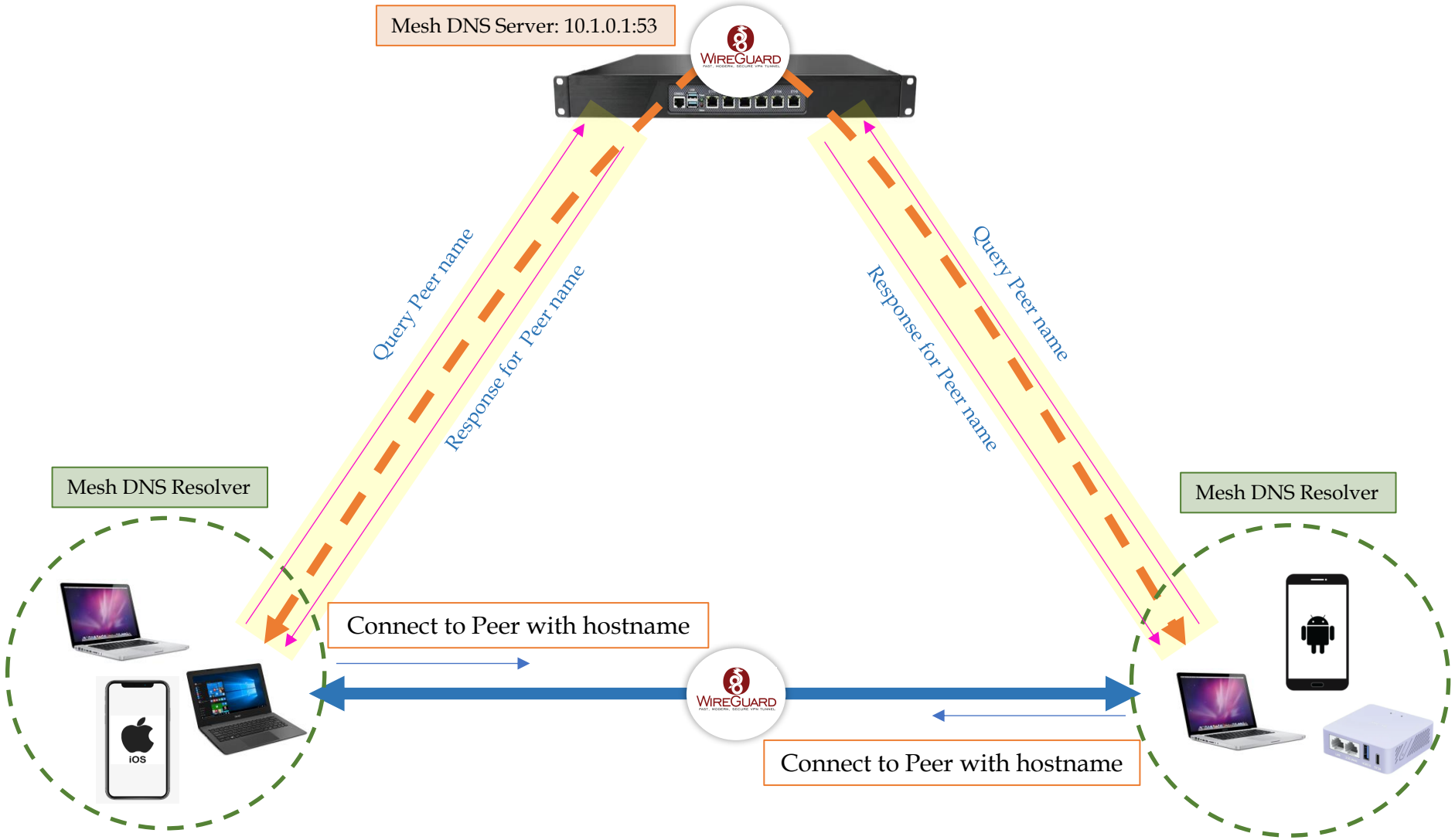
1. Zero Trust P2P Security Platform(4) – Mesh(3)



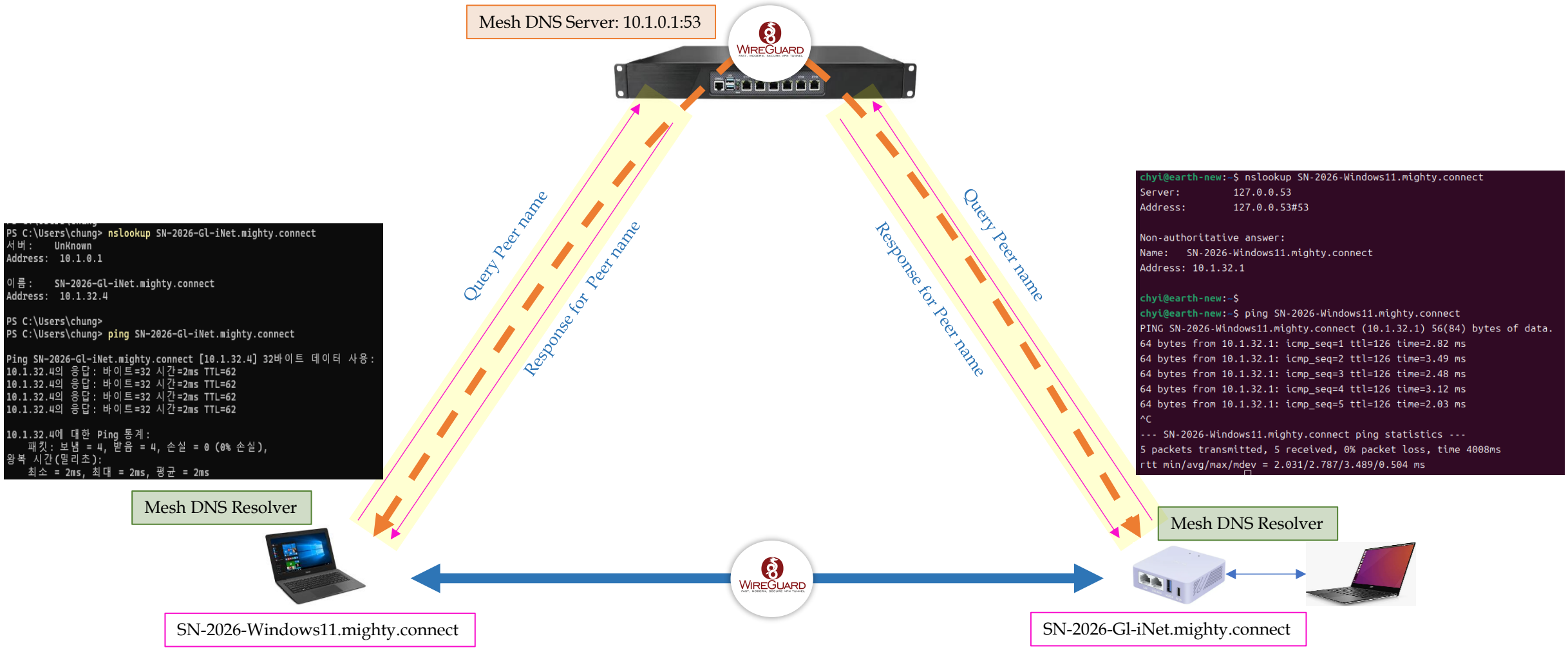
1. Zero Trust P2P Security Platform(4) - Mesh(4)



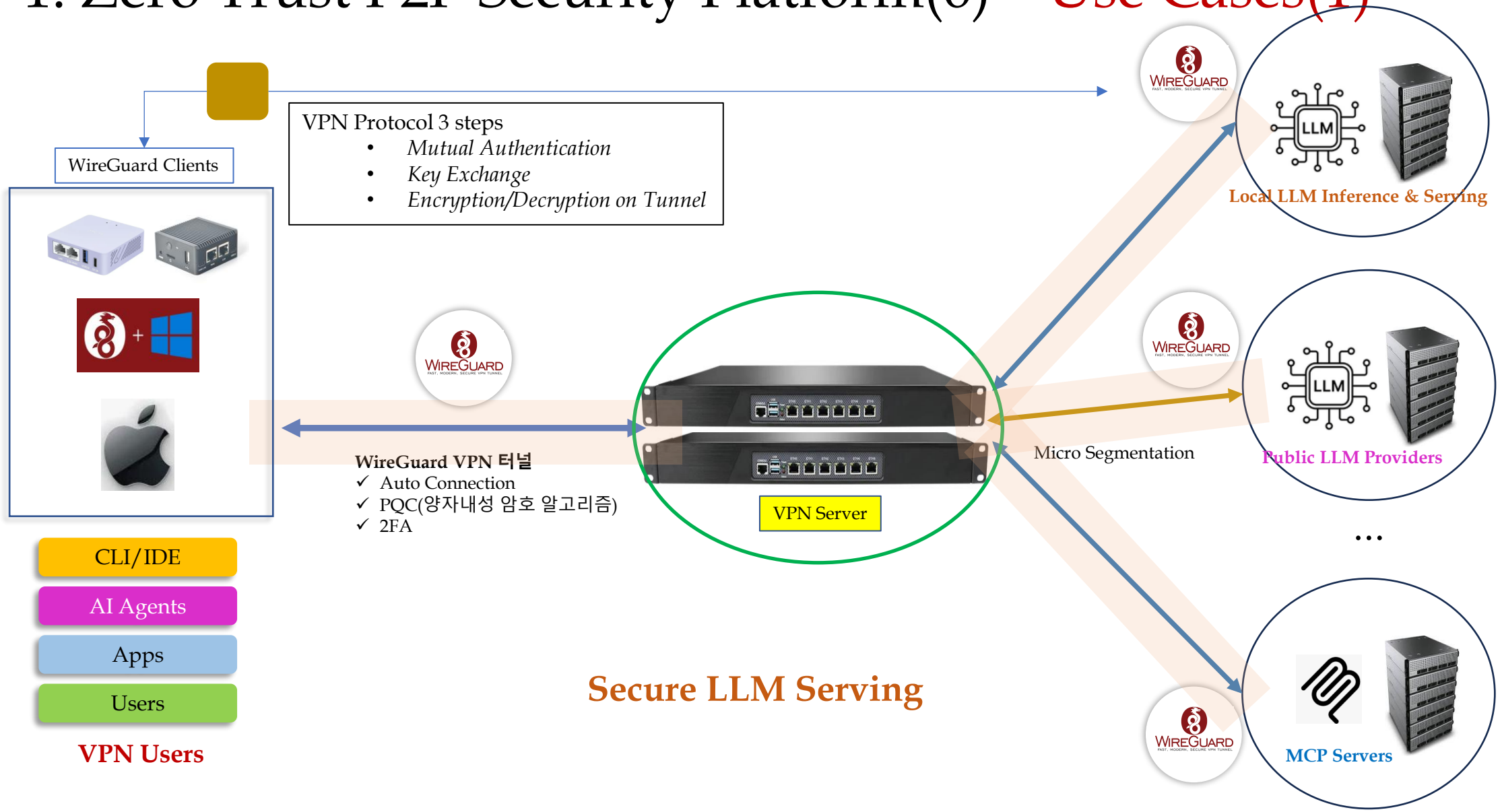
1. Zero Trust P2P Security Platform(5) – Mesh DNS(1)



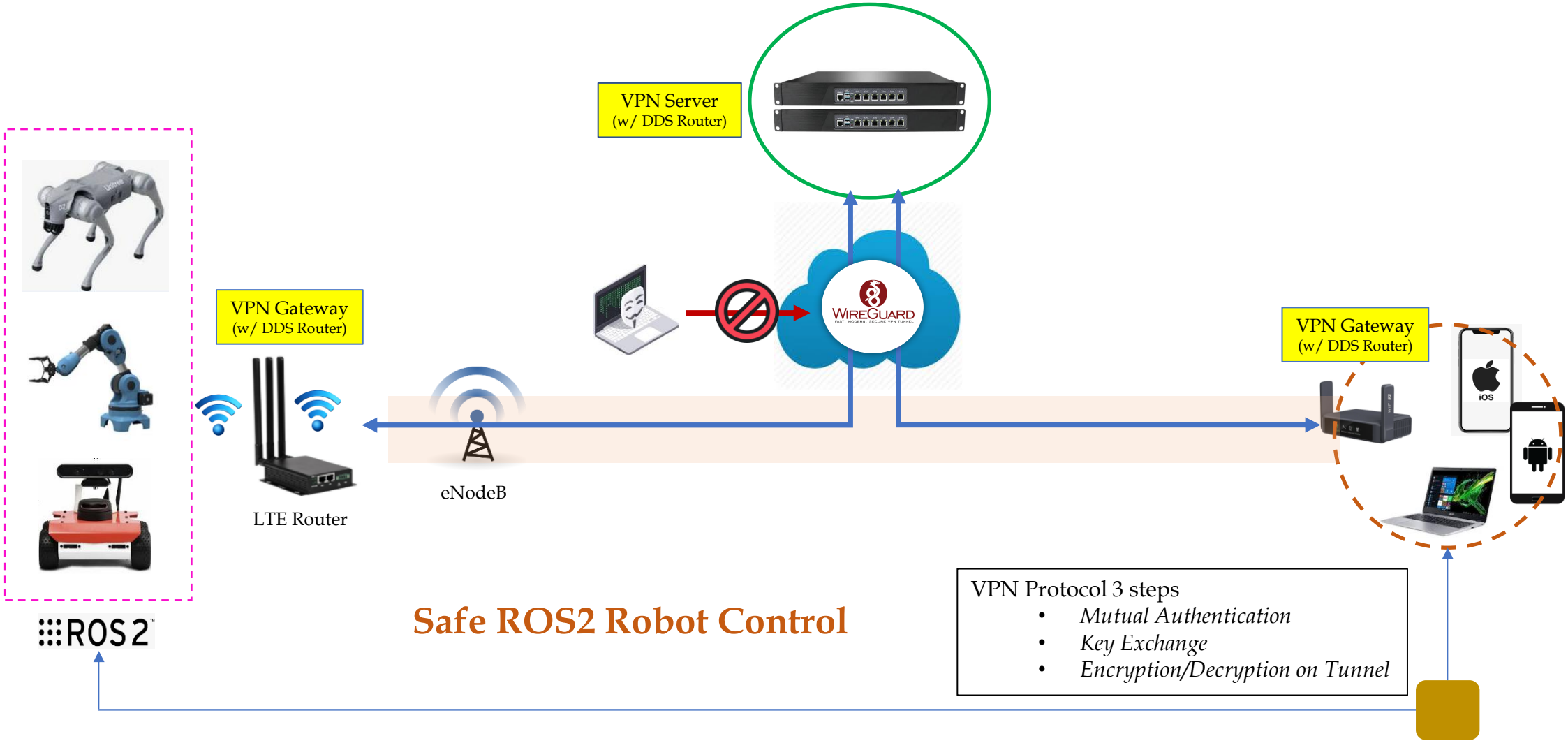
1. Zero Trust P2P Security Platform(5) – Mesh DNS(2)



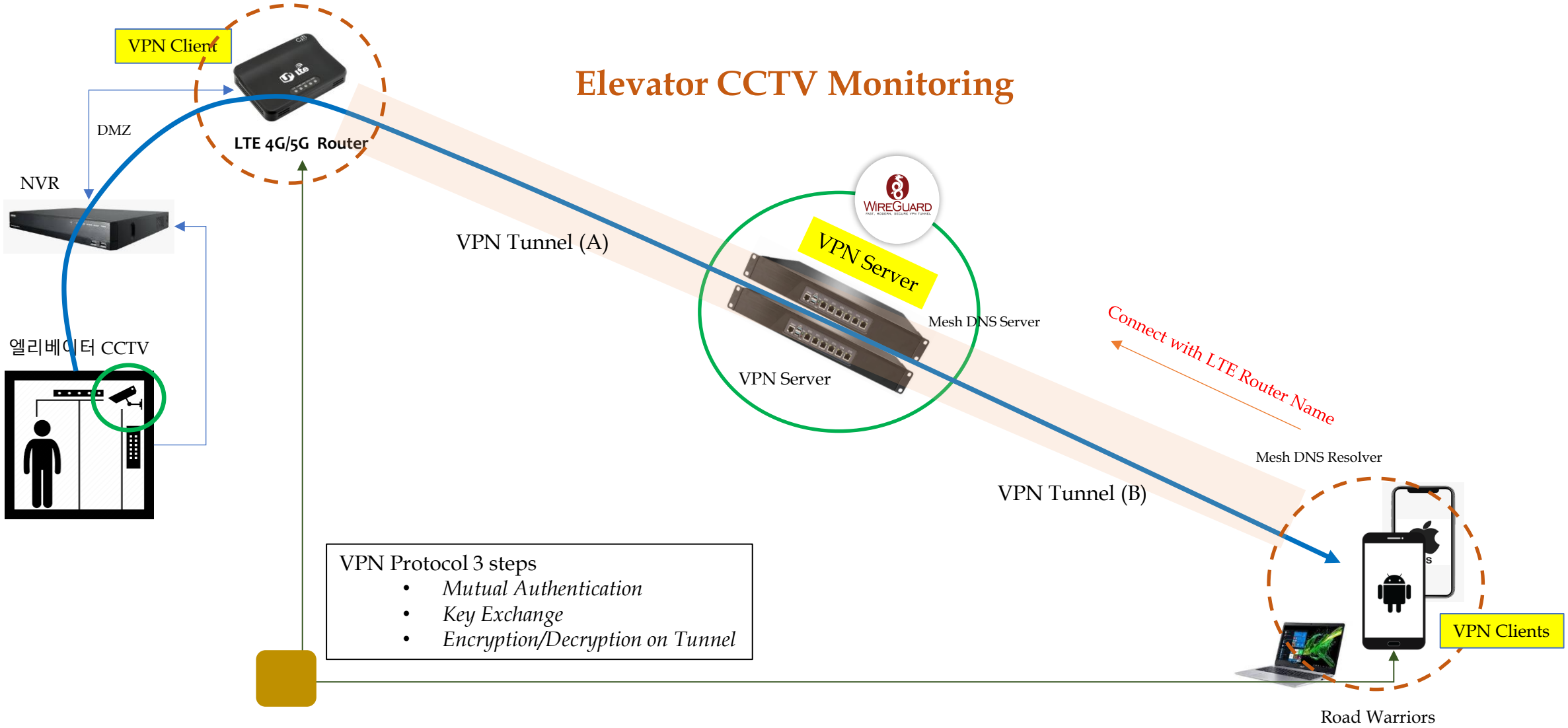
1. Zero Trust P2P Security Platform(6) – Use Cases(1)



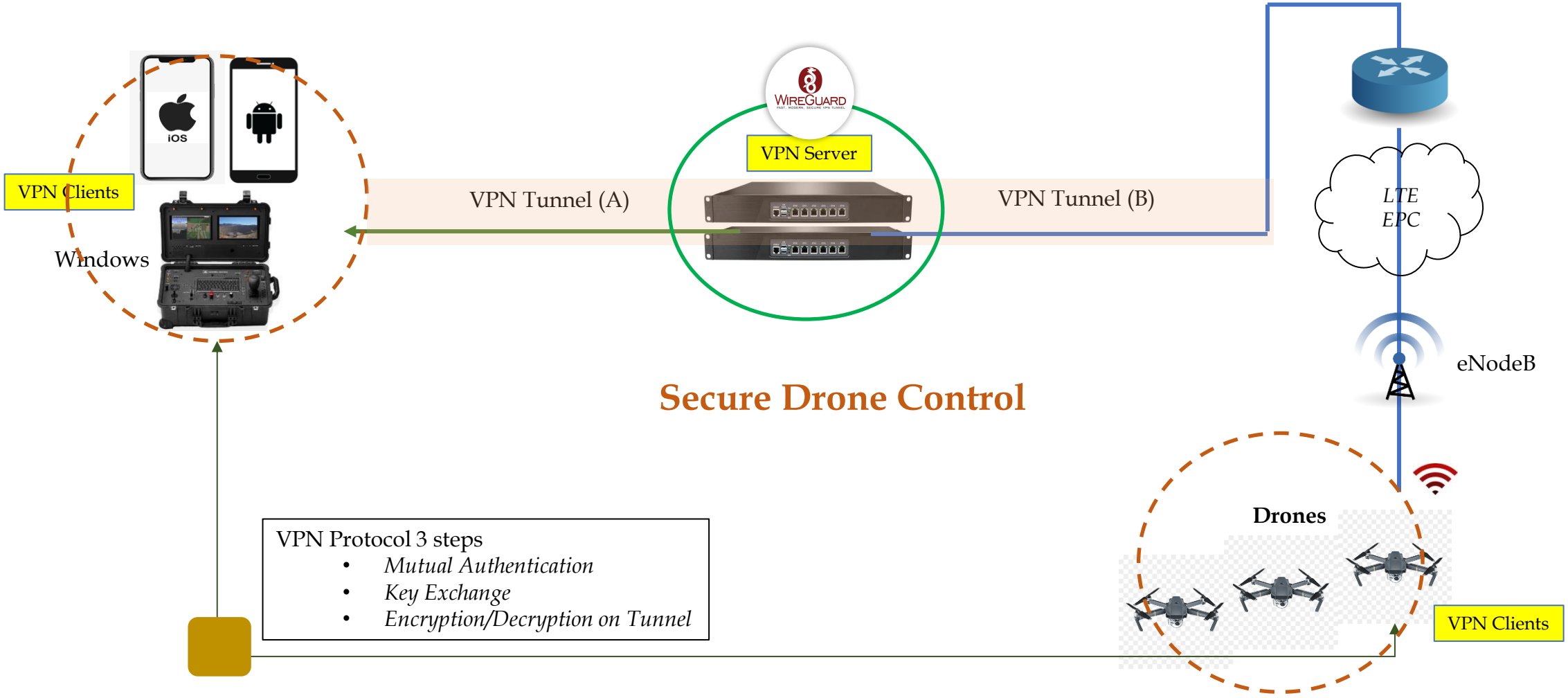
1. Zero Trust P2P Security Platform(6) - Use Cases(2)



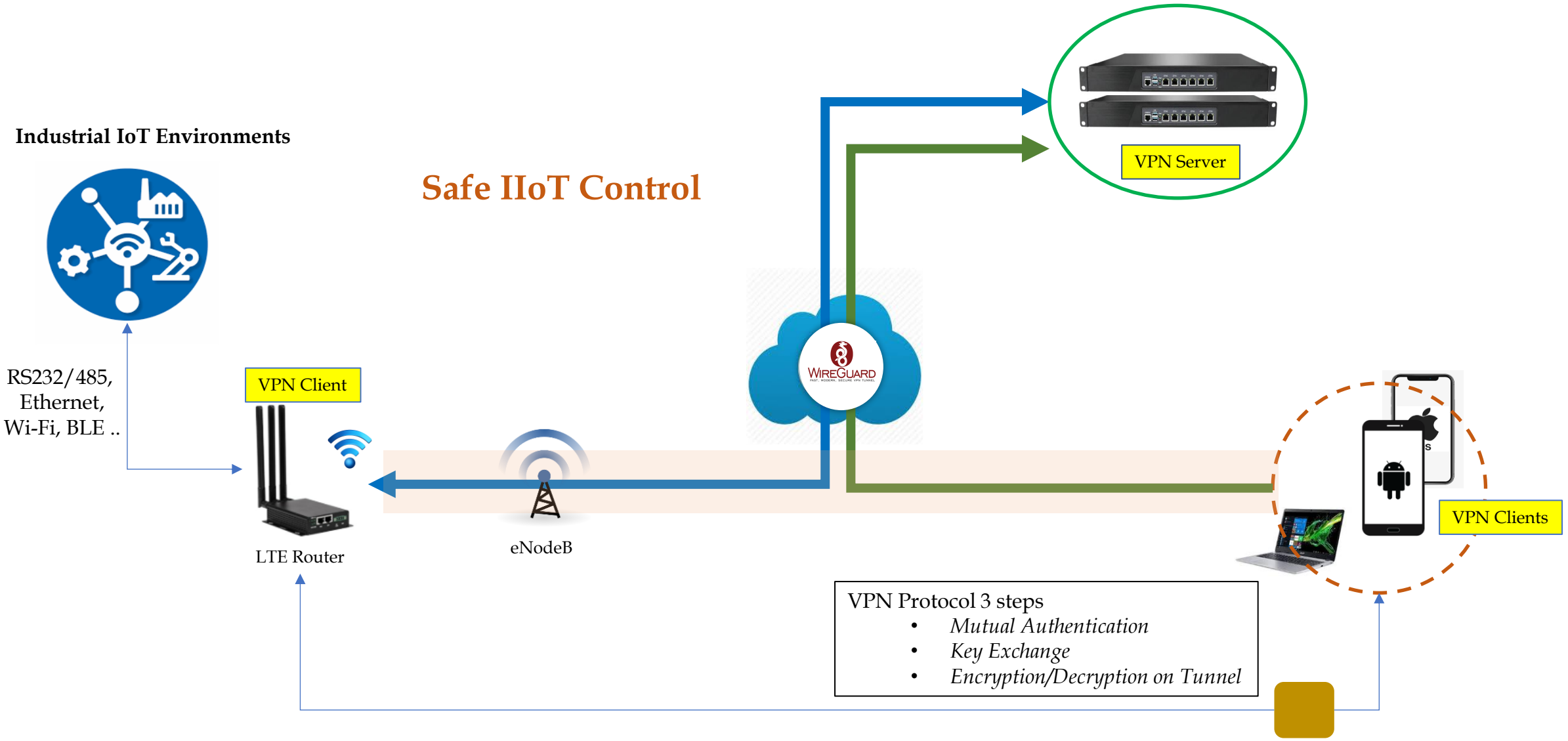
1. Zero Trust P2P Security Platform(6) - Use Cases(3)



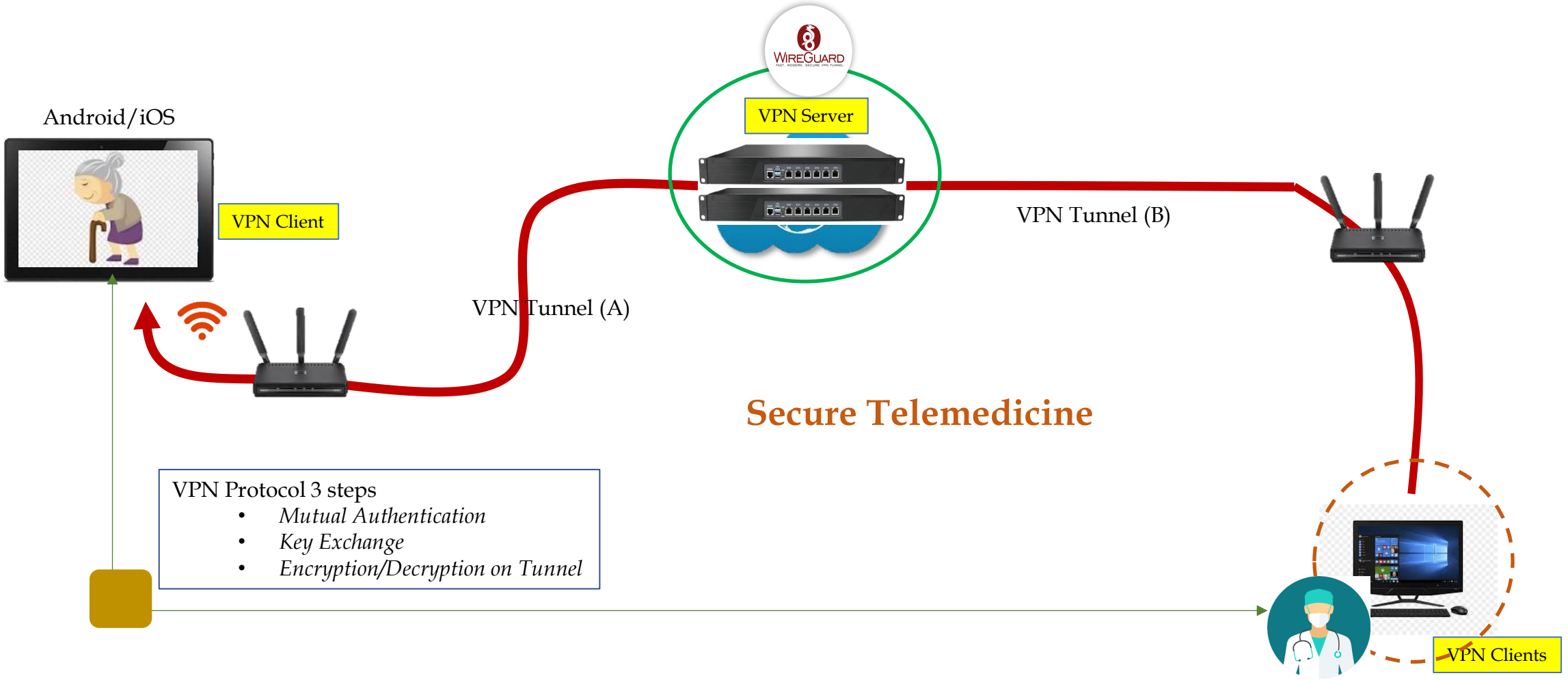
1. Zero Trust P2P Security Platform(6) - Use Cases(4)



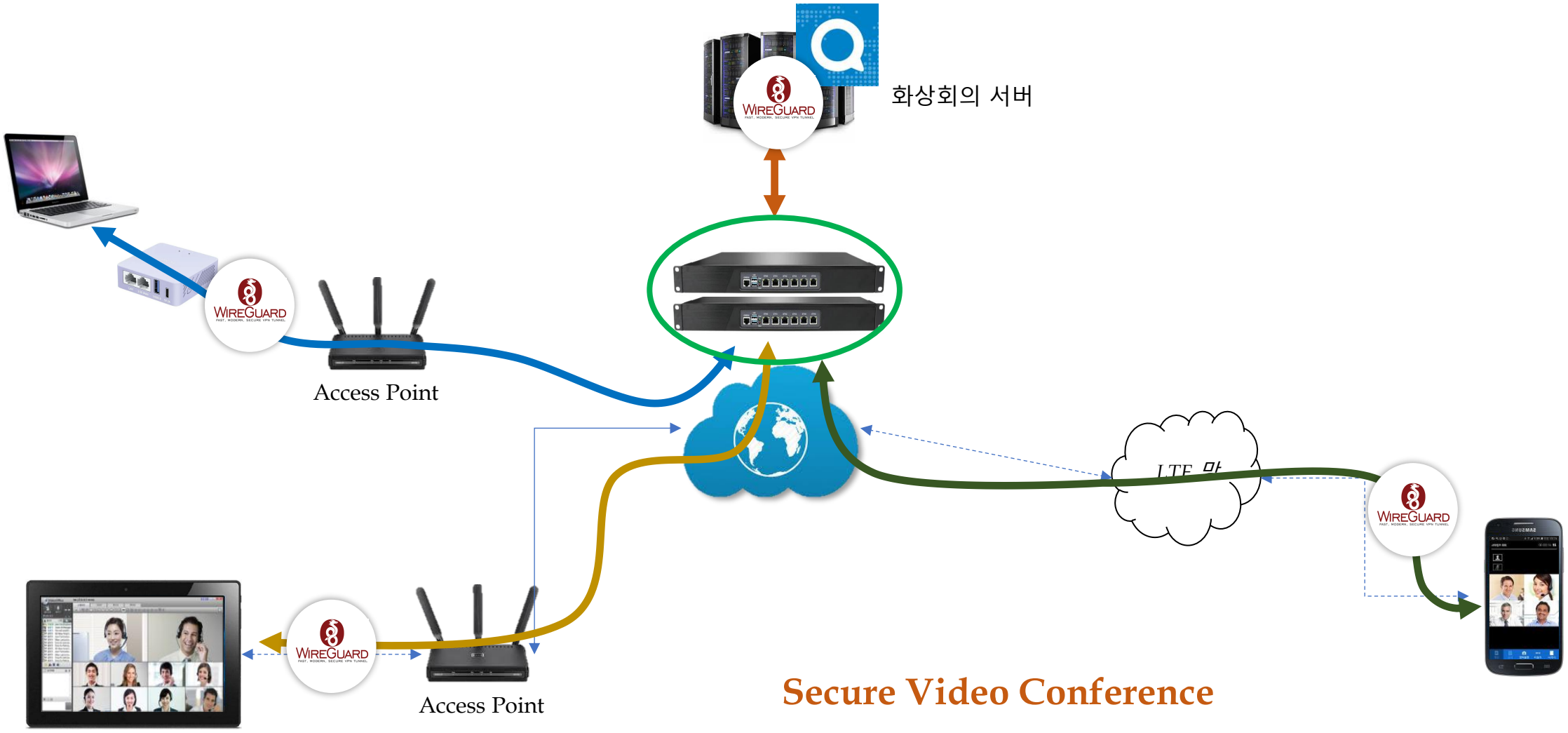
1. Zero Trust P2P Security Platform(6) - Use Cases(5)



1. Zero Trust P2P Security Platform(6) - Use Cases(6)



1. Zero Trust P2P Security Platform(6) - Use Cases(7)



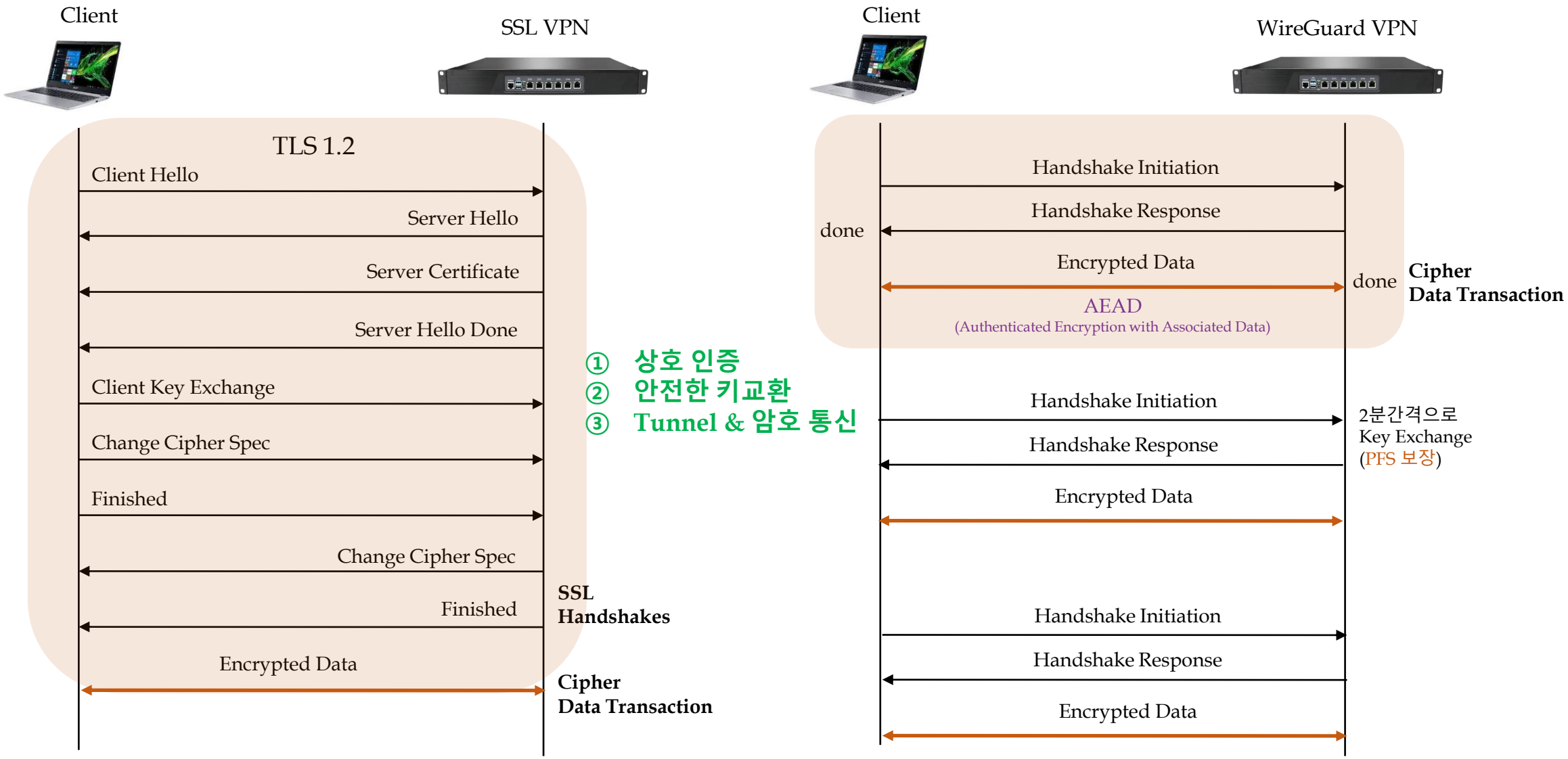
2. WireGuard

New Generation VPN Protocol



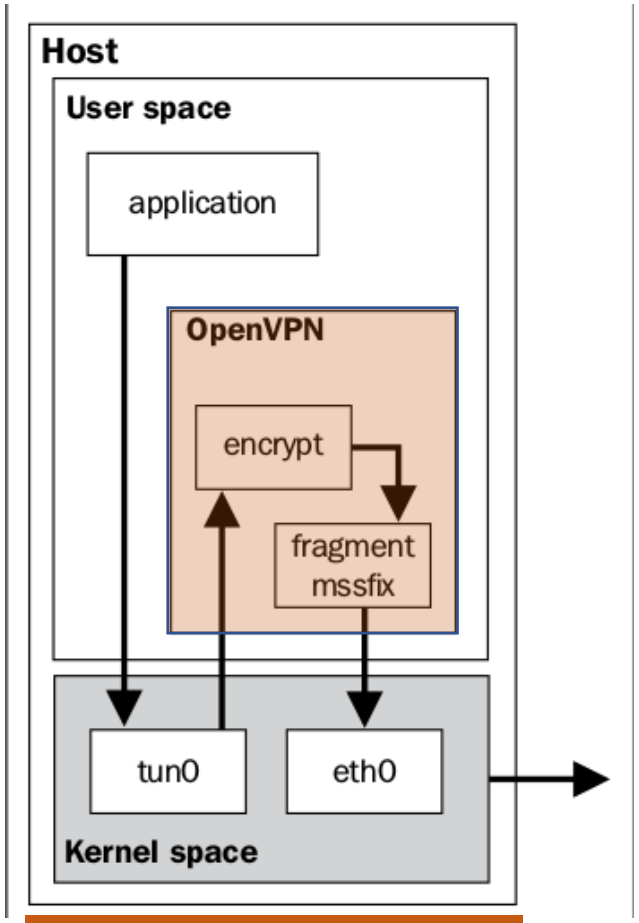
"WireGuard" and the "WireGuard" logo are registered trademarks of Jason A. Donenfeld

2. WireGuard(1) – Architecture(1)



2. WireGuard(1) - Architecture(2)

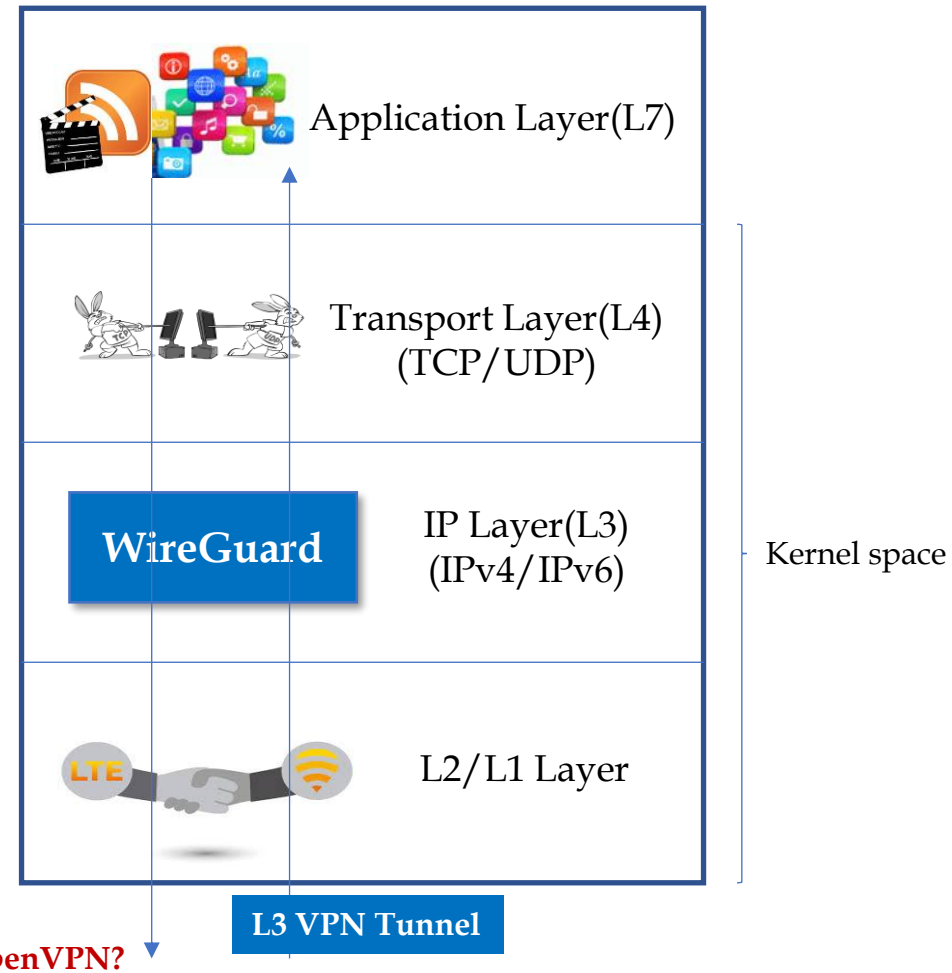
Limitations of OpenVPN



SSL VPN Tunnel(Before DCO)

vs

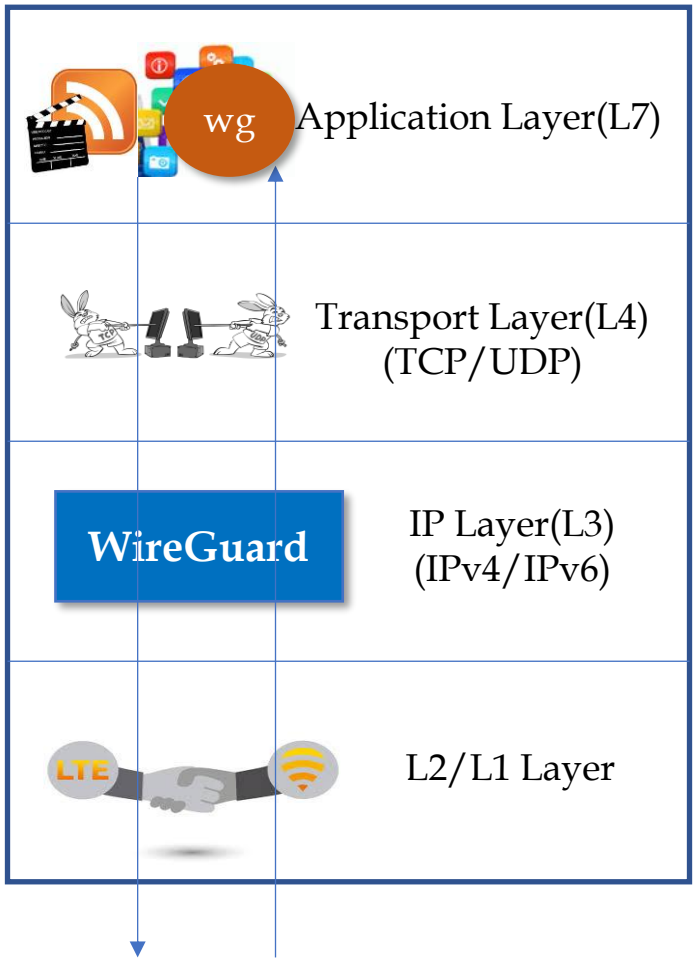
No Limitation. Cryptokey Routing

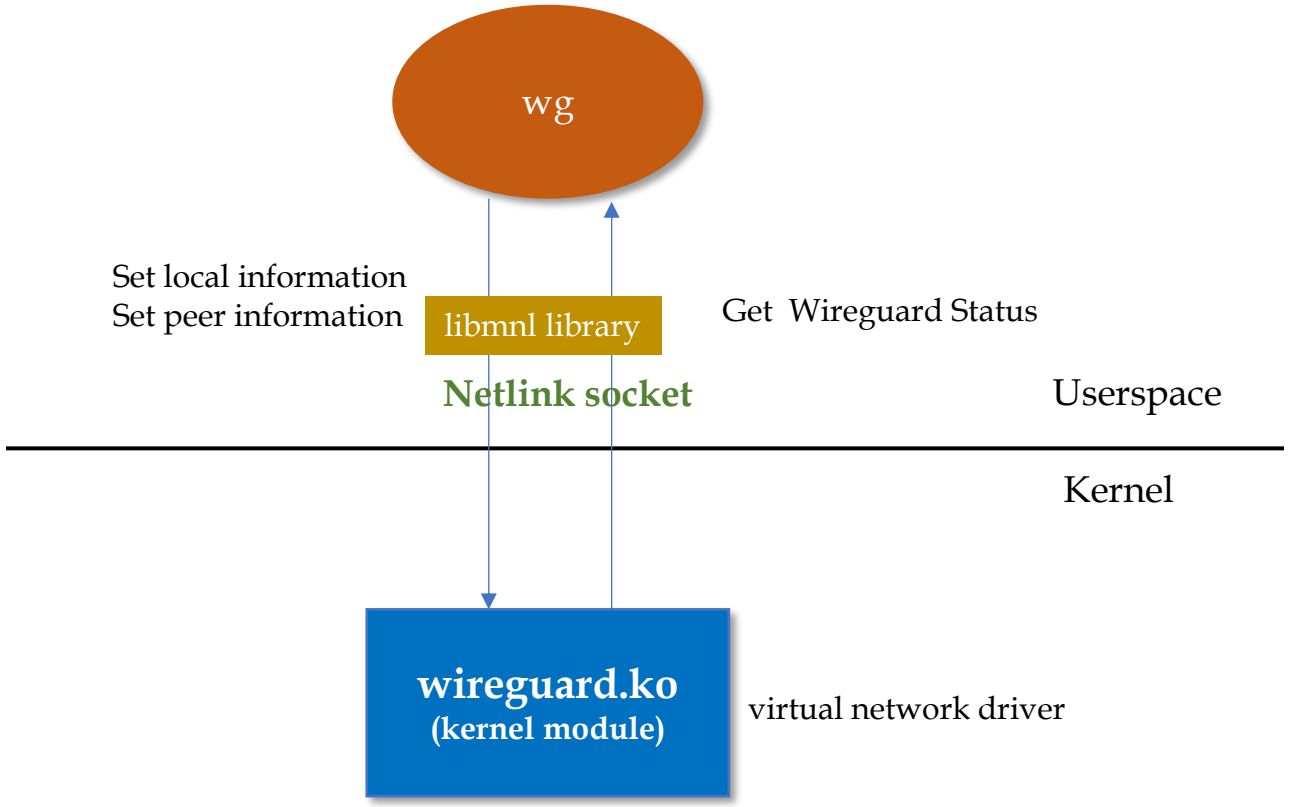


L3 VPN Tunnel

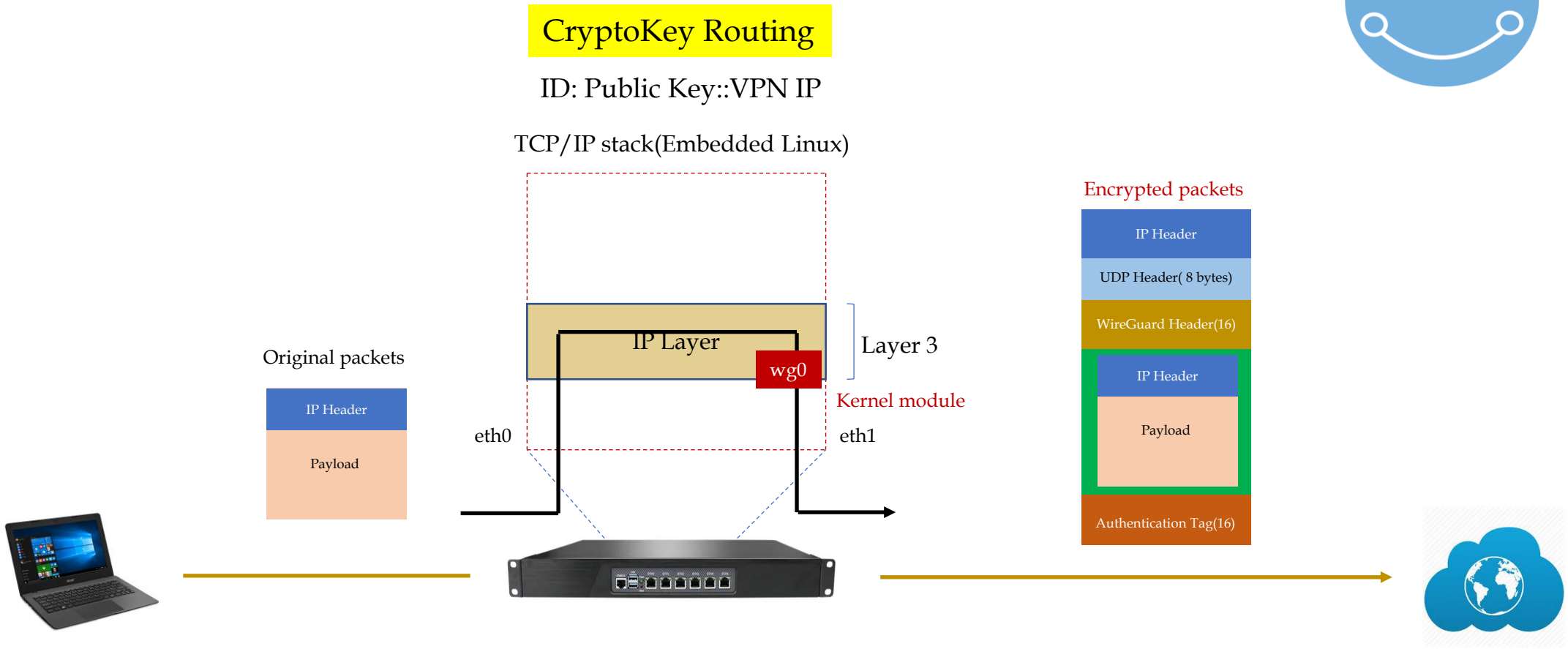
Why is WireGuard faster than OpenVPN?

2. WireGuard(1) - Architecture(3)

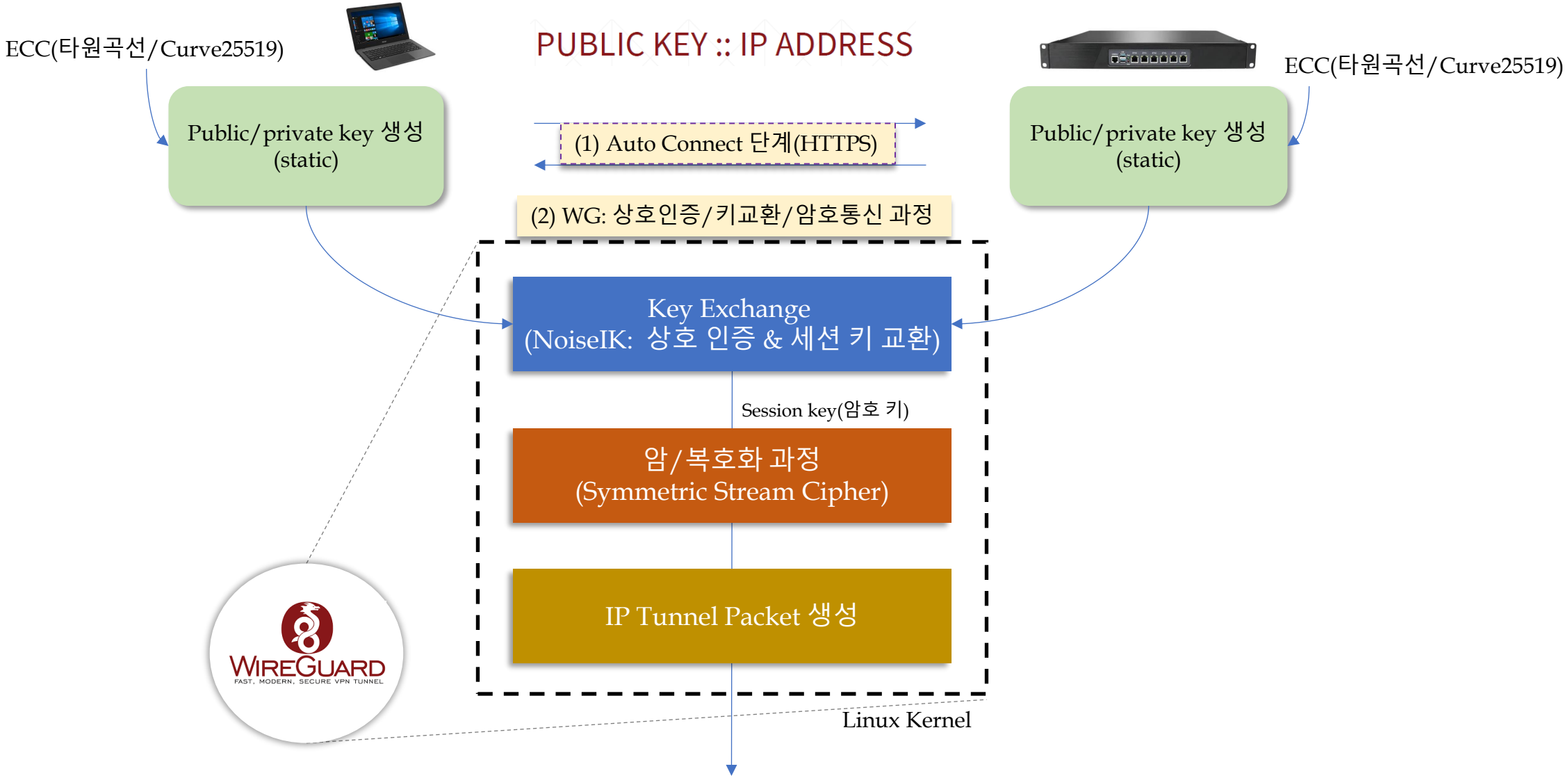




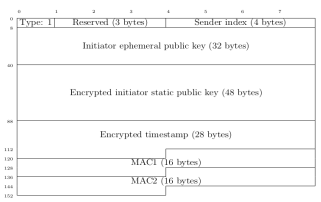
2. WireGuard(1) - Architecture(4)

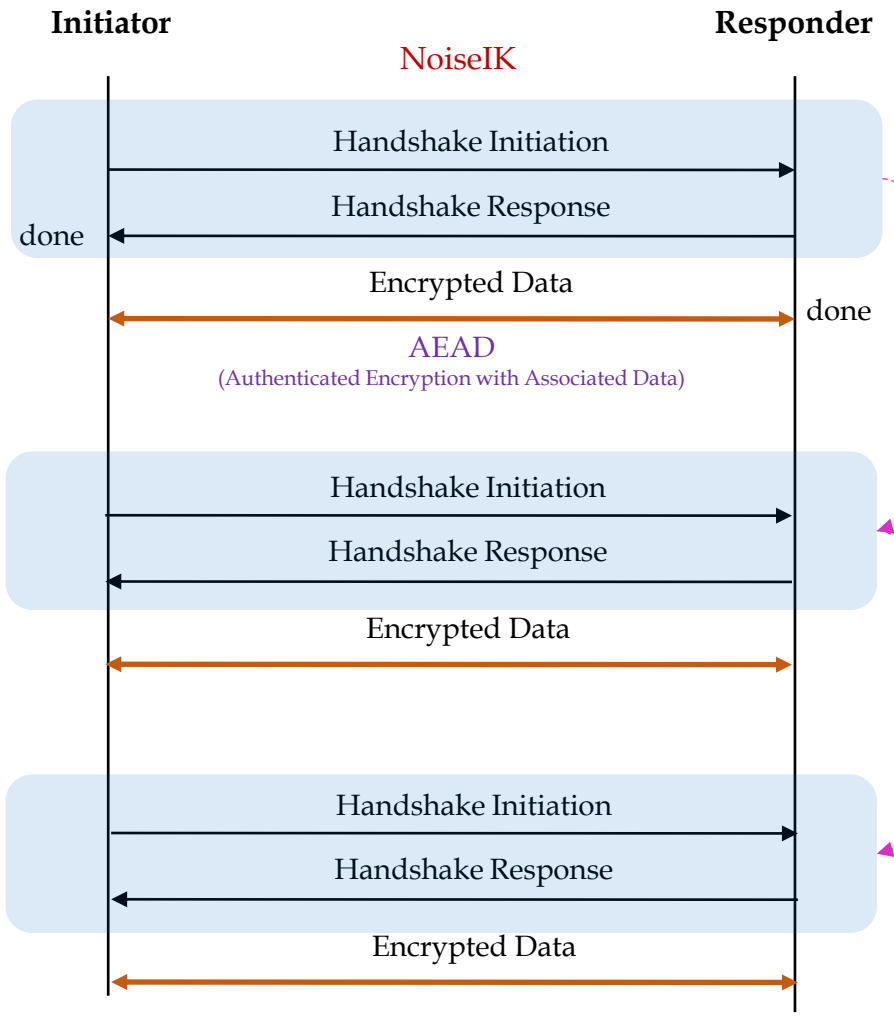


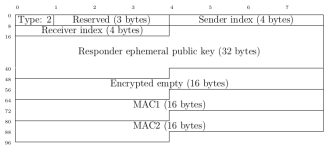
2. WireGuard(2) - Protocol(1)



2. WireGuard(2) - Protocol(2)





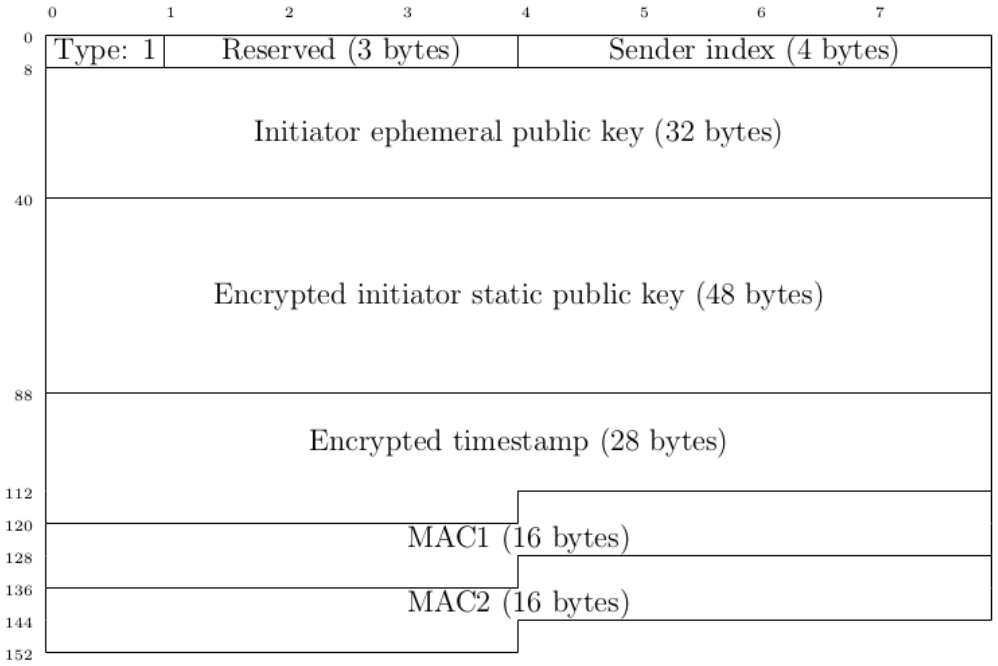


2분간격으로
Key Exchange
(PFS 보장)

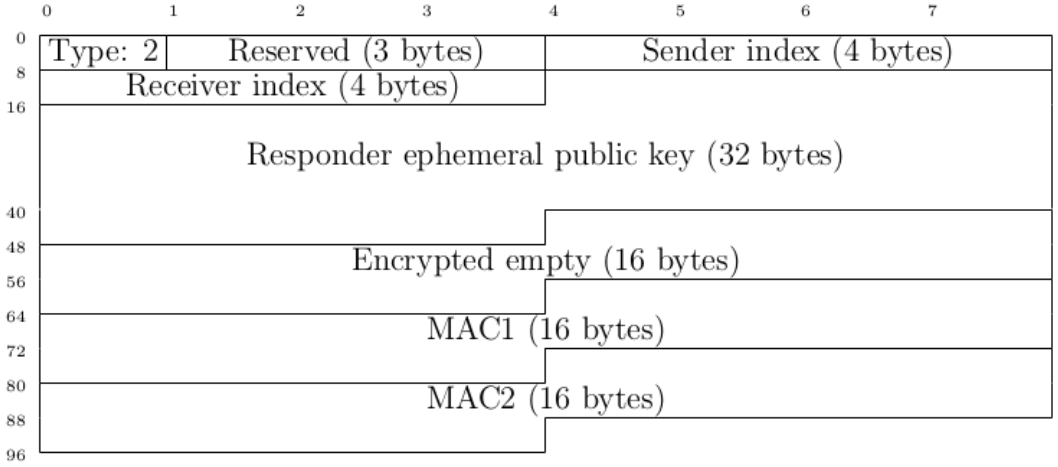
2분간격으로
Key Exchange
(PFS 보장)

2. WireGuard(2) - Protocol(3)

Initiation Packet Format



Response Packet Format



Type	Message Name	Length
1	Handshake Initiation	148 bytes
2	Handshake Response	92 bytes
3	Cookie Reply	64 bytes
4	Transport Data	$\geq$ 32 bytes

2. WireGuard(2) - Protocol(4)

Session keys = **Noise**(

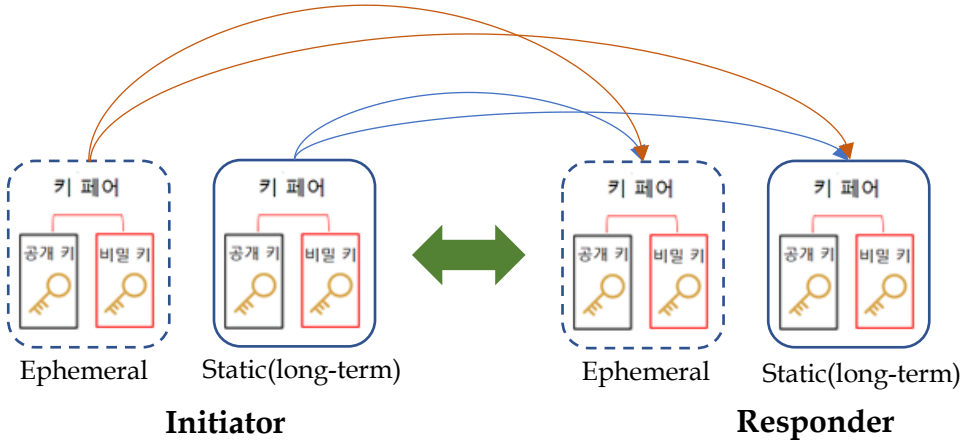
ECDH(ephemeral, static),
ECDH(static, ephemeral),
ECDH(ephemeral, ephemeral),
ECDH(static, static)

)

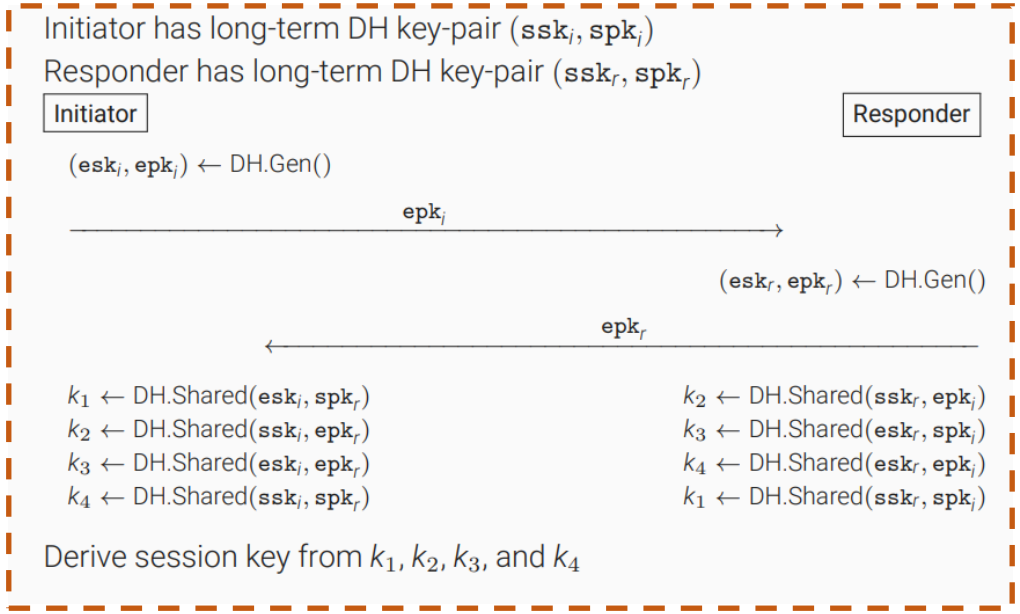
256bits(32bytes)



AEAD



상호 인증
및
키 교환



2. WireGuard(3) – Algorithms(1)

Layer 3(IPv4, IPv6 지원) VPN Protocol

handshake pattern	IKpsk2 (Noise Protocol Framework 기반)
DH function	X25519
cipher function	ChaCha20-Poly1305
hash function	BLAKE2s
prologue	`WireGuard v1 zx2c4 Jason@zx2c4.com` (34 bytes)

2. WireGuard(3) - Algorithms(2)

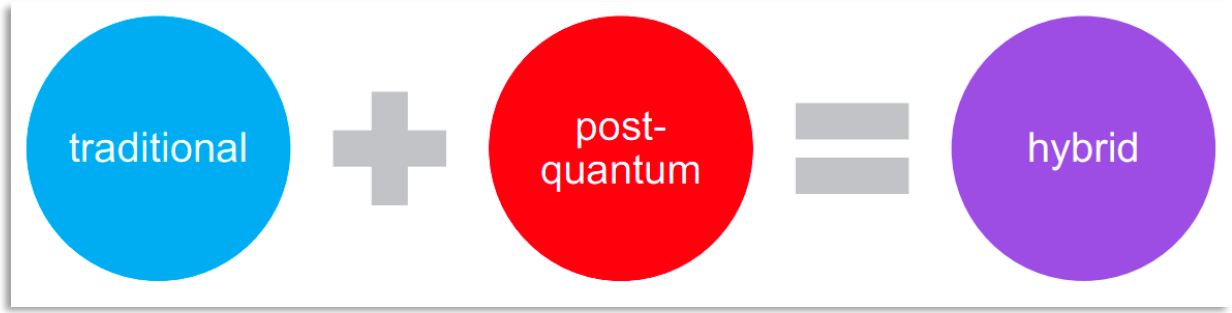
보안 알고리즘	상세 내용
NoiseIK **Key 교환 방식** **및** **상호 인증**	**NoiseIK handshake 방식(Noise IKpsk2)** ▪ ECDH(Diffie-Hellman) 기반 ▪ **Curve25519** Public key(32 byte)를 교환 후, 이를 통해 안전하게 shared secret 생성 ▪ Static/Ephemeral private/public key(2 쌍) 이용 ▪ Key 교환 시 아래 기능 보장 ▪ *Perfect forward secrecy 보장, replay attack 방지 기능* ▪ *Identity 감춤 기능 제공, DoS 공격 완화 기능(Cookie),* ▪ *PSK 사용 시, post-quantum security 보장* **Hash 알고리즘** ▪ **BLAKE2** – fast secure hashing 알고리즘 ▪ SHA series 보다 빠름. 즉 MD5 수준임. SHA-3 보안 강도. **PRF & MAC 및 Key Derivation 함수** ▪ **SipHash4** (Hash table lookup 용으로 사용) ▪ **HKDF** (HMAC-based Key Derivation Function) 사용
암호 알고리즘 **Authenticator 알고리즘** **(ChaCha20-Poly1305 AEAD)**	**ChaCha20** – 256 bits stream cipher(20 round cipher Salsa20 기반) ▪ Stream cipher는 일반적인 block cipher(예: AES-256-CBC)에 비해 속도가 빠름 ▪ key(32 bytes)는 대칭키를 사용(즉, 암호화 용 키와 복호화 용 키 동일) ▪ **Video/Audio 등 stream 암호화에 적합**
	Poly1305 - message authentication code 알고리즘(16 bytes output 생성)

Noise_IKpsk2_25519_ChaChaPoly_BLAKE2s

2. WireGuard(4) – PQC(1)



Post Quantum Cryptography



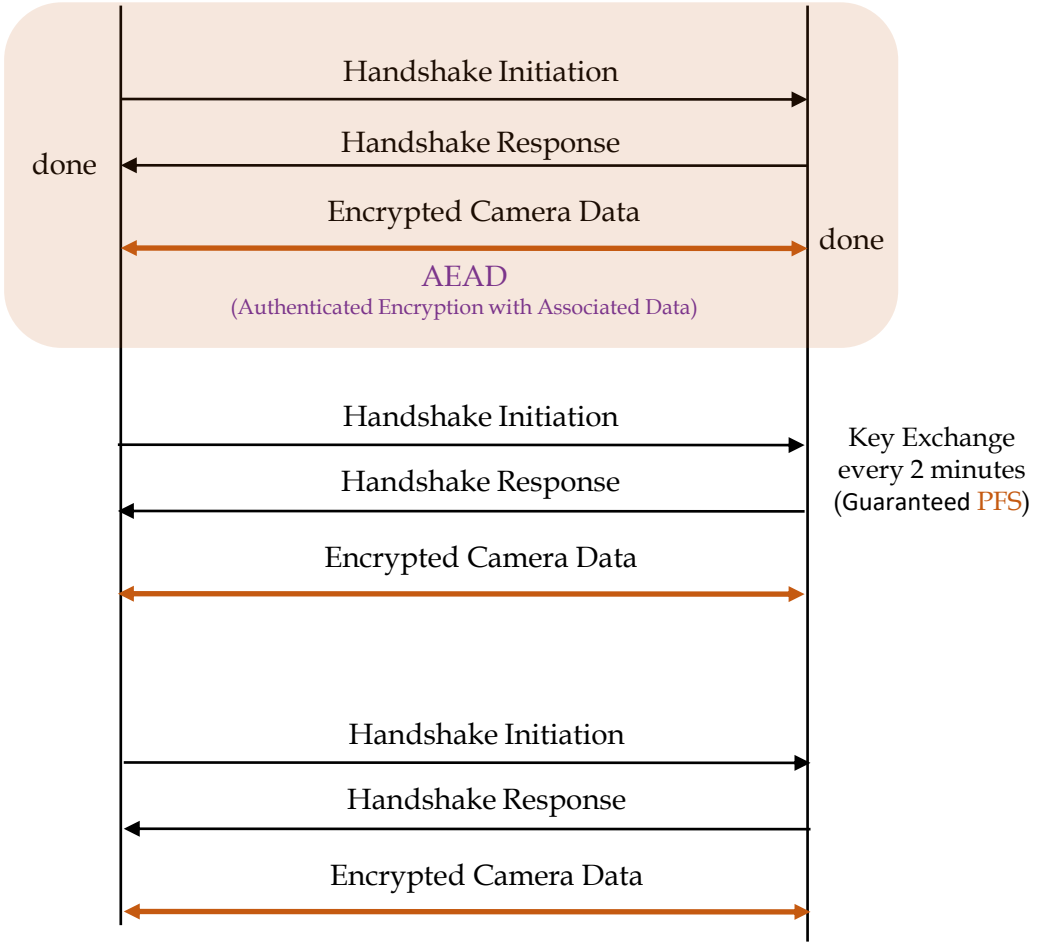
WireGuard **ML-KEM 1024** **Quantum Safe WireGuard**

PQC 알고리즘이 통합된 WireGuard Protocol



WireGuard



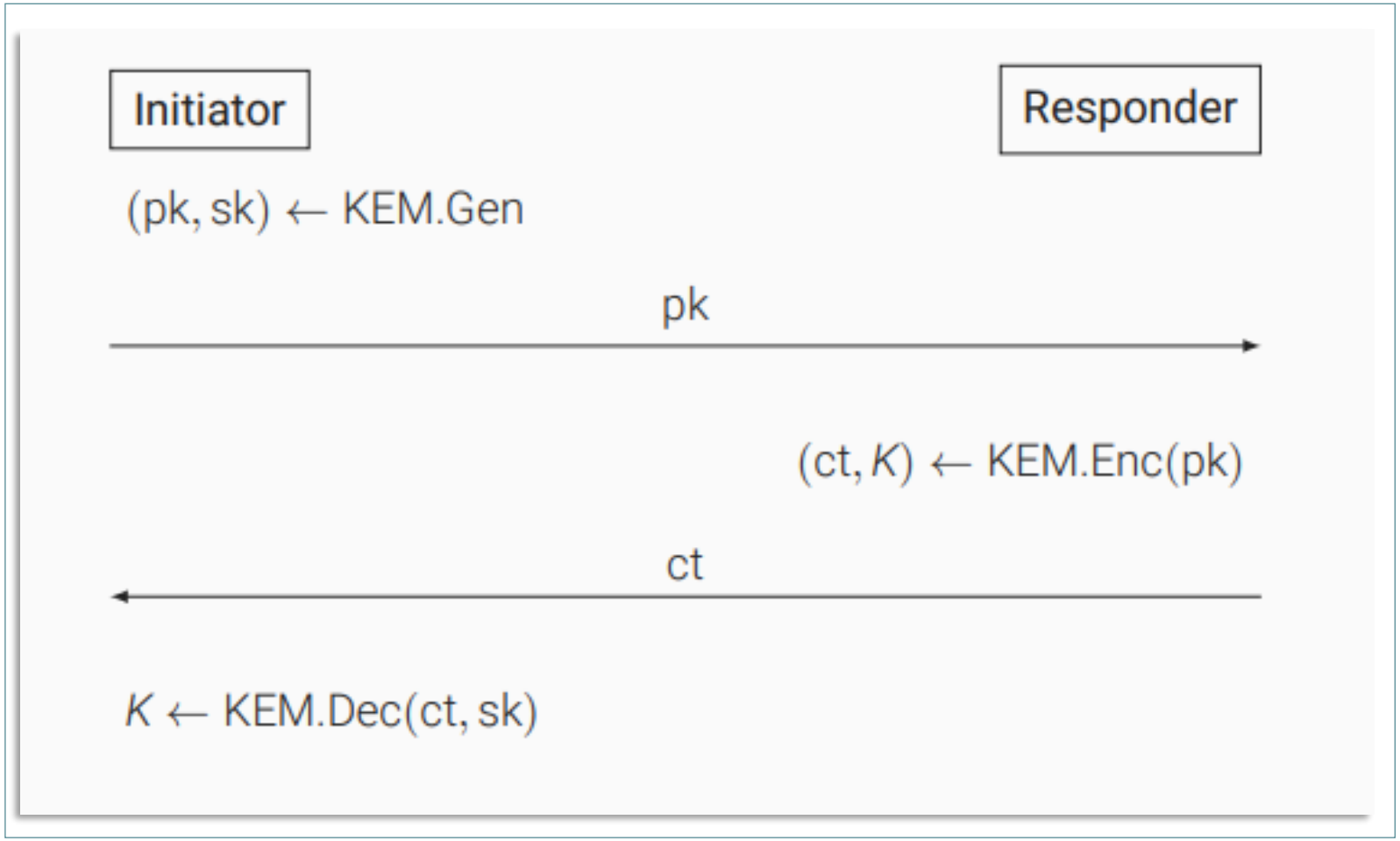


2. WireGuard(4) – PQC(2)

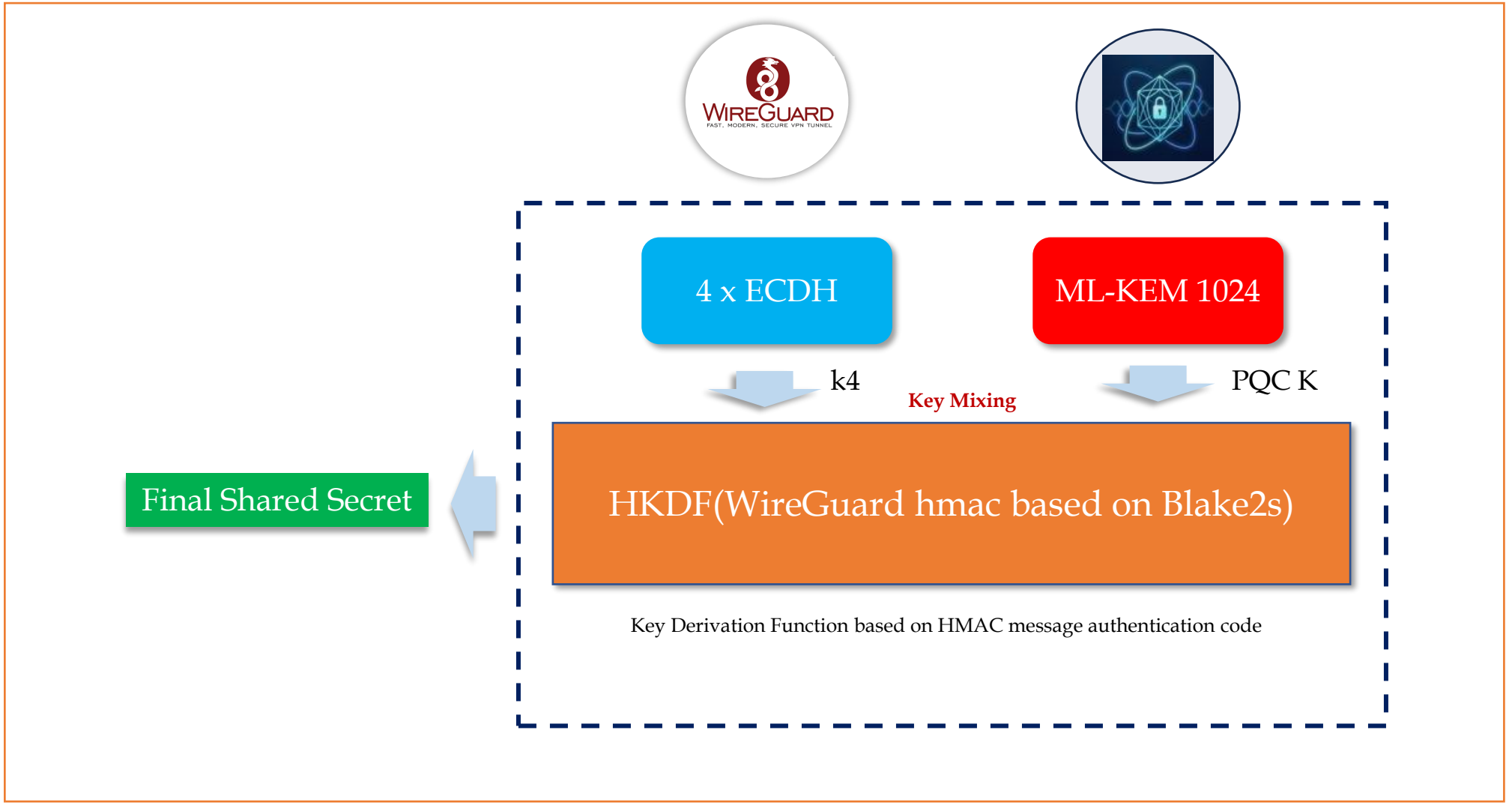
Post Quantum Cryptography KEM



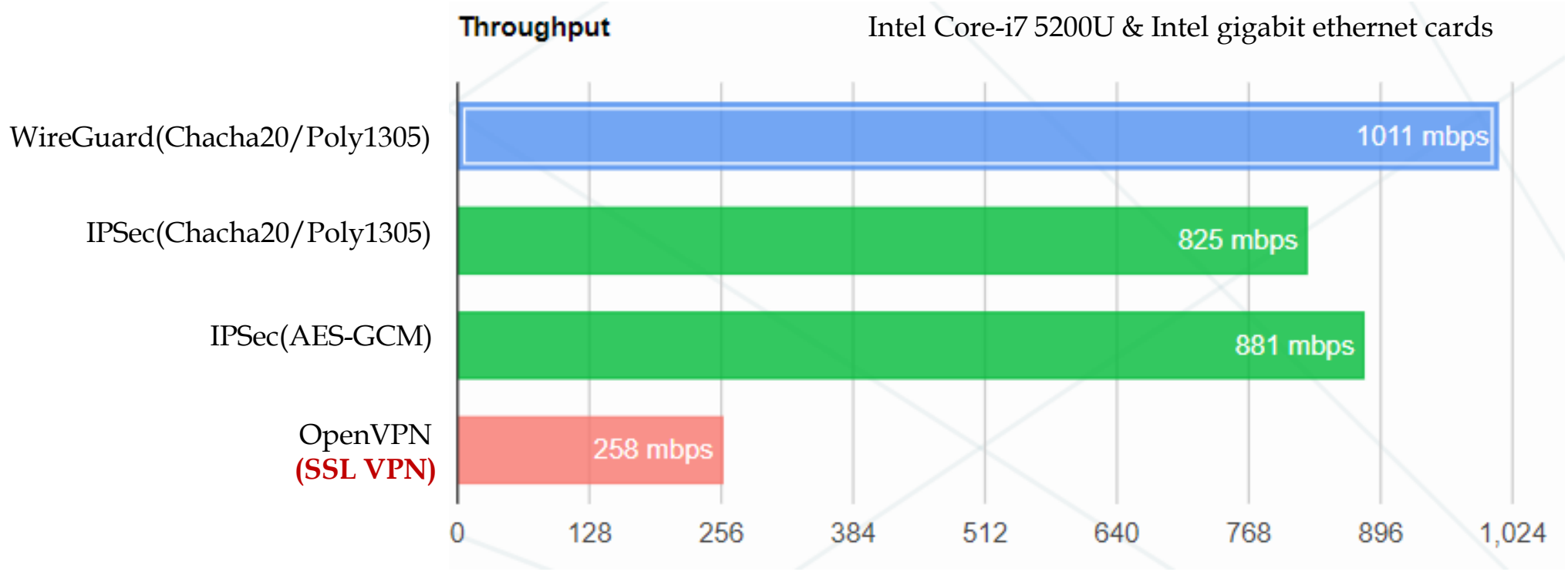
PQC KEM



2. WireGuard(4) – PQC(3)



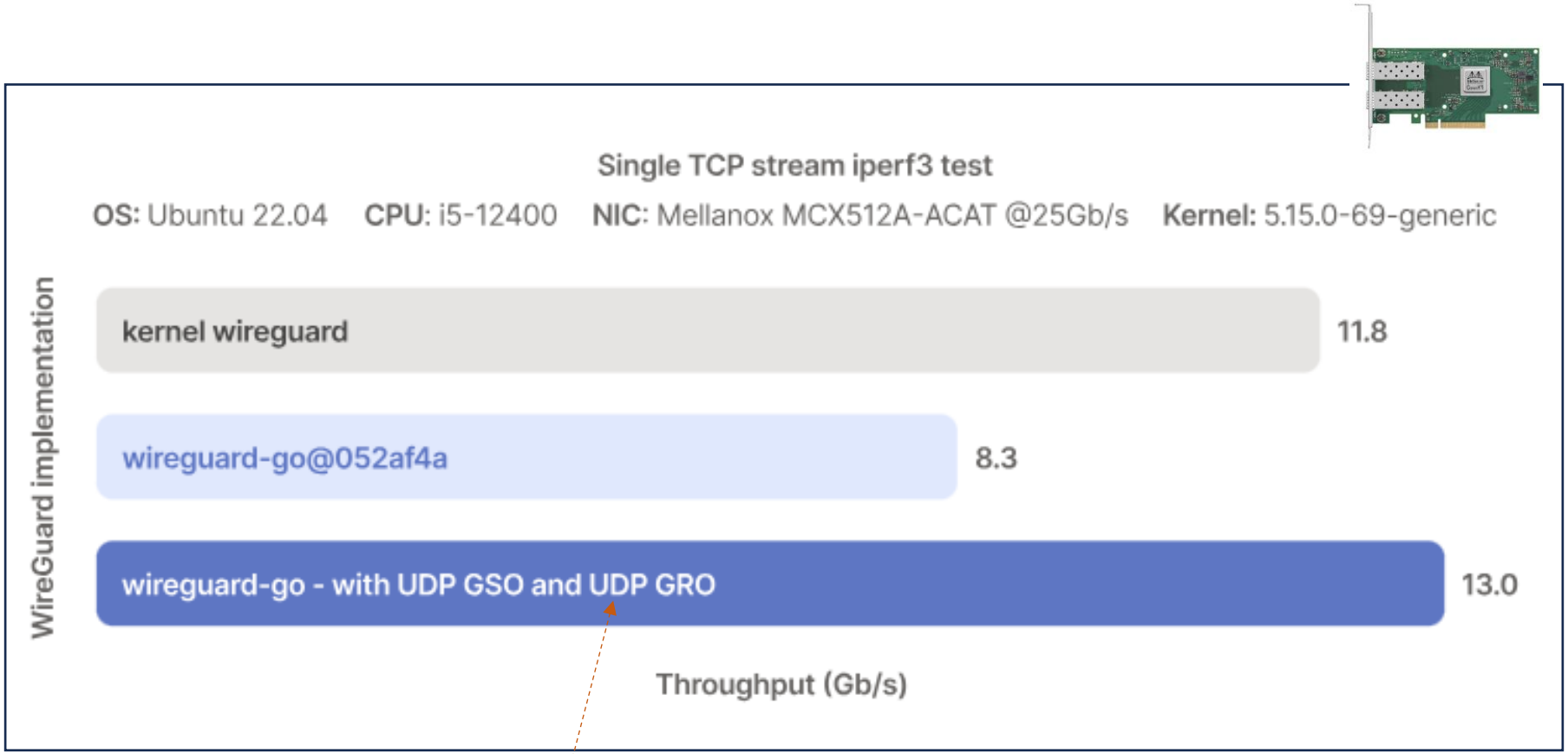
2. WireGuard(5) – Performance(1)



[출처] https://www.wireguard.com/performance/

[주의] OpenVPN DCO(Data Channel Offload)는 위의 그림보다는 훨씬 빠르다.

2. WireGuard(5) - Performance(2)



Packet Vector Processing
UDP GRO/GSO
Checksum Offload

그림 출처: https://tailscale.com/blog/more-throughput

2. WireGuard(5) - Performance(3)

- **WireGuard VPN의 우수한 성능 (Multi-Core 환경)**

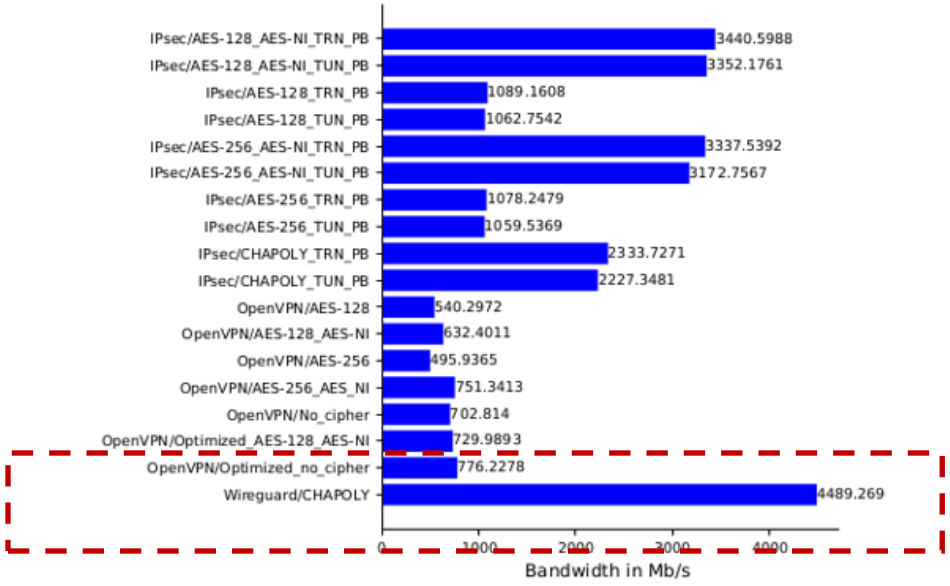
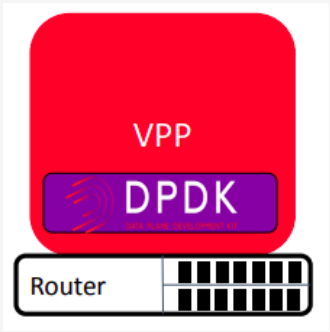


그림 출처: Performance Comparison of VPN Solutions, Lukas Osswald

- **Signature 없이 인증 & 키 교환하는 AKE 기법 도입**
- **최신 암호 알고리즘 및 암호 가속 기능으로 무장**
  - Curve25519, Blake2s, ChaCha20, Poly1305, SipHash4
  - X86_64 AVX2, AVX512
- **Perfect Forward Secrecy 보장(2분 간격 Handshaking)**
  - 1-RTT만에 Handshaking 완료
- **Replay Attack 방지 기능(Counter 필드 유지/관리)**
- **Identity Hiding 기능**
- **DoS 공격 완화 기능(Cookie 활용)**

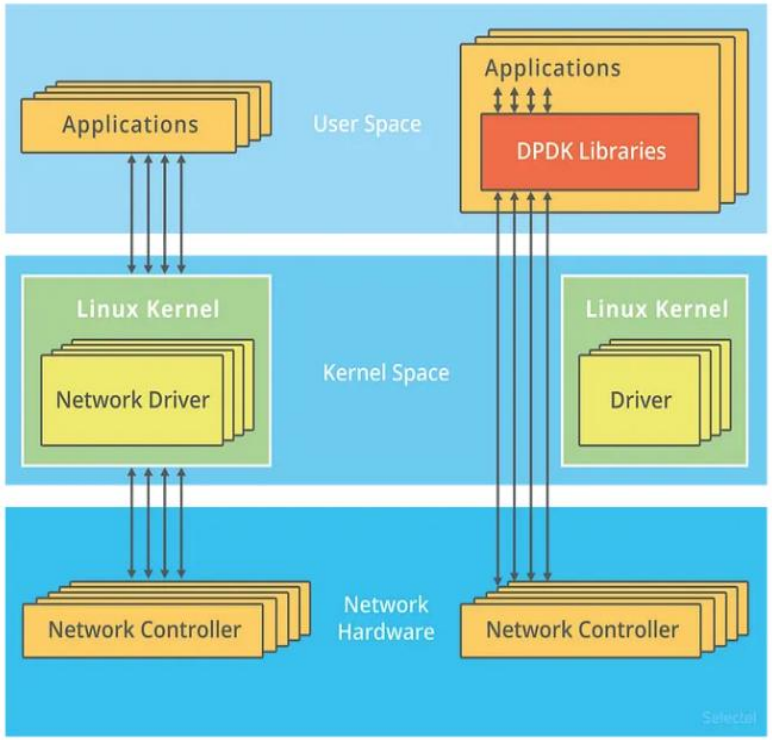
3. DPDK & FD.io VPP

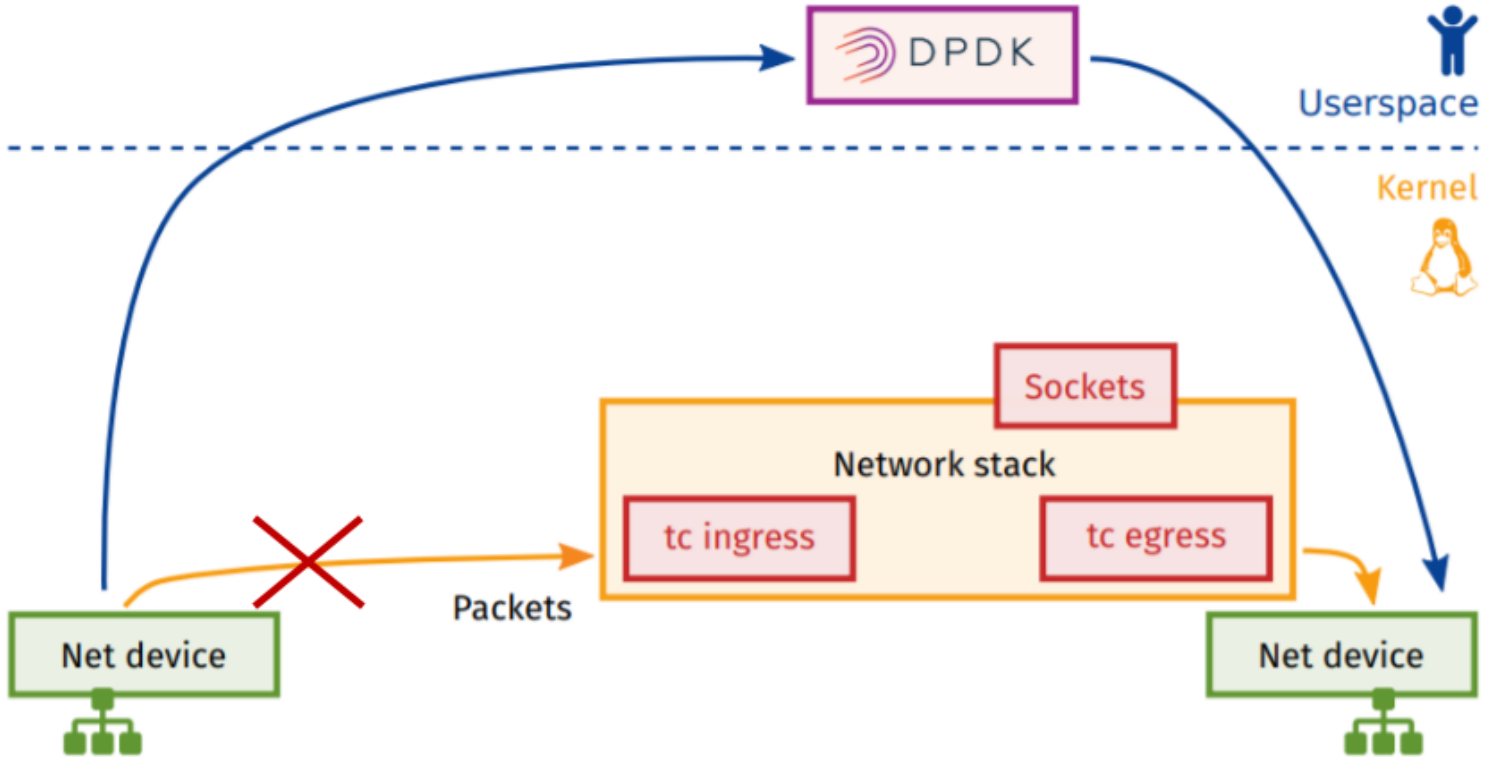


Software ASIC

FD.io VPP and FD.io logo are registered trademarks of The Fast Data Project, a series of LF Projects, LLC

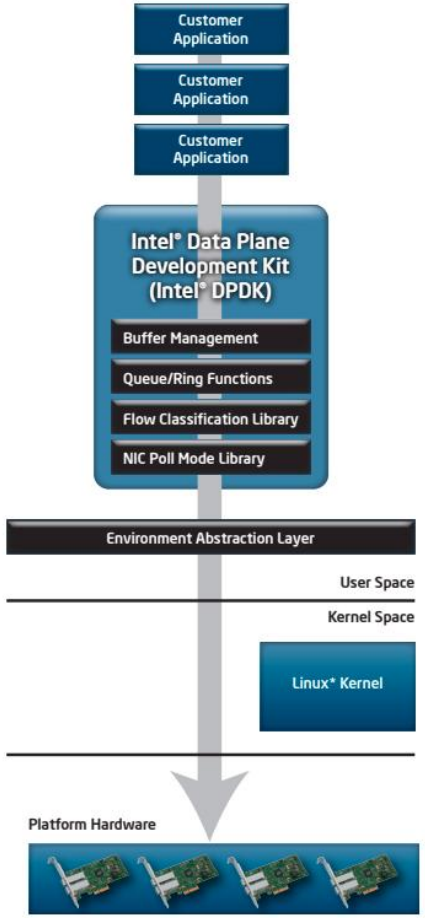
3. DPDK & FD.io VPP(1) - DPDK Overview(1)

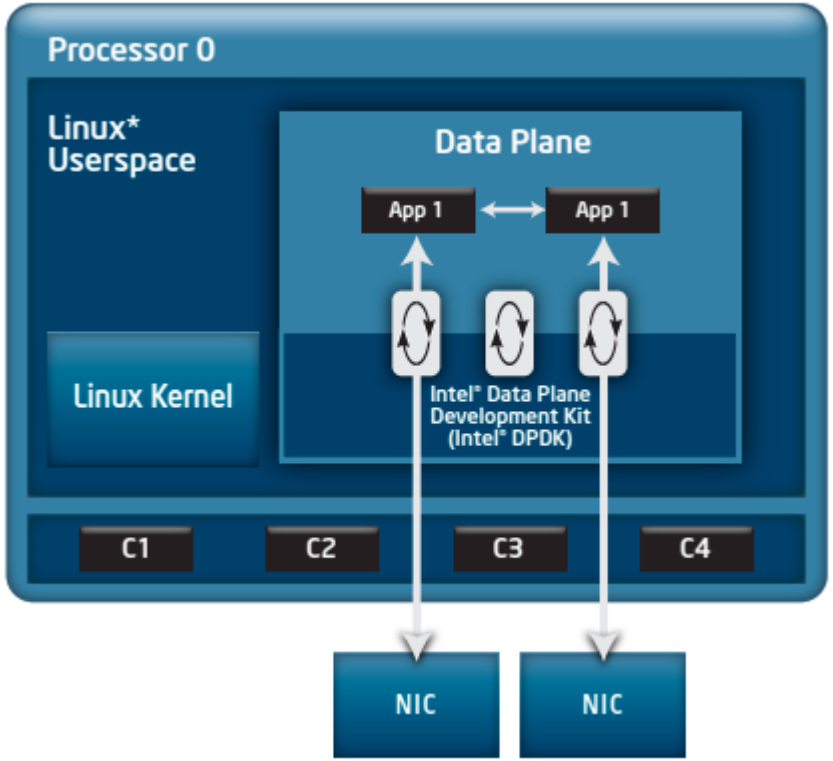




Ref) https://media.frnog.org/FRnOG_28/FRnOG_28-3.pdf

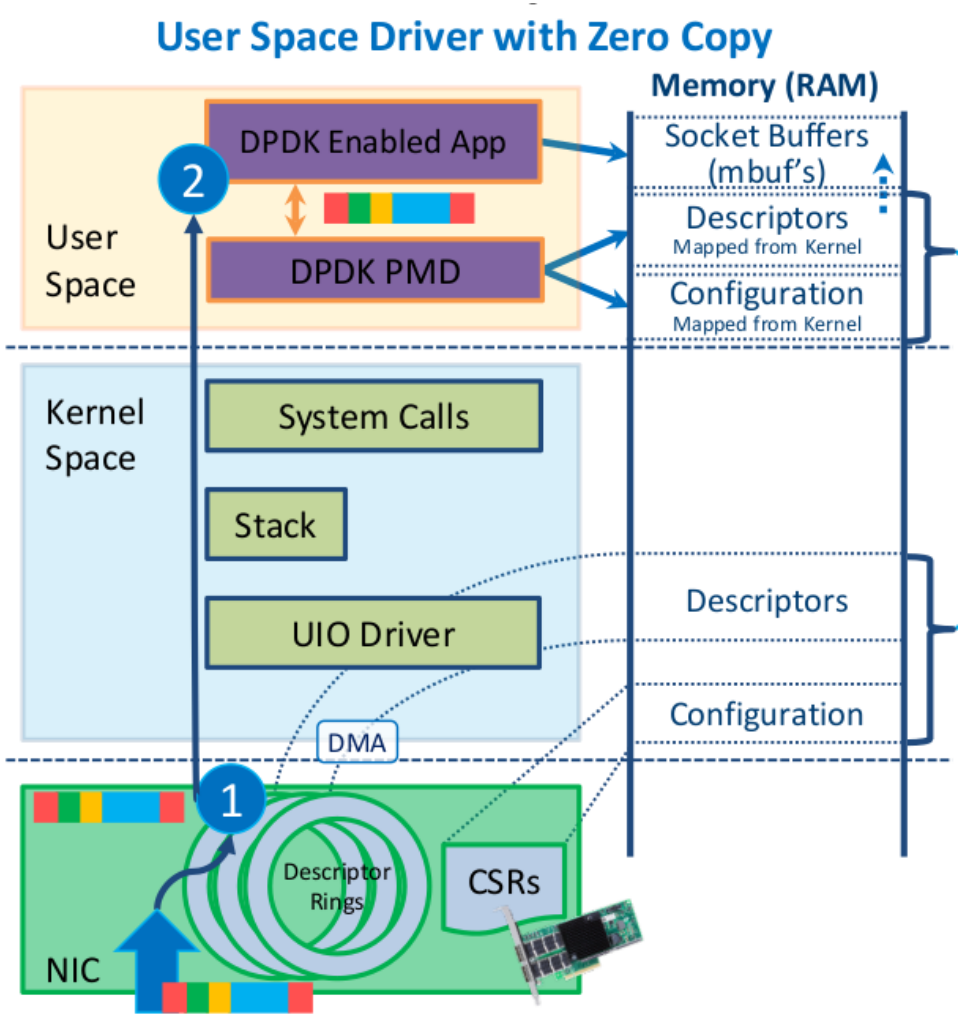
3. DPDK & FD.io VPP(1) - DPDK Overview(2)

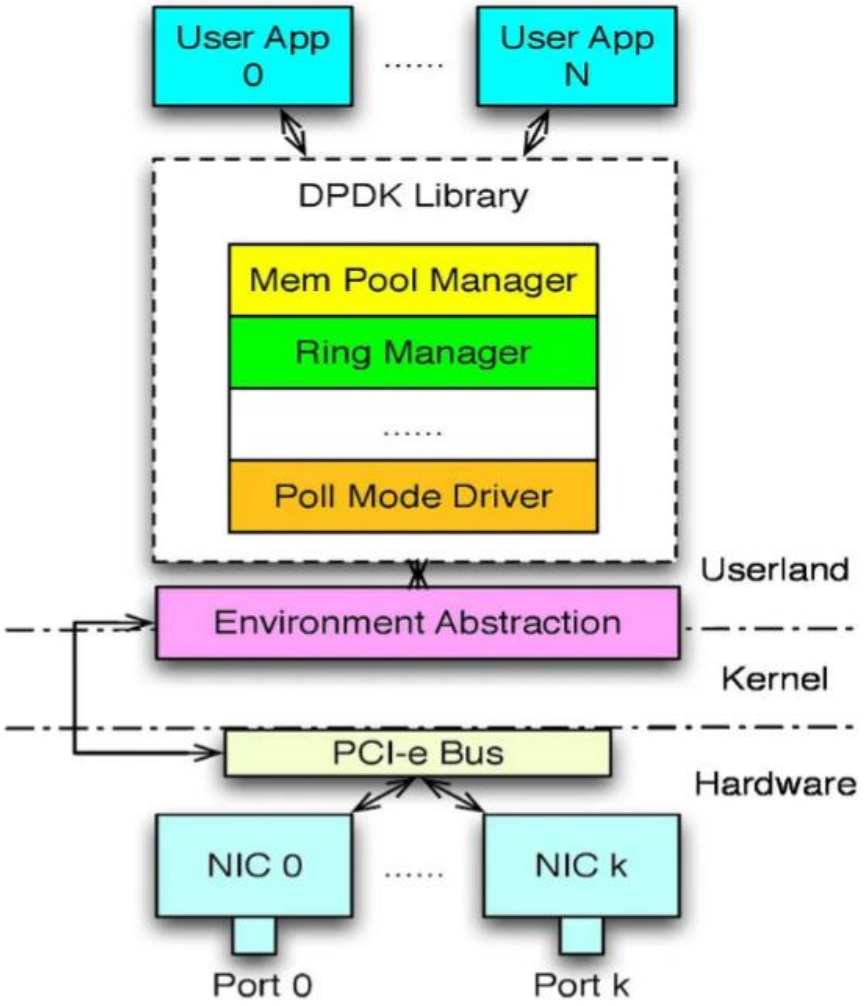




[출처] Impressive Packet Processing Performance Enables Greater Workload Consolidation

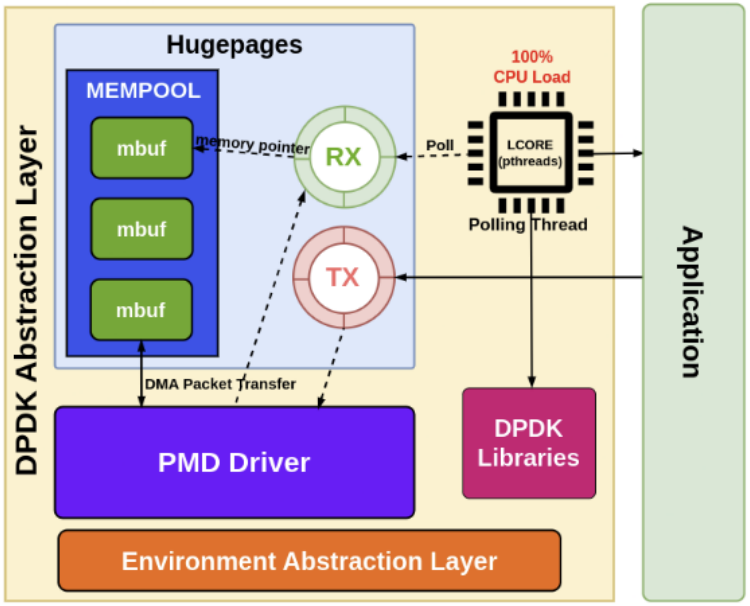
3. DPDK & FD.io VPP(1) - DPDK Overview(3)

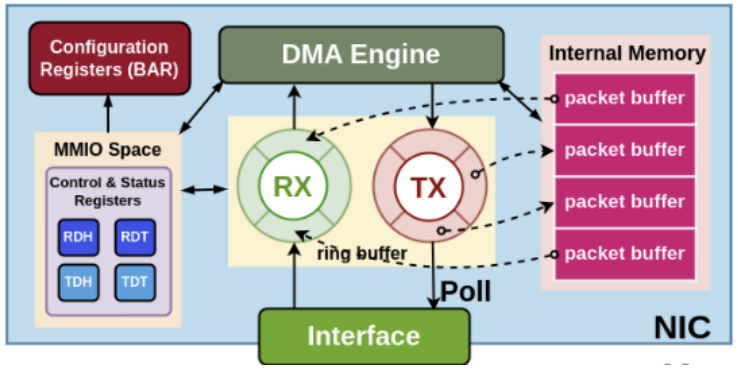


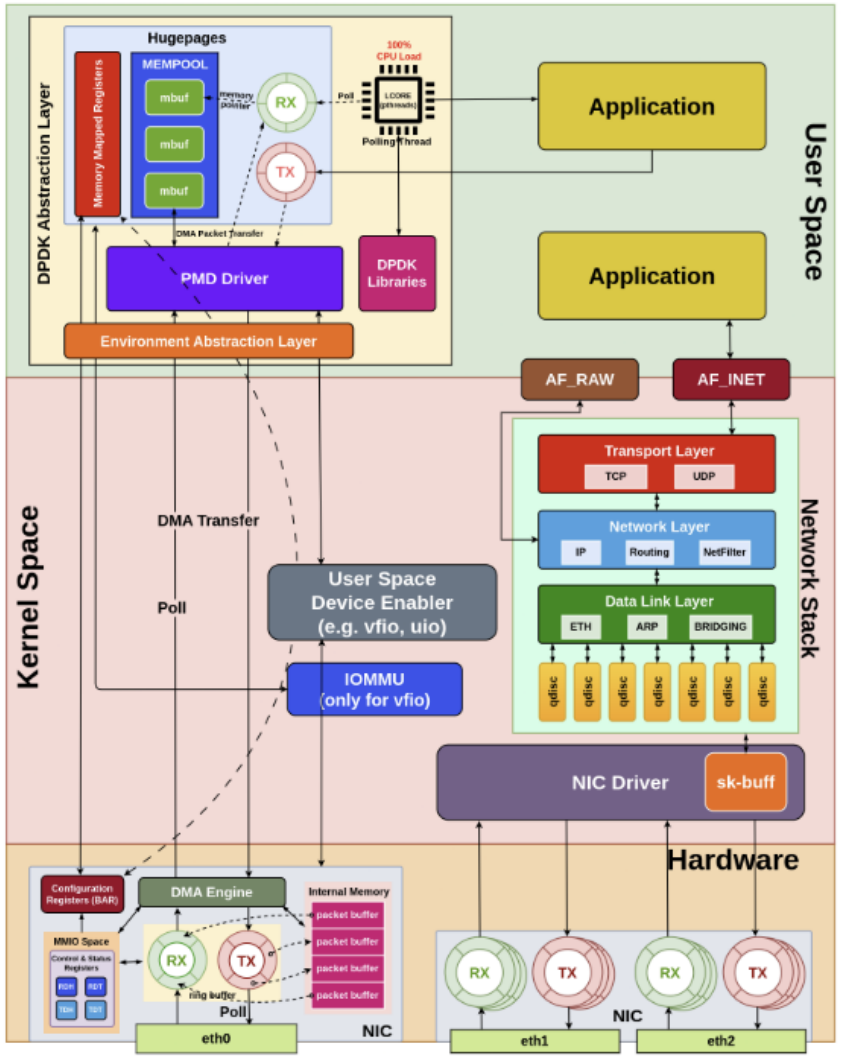


Ref) https://medium.com/google-cloud/forwarding-over-100-mpps-with-fd-io-vpp-on-x86-62b9447da554

3. DPDK & FD.io VPP(1) - DPDK Overview(4)



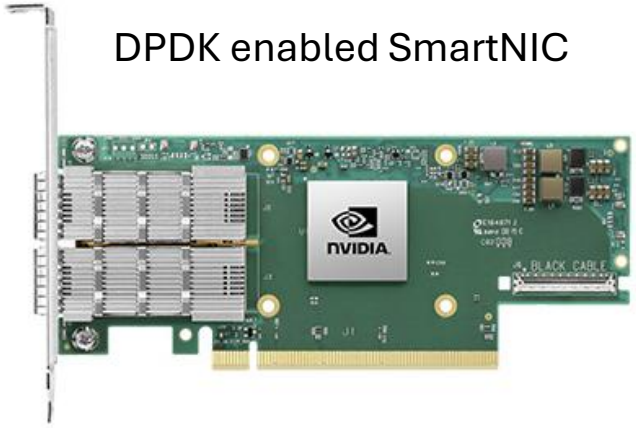




Ref) https://docs.ngkore.com/blogs/dpdk/dpdk.html#hardware-requirements

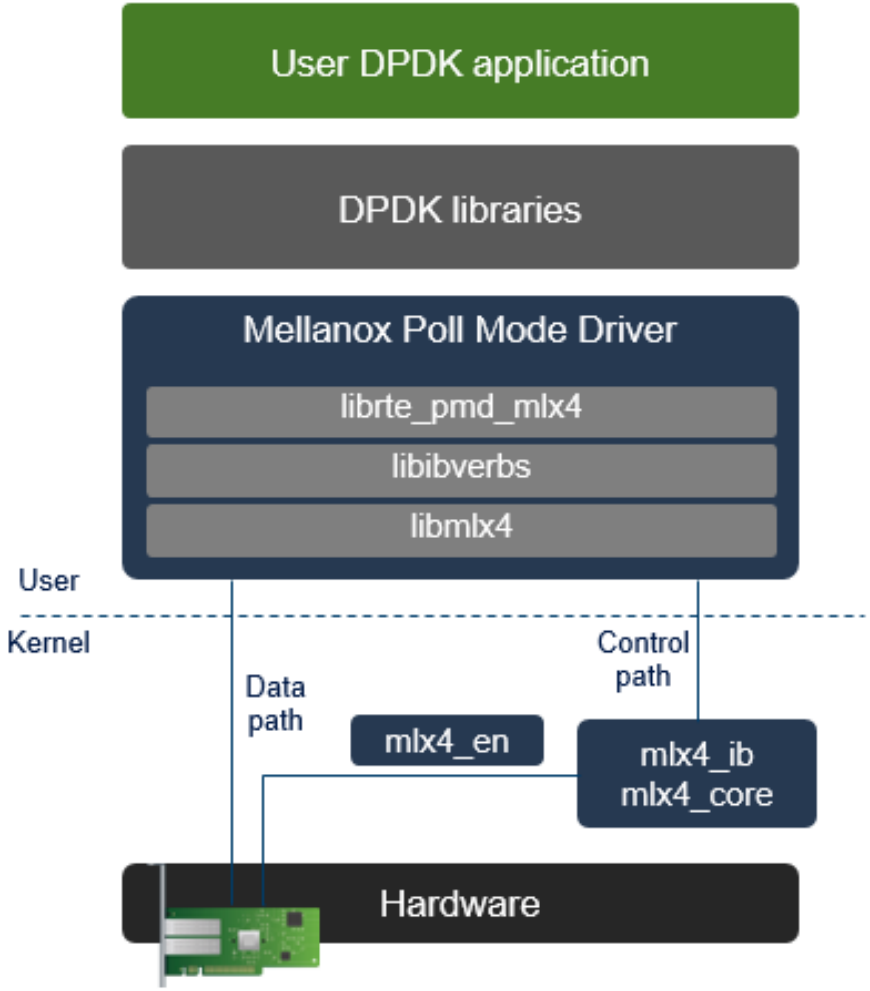
3. DPDK & FD.io VPP(1) - DPDK Overview(5)

DPDK enabled SmartNIC

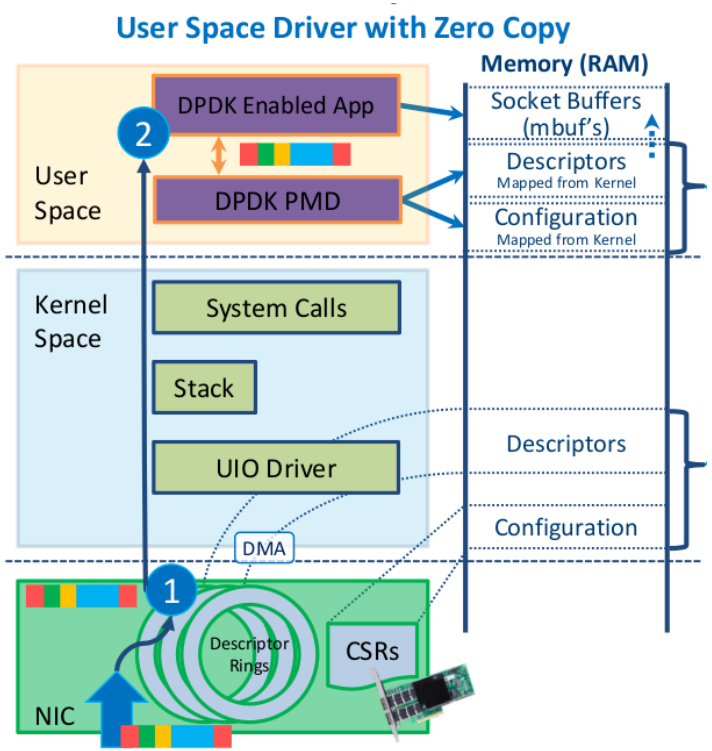


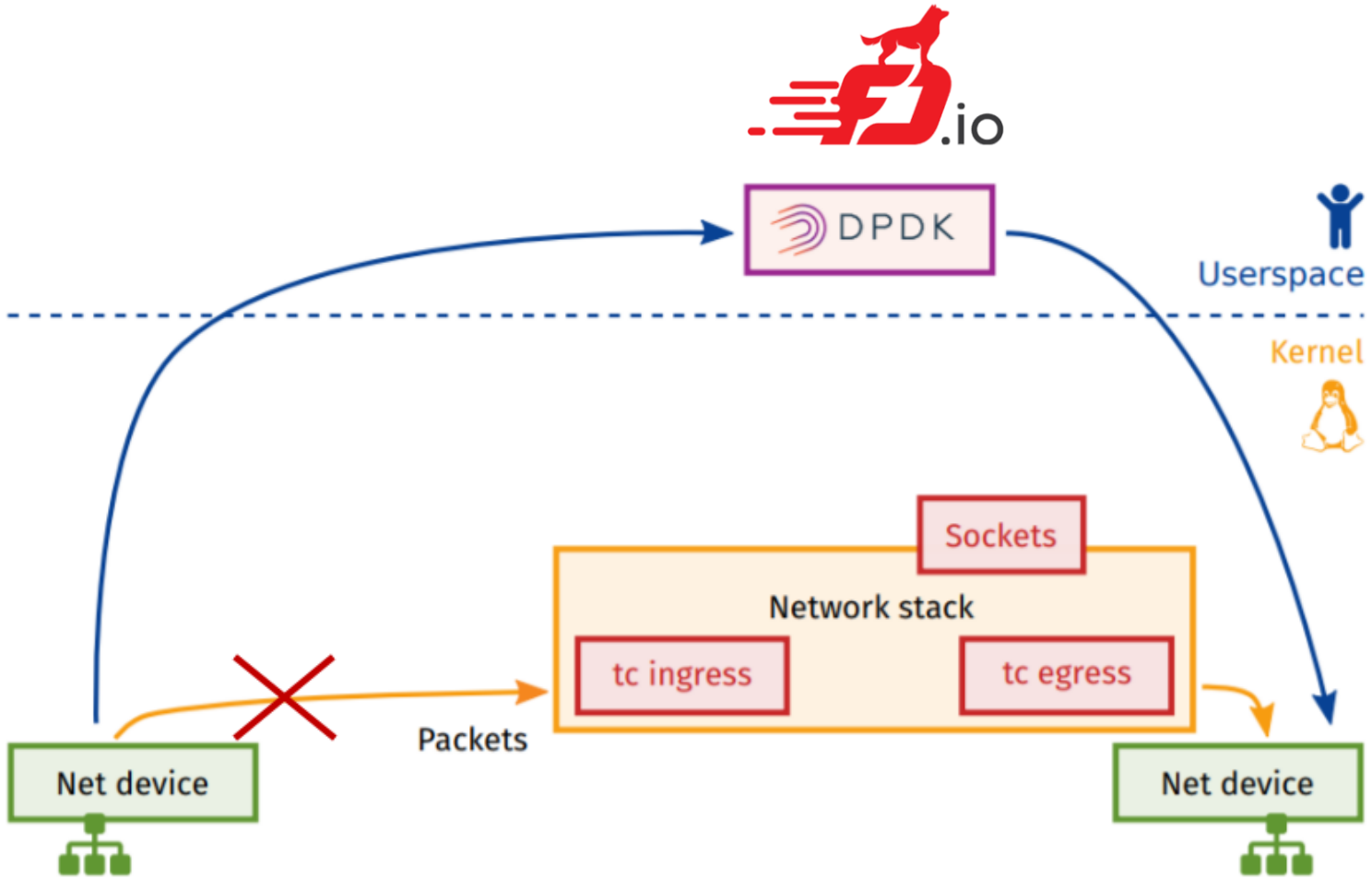
NVIDIA (Old Mellanox) SmartNIC

Ref) https://www.nvidia.com/ko-kr/networking/ethernet/connectx-6/



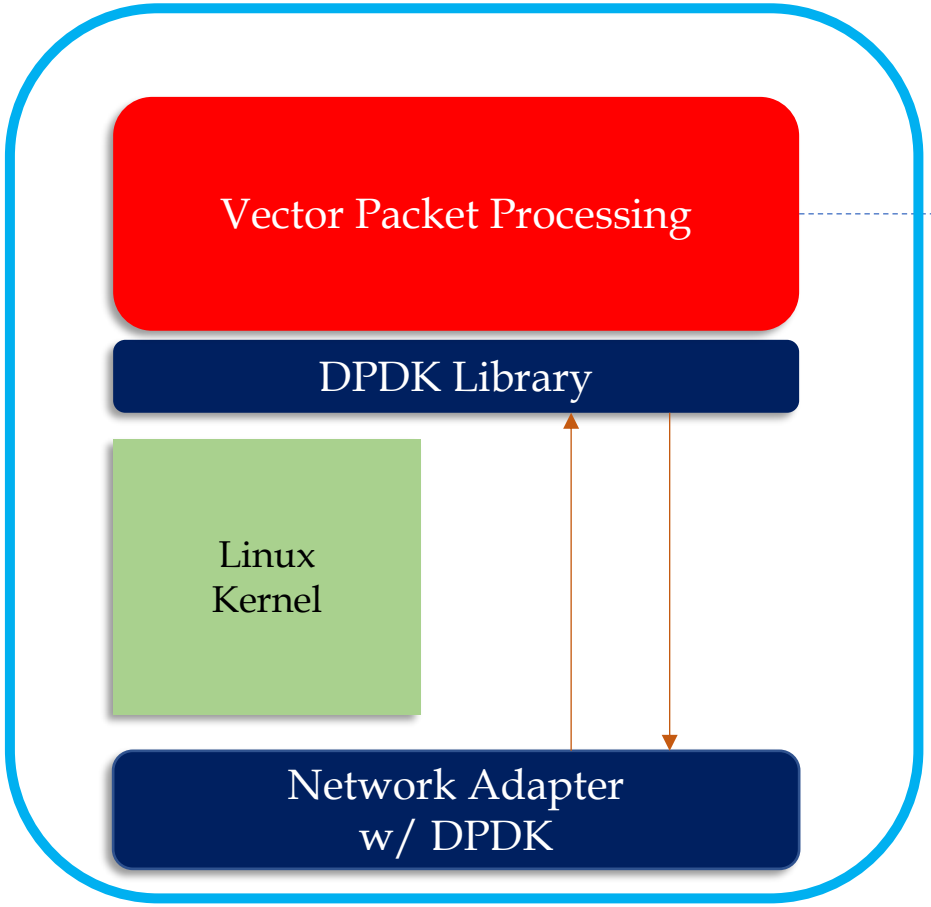
3. DPDK & FD.io VPP(2) - VPP Overview(1)



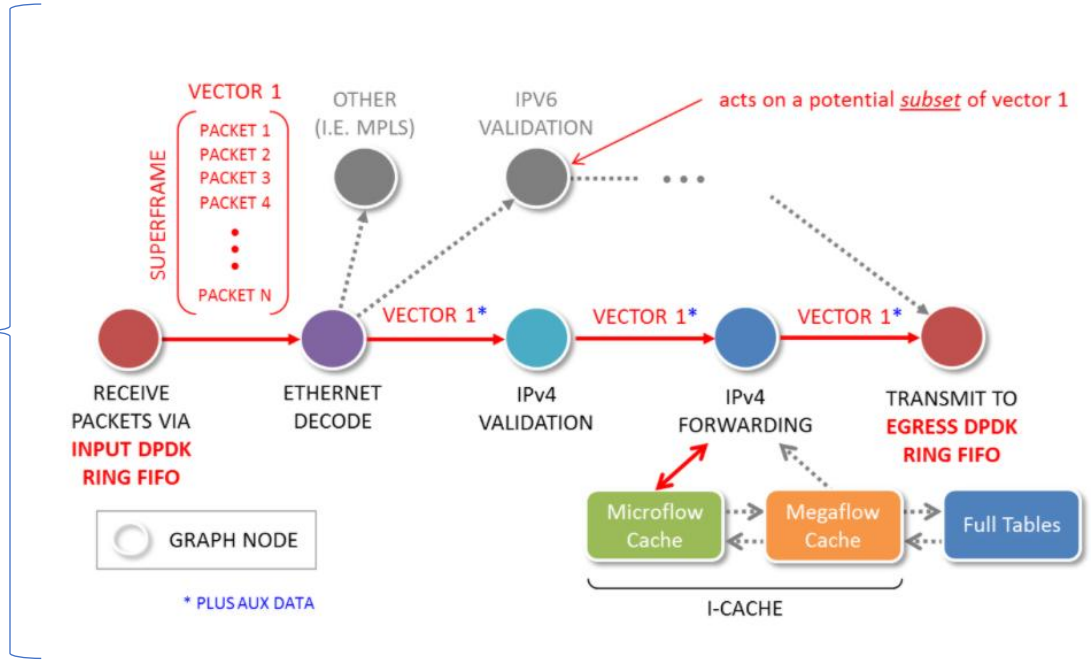


Ref) https://media.frnog.org/FRnOG_28/FRnOG_28-3.pdf

3. DPDK & FD.io VPP(2) - VPP Overview(2)







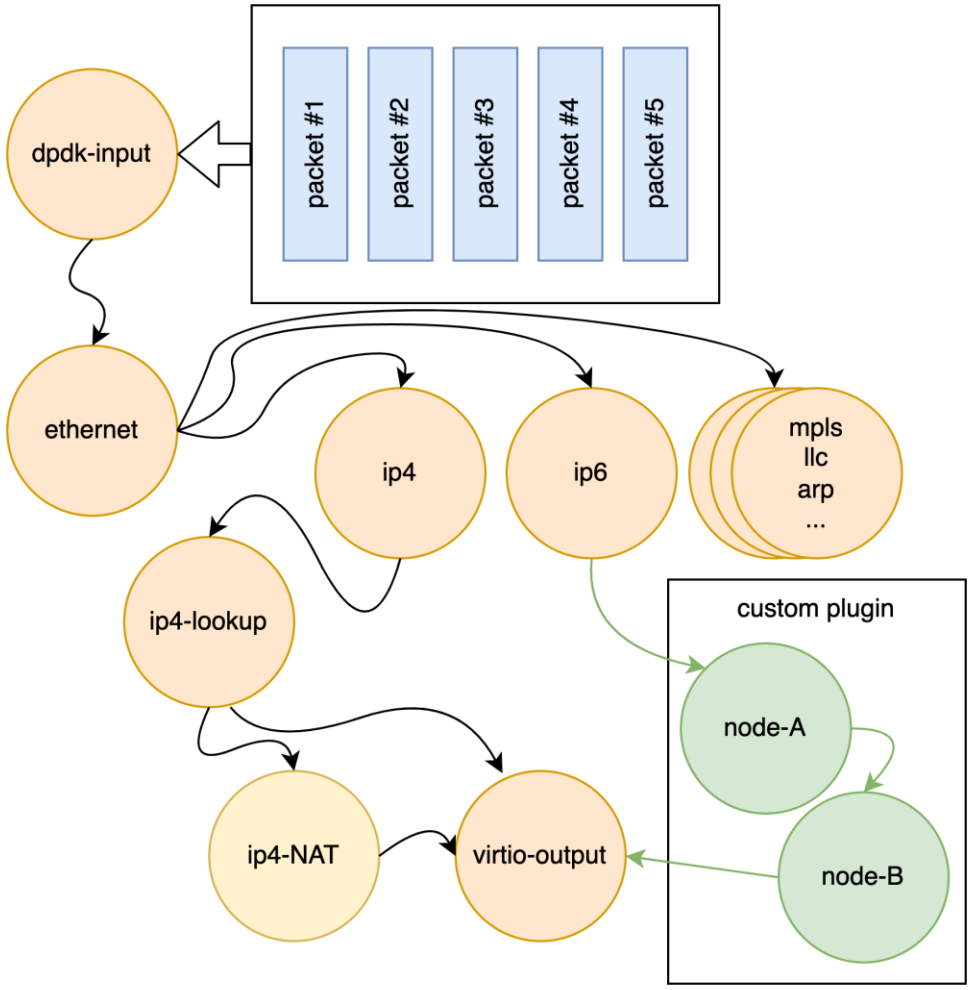


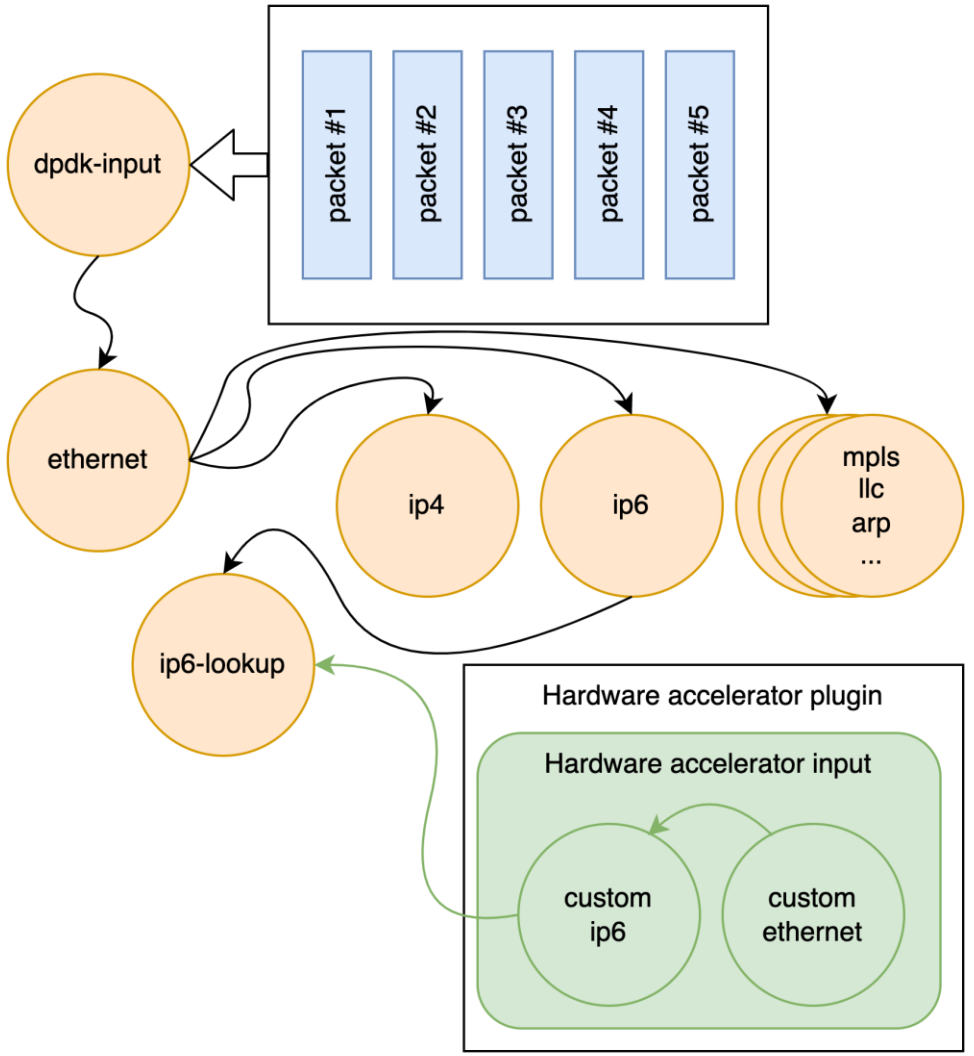
VPP

Ref) https://fd.io/technology/

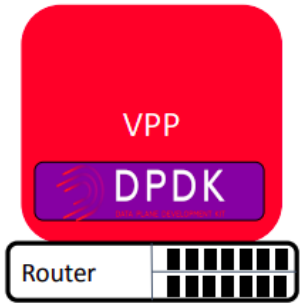
3. DPDK & FD.io VPP(2) - VPP Overview(3)

Ref) https://fd.io/technology/

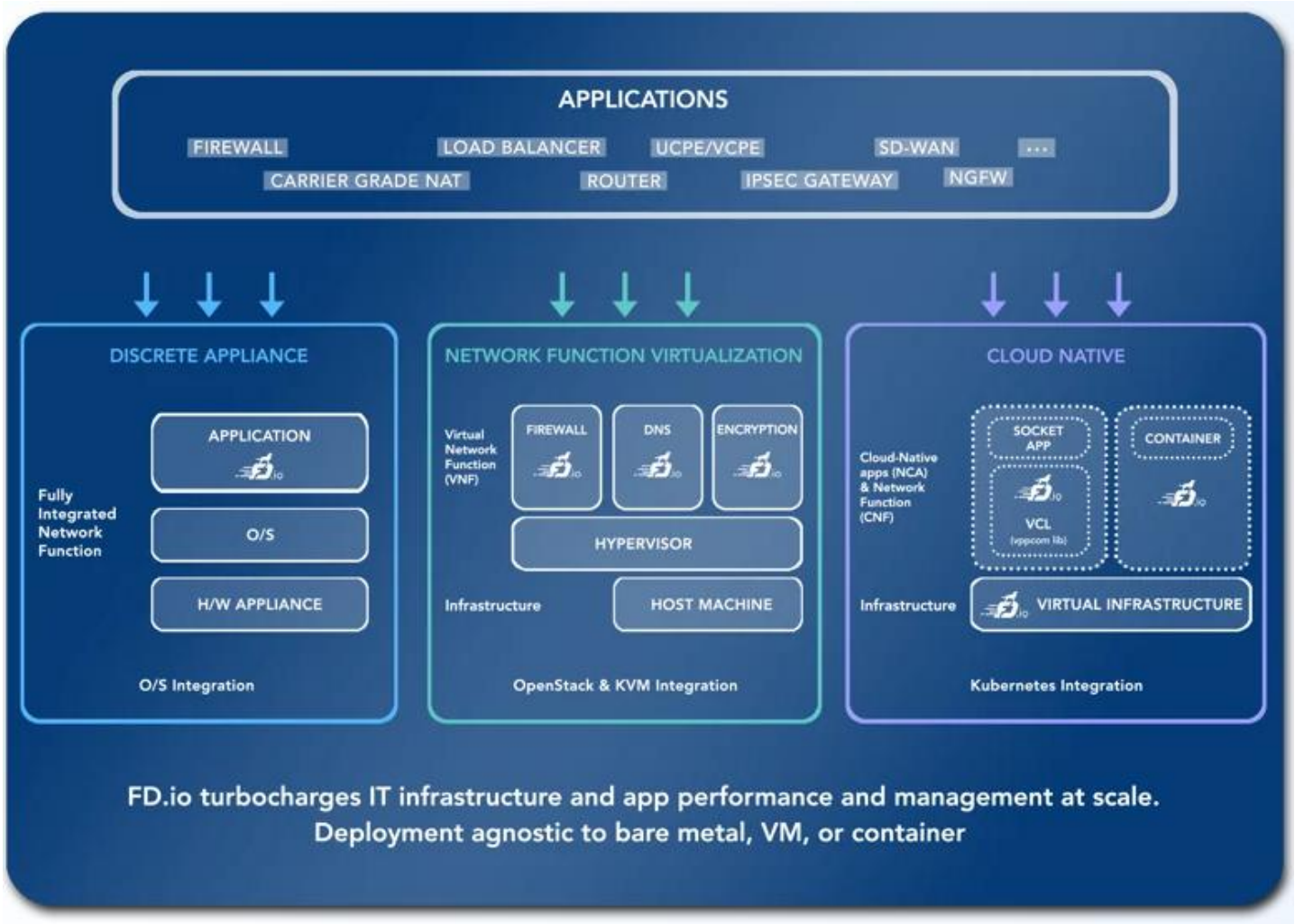




3. DPDK & FD.io VPP(2) - VPP Overview(4)



- L2 Switch
- L3 Router
- CGNAT
- L4 Switch
- WireGuard VPN
- IPSec VPN
- IPS
- VxLAN Gateway
- …

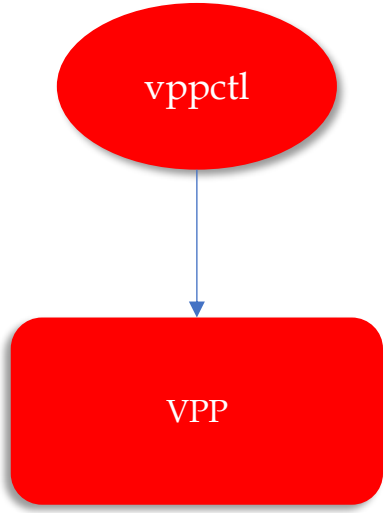


Ref) https://www.netgate.com/resources/articles-vector-packet-processing

3. DPDK & FD.io VPP(2) - VPP Overview(5)

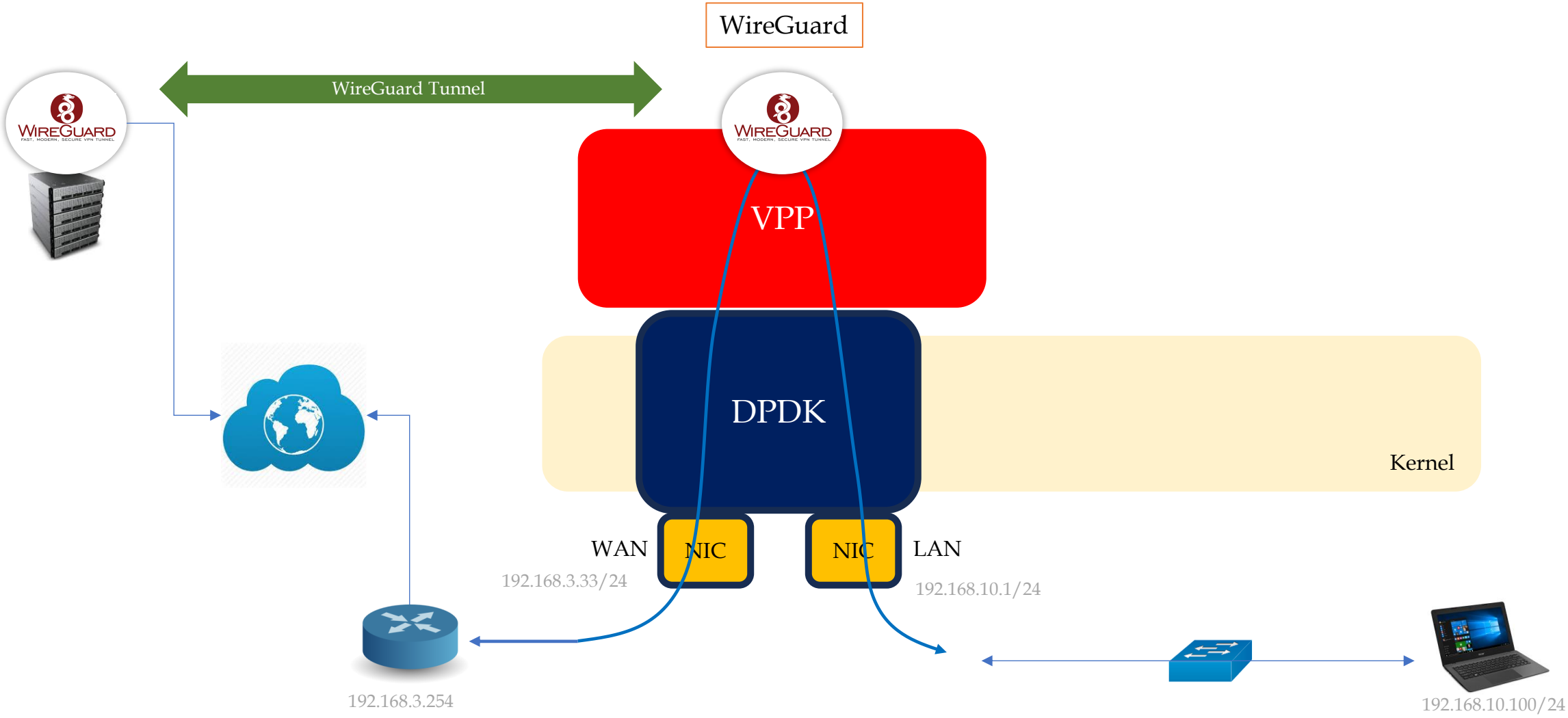
```
chyi@mightycn:~/workspace/mightycn$ sudo vppctl
    _______    _        _   _____  ___
 __/ __/ _ \  (_)__    | | / / _ \/ _ \
 _/ _// // / / / _ \   | |/ / ___/ ___/
 /_/ /____(_)_/\___/   |___/_/  /_/

vpp#
vpp# show interface
              Name               Idx    State  MTU (L3/IP4/IP6/MPLS)     Counter          Count
enp2s0                            1      up          9000/0/0/0     rx packets              1532
                                                                    rx bytes              204225
                                                                    tx packets              1398
                                                                    tx bytes              162174
                                                                    drops                    165
                                                                    punt                     219
                                                                    ip4                     1296
                                                                    ip6                       90
                                                                    tx-error                   1
local0                            0     down          0/0/0/0     drops                      2
tap4096                           2      up          9000/0/0/0     rx packets              157
                                                                    rx bytes              11050
                                                                    tx packets              198
                                                                    tx bytes              24133
                                                                    drops                     14
                                                                    ip4                       91
                                                                    ip6                       15
tun4097                           4      up          1420/0/0/0     rx packets              882
                                                                    rx bytes              50308
                                                                    tx packets              869
                                                                    tx bytes              67456
                                                                    ip4                     871
                                                                    ip6                      11
wg0                               3      up          1420/0/0/0     rx packets              993
```

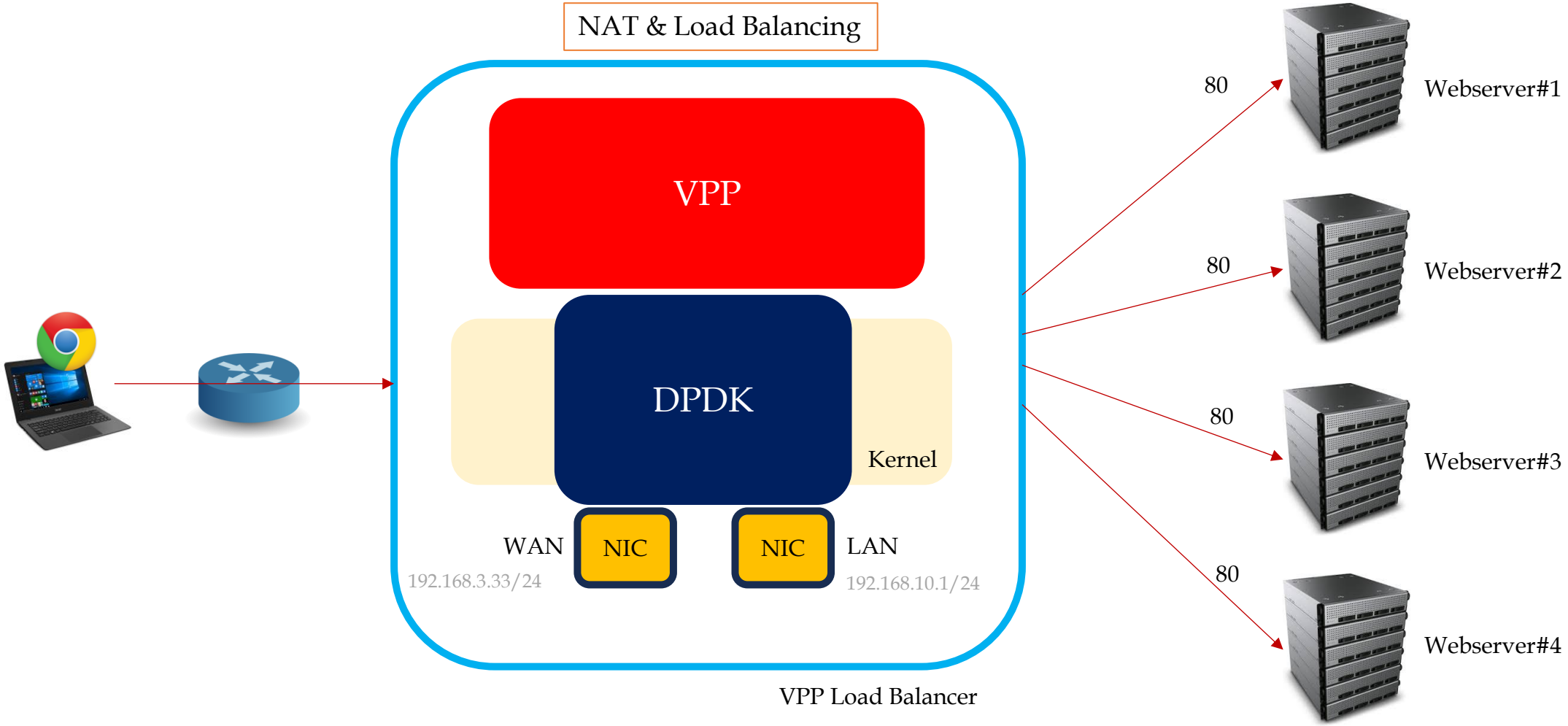


/usr/bin/vpp –c /etc/vpp/startup.conf

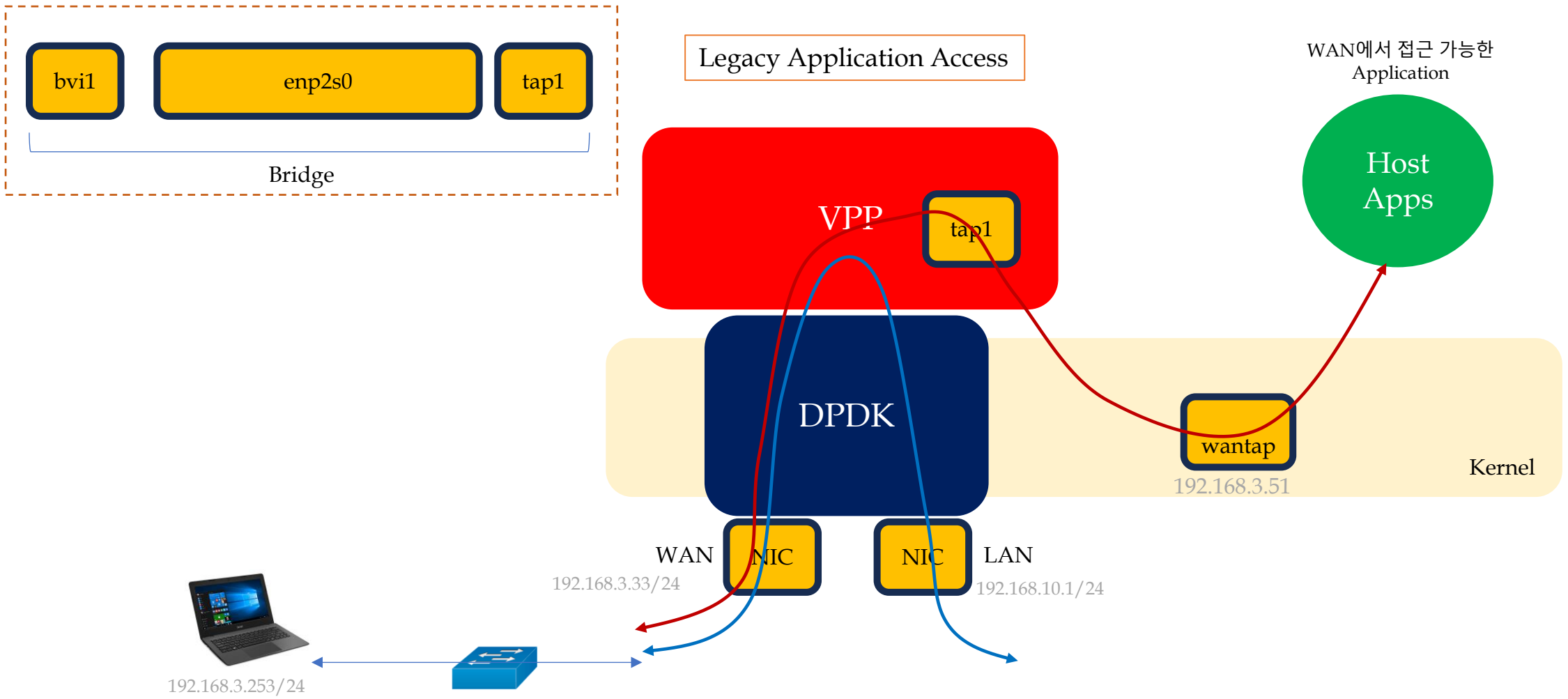
3. DPDK & FD.io VPP(2) - VPP Overview(6)



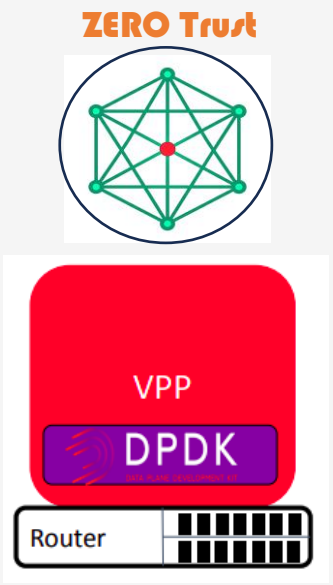
3. DPDK & FD.io VPP(2) - VPP Overview(7)



3. DPDK & FD.io VPP(2) - VPP Overview(8)

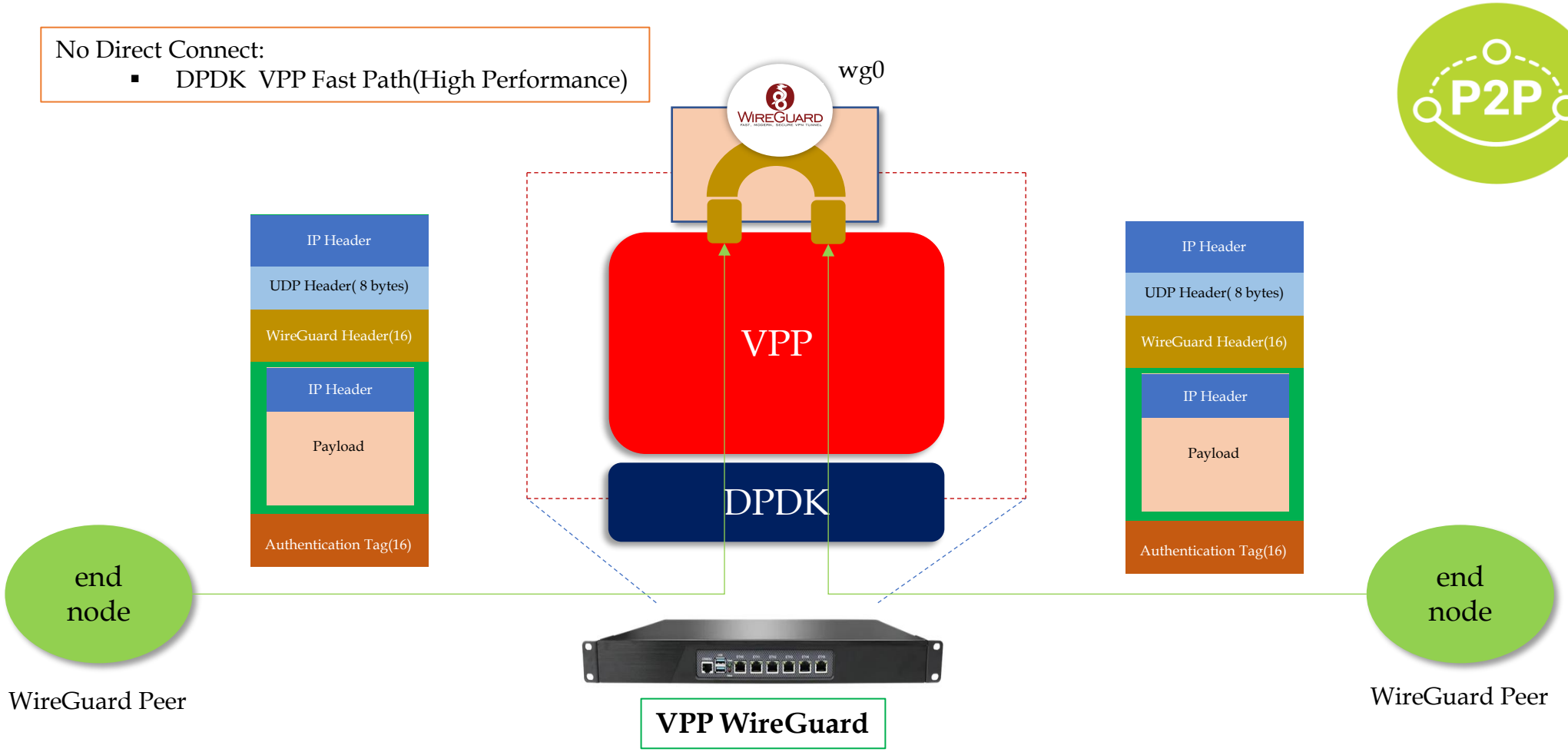


4. MightyConnect Security Box

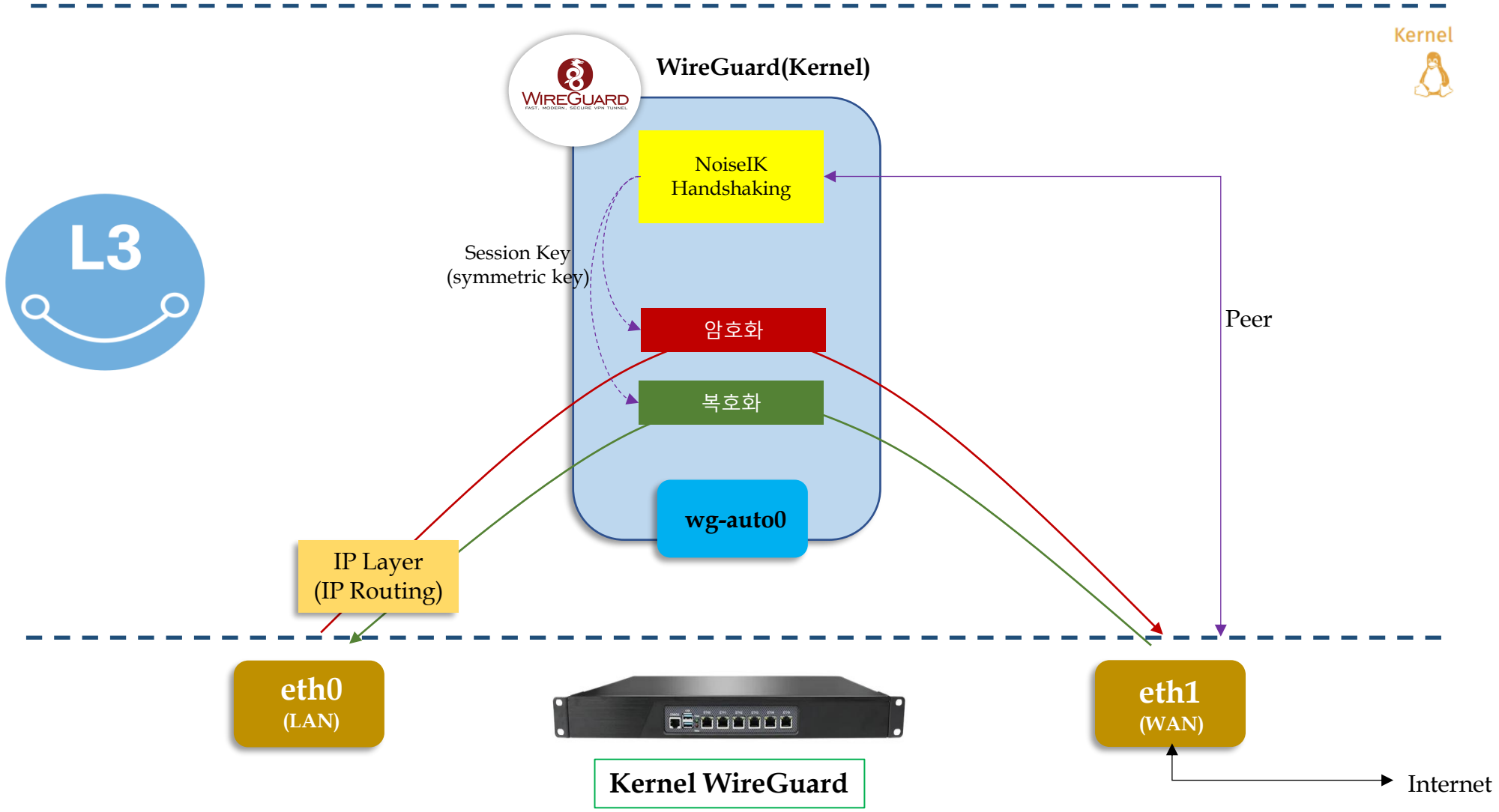


Software ASIC

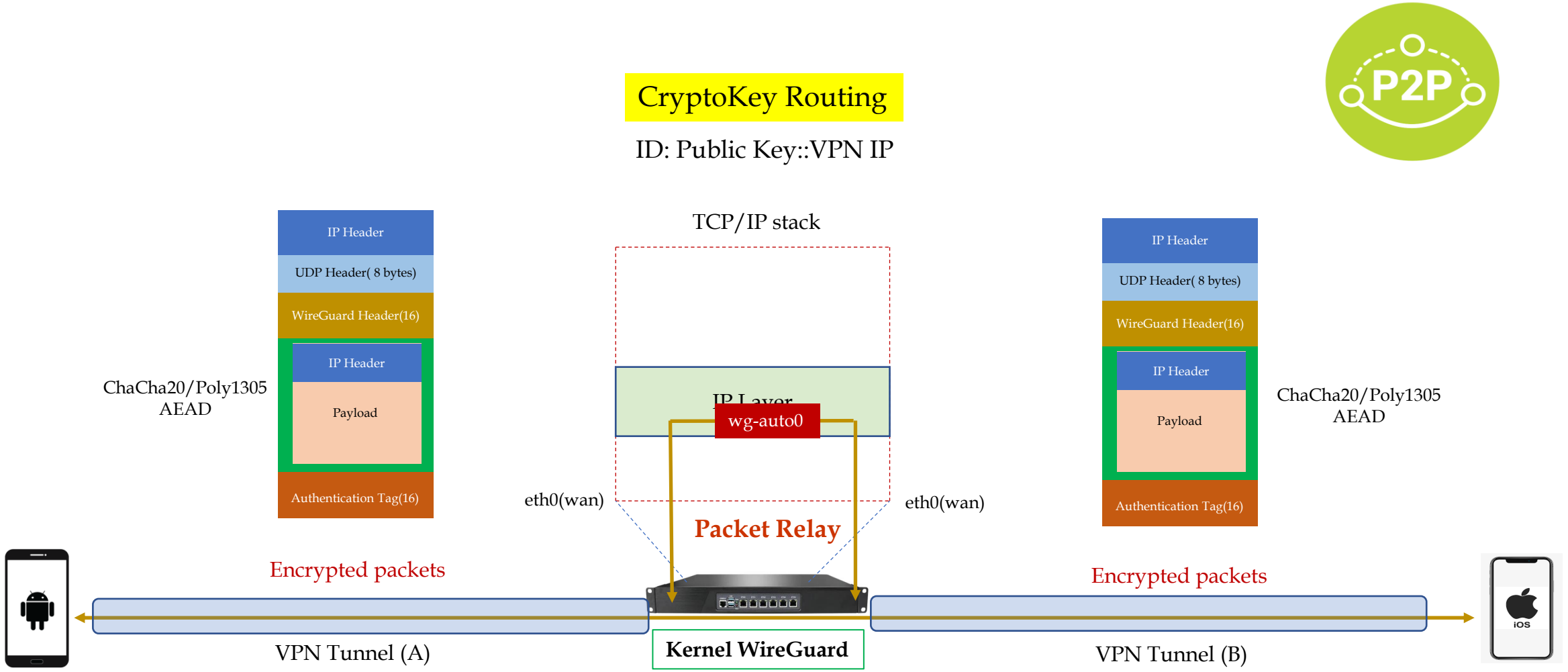
4. MightyConnect Security Box(1) – WireGuard(1)



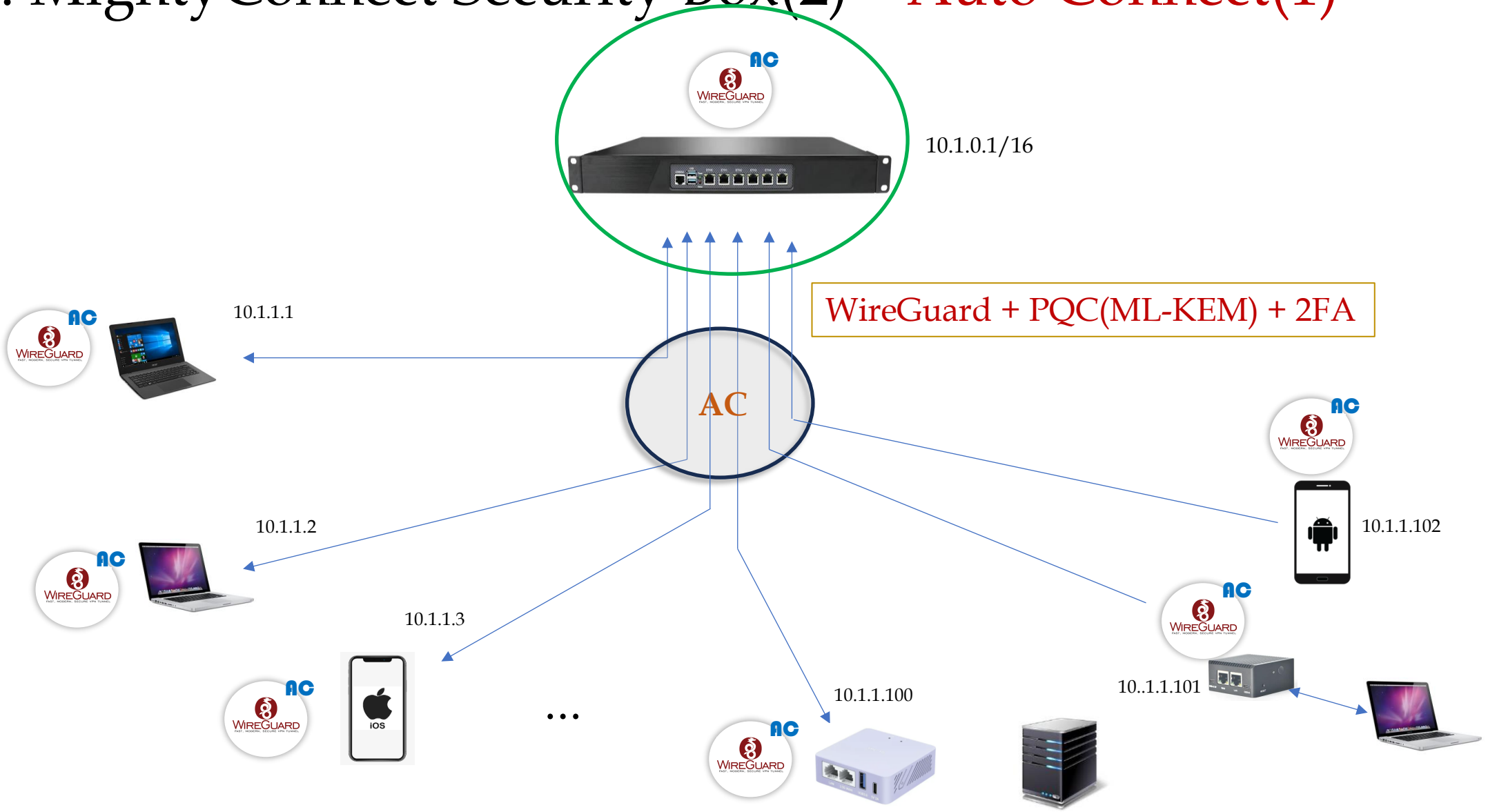
4. MightyConnect Security Box(1) – WireGuard(2-1)



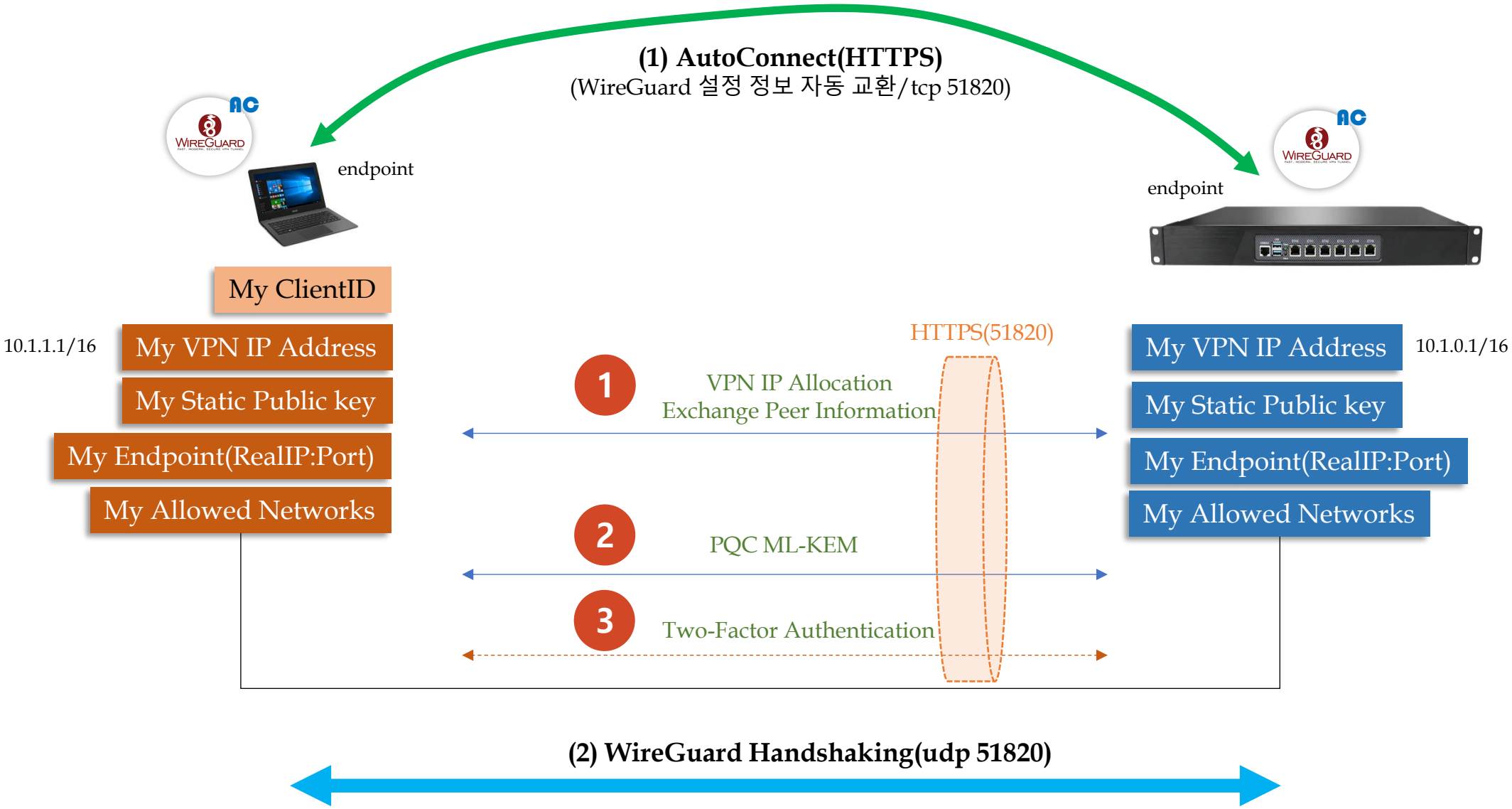
4. MightyConnect Security Box(1) – WireGuard(2-2)



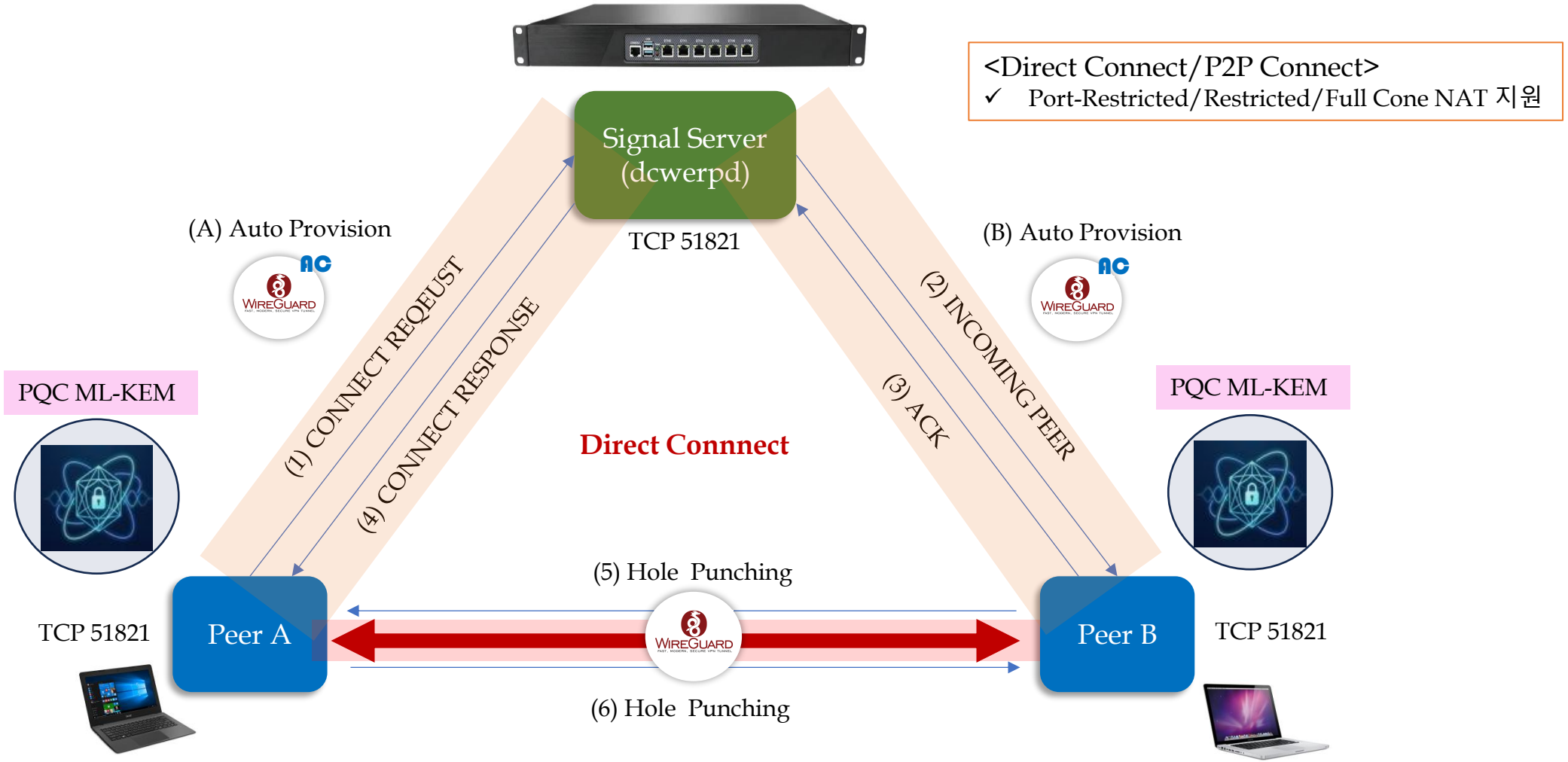
4. MightyConnect Security Box(2) – Auto Connect(1)



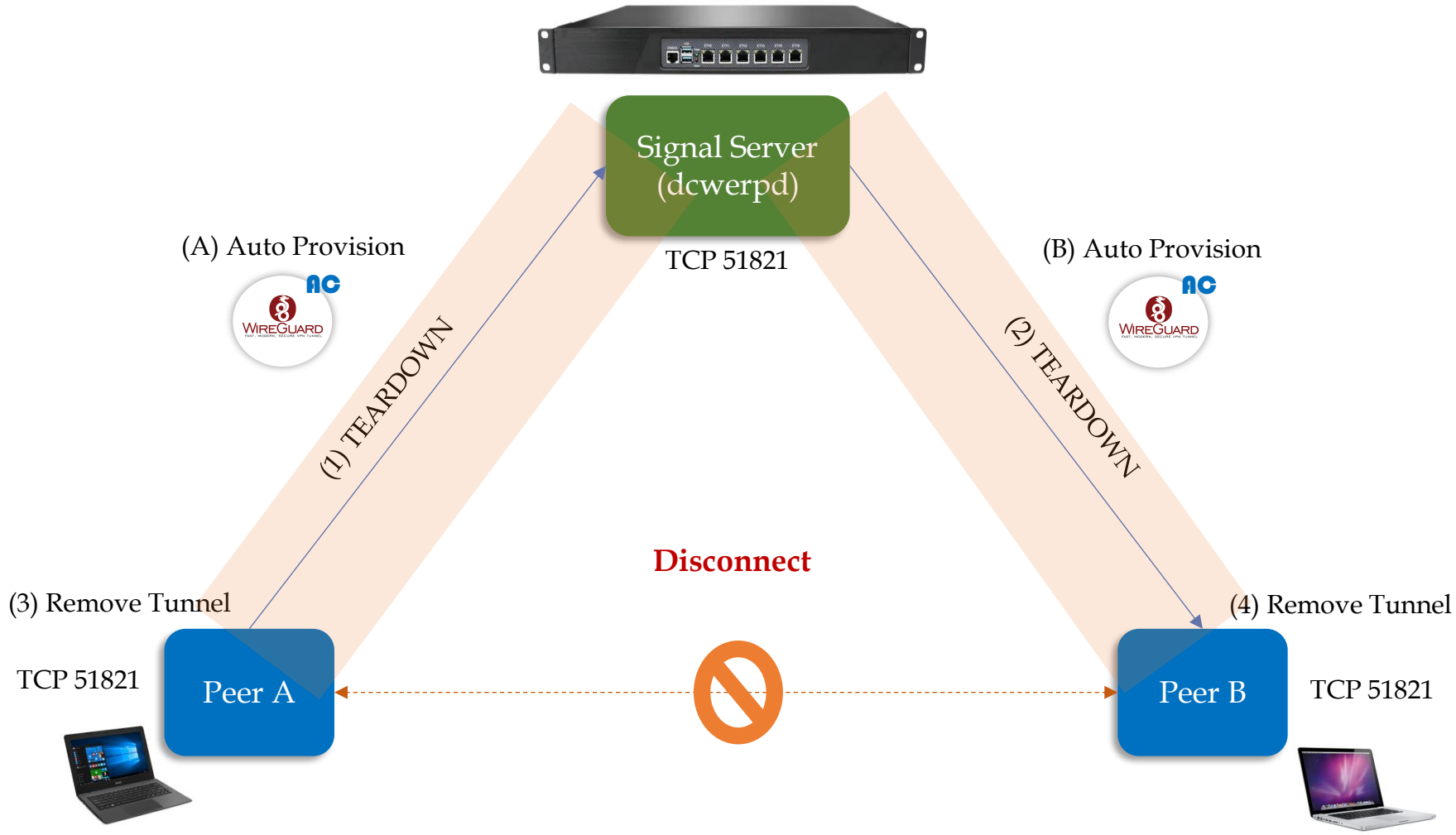
4. MightyConnect Security Box(2) – Auto Connect(2)



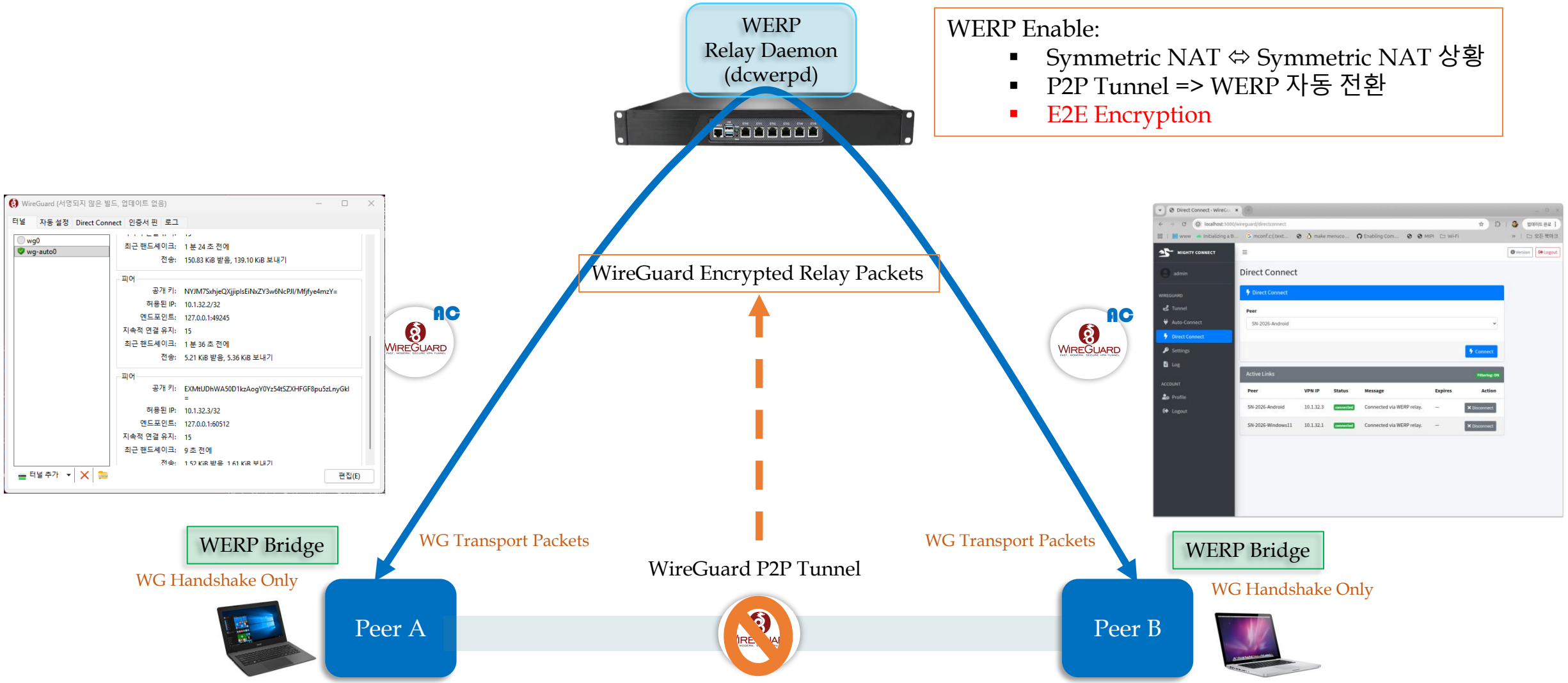
4. MightyConnect Security Box(3) – Direct Connect(1)



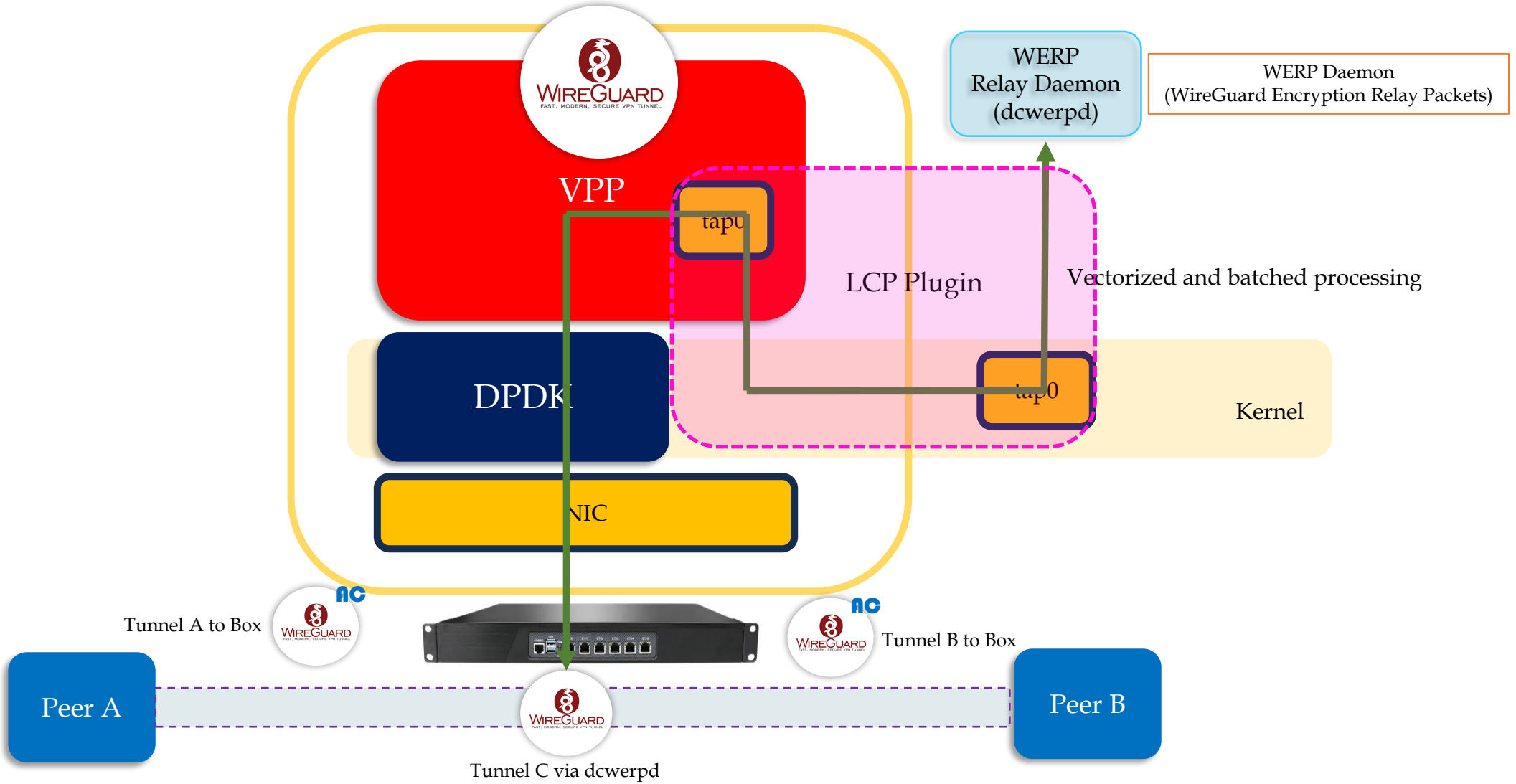
4. MightyConnect Security Box(3) – Direct Connect(2)



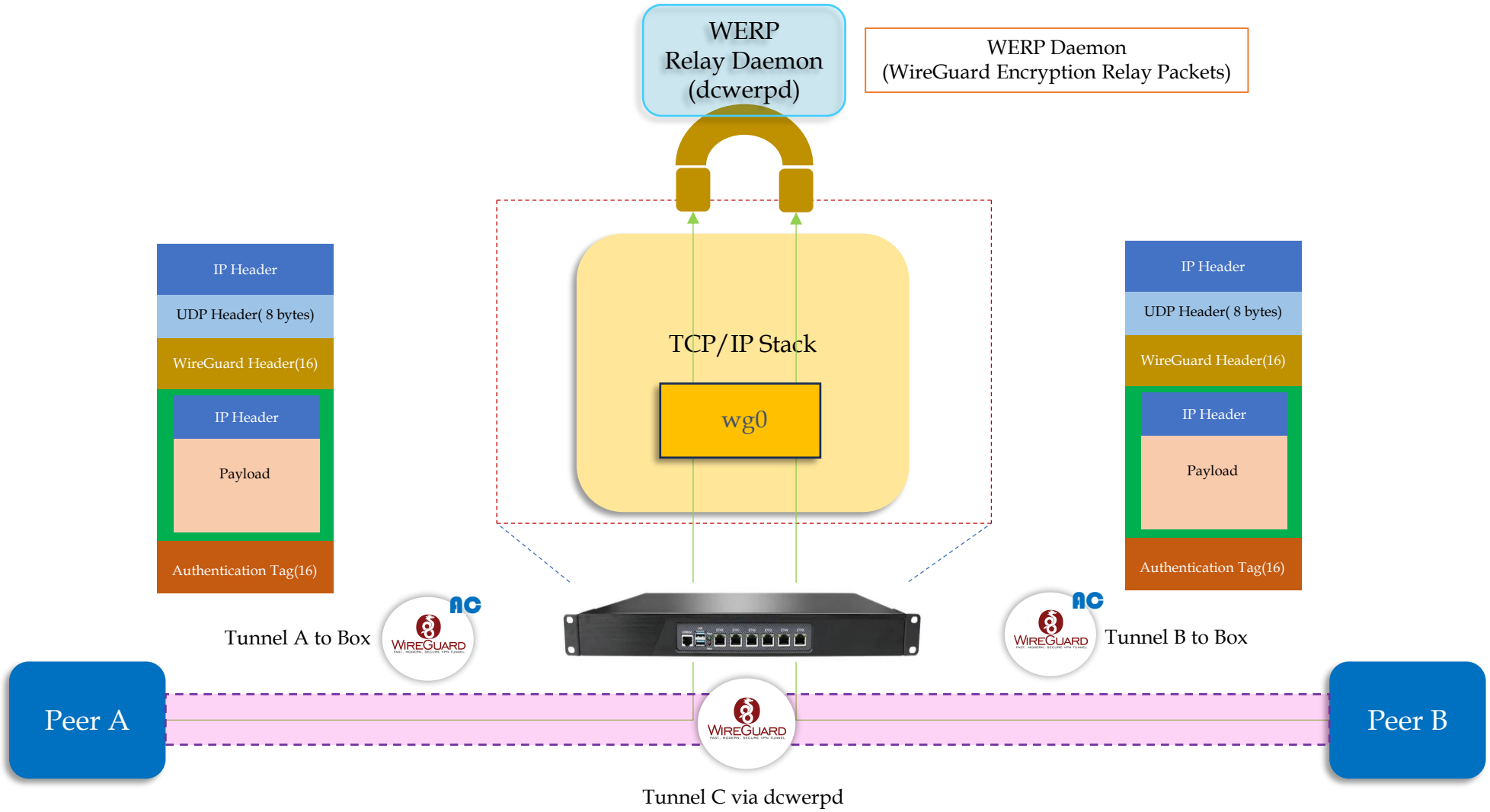
4. MightyConnect Security Box(3) – Direct Connect(3-1)



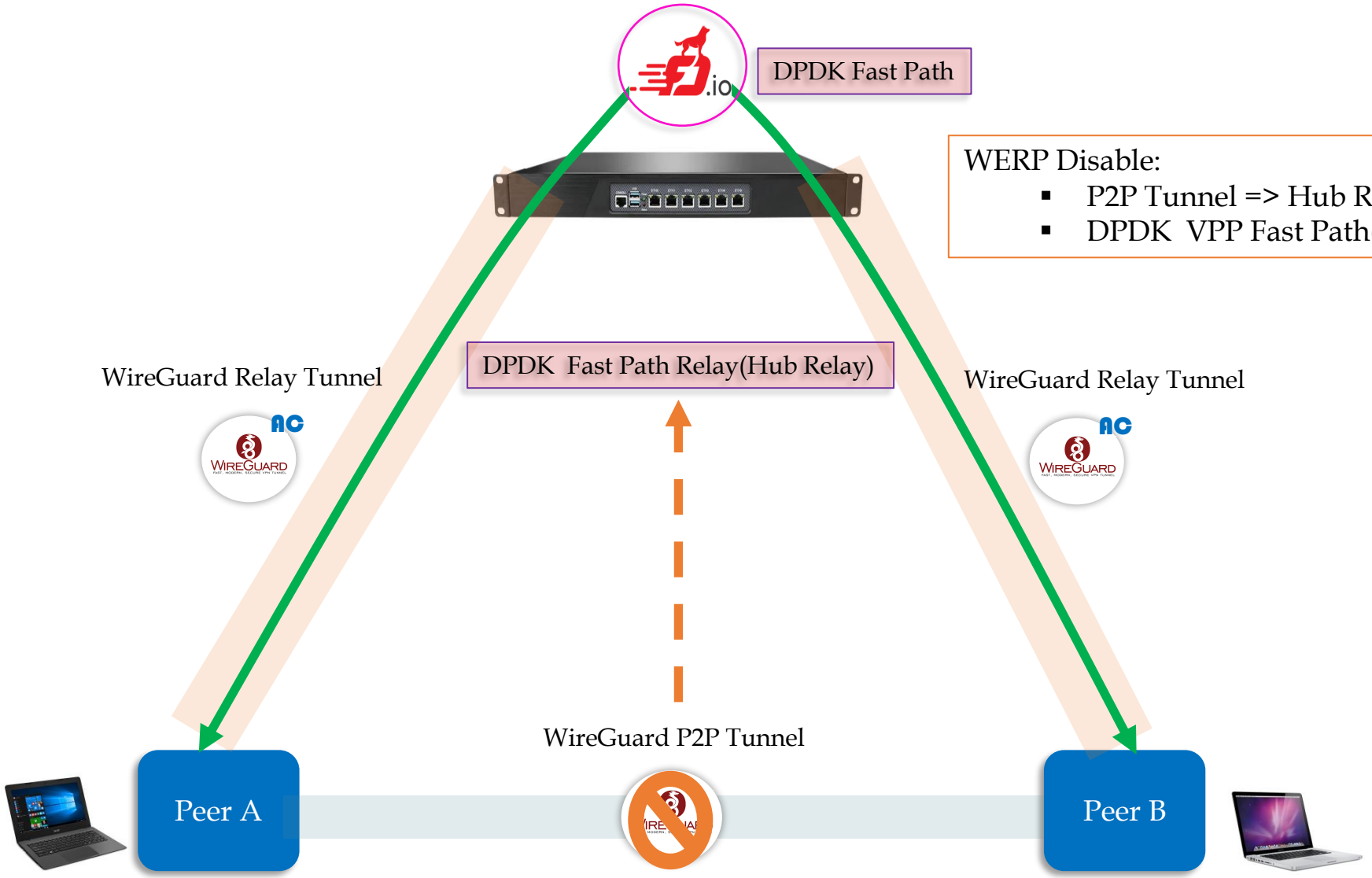
4. MightyConnect Security Box(3) – Direct Connect(3-2)



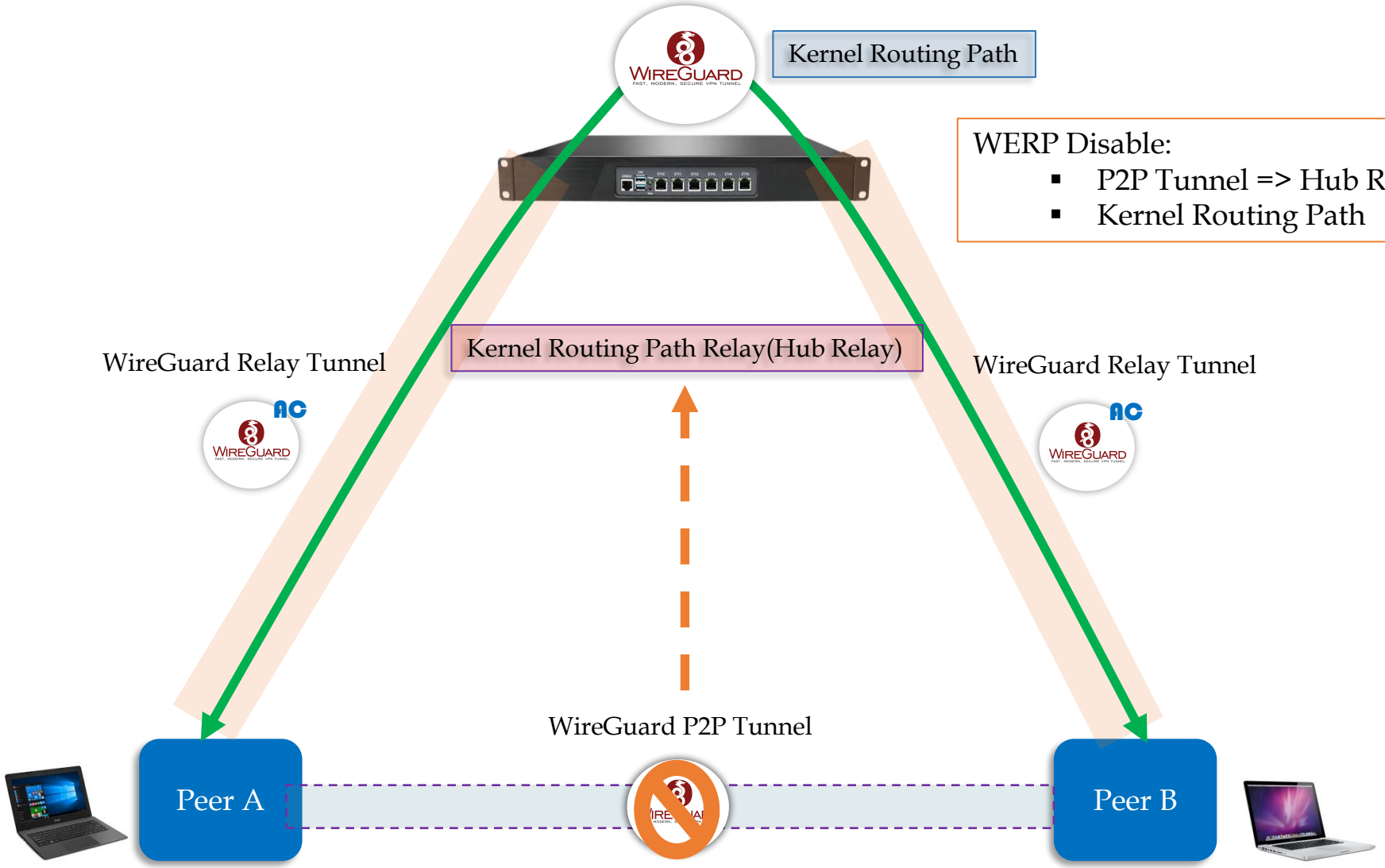
4. MightyConnect Security Box(3) – Direct Connect(3-3)



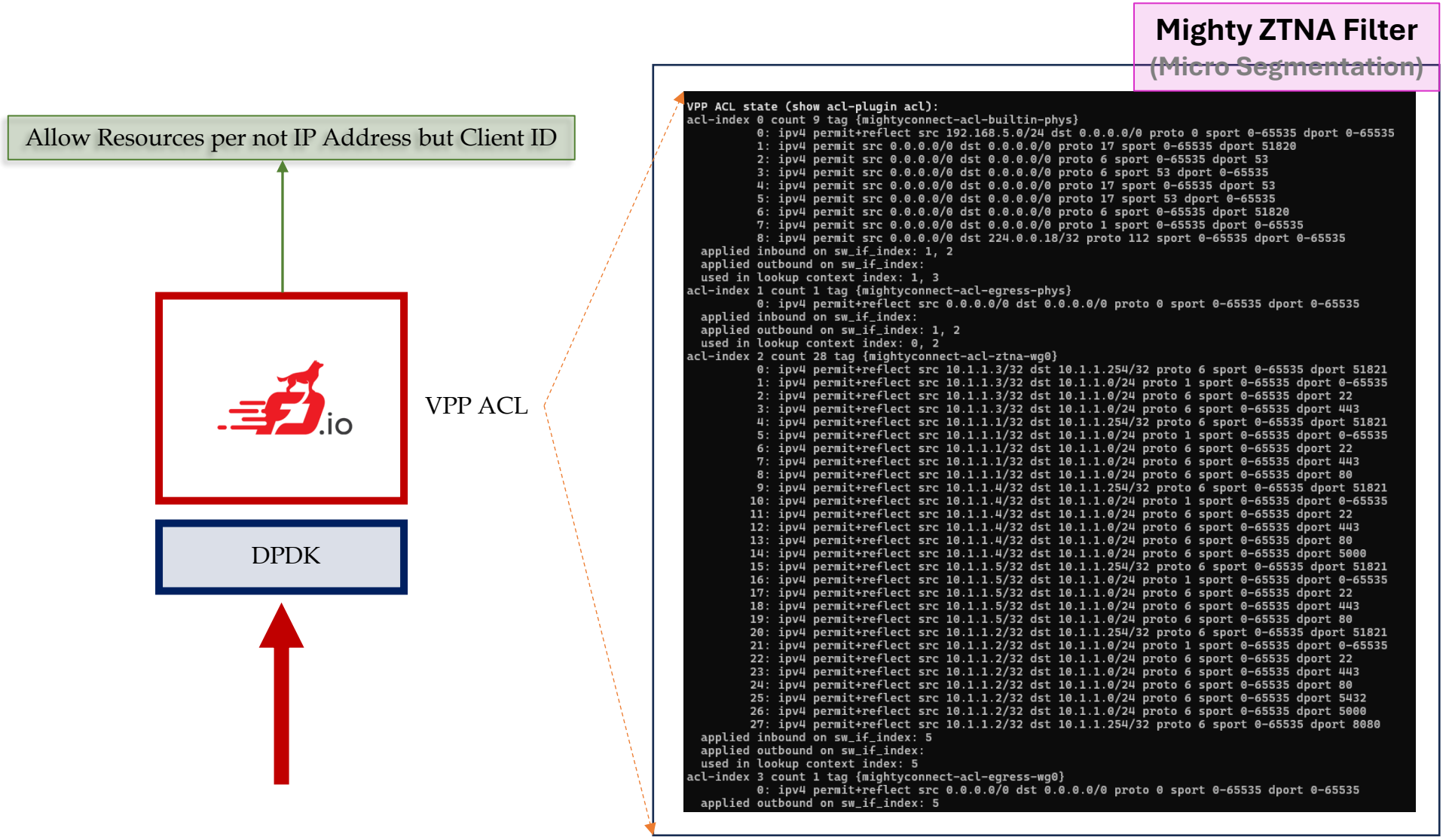
4. MightyConnect Security Box(4) – Fast Hub Relay(1)



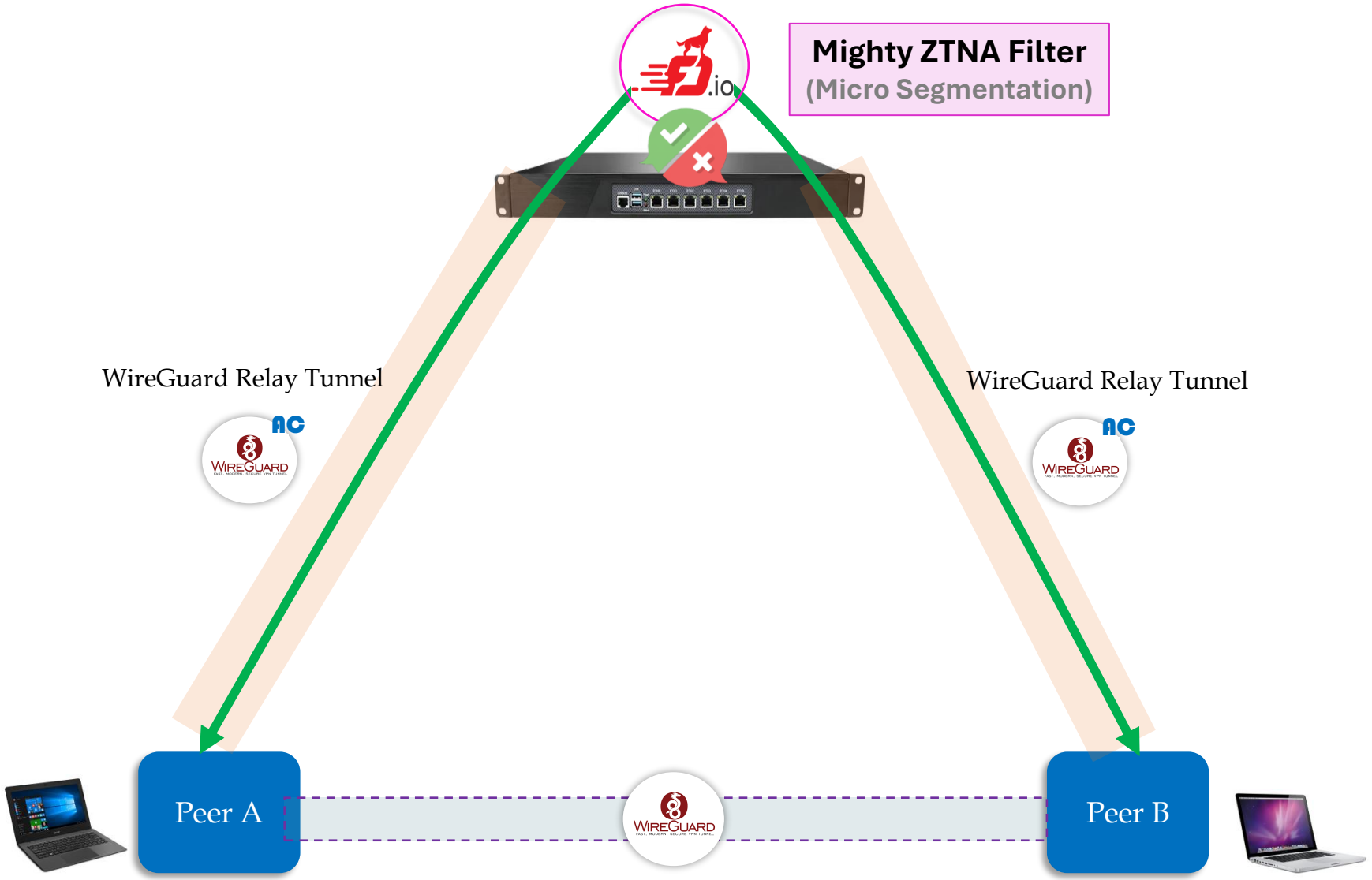
4. MightyConnect Security Box(4) – Fast Hub Relay(2)



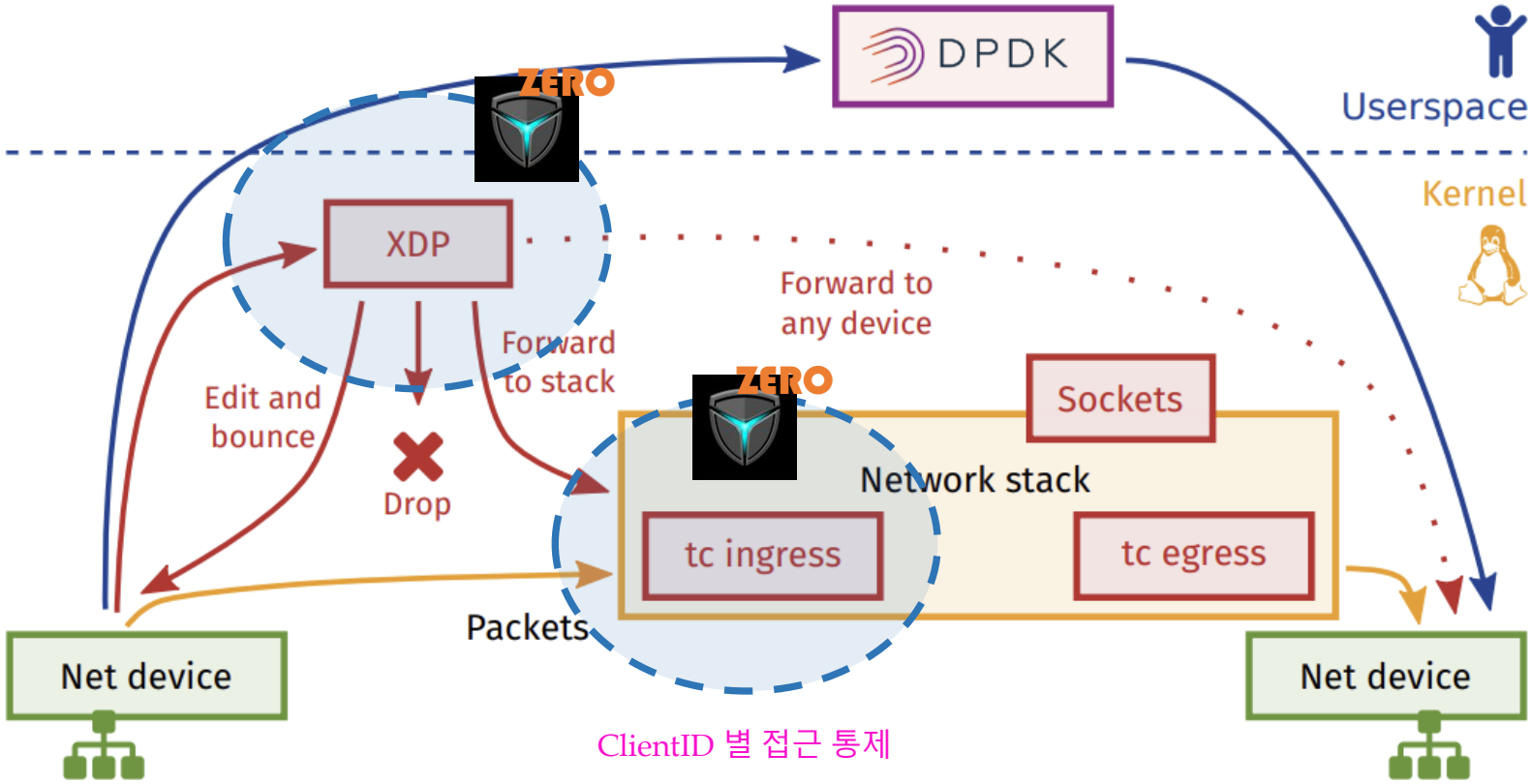
4. MightyConnect Security Box(5) – VPP ZTNA(1)

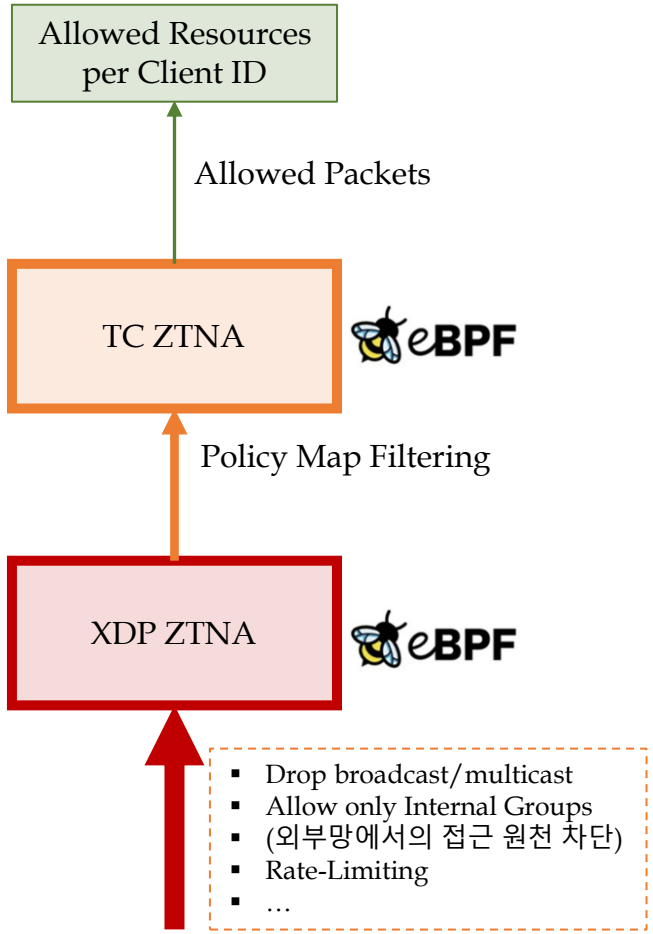


4. MightyConnect Security Box(5) – VPP ZTNA(2)

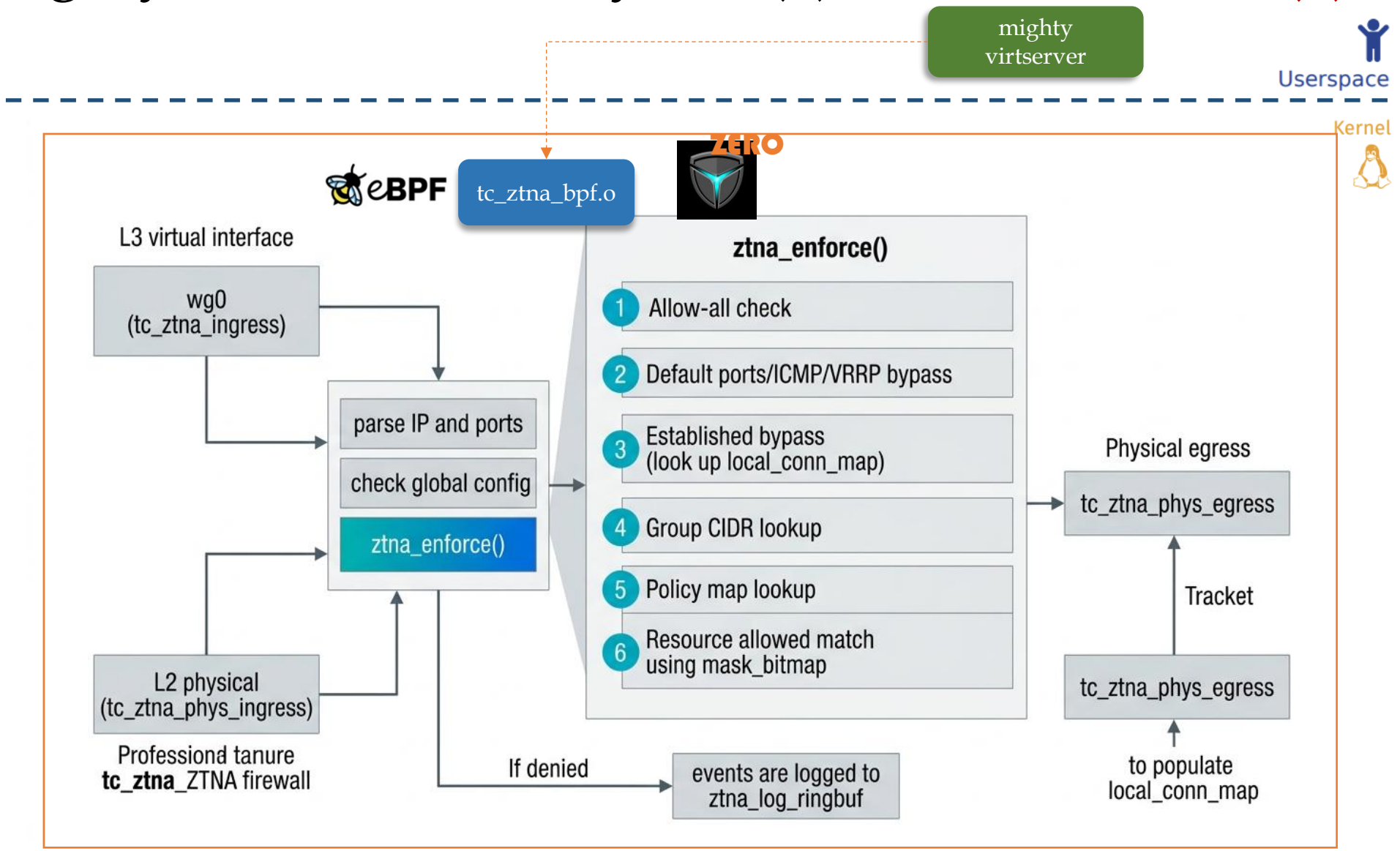


4. MightyConnect Security Box(6) – Kernel ZTNA(1)

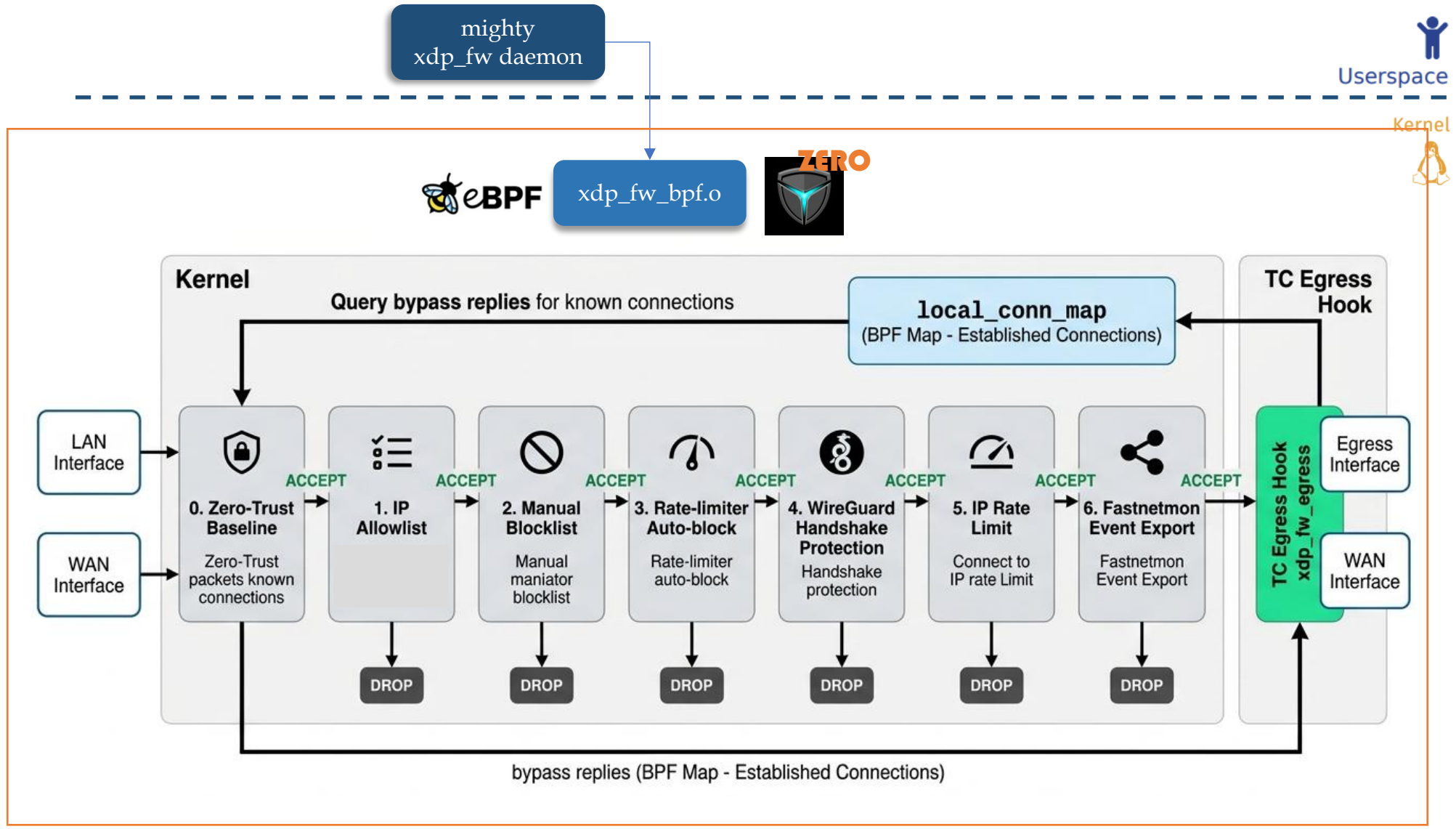




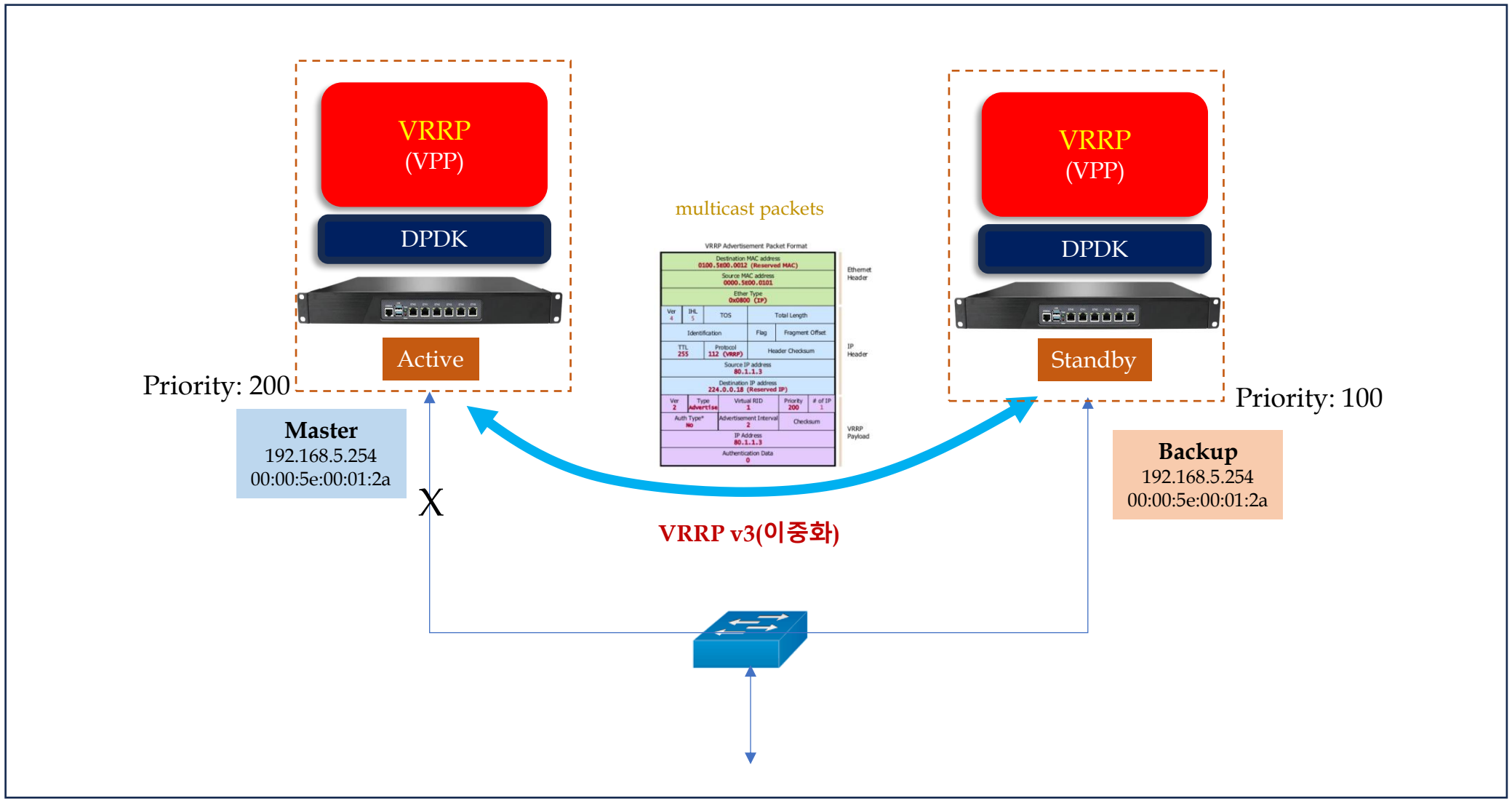
4. MightyConnect Security Box(6) – Kernel ZTNA(2)



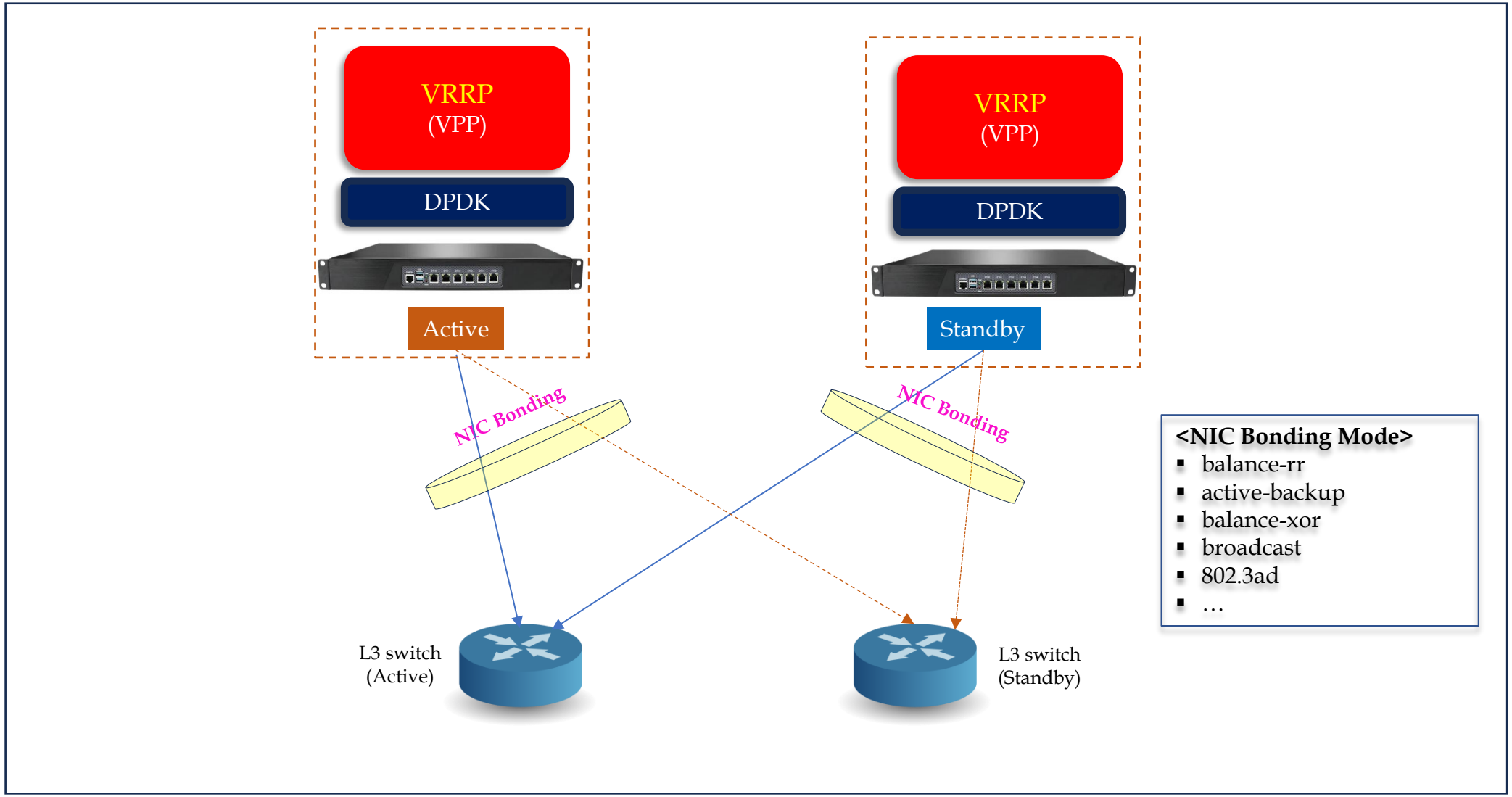
4. MightyConnect Security Box(6) – Kernel ZTNA(3)



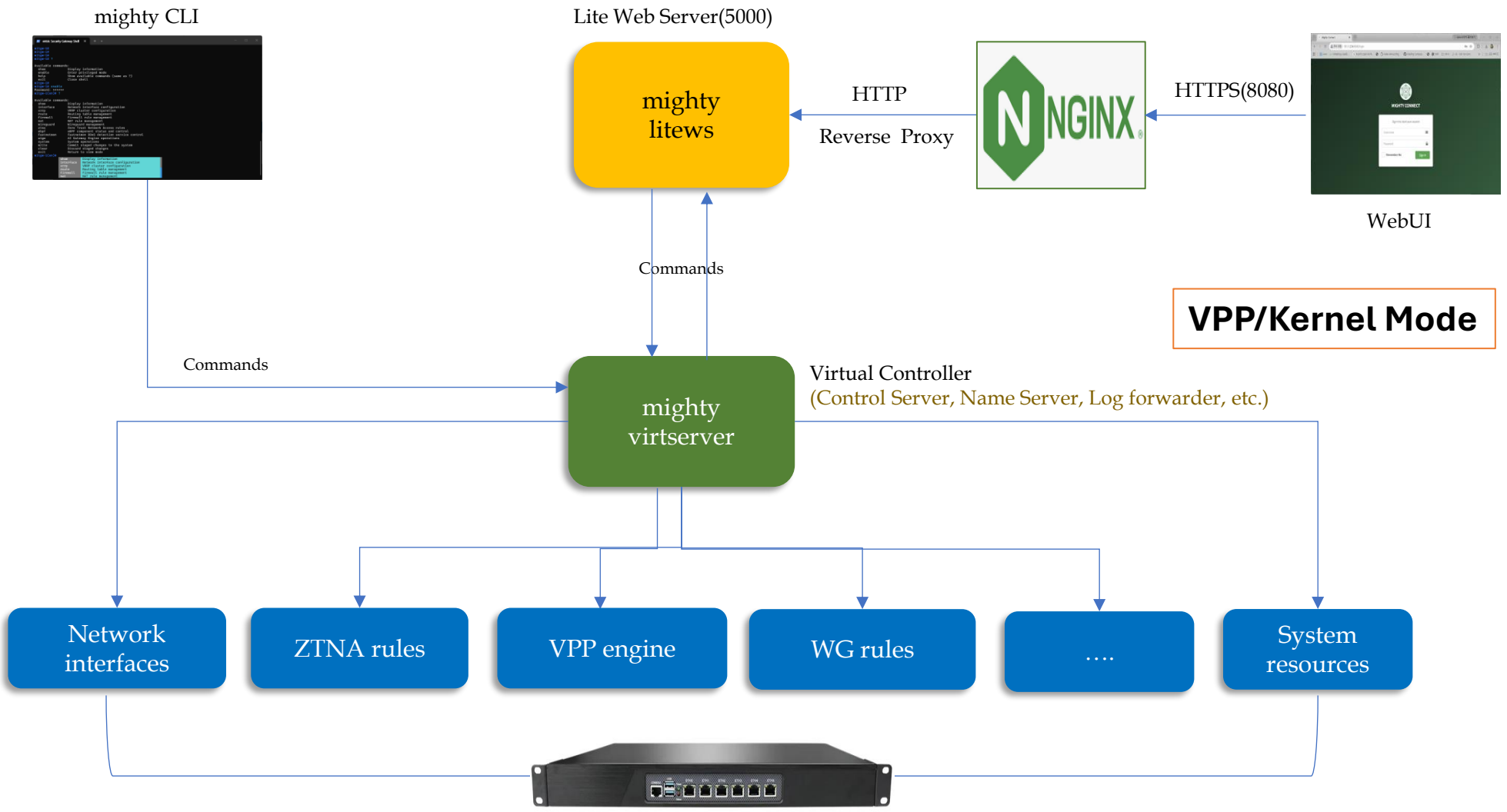
4. MightyConnect Security Box(7) – Redundancy(1)



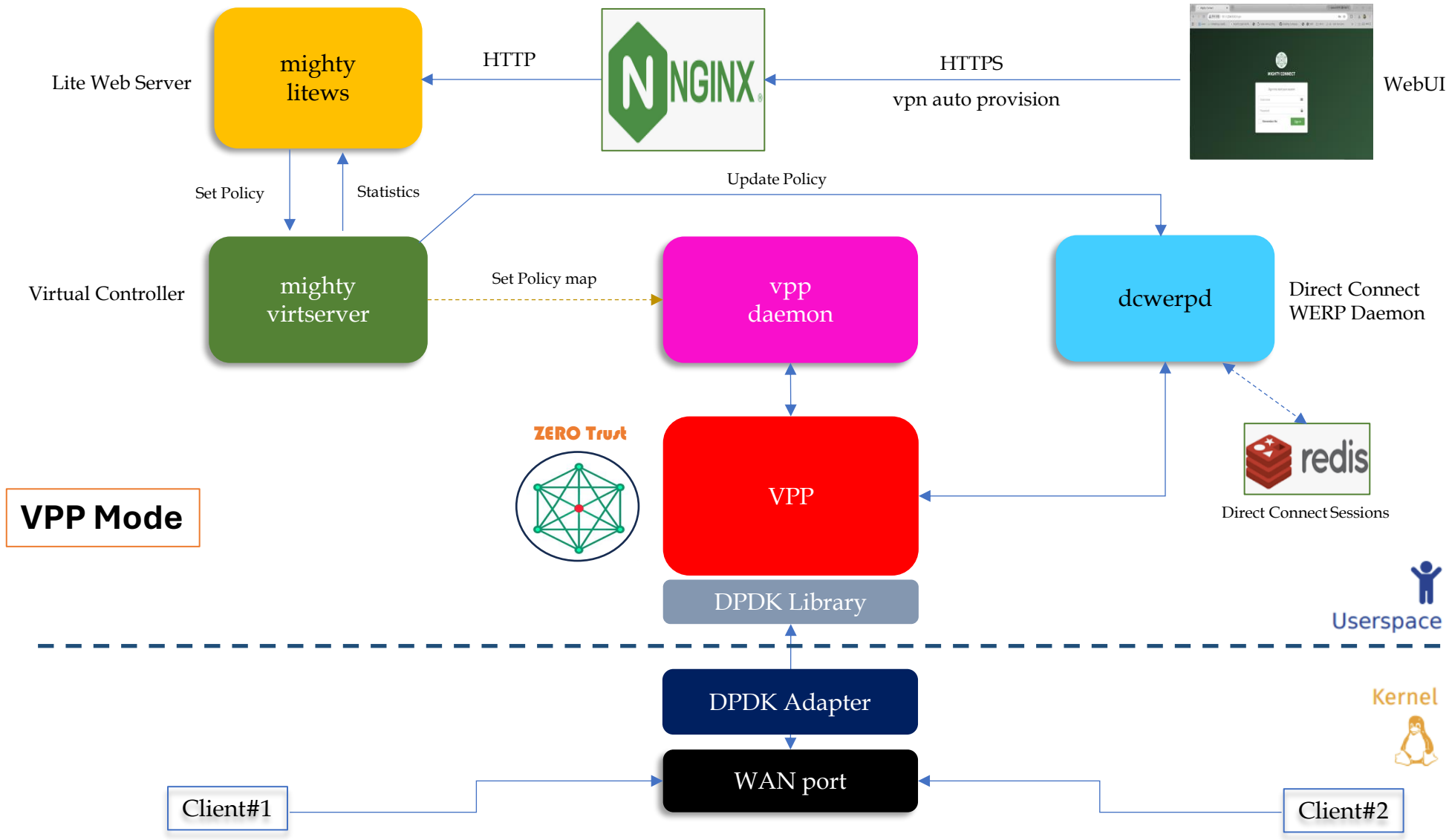
4. MightyConnect Security Box(7) – Redundancy(2)



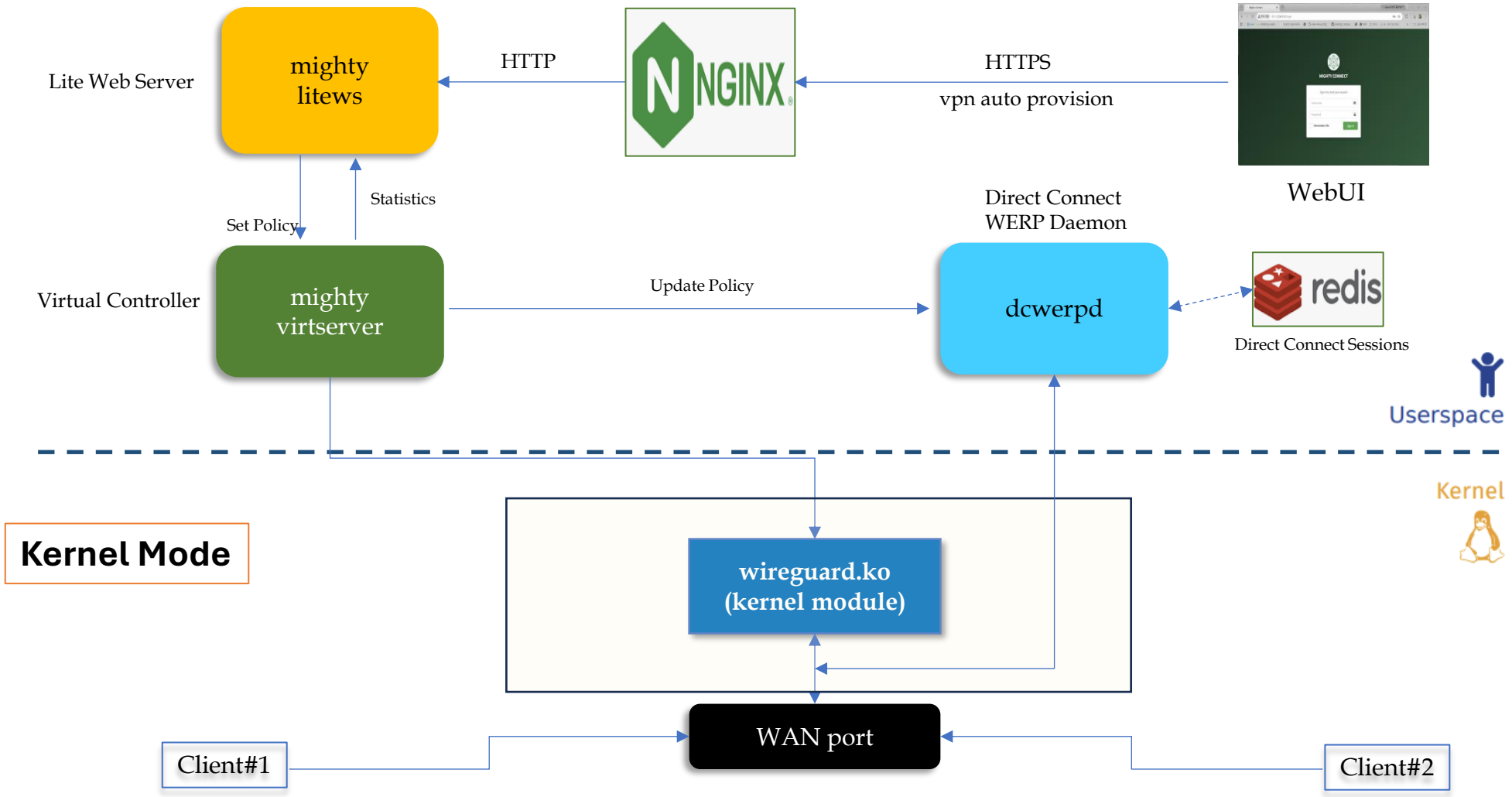
4. MightyConnect Security Box(8) – Mighty Engine(1)



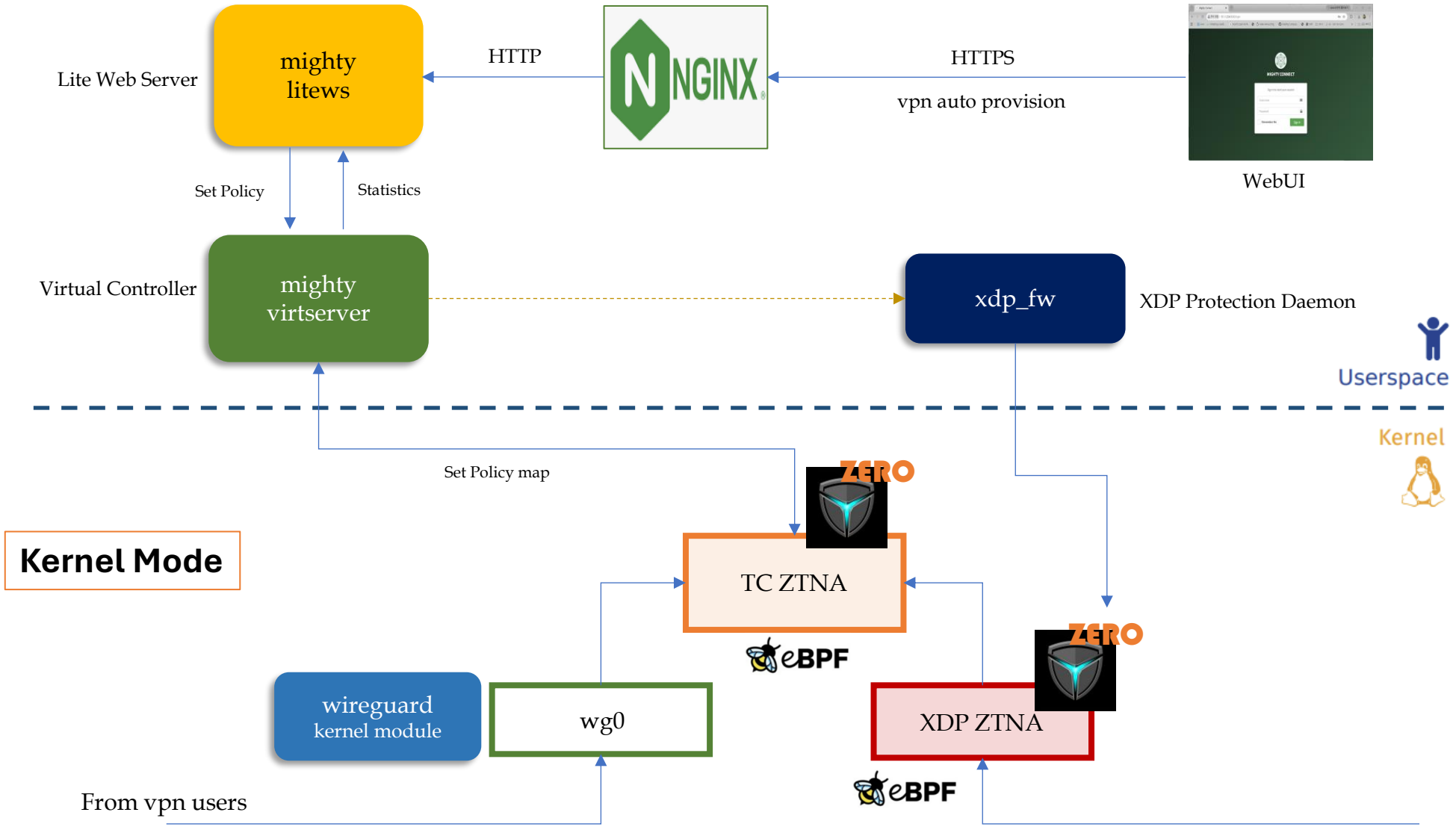
4. MightyConnect Security Box(8) – Mighty Engine(2)



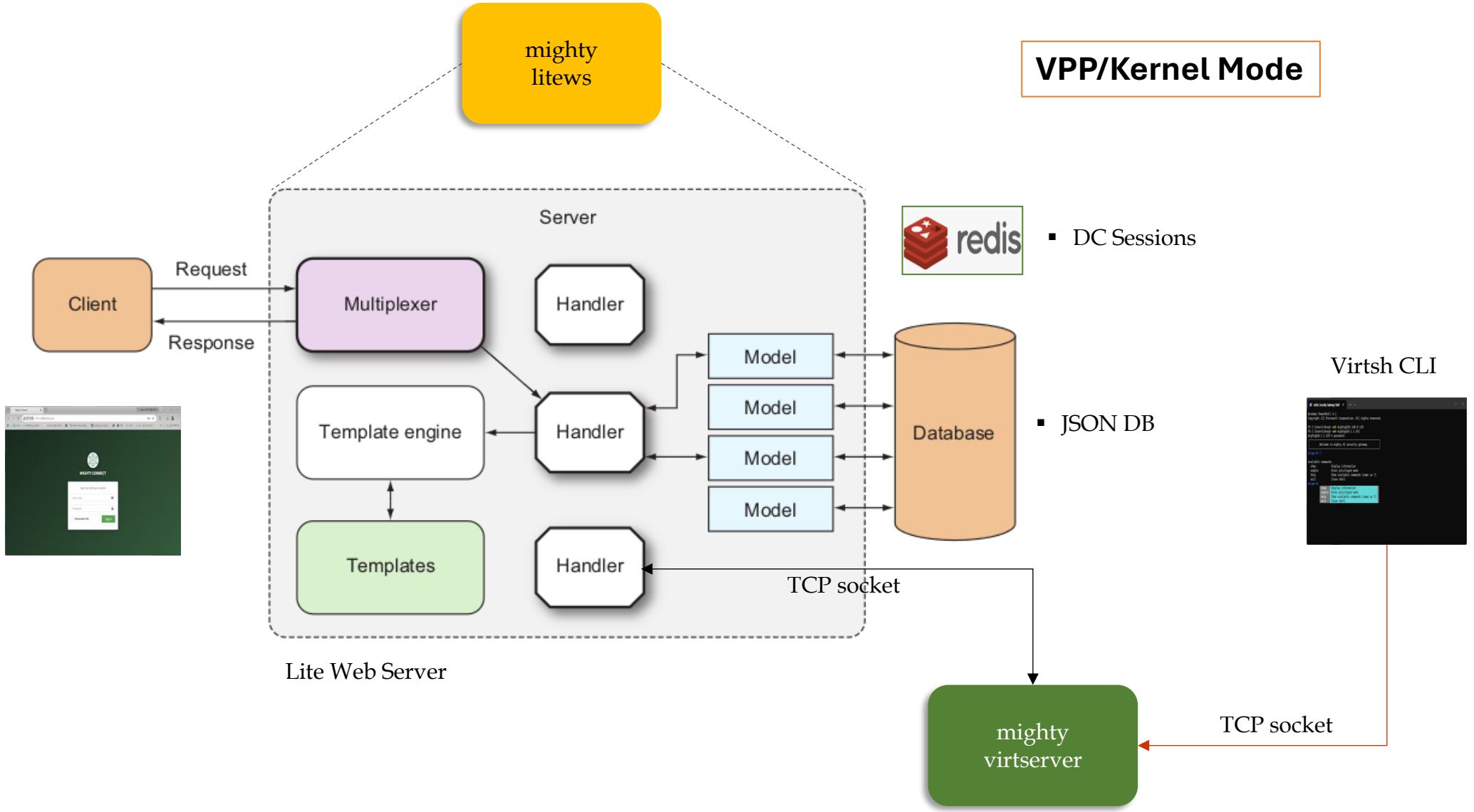
4. MightyConnect Security Box(8) – Mighty Engine(3-1)



4. MightyConnect Security Box(8) – Mighty Engine(3-2)

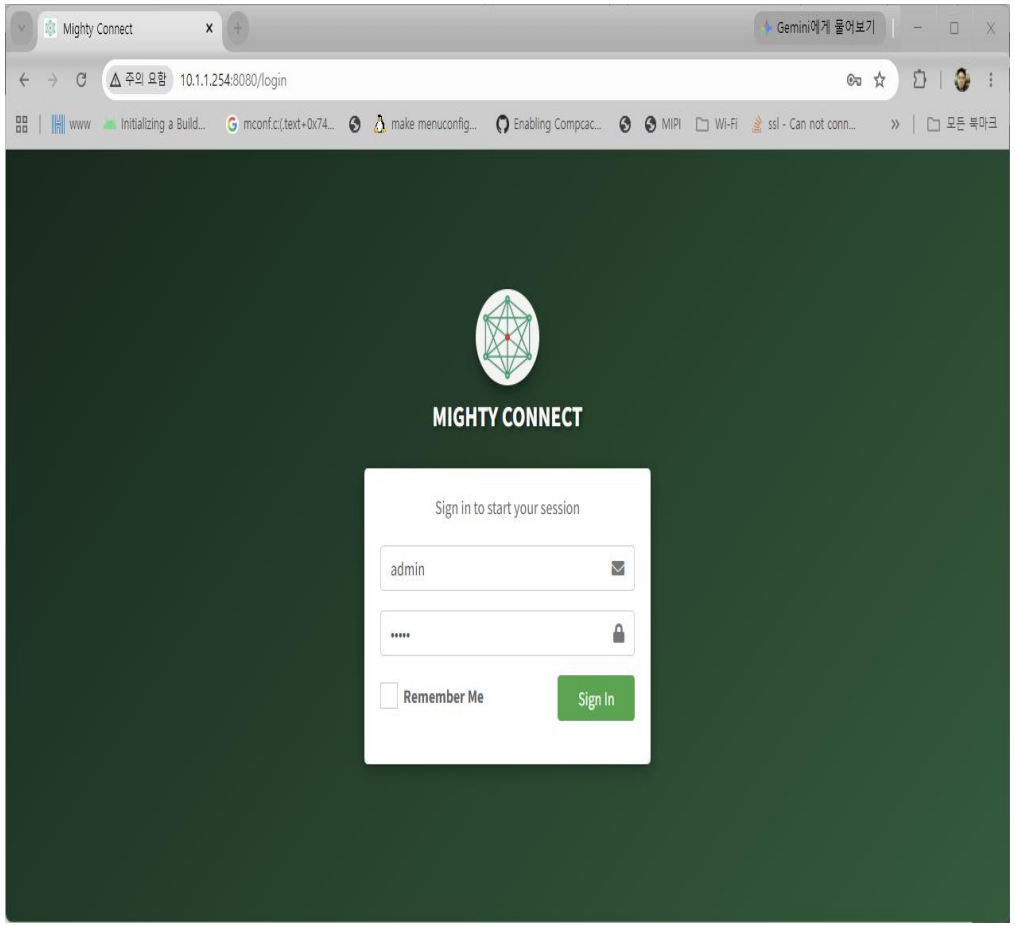


4. MightyConnect Security Box(8) – Mighty Engine(4)

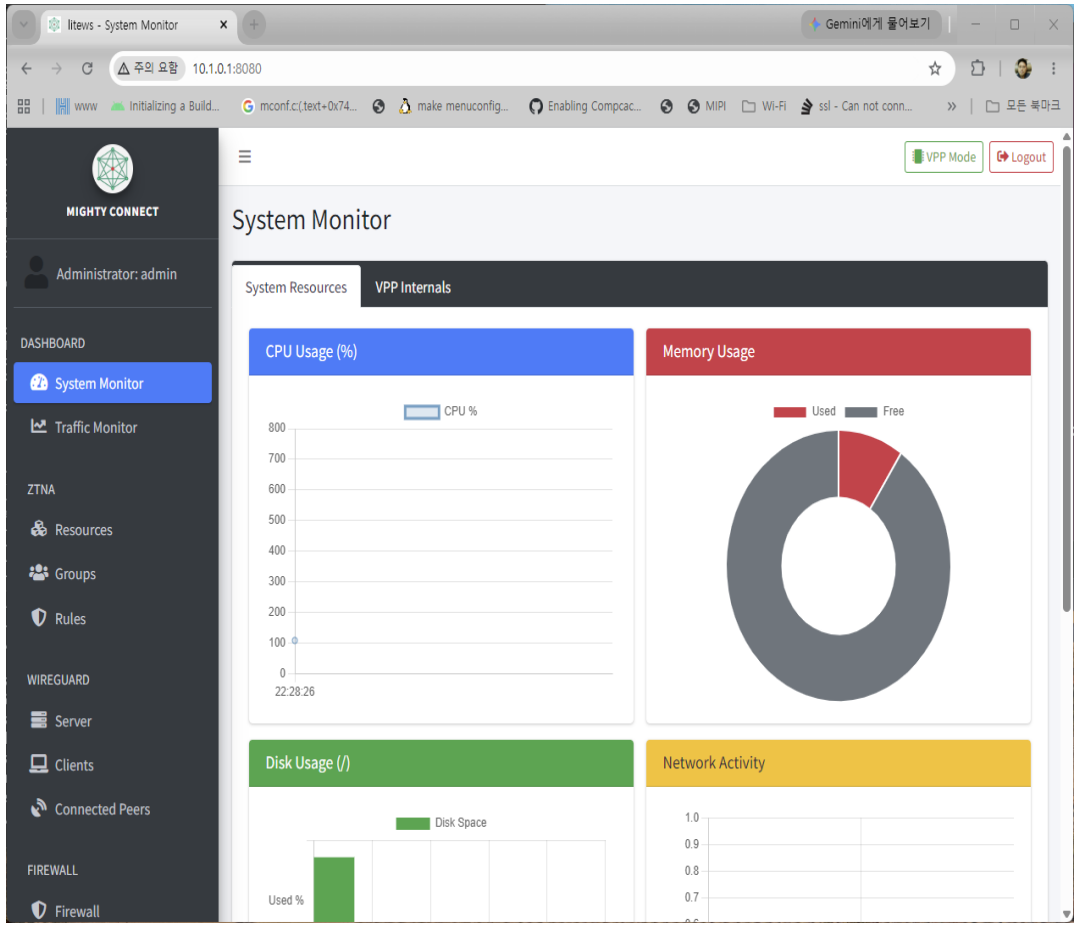


4. MightyConnect Security Box(9) – WebUI(1)

- WebUI Login Page

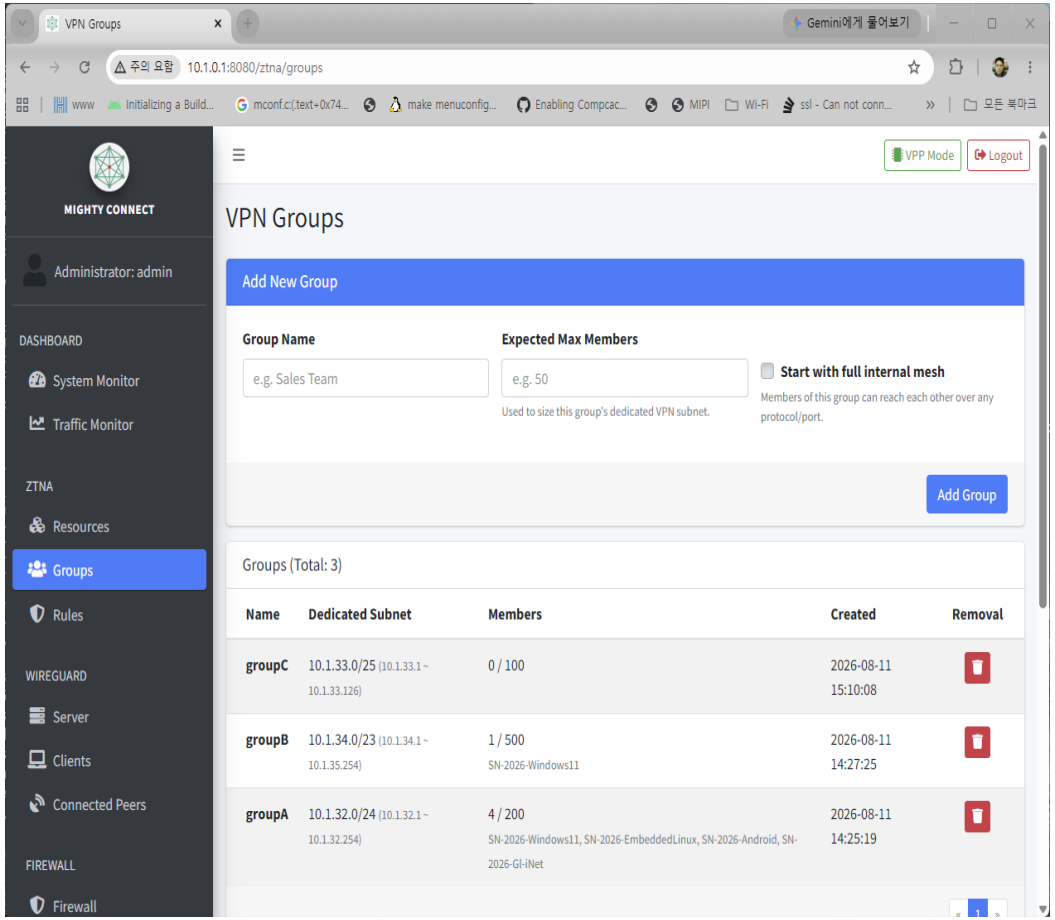


- Dashboard Page

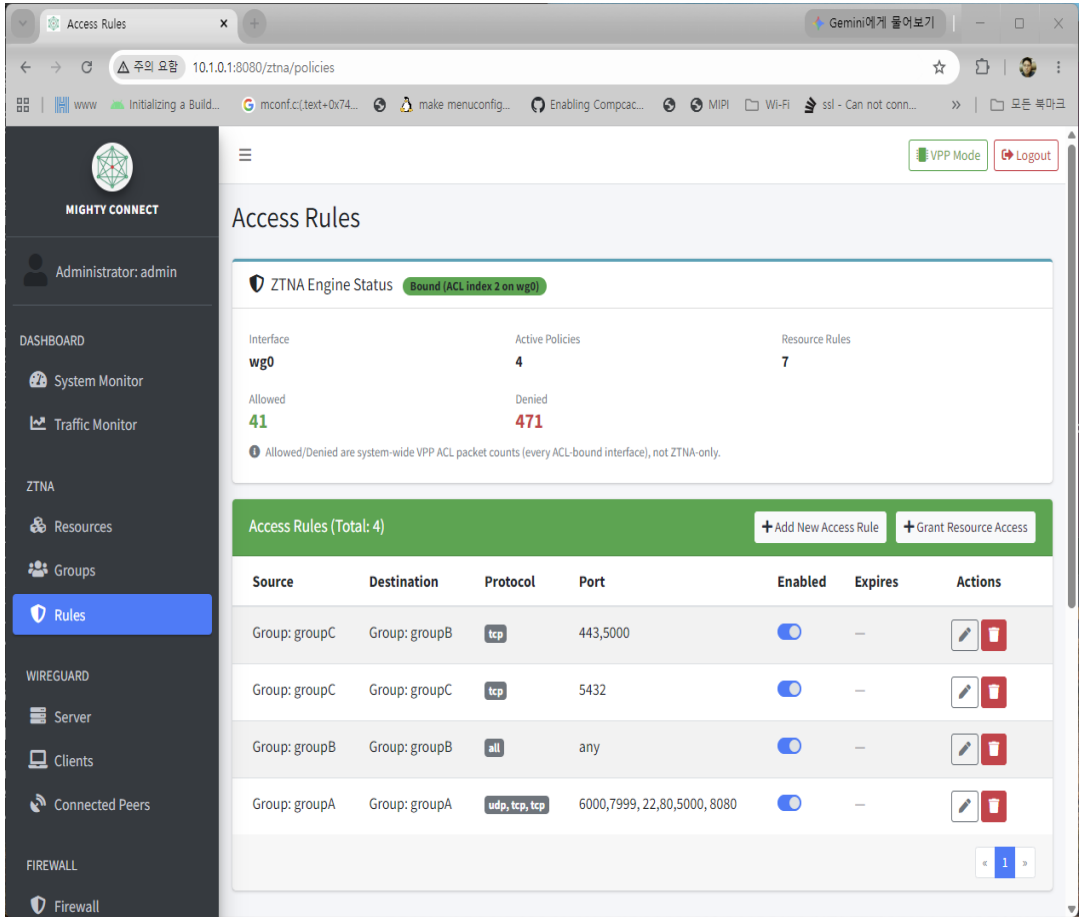


4. MightyConnect Security Box(9) – WebUI(2)

- ZTNA VPN Groups Page

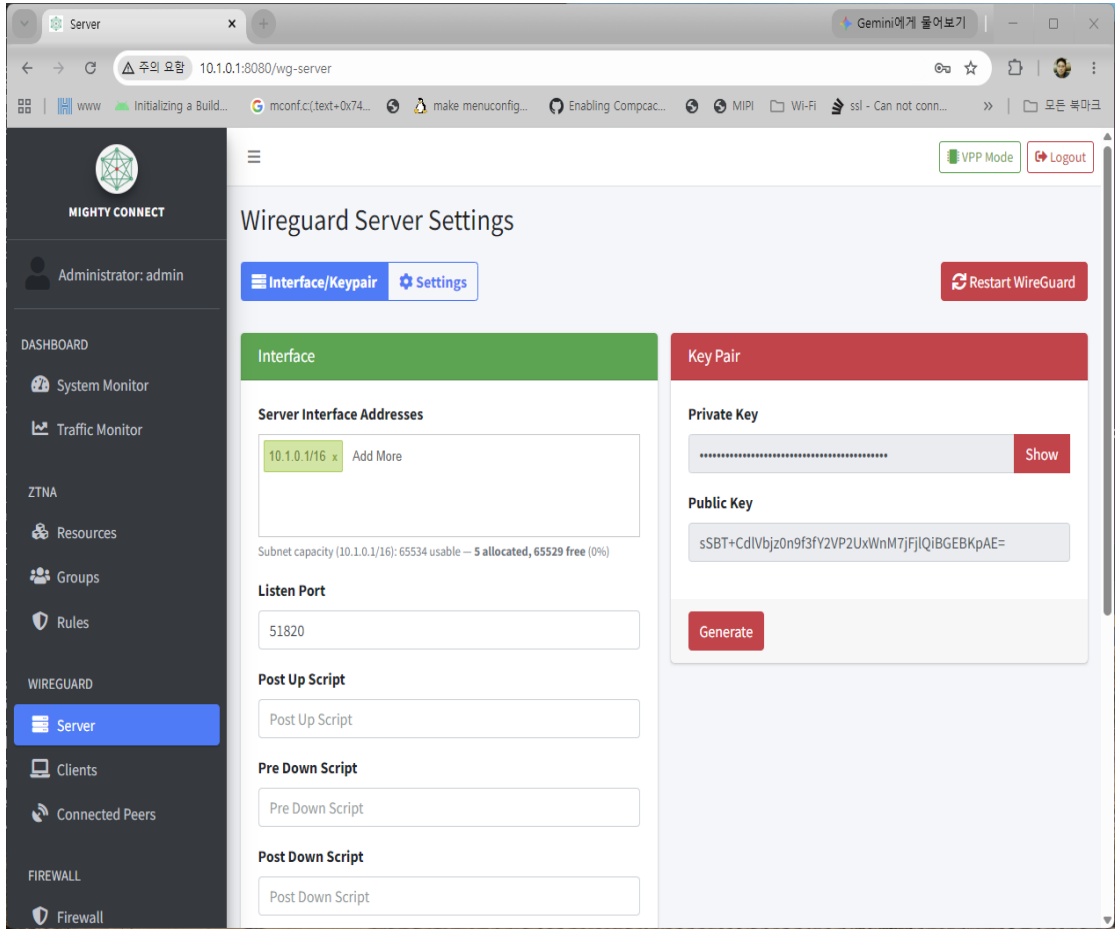


- ZTNA Access Rules Page

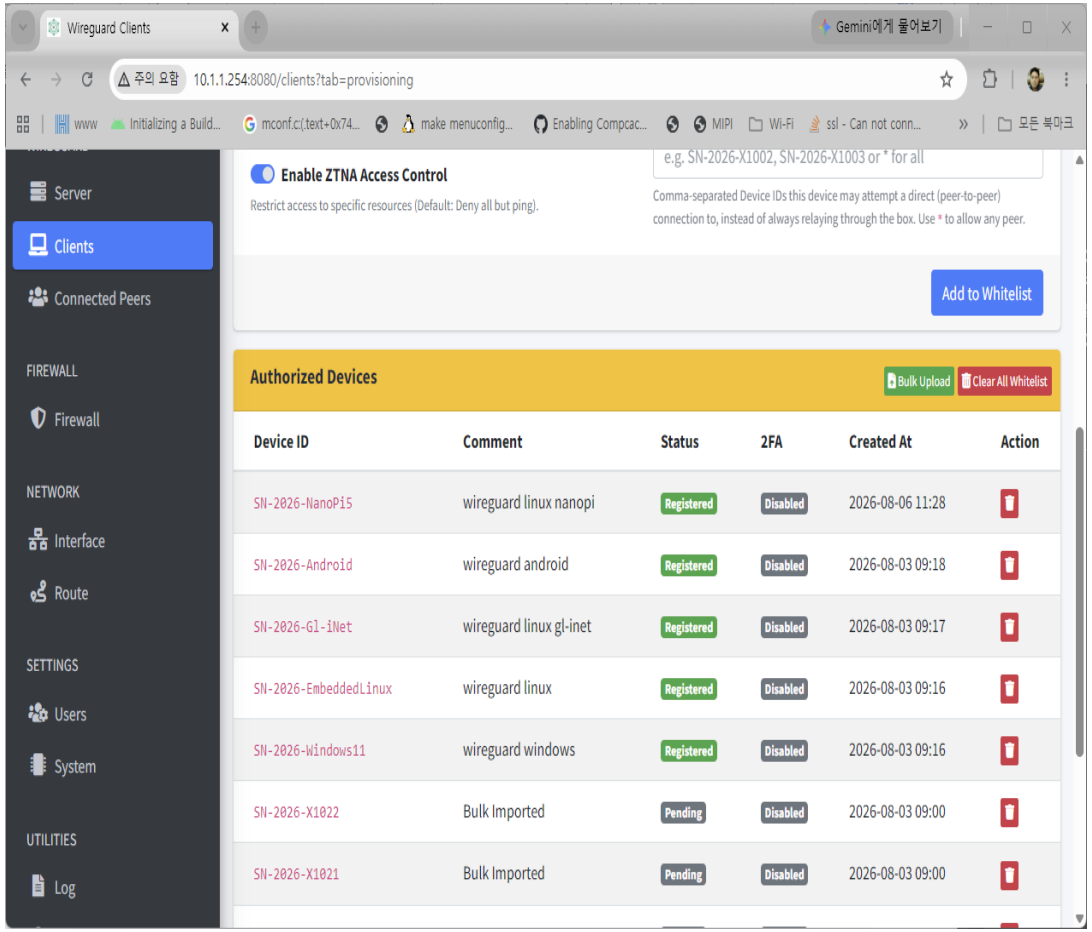


4. MightyConnect Security Box(9) – WebUI(3)

- WireGuard Server Page

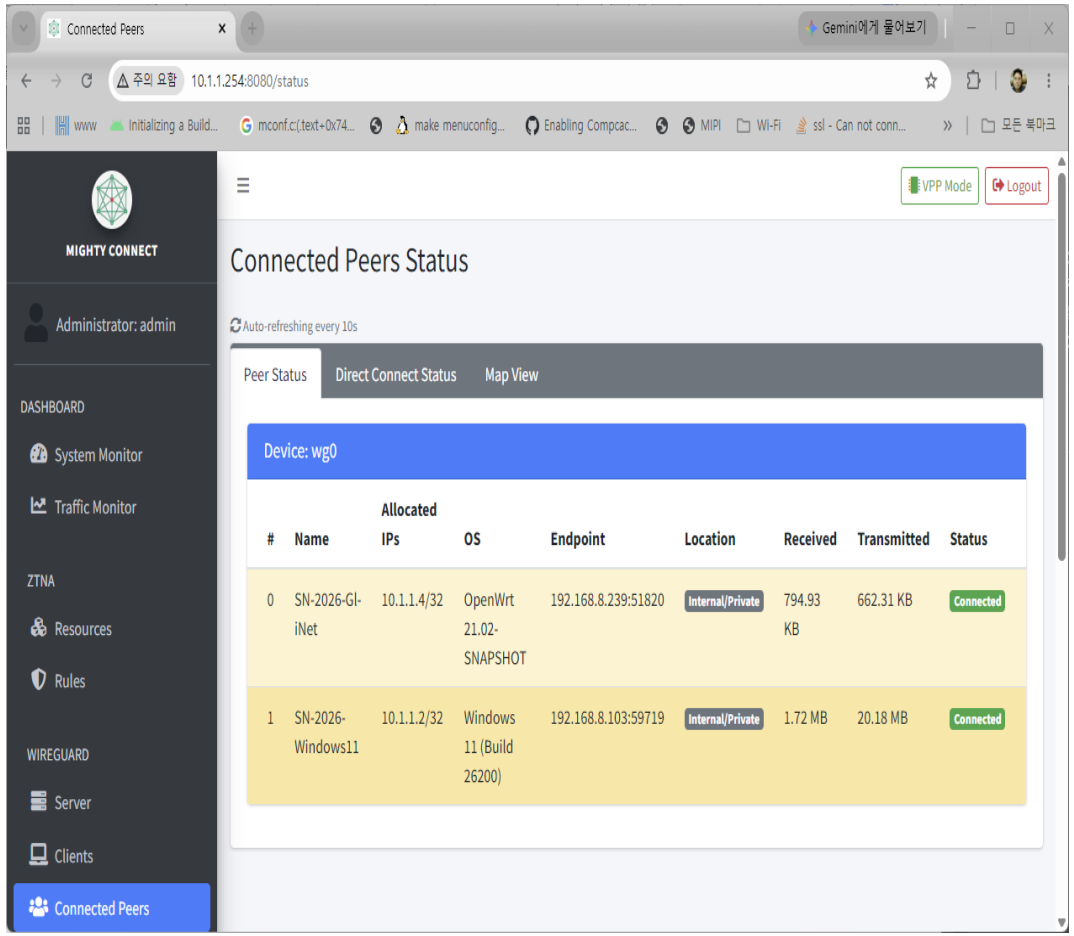


- WireGuard Client Auto Provision Page

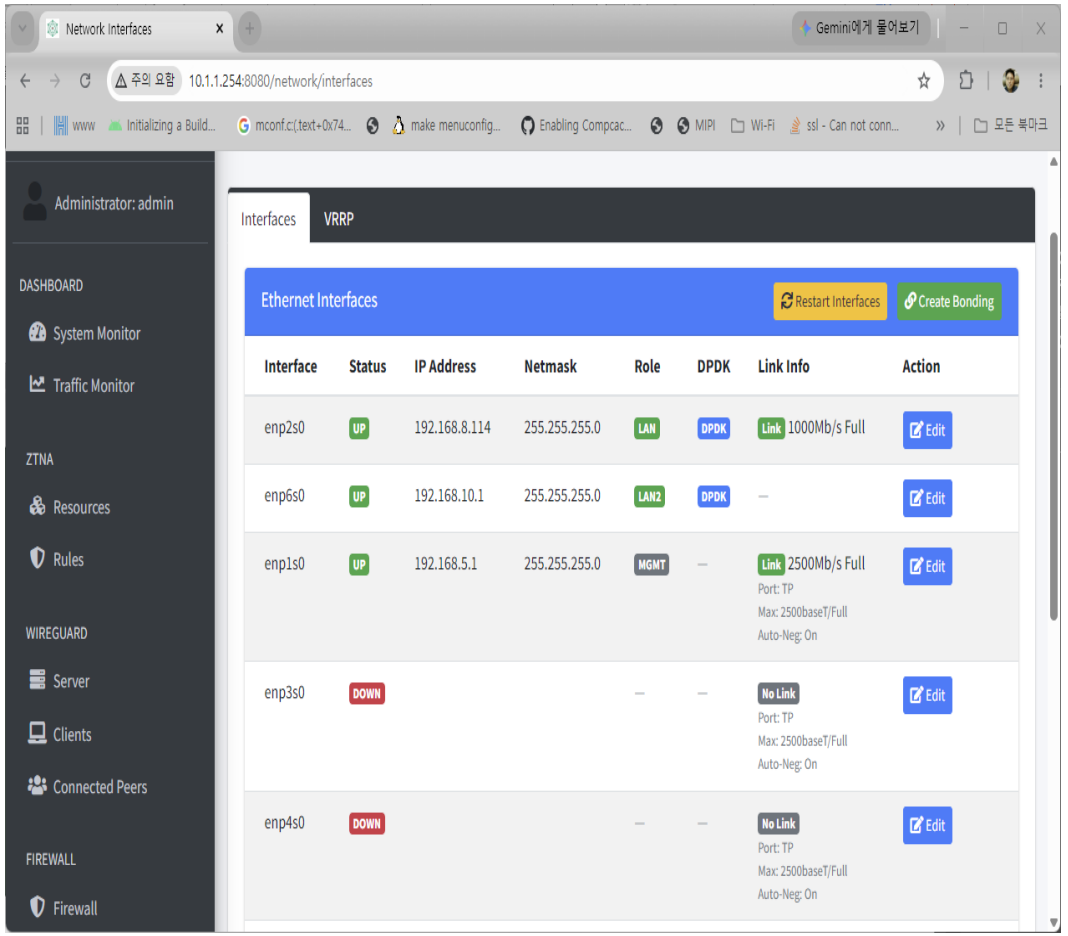


4. MightyConnect Security Box(9) – WebUI(4)

- WireGuard Connected Peers Page

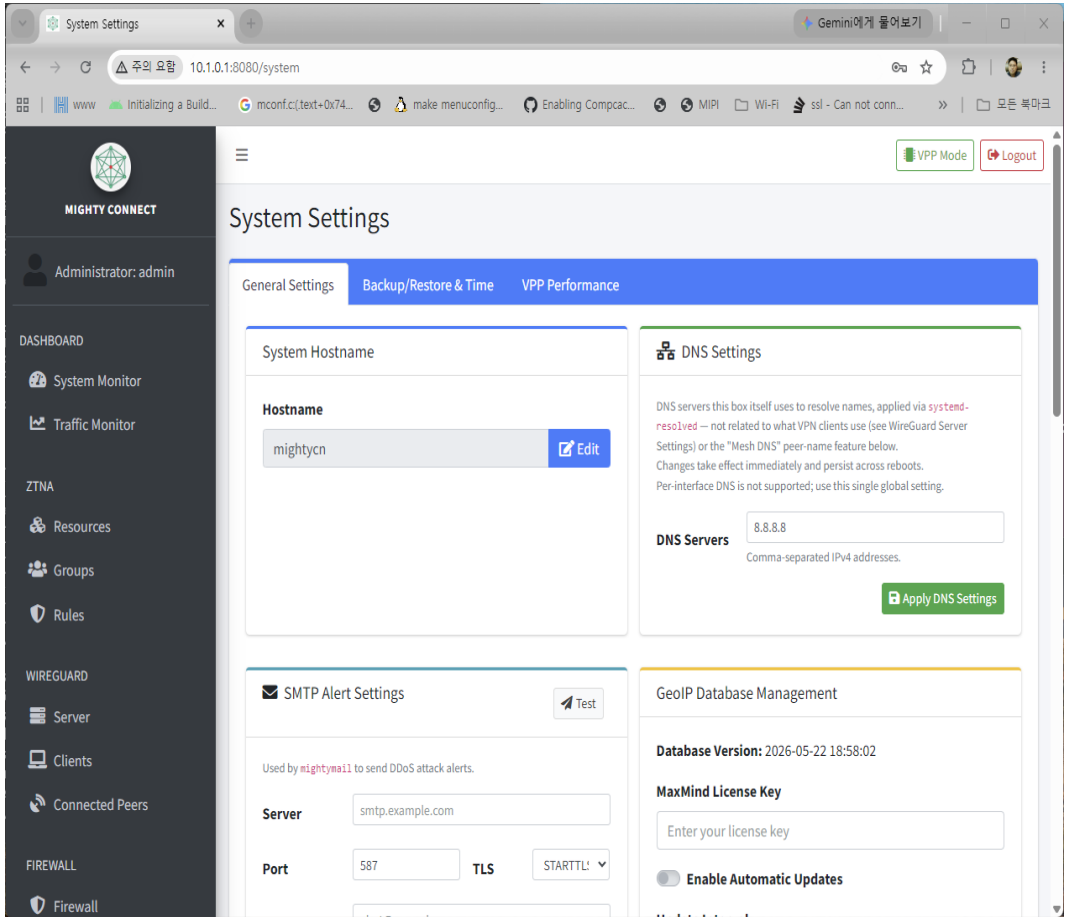


- Network Interfaces Page

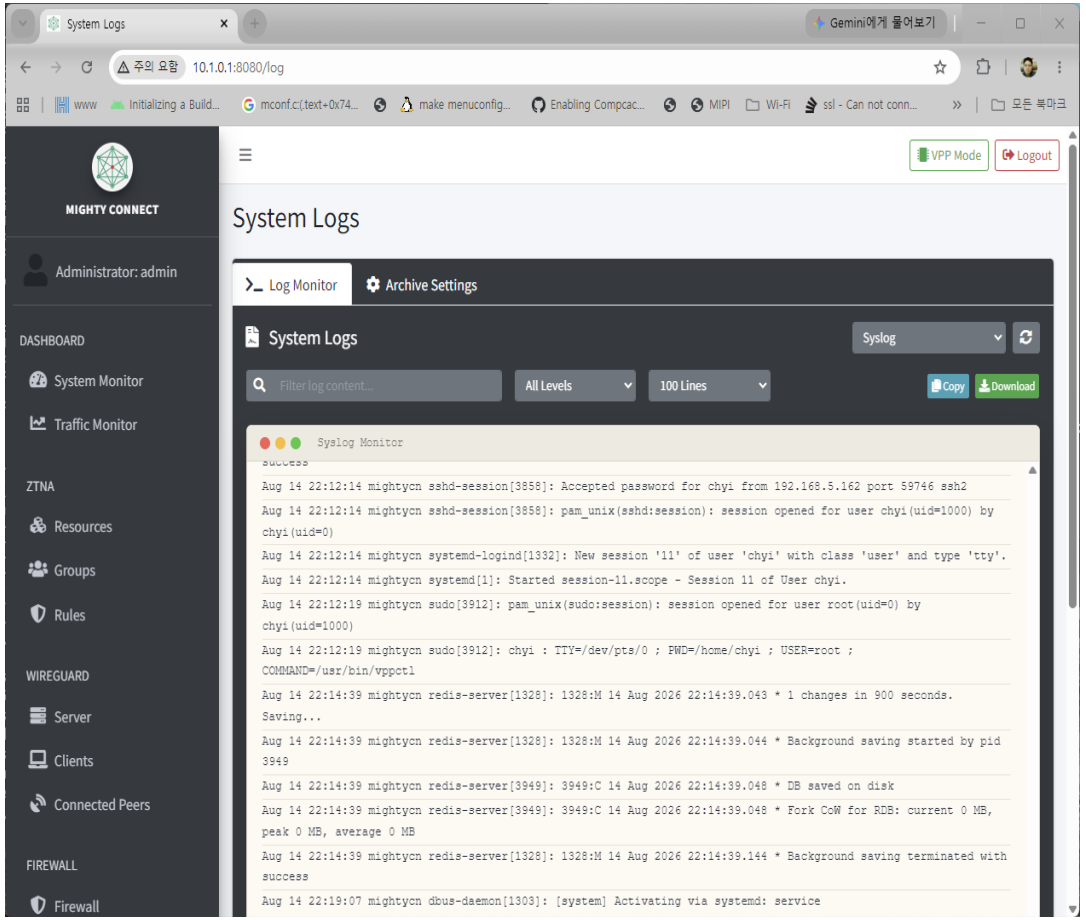


4. MightyConnect Security Box(9) – WebUI(5)

- System Settings Page



- Logs Page



4. MightyConnect Security Box(10) – CLI

```
PS C:\Users\chung> ssh mighty@10.1.1.254
mighty@10.1.1.254's password:

        Welcome to mighty AI security gateway

mightycn# ?

Available commands:
  show            Display information
  enable          Enter privileged mode
  help            Show available commands (same as ?)
  exit            Close shell
mightycn# show interface
Interface  Status  IP Address                  DPDK
-------------------------------------------------
enp2s0     UP      192.168.8.114/255.255.255.0  yes
enp6s0     UP      192.168.10.1/255.255.255.0   yes
tap4096    UP      -                            yes
tap4097    UP      -                            yes
tun4098    UP      -                            yes
wg0        UP      10.1.1.254/255.255.255.0     yes
enp1s0     UP      192.168.5.1/255.255.255.0    -
enp3s0     DOWN    -                            -
enp4s0     DOWN    -                            -
enp5s0     DOWN    192.168.20.1/255.255.255.0   -

VPP interfaces (show interface):
Name               Idx    State  MTU (L3/IP4/IP6/MPLS)     Counter          Count
enp2s0               1      up          9000/0/0/0     rx packets                83438
                                                       rx bytes                31067596
                                                       tx packets                96924
                                                       tx bytes                69994133
                                                       drops                      1197
                                                       punt                       2101
                                                       ip4                       82166
                                                       ip6                         286
                                                       tx-error                      2
enp6s0               2      up          9000/0/0/0     tx-error                      1
local0               0     down          0/0/0/0       drops                         4
tap4096              3      up          9000/0/0/0     rx packets                  547
                                                       rx bytes                 163088
                                                       tx packets                 862
                                                       tx bytes                 191013
                                                       drops                       16
                                                       ip4                        459
                                                       ip6                         17
tap4097              4      up          9000/0/0/0
tun4098              6      up          1420/0/0/0     rx packets                41921
                                                       rx bytes                40105173
                                                       tx packets                27211
                                                       tx bytes                2196120
                                                       ip4                       41903
                                                       ip6                          18
wg0                  5      up          1420/0/0/0     rx packets                79471
```

```
mightycn# show ztna
Network ZTNA Rules (L4 Access Control):
Client Name (Label)    Client IP     Authorized Resources                                                                    Status
-------------------------------------------------------------------------------------------------------------------------------
SN-2026-Android        10.1.1.3/32   default-ping (icmp/10.1.1.0/24/any),                                                    Enabled
                                     ssh (tcp/10.1.1.0/24/22), https (tcp/10.1.1.0/24/443)
SN-2026-EmbeddedLinux  10.1.1.1/32   default-ping (icmp/10.1.1.0/24/any),                                                    Enabled
                                     ssh (tcp/10.1.1.0/24/22), https (tcp/10.1.1.0/24/443),
                                     http (tcp/10.1.1.0/24/80)
SN-2026-Gl-iNet        10.1.1.4/32   default-ping (icmp/10.1.1.0/24/any),                                                    Enabled
                                     ssh (tcp/10.1.1.0/24/22), https (tcp/10.1.1.0/24/443),
                                     http (tcp/10.1.1.0/24/80), litews (tcp/10.1.1.0/24/5000)
SN-2026-NanoPi5        10.1.1.5/32   default-ping (icmp/10.1.1.0/24/any),                                                    Enabled
                                     ssh (tcp/10.1.1.0/24/22), https (tcp/10.1.1.0/24/443),
                                     http (tcp/10.1.1.0/24/80)
SN-2026-Windows11      10.1.1.2/32   default-ping (icmp/10.1.1.0/24/any),                                                    Enabled
                                     ssh (tcp/10.1.1.0/24/22), https (tcp/10.1.1.0/24/443),
                                     http (tcp/10.1.1.0/24/80), postgresql (tcp/10.1.1.0/24/5432),
                                     litews (tcp/10.1.1.0/24/5000),
                                     boxlitews (tcp/10.1.1.254/32/8080)

Always Allowed (all clients, bypasses policy_map entirely): DNS tcp/53, DNS udp/53, ICMP

VPP ACL state (show acl-plugin acl):
acl-index 0 count 9 tag {mightyconnect-acl-builtin-phys}
         0: ipv4 permit+reflect src 192.168.5.0/24 dst 0.0.0.0/0 proto 0 sport 0-65535 dport 0-65535
         1: ipv4 permit src 0.0.0.0/0 dst 0.0.0.0/0 proto 17 sport 0-65535 dport 51820
         2: ipv4 permit src 0.0.0.0/0 dst 0.0.0.0/0 proto 6 sport 0-65535 dport 53
         3: ipv4 permit src 0.0.0.0/0 dst 0.0.0.0/0 proto 6 sport 53 dport 0-65535
         4: ipv4 permit src 0.0.0.0/0 dst 0.0.0.0/0 proto 17 sport 0-65535 dport 53
         5: ipv4 permit src 0.0.0.0/0 dst 0.0.0.0/0 proto 17 sport 53 dport 0-65535
         6: ipv4 permit src 0.0.0.0/0 dst 0.0.0.0/0 proto 6 sport 0-65535 dport 51820
         7: ipv4 permit src 0.0.0.0/0 dst 0.0.0.0/0 proto 1 sport 0-65535 dport 0-65535
         8: ipv4 permit src 0.0.0.0/0 dst 224.0.0.18/32 proto 112 sport 0-65535 dport 0-65535
  applied inbound on sw_if_index: 1, 2
  applied outbound on sw_if_index:
  used in lookup context index: 1, 3
acl-index 1 count 1 tag {mightyconnect-acl-egress-phys}
         0: ipv4 permit+reflect src 0.0.0.0/0 dst 0.0.0.0/0 proto 0 sport 0-65535 dport 0-65535
  applied inbound on sw_if_index:
  applied outbound on sw_if_index: 1, 2
  used in lookup context index: 0, 2
acl-index 2 count 28 tag {mightyconnect-acl-ztna-wg0}
         0: ipv4 permit+reflect src 10.1.1.3/32 dst 10.1.1.254/32 proto 6 sport 0-65535 dport 51821
         1: ipv4 permit+reflect src 10.1.1.3/32 dst 10.1.1.0/24 proto 1 sport 0-65535 dport 0-65535
         2: ipv4 permit+reflect src 10.1.1.3/32 dst 10.1.1.0/24 proto 6 sport 0-65535 dport 22
         3: ipv4 permit+reflect src 10.1.1.3/32 dst 10.1.1.0/24 proto 6 sport 0-65535 dport 443
         4: ipv4 permit+reflect src 10.1.1.1/32 dst 10.1.1.254/32 proto 6 sport 0-65535 dport 51821
         5: ipv4 permit+reflect src 10.1.1.1/32 dst 10.1.1.0/24 proto 1 sport 0-65535 dport 0-65535
         6: ipv4 permit+reflect src 10.1.1.1/32 dst 10.1.1.0/24 proto 6 sport 0-65535 dport 22
         7: ipv4 permit+reflect src 10.1.1.1/32 dst 10.1.1.0/24 proto 6 sport 0-65535 dport 443
         8: ipv4 permit+reflect src 10.1.1.1/32 dst 10.1.1.0/24 proto 6 sport 0-65535 dport 80
         9: ipv4 permit+reflect src 10.1.1.4/32 dst 10.1.1.254/32 proto 6 sport 0-65535 dport 51821
        10: ipv4 permit+reflect src 10.1.1.4/32 dst 10.1.1.0/24 proto 1 sport 0-65535 dport 0-65535
        11: ipv4 permit+reflect src 10.1.1.4/32 dst 10.1.1.0/24 proto 6 sport 0-65535 dport 22
        12: ipv4 permit+reflect src 10.1.1.4/32 dst 10.1.1.0/24 proto 6 sport 0-65535 dport 443
--More--
```

4. MightyConnect Security Box(11) – H/W Spec(1)

고급(상급) 모델

항목	사양
모델명 / 베이슬롯	mighty-connect-5000 / 2U Rackmount
CPU	Intel® Xeon® Scalable Processor Gold 5218 2CPU(32 cores, 64 threads), 2.3GHz
RAM	DDR4 64GB
Storage(SSD/HDD)	480GB/4TB (OS/Log 저장용)
Ethernet Ports	1 GbE RJ45 2 ports + 10 GbE RJ45 2 ports
Power Input	1+1 1U 800W Redundant Power Supply 76mm width, W/PMbus, Titanium

H/W 스펙은 수시로 변경될 수 있음. 특히. CPU Core 종류는 단종 등의 이유로 변경될 수 있음.

4. MightyConnect Security Box(11) – H/W Spec(2)

중급 모델

항목	사양
모델명 / 베이슬롯	mighty-connect-2500 / 2U Rackmount
CPU	Intel® Xeon® D-2779 1 CPU(16 core, 32 threads)
RAM	DDR4 64GB
Storage(SSD/HDD)	480GB/1TB (OS/Log 저장용)
Ethernet Ports	1 GbE RJ45 2 ports + 10 GbE RJ45 2 ports
Power Input	1+1 1U 800W Redundant Power Supply 76mm width, W/PMbus, Titanium

H/W 스펙은 수시로 변경될 수 있음. 특히. CPU Core 종류는 단종 등의 이유로 변경될 수 있음.

4. MightyConnect Security Box(11) – H/W Spec(3)

하급 모델

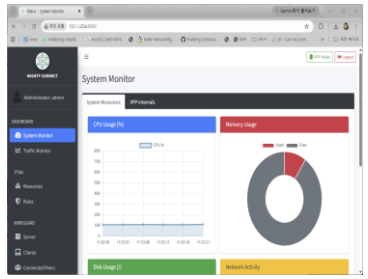
항목	사양
모델명 / 베이슬롯	mighty-connect-1000 / 1U Rackmount
CPU	Intel® Xeon® D-2146NT 1 CPU(8 core, 16 threads)
RAM	DDR4 32GB
Storage(SSD/HDD)	480GB/1TB (OS/Log 저장용)
Ethernet Ports	1 GbE RJ45 2 ports + 1 GbE RJ45 2 ports
Power Input	200W Low-noise AC-DC power supply

H/W 스펙은 수시로 변경될 수 있음. 특히. CPU Core 종류는 단종 등의 이유로 변경될 수 있음.

5. MightyConnect Devices

MightyConnect Devices

5. MightyConnect Devices





MightyConnect™ Box



MightyConnect™ Mini



MightyConnect™ Windows



MightyConnect™ Ubuntu



MightyConnect™ macOS/iOS



MightyConnect™ Android

5.1 MightyConnect Mini

MightyConnect Mini Security Gateway



GL-iNet Series
https://www.gl-inet.com/

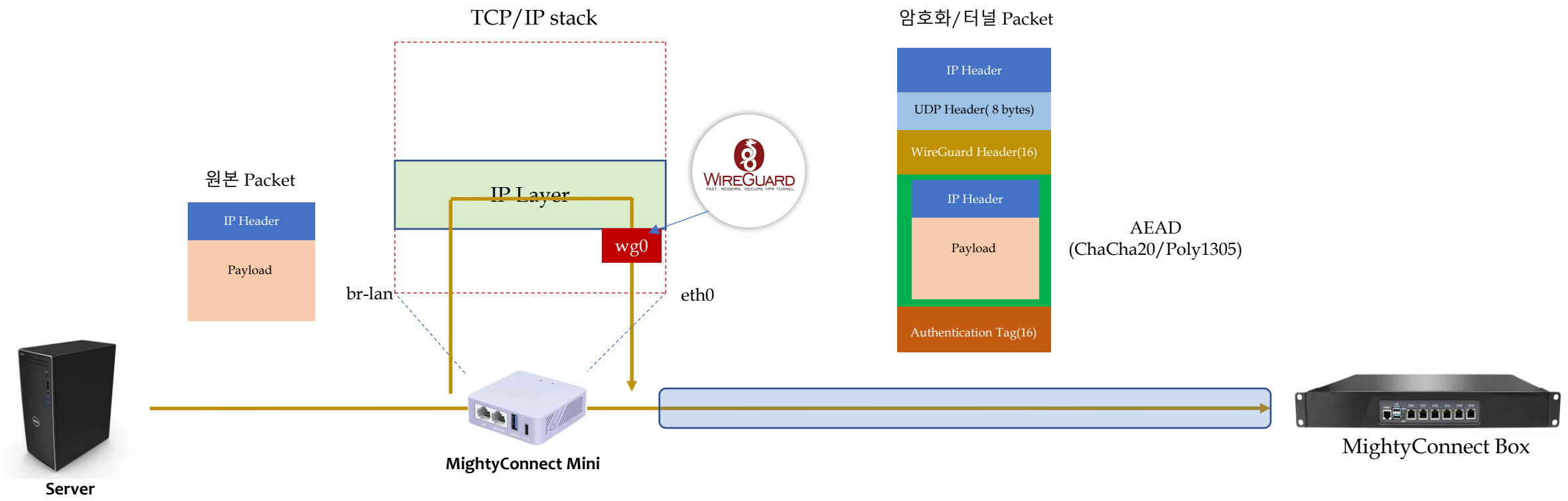


NanoPi Series
https://www.friendlyelec.com/

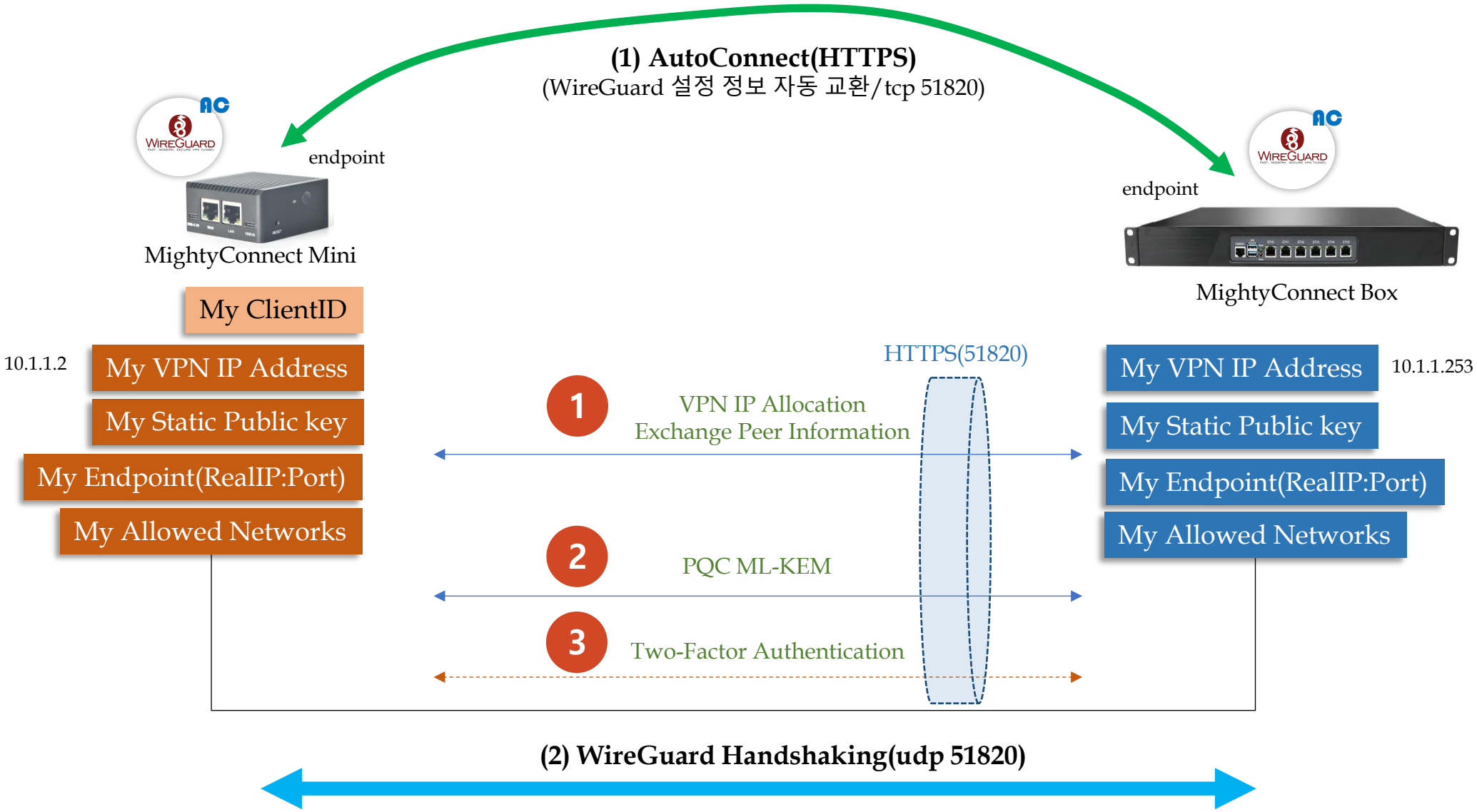
5.1 MightyConnect Mini(1) – Overview

CryptoKey Routing

ID: Public Key::VPN IP

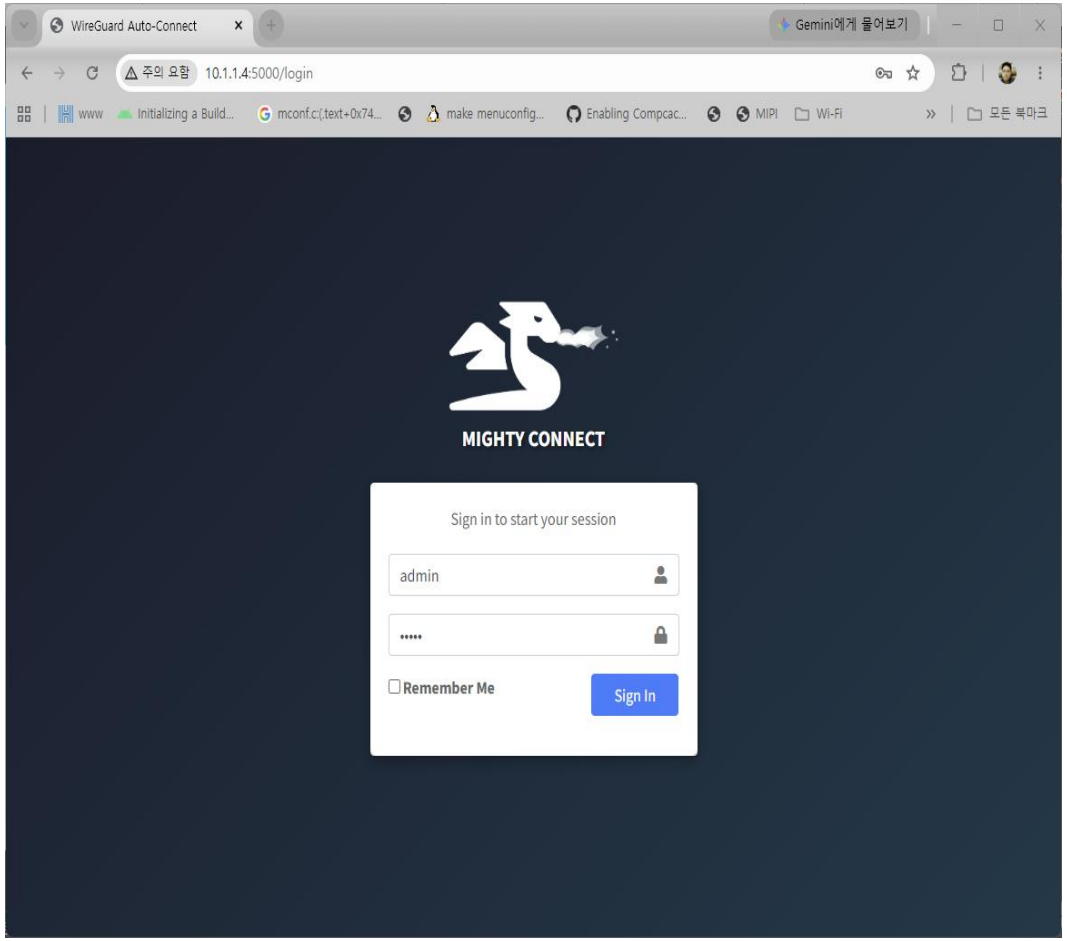


5.1 MightyConnect Mini(2) – Auto Connection

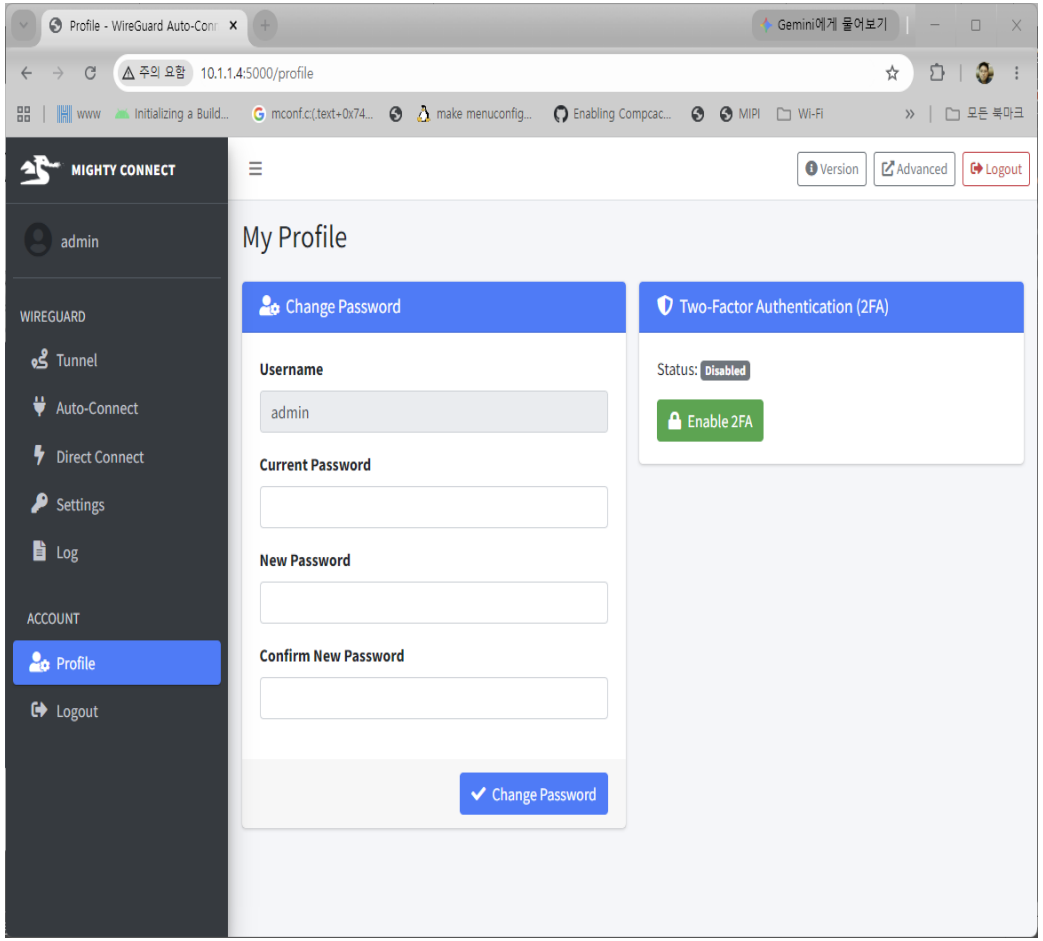


5.1 MightyConnect Mini(3) – WebUI(1)

- Login Page

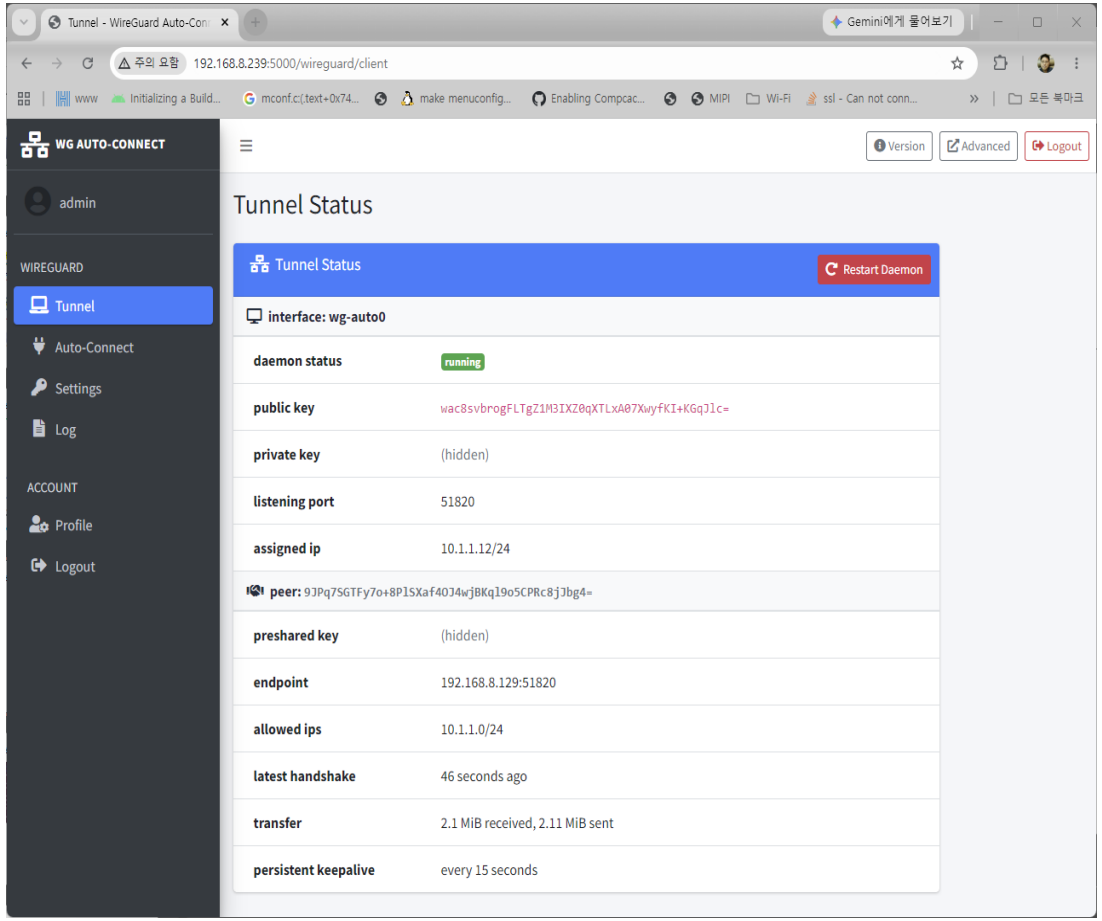


- Admin User Page

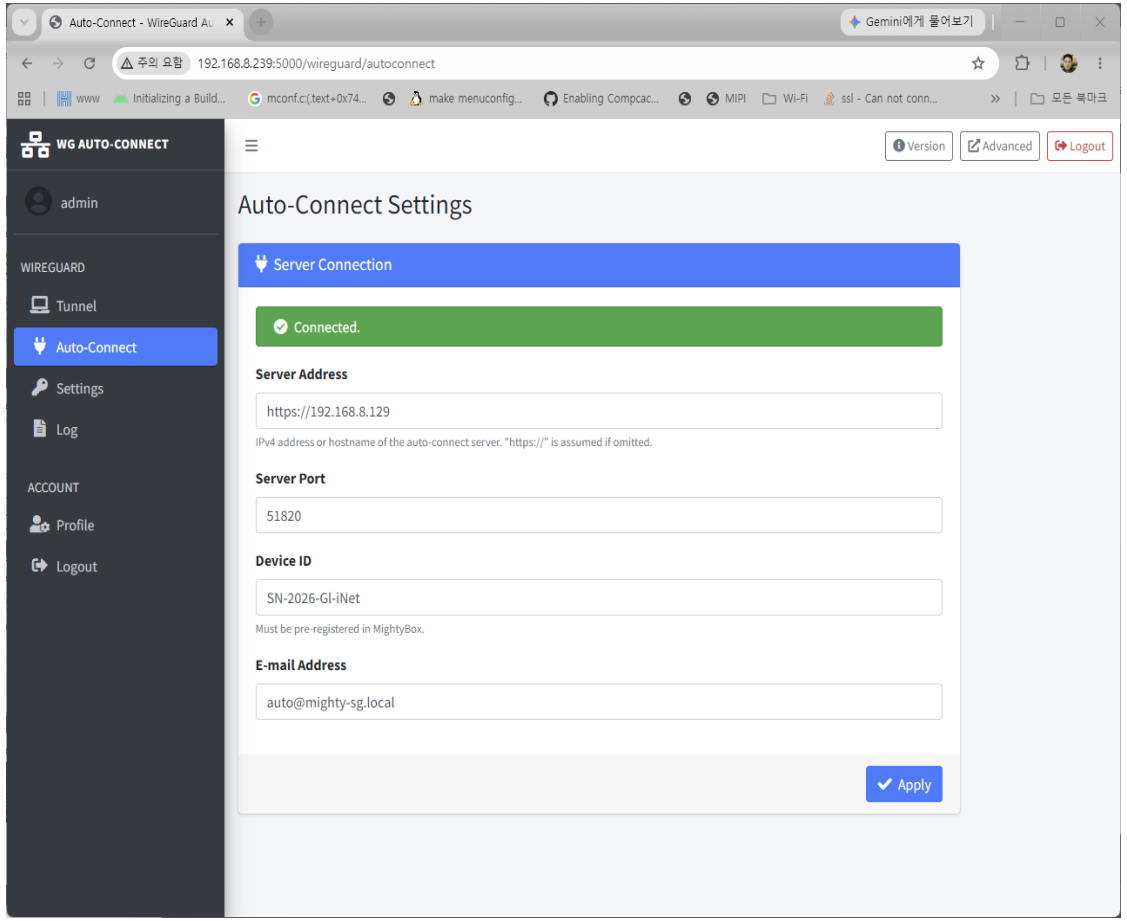


5.1 MightyConnect Mini(3) – WebUI(2)

- Tunnel Status Page

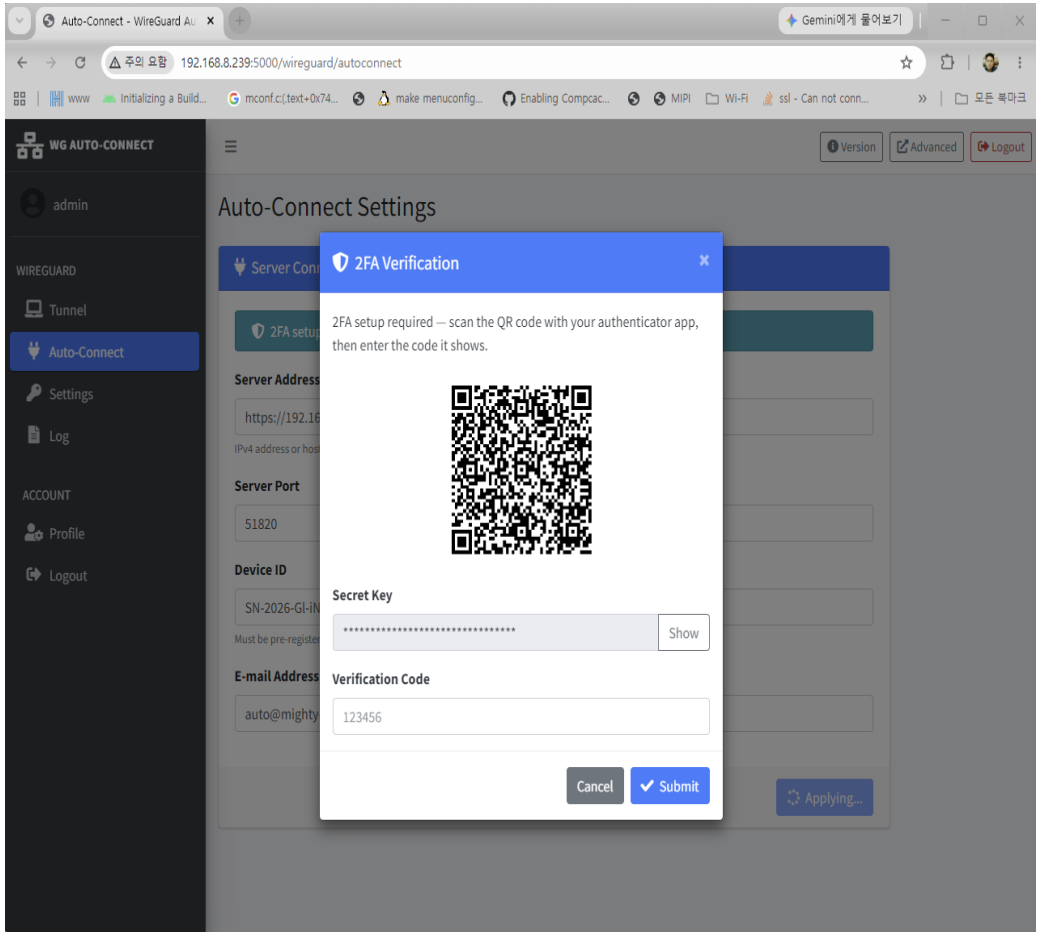


- Auto Connection Page

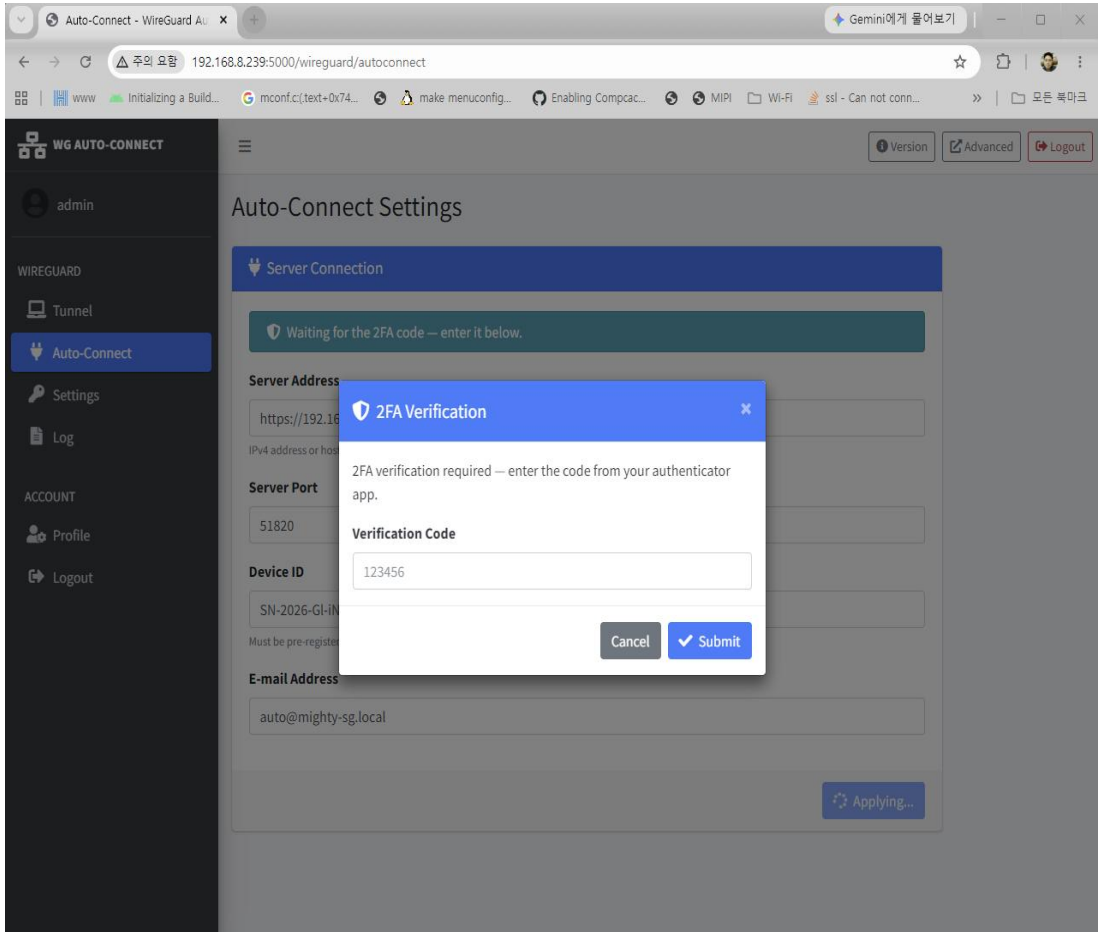


5.1 MightyConnect Mini(3) – WebUI(3)

- Auto Connection 2FA Step1 Page

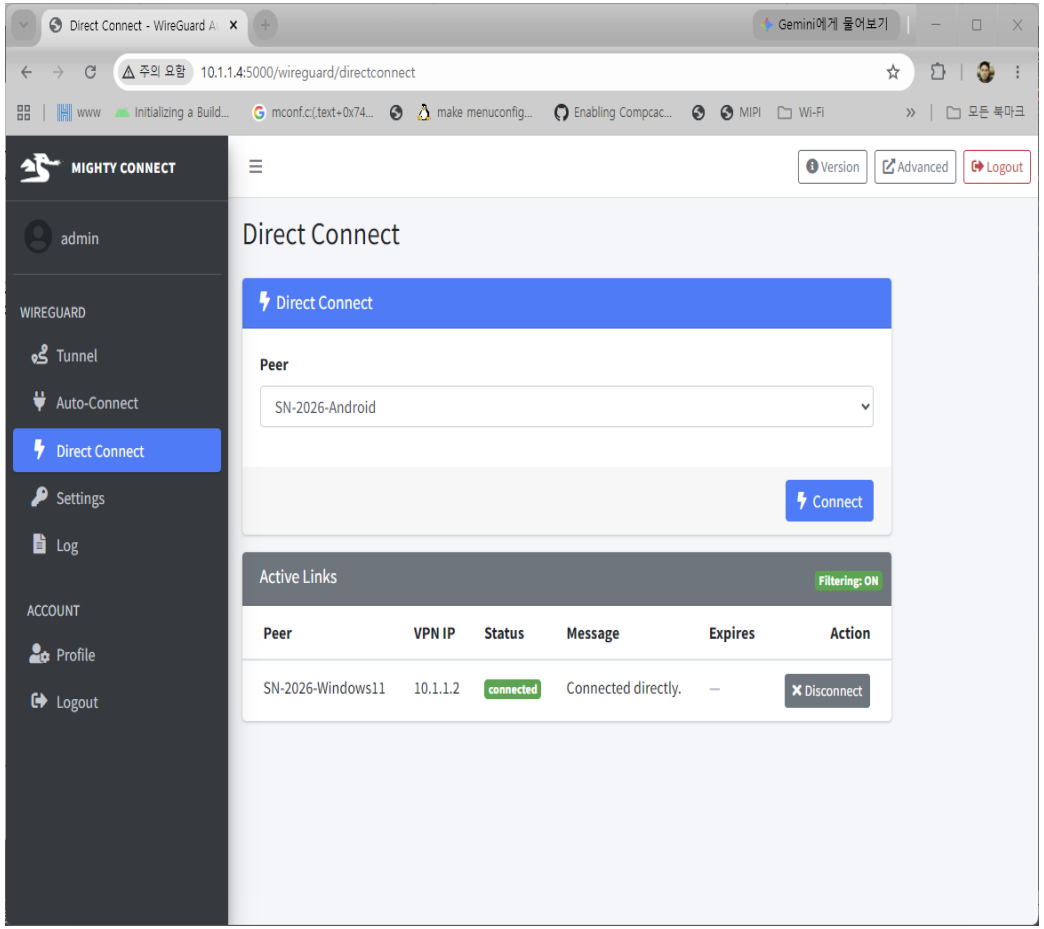


- Auto Connection 2FA Step2 Page

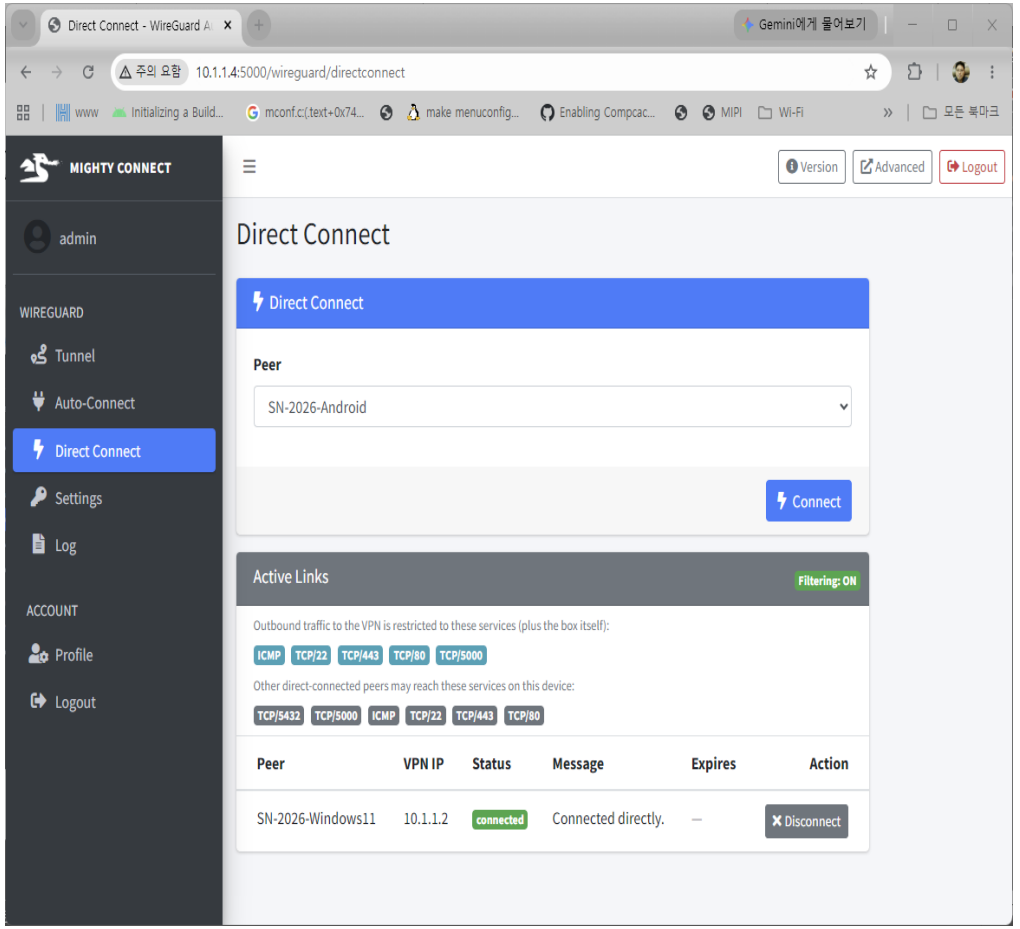


5.1 MightyConnect Mini(3) – WebUI(4)

- Direct Connect Page

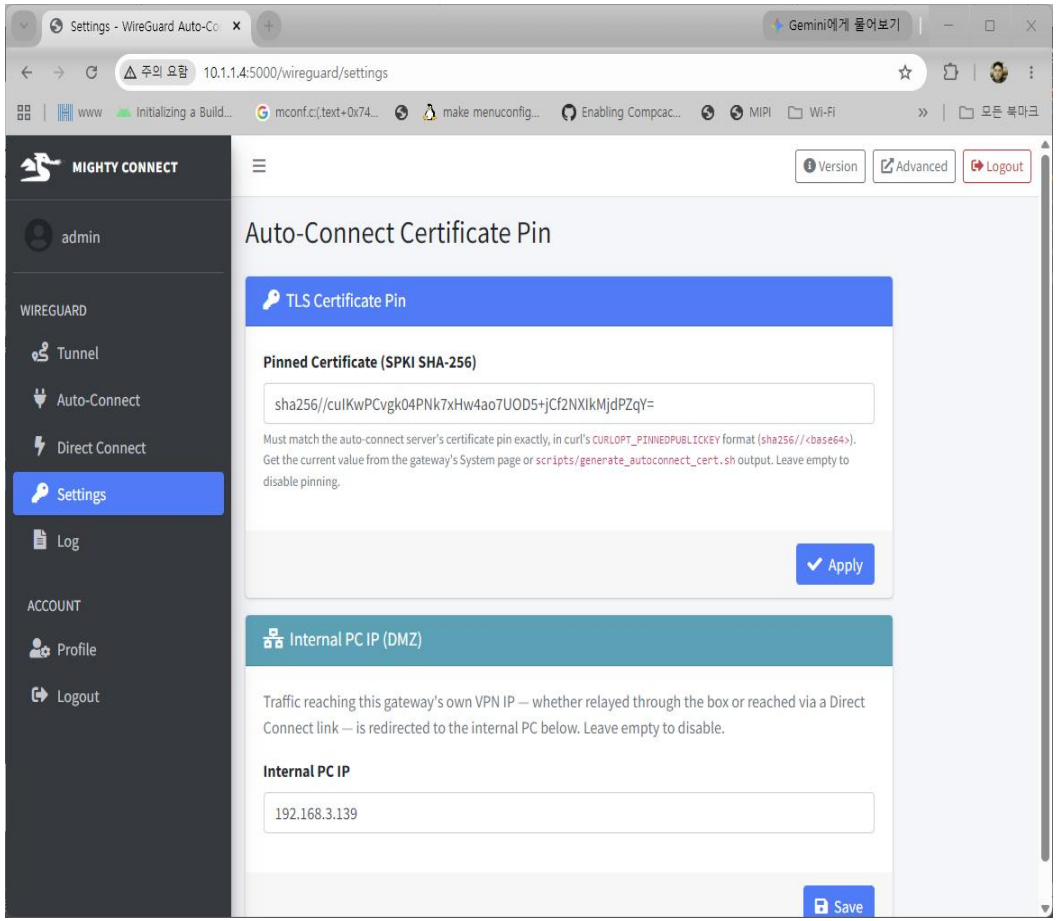


- Direct Connect Filter Rules

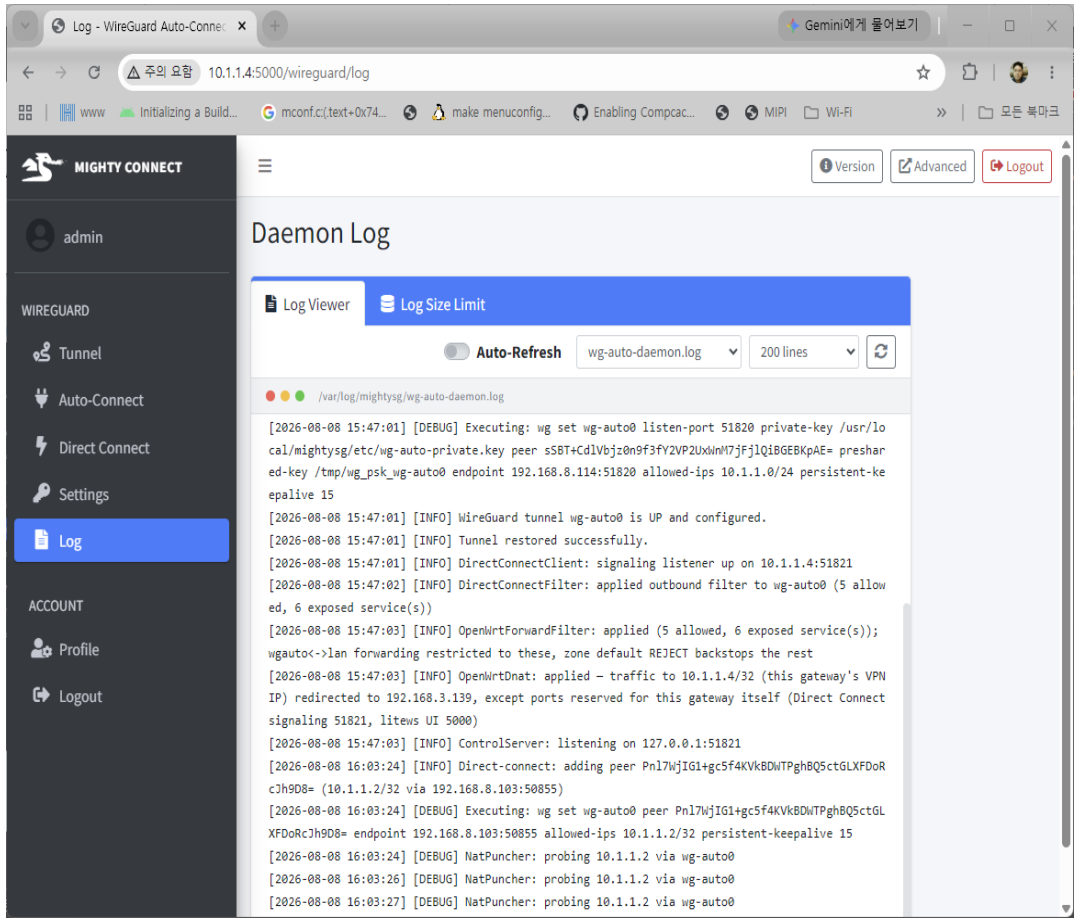


5.1 MightyConnect Mini(3) – WebUI(5)

- Settings Page



- Log Page

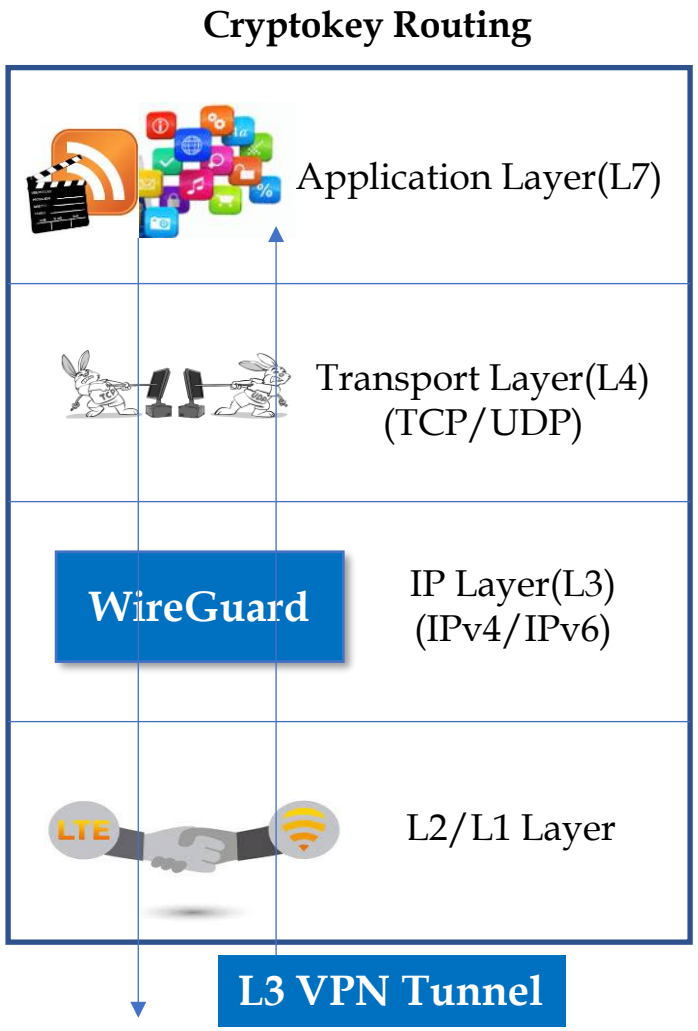


5.2 MightyConnect Ubuntu





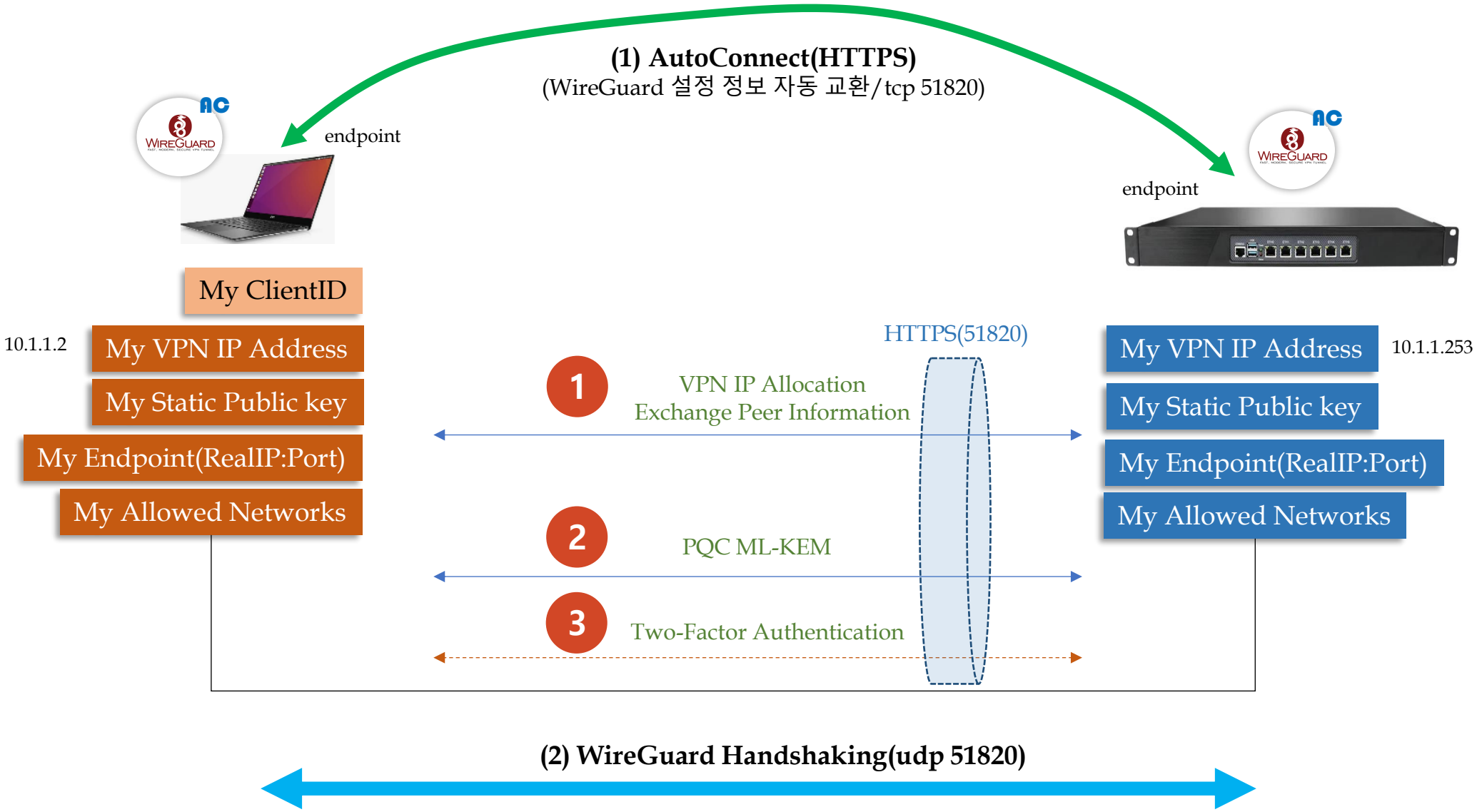
5.2 MightyConnect Ubuntu(1) – WireGuard




```
chyi@earth-new:/mnt/hdd/workspace/mighty_sg_prj/07162026$ sudo wg show
interface: wg-auto0
  public key: NYJM7SxhjeQXjjipIsEiNxZY3w6NcPJl/Mfjfye4mzY=
  private key: (hidden)
  listening port: 51820

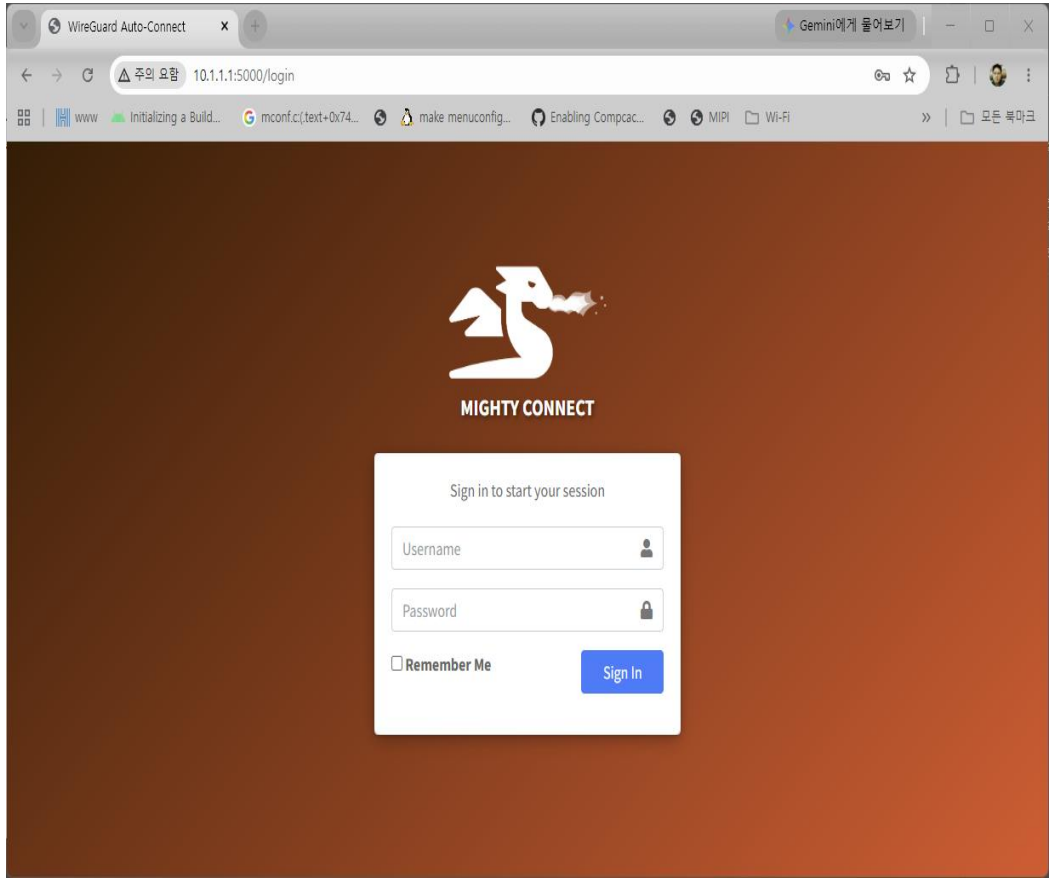
peer: 9JPq7SGTFy7o+8PlSXaf4OJ4wjBKql9o5CPRc8jJbg4=
  preshared key: (hidden)
  endpoint: 192.168.8.129:51820
  allowed ips: 10.1.1.0/24
  latest handshake: 1 minute, 40 seconds ago
  transfer: 316 B received, 180 B sent
  persistent keepalive: every 15 seconds
chyi@earth-new:/mnt/hdd/workspace/mighty_sg_prj/07162026$ ping 10.1.1.253
PING 10.1.1.253 (10.1.1.253) 56(84) bytes of data.
64 bytes from 10.1.1.253: icmp_seq=1 ttl=64 time=2.70 ms
64 bytes from 10.1.1.253: icmp_seq=2 ttl=64 time=2.94 ms
^C
--- 10.1.1.253 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1002ms
rtt min/avg/max/mdev = 2.698/2.817/2.936/0.119 ms
```

5.2 MightyConnect Ubuntu(2) – Auto Connection

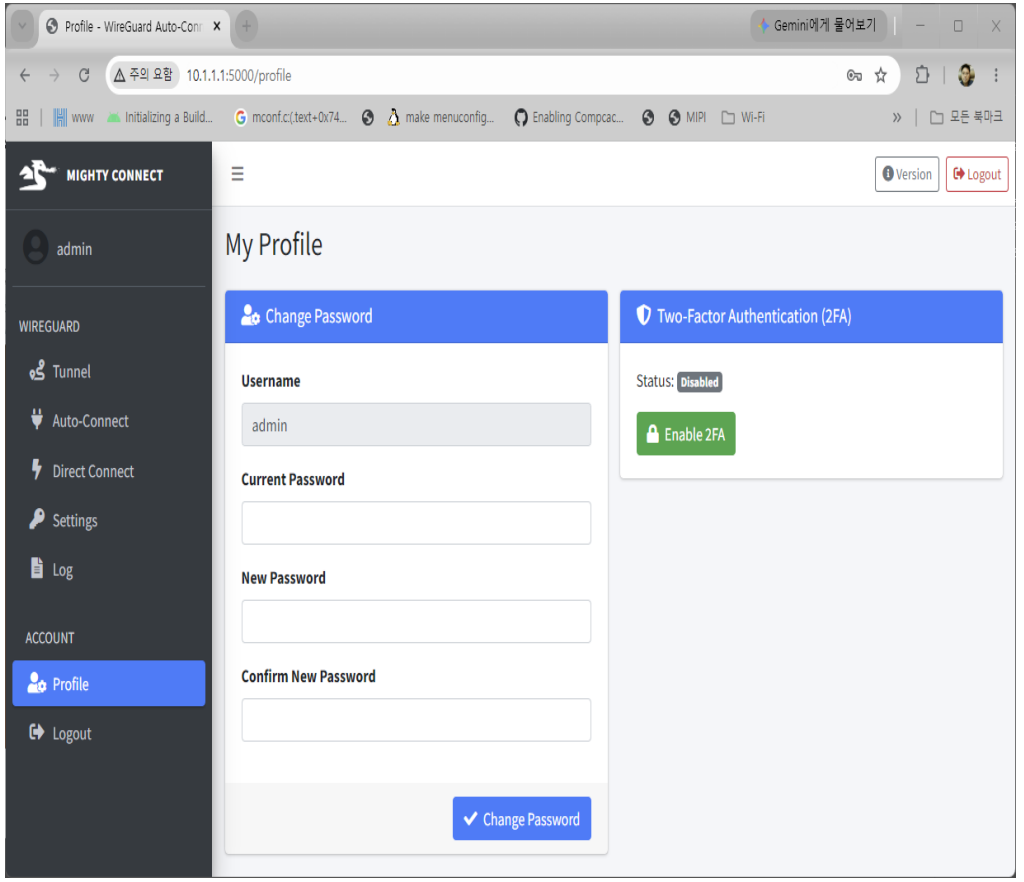


5.2 MightyConnect Ubuntu(3) – WebUI(1)

- Login Page

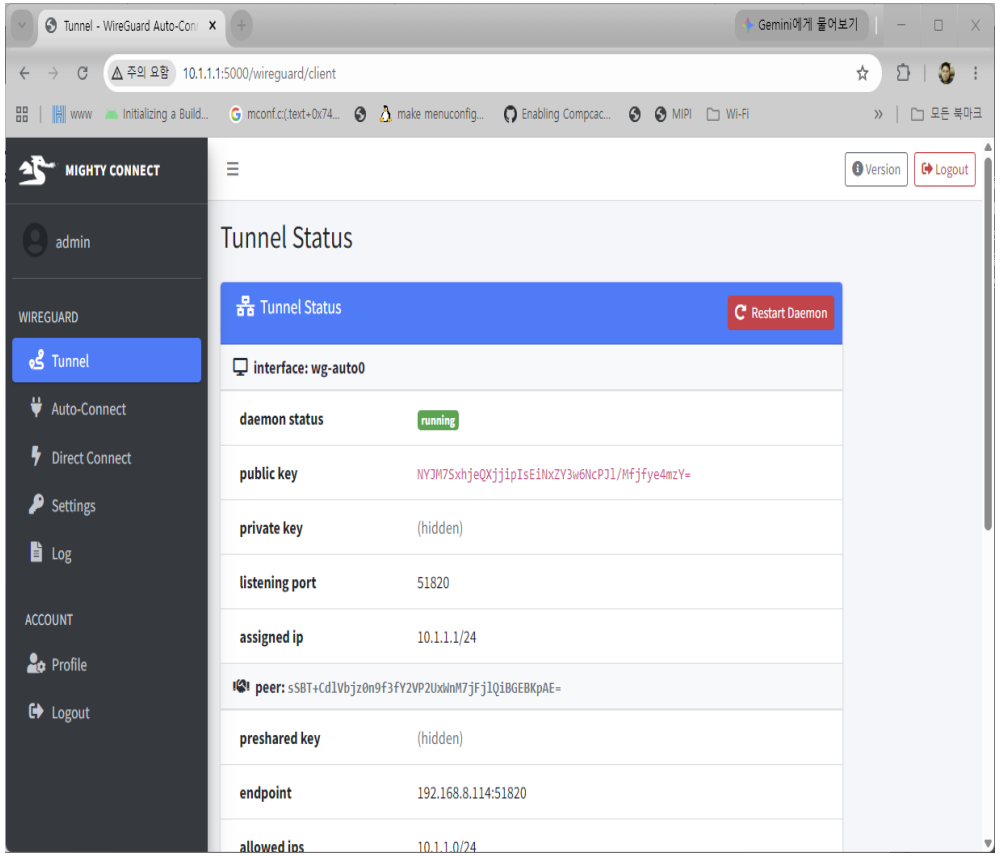


- Admin User Page

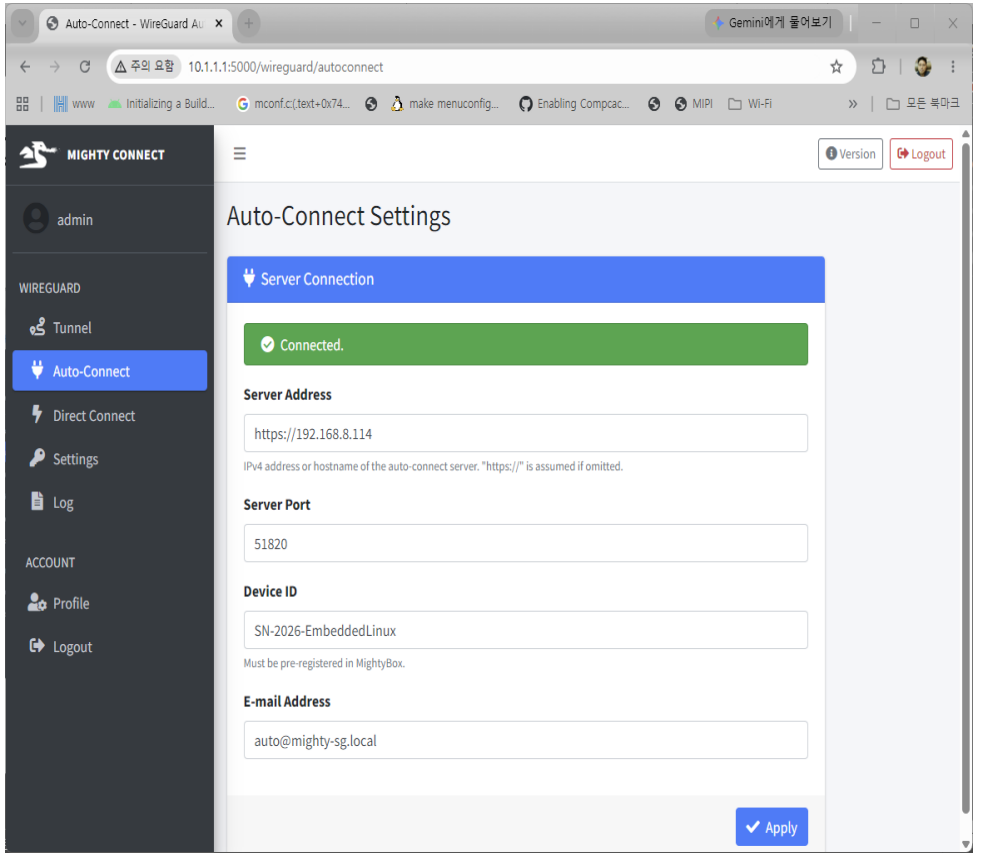


5.2 MightyConnect Ubuntu(3) – WebUI(2)

- Tunnel Status Page

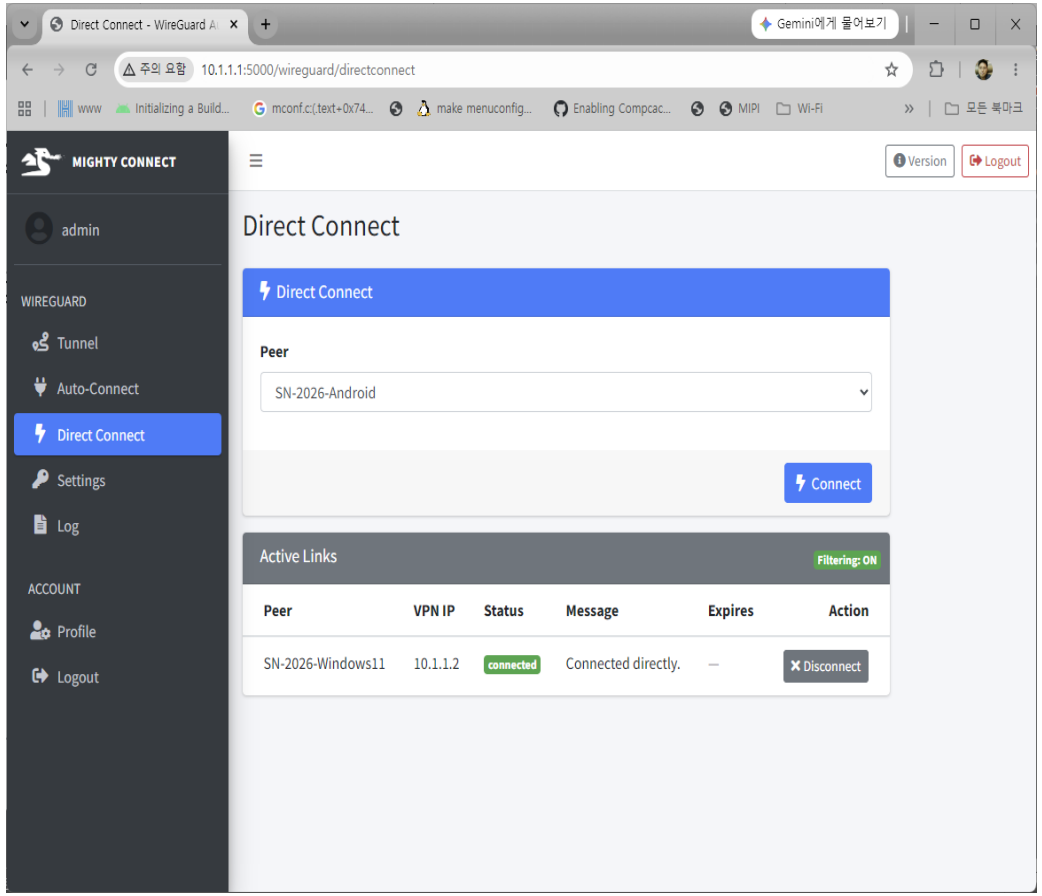


- Auto Connection Page

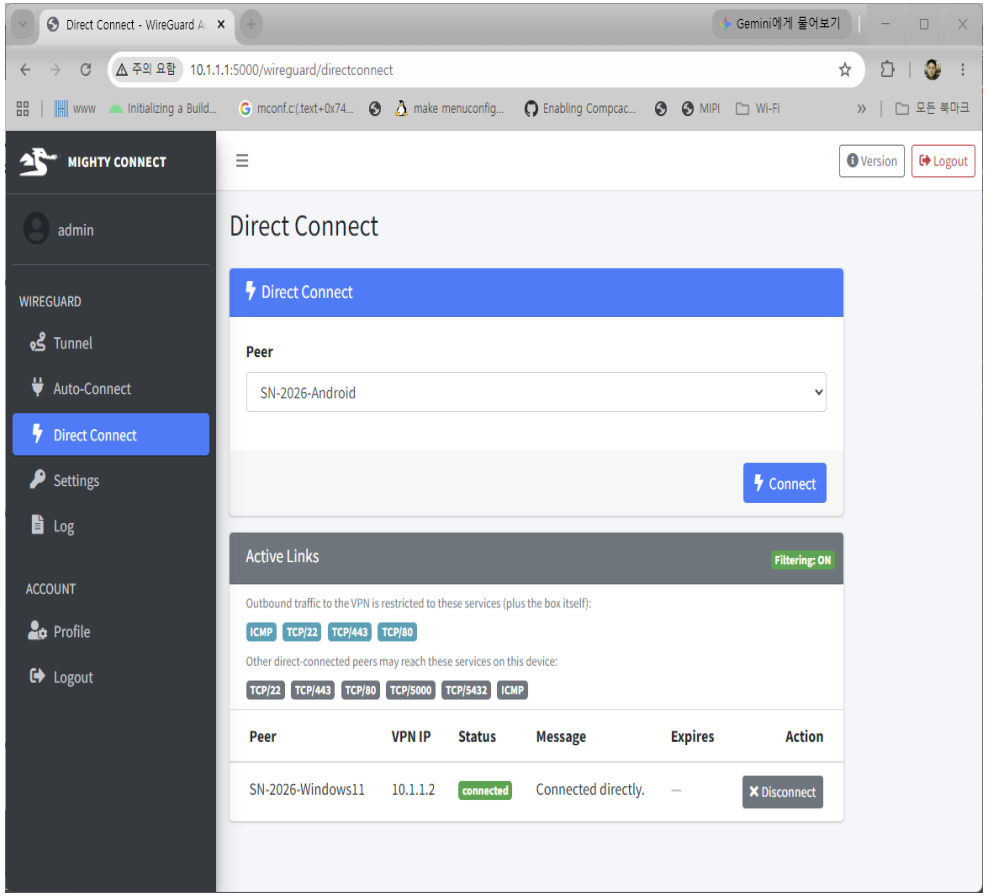


5.2 MightyConnect Ubuntu(3) – WebUI(3)

- Direct Connect Page



- Direct Connect Filtering



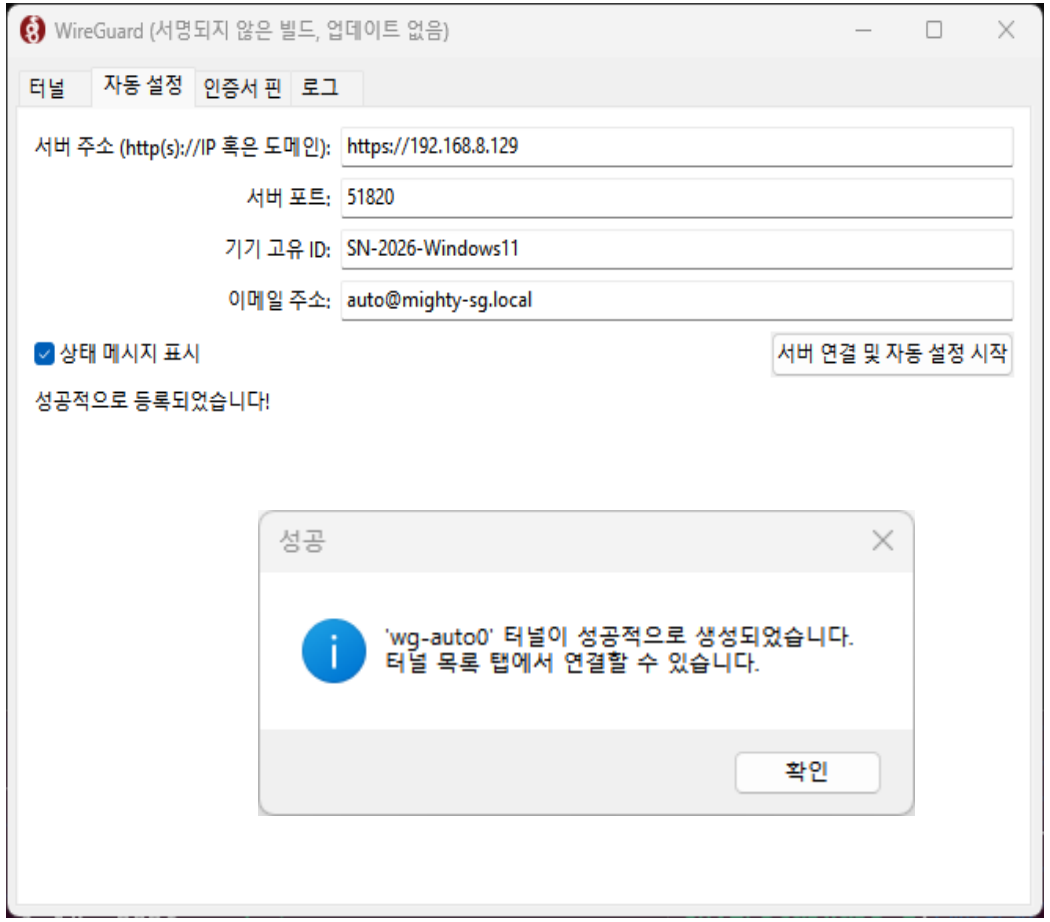
5.3 MightyConnect Windows

MightyConnect Windows Agent

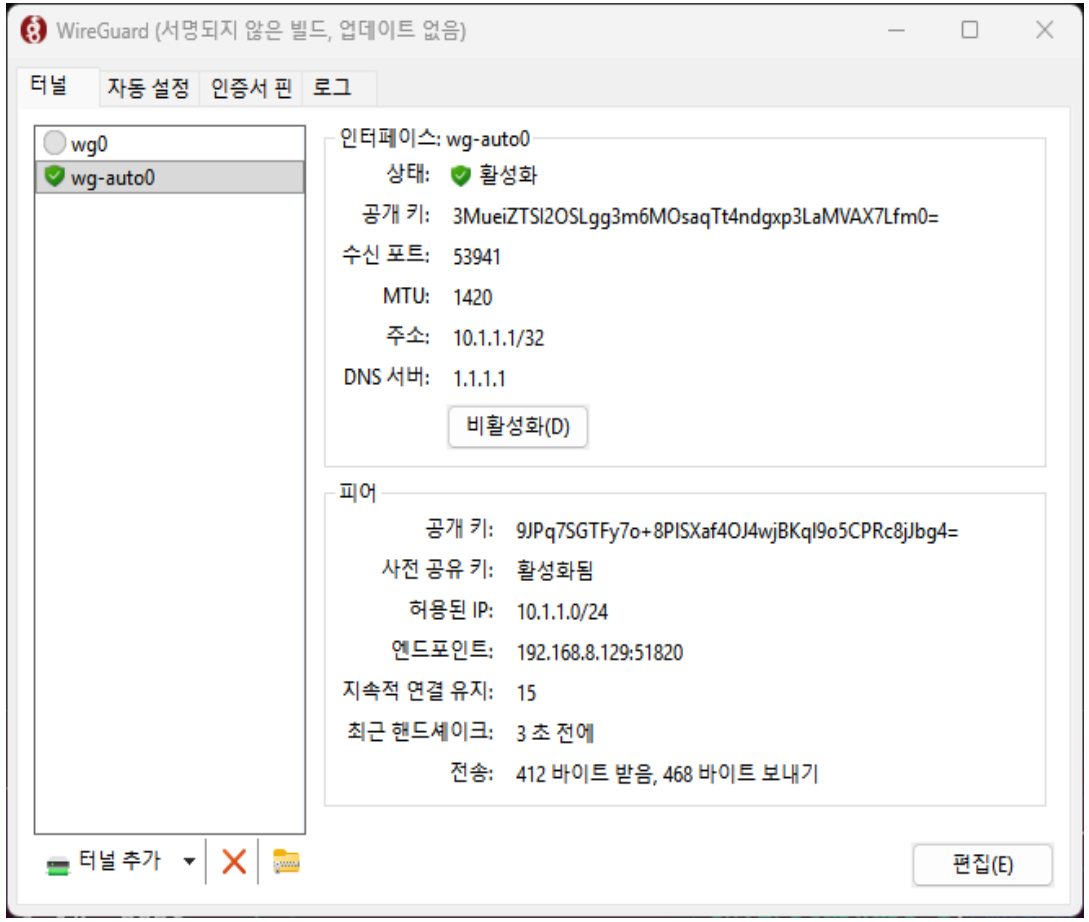


5.3 MightyConnect Windows(1) – User Interface(1)

- Auto Connection Page

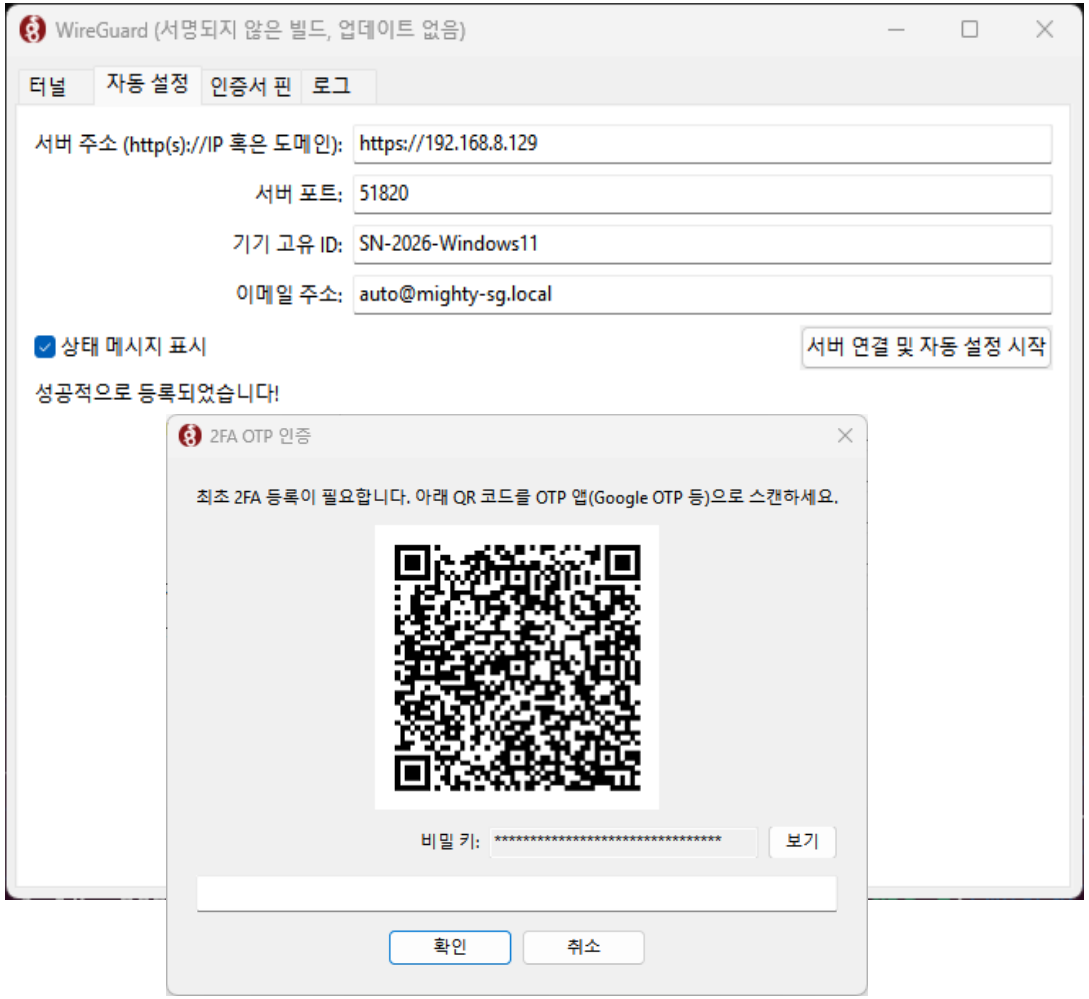


- wg-auto0 interface 생성 및 터널 활성화 Page

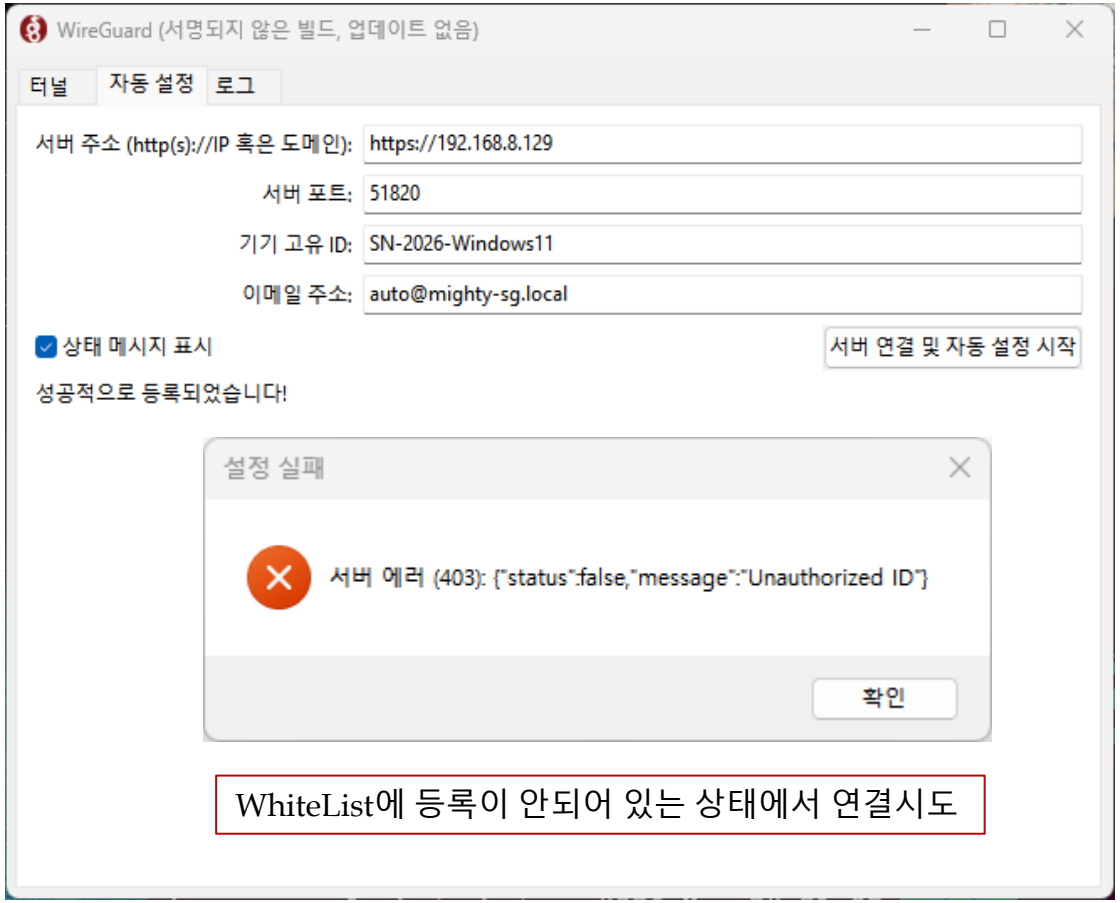


5.3 MightyConnect Windows(1) – User Interface(2)

- Auto Connection – 2FA 인증 Page



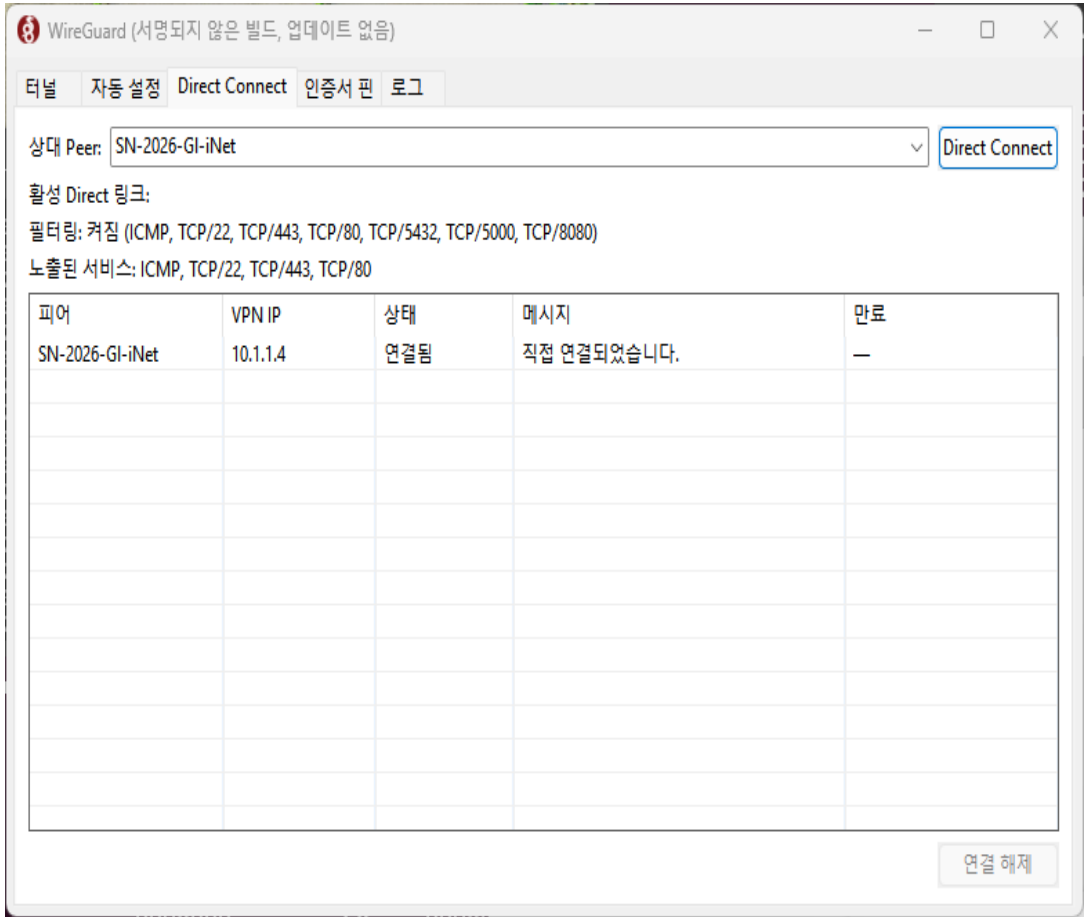
- Auto Connection – 인증 실패 예시



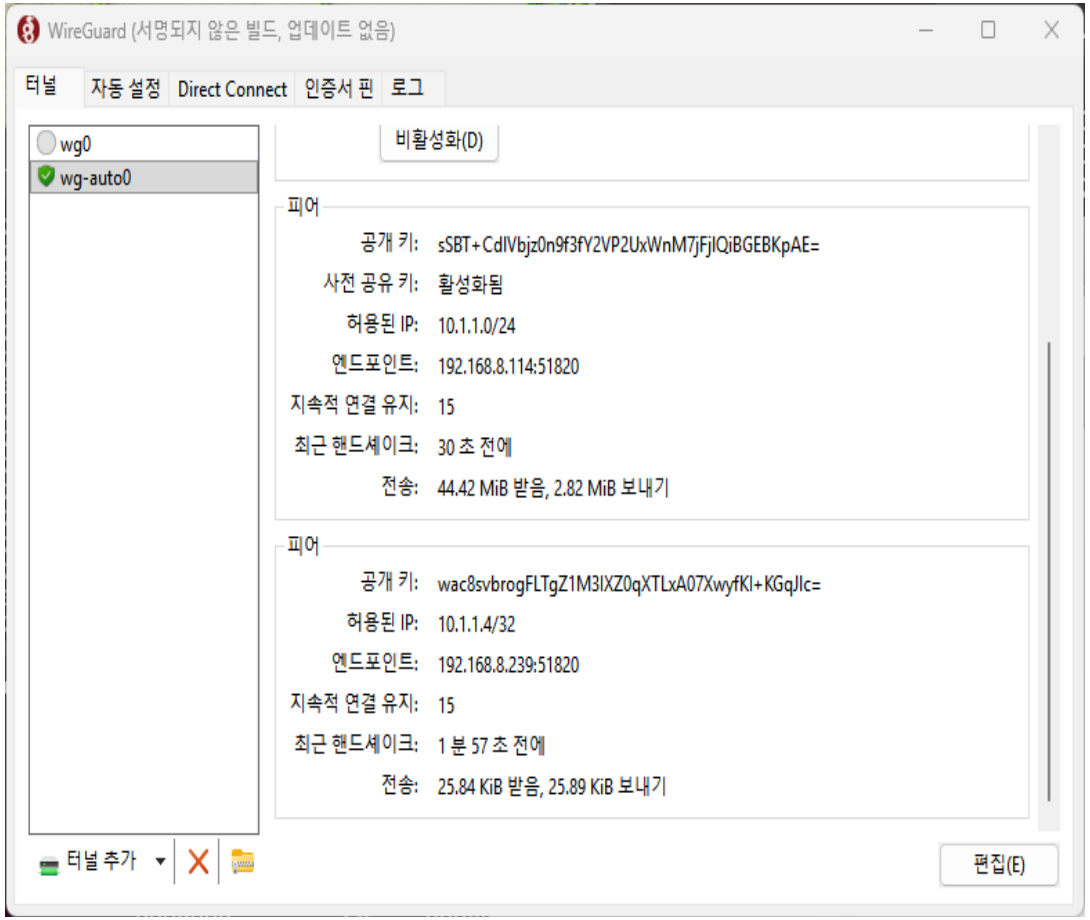
WhiteList에 등록이 안되어 있는 상태에서 연결시도

5.3 MightyConnect Windows(1) – User Interface(3)

- Direct Connect Page

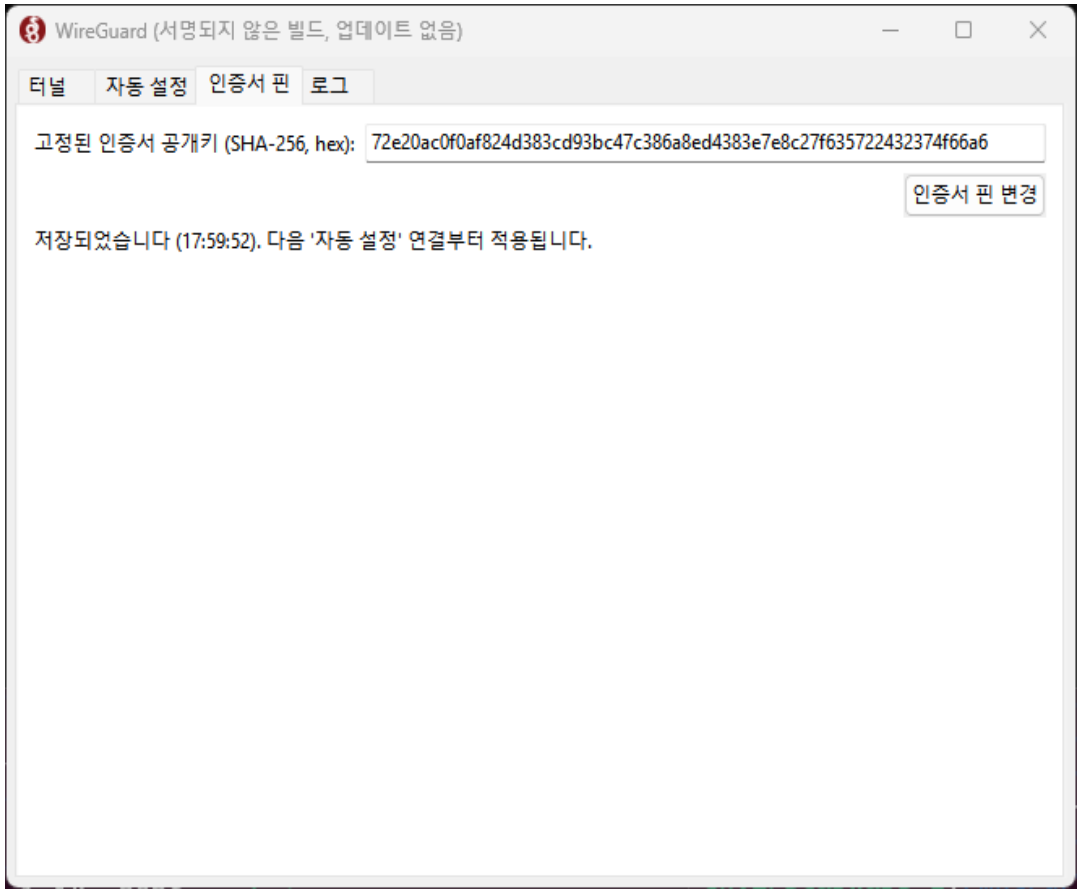


- Direct Connect Peer 확인

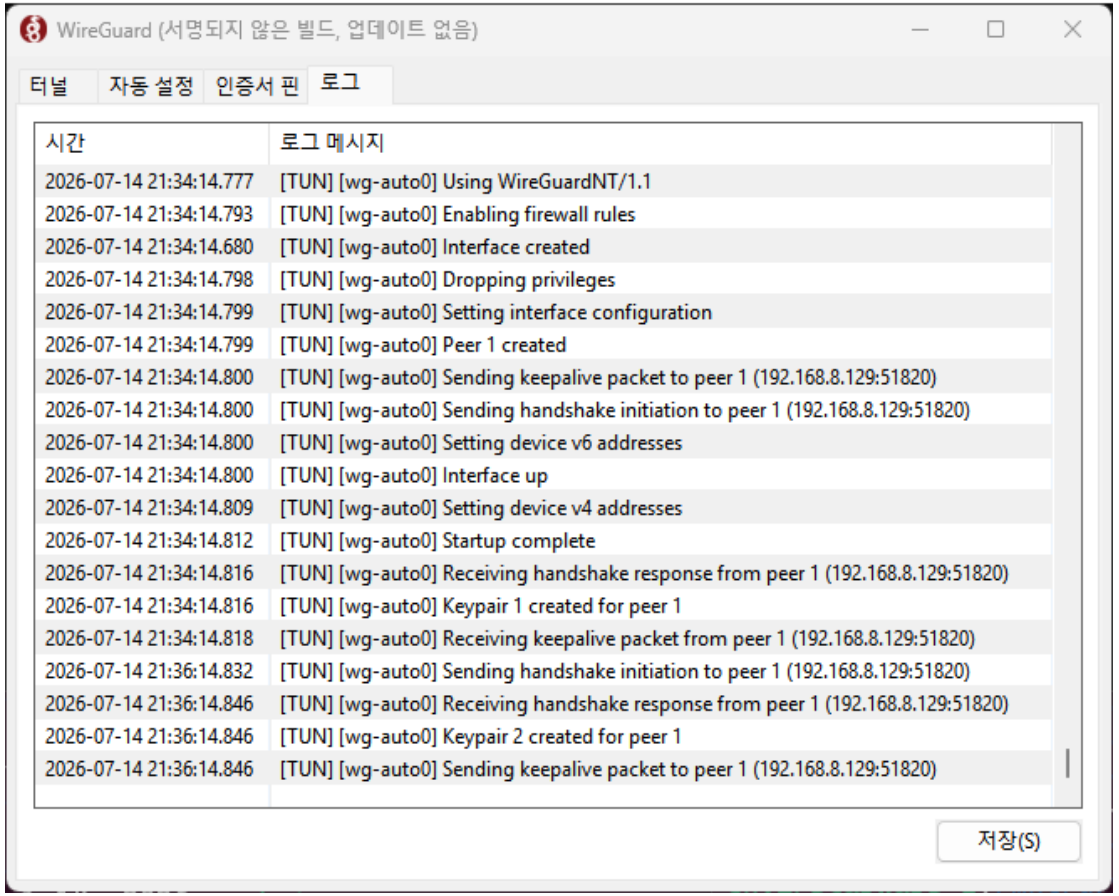


5.3 MightyConnect Windows(1) – User Interface(4)

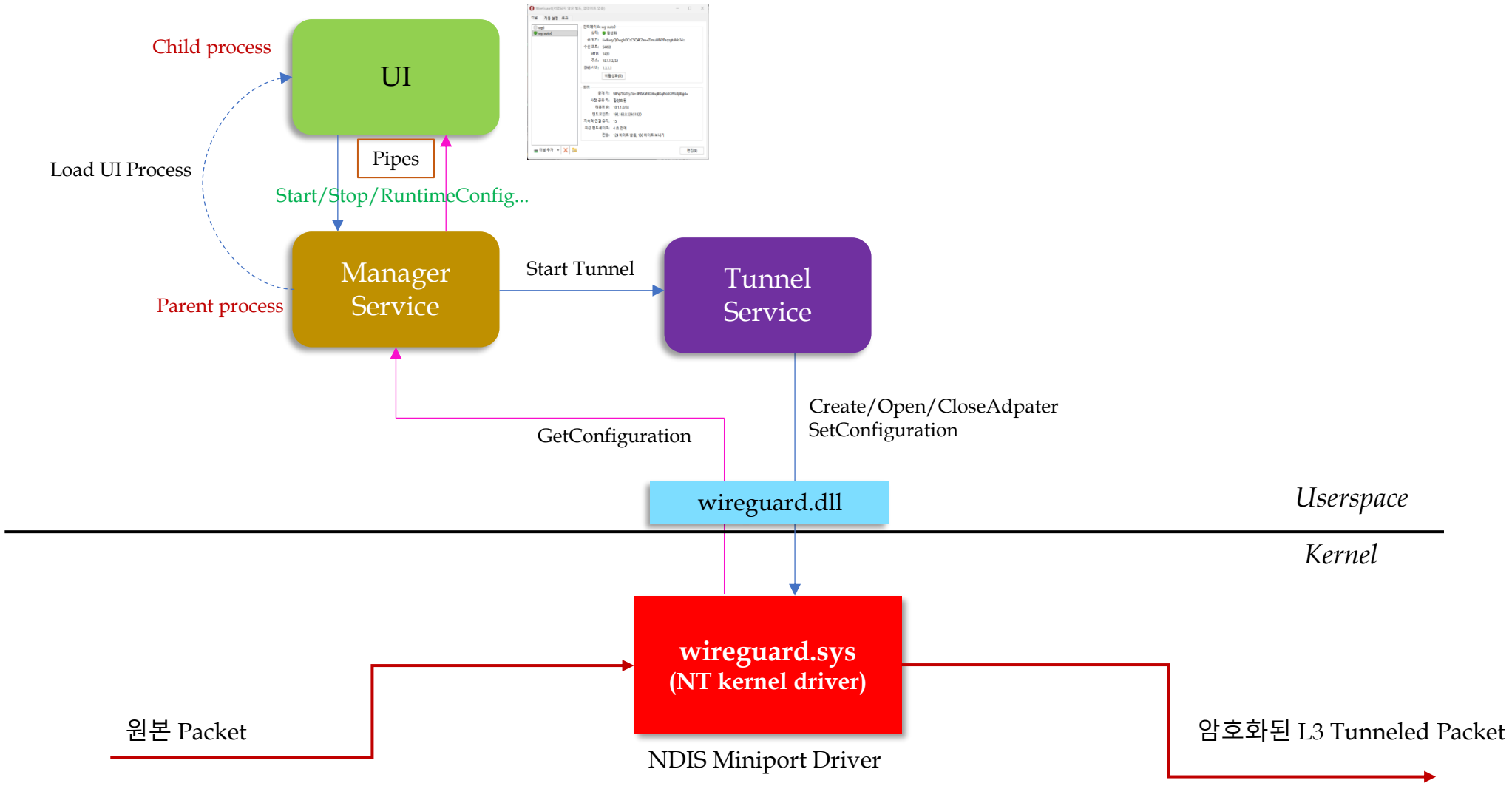
- X.509 인증서 Pin 값 변경 Page



- Log Page



5.3 MightyConnect Windows(2) – Architecture

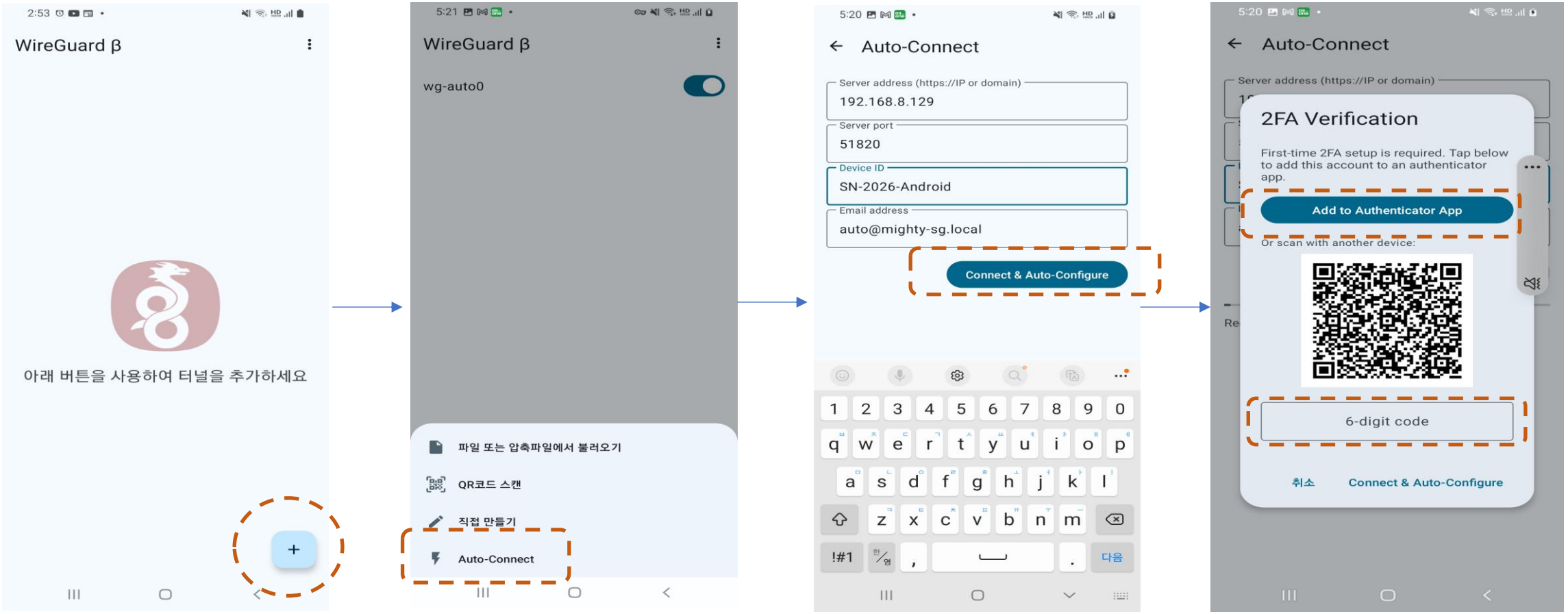


5.4 MightyConnect Android

Android App

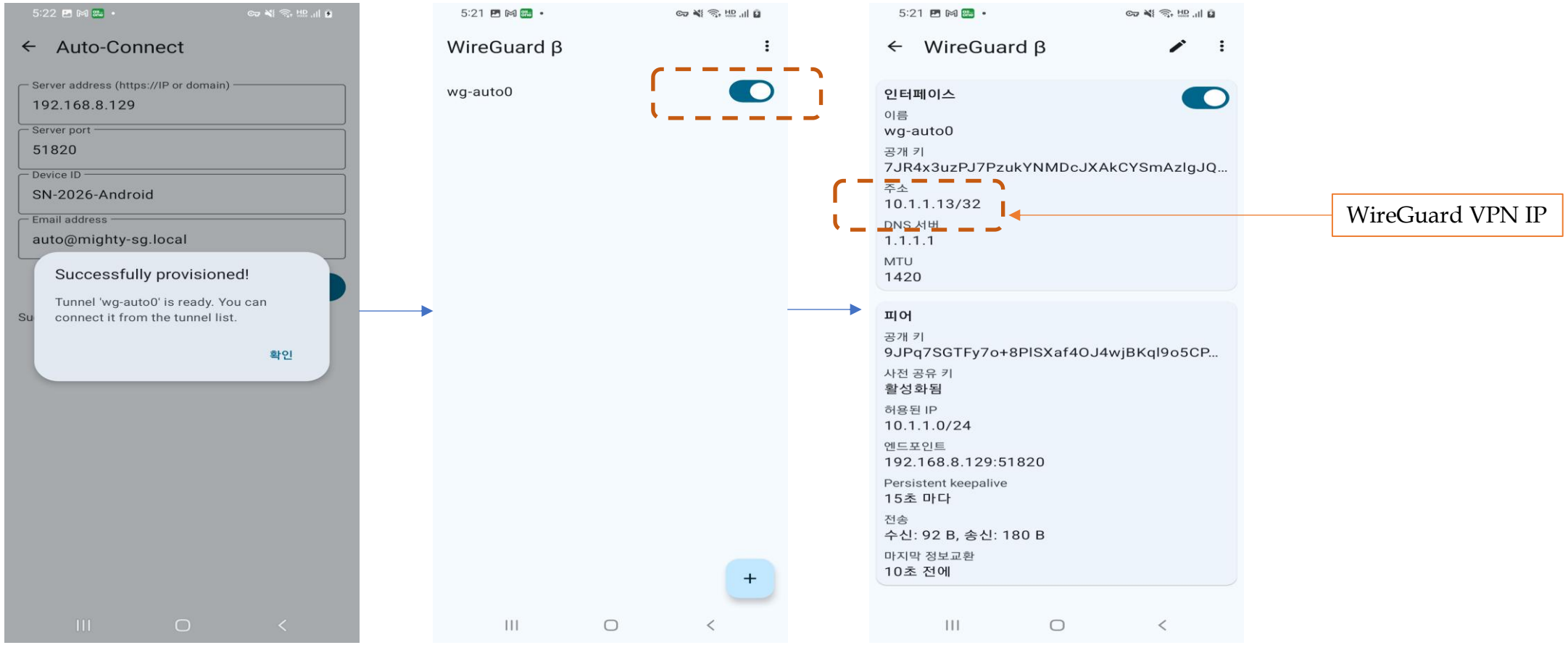


5.4 MightyConnect Android(1) – Auto Connect(1)

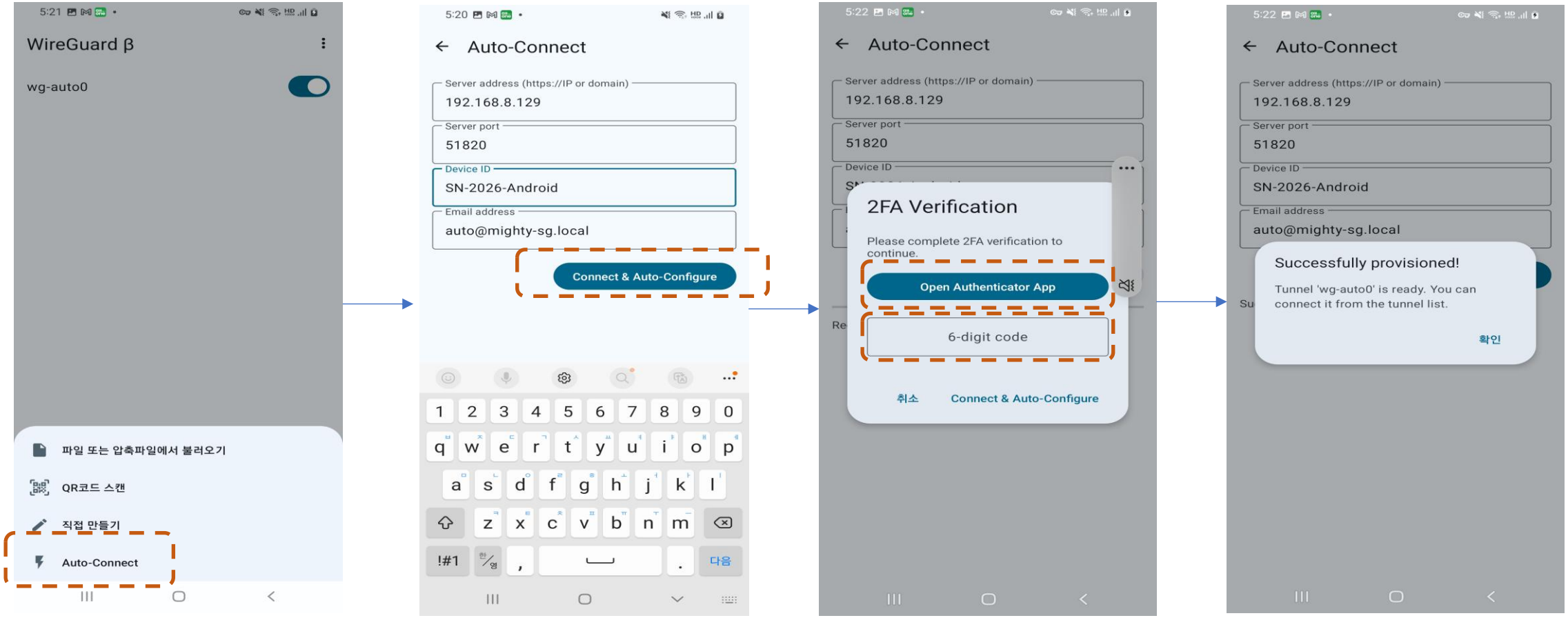


터널 추가 버튼

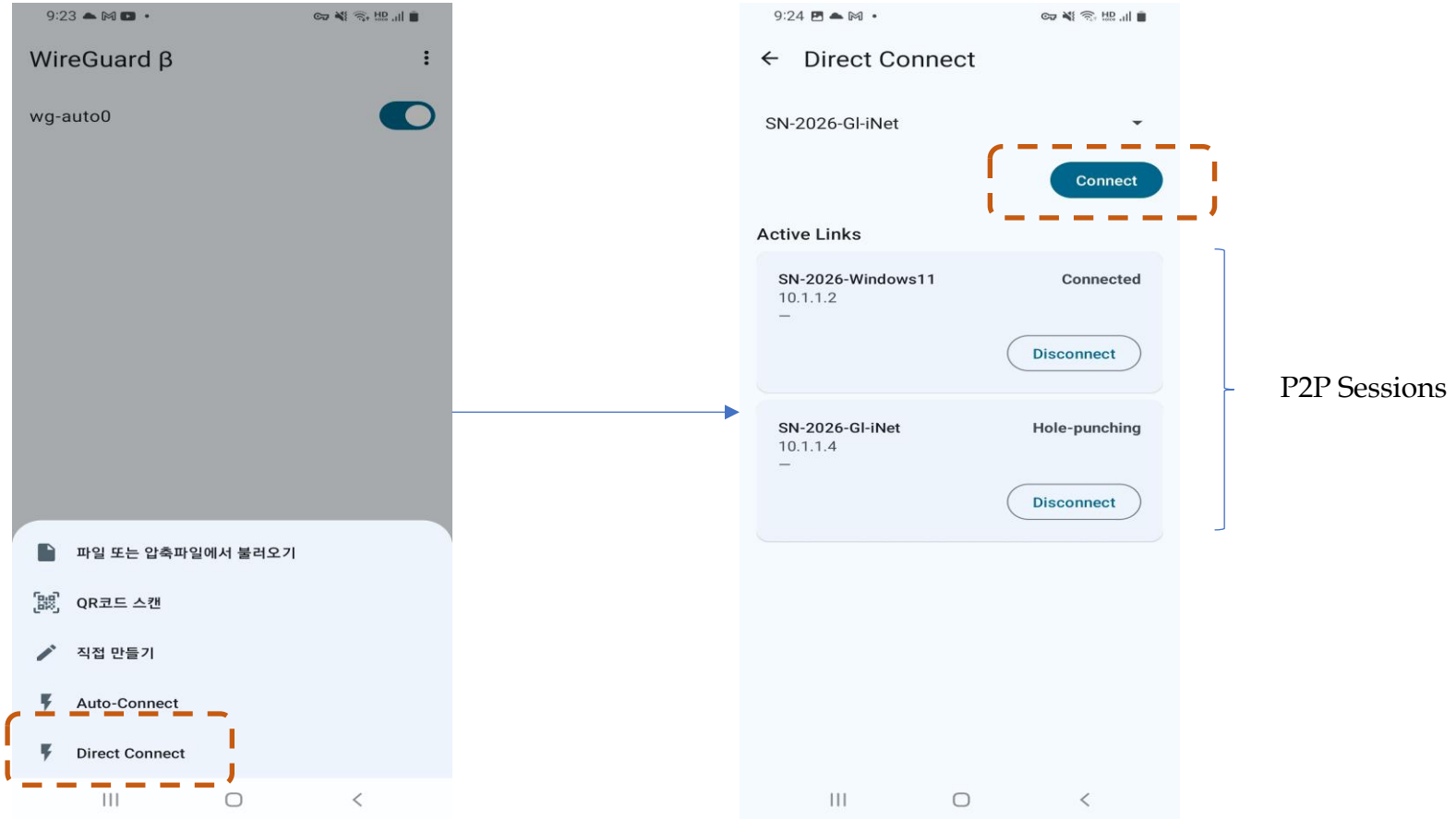
5.4 MightyConnect Android(1) – Auto Connect(2)



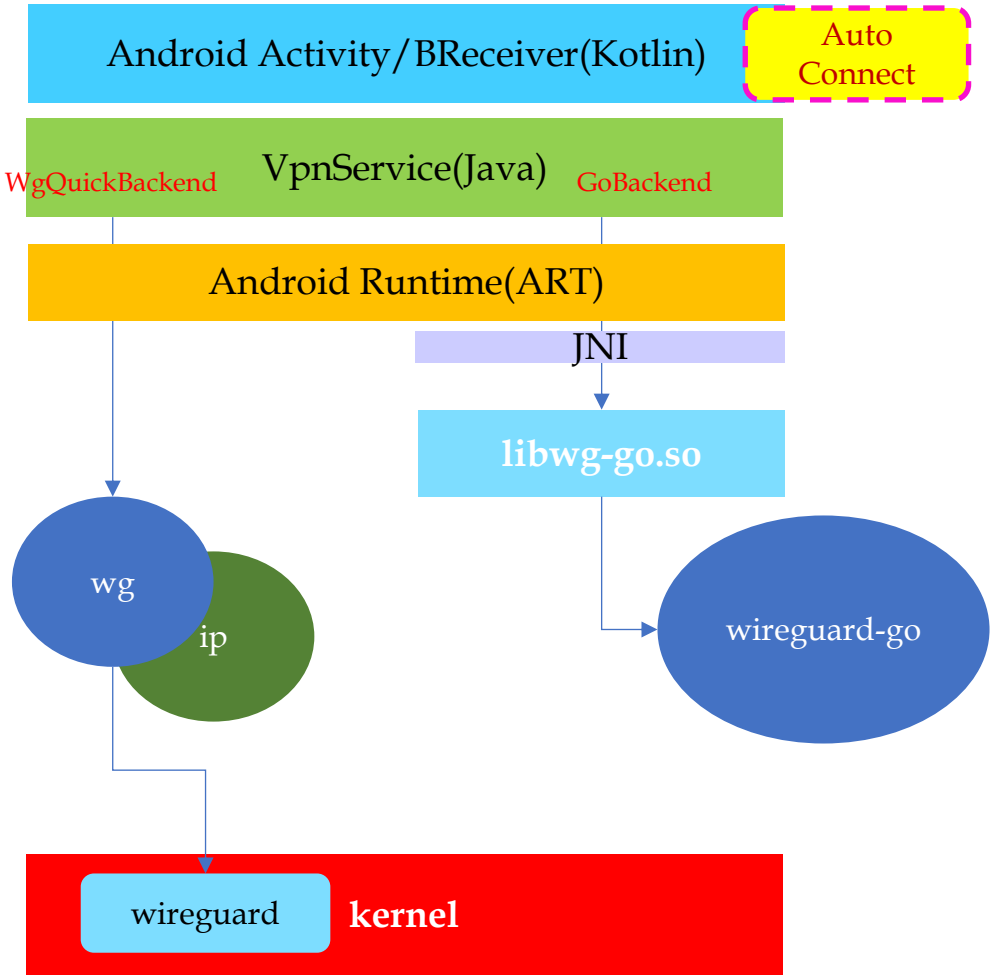
5.4 MightyConnect Android(1) – Auto Connect(3)

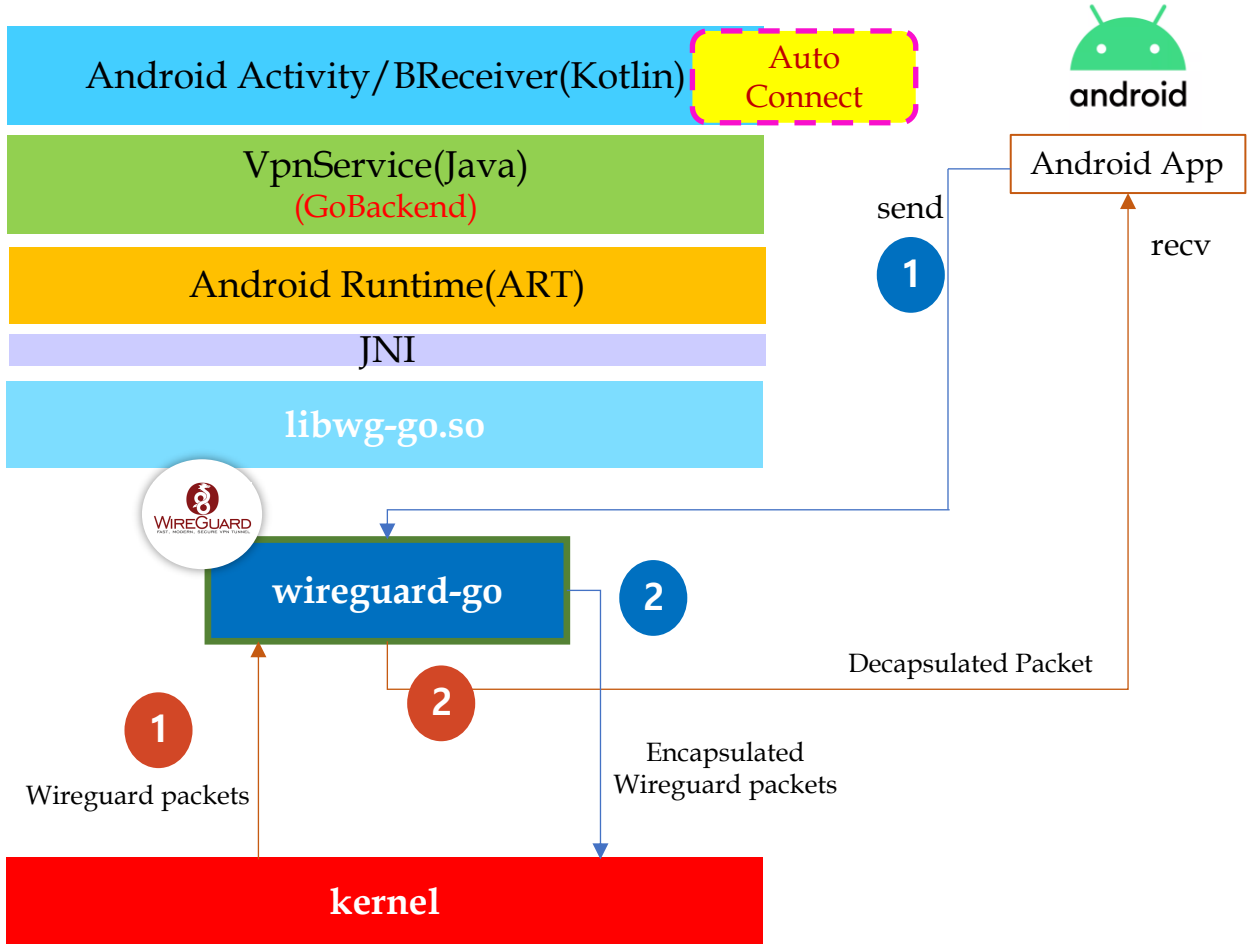


5.4 MightyConnect Android(2) – Direct Connect



5.4 MightyConnect Android(3) – Architecture



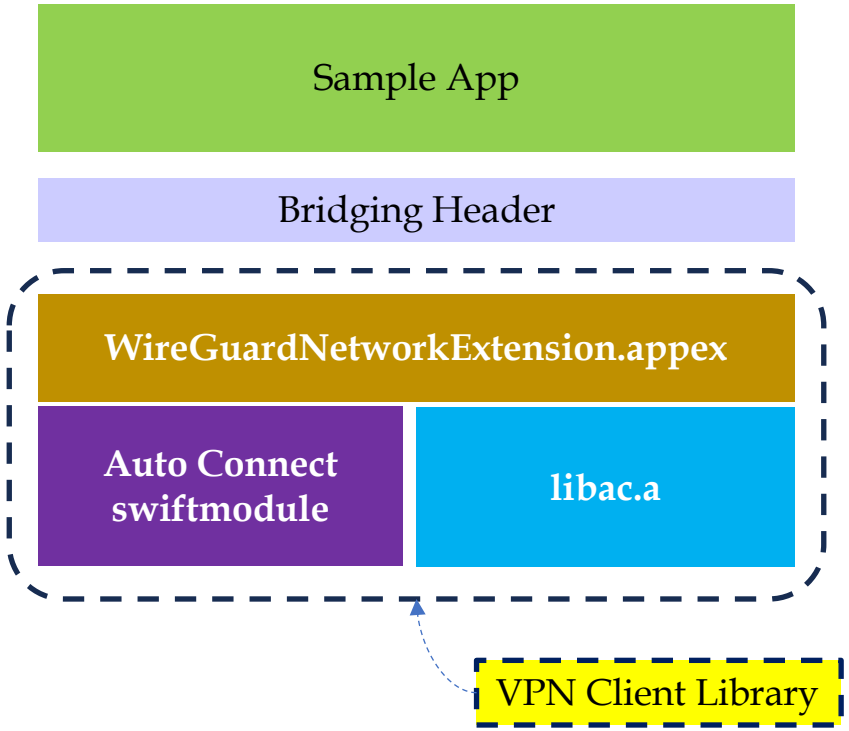


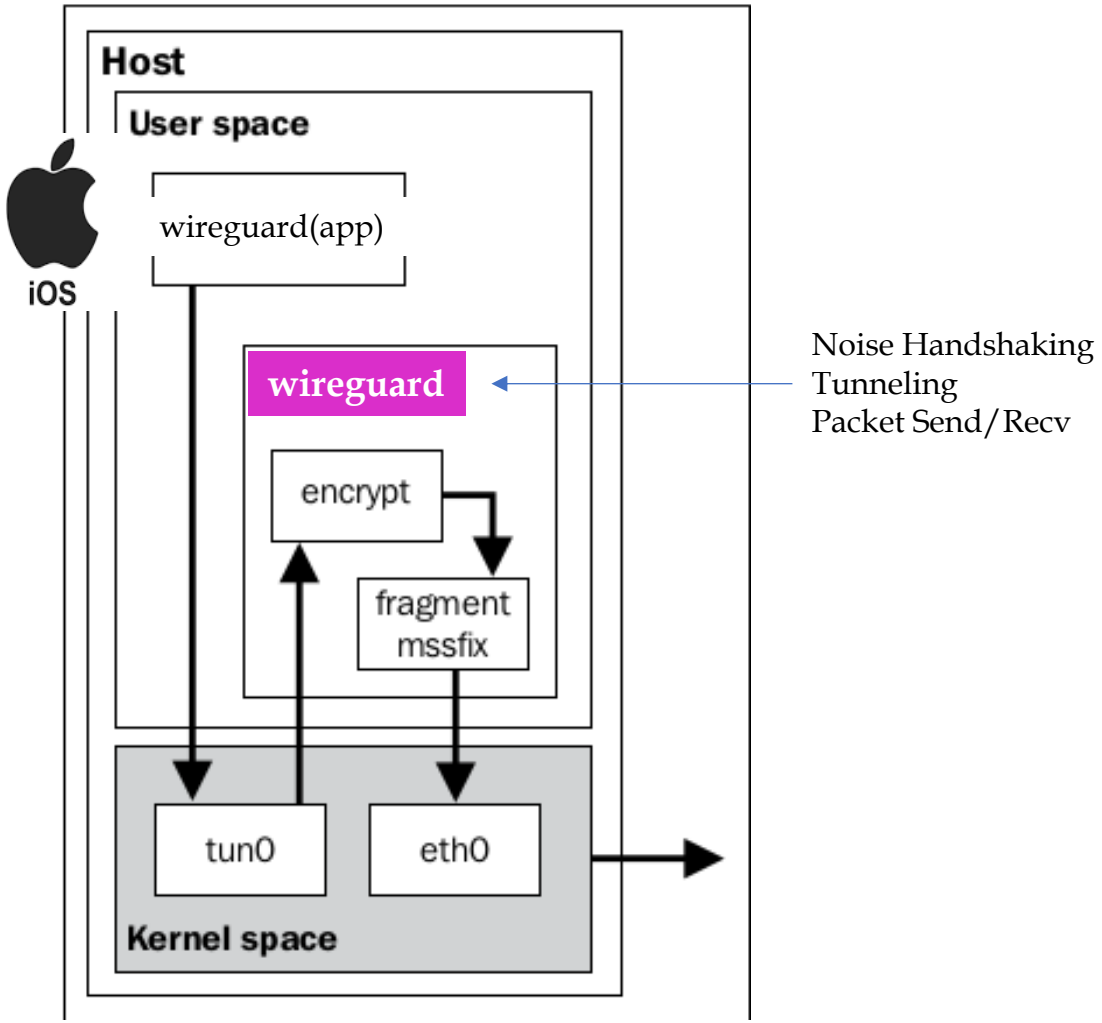
5.5 MightyConnect macOS/iOS

MightyConnect macOS/iOS Agent



5.5 MightyConnect macOS/iOS – Architecture





6. References

6. References

- [1] https://www.wireguard.com
- [2] https://fd.io/
- [3] https://www.dpdk.org/
- [4] https://www.gl-inet.com/collections/all-products
- [5] https://www.friendlyelec.com/
- [6] https://ferrumgate.com/
- [7] https://tailscale.com/blog/more-throughput
- [8] https://media.frnog.org/FRnOG_28/FRnOG_28-3.pdf
- [9] Performance Comparison of VPN Solutions, Lukas Osswald
- [10] Impressive Packet Processing Performance Enables Greater Workload Consolidation
- [11] https://www.nvidia.com/ko-kr/networking/ethernet/connectx-6/
- [12] https://www.netgate.com/resources/articles-vector-packet-processing
- [13] https://docs.ngkore.com/blogs/dpdk/dpdk.html#hardware-requirements
- [14] Windows is a registered trademark of Microsoft.
- [15] Android is a registered trademark of Google.
- [16] macOS & iOS is a registered trademark of Apple.
- [17] Ubuntu is a registered trademark of Canonical Ltd.
- [18] And Google~

Faster Connectivity, Flawless Security

©opyright 2026 MightyLink Lab., All rights reserved.

